Bases scientifiques pour la transition énergétique

École Nationale des Ponts et Chaussées
Département de première année

Laurent Brochard

2026

Contents

Syllabus **7**

1 Énergie, ressources et climat : la nécessité d'une transition **11**
Foreword . . . 11
1.1 What is energy? . . . 11
1.1.1 Noether's theorem . . . 11
1.1.2 Fundamental forces . . . 13
1.1.3 Energy-mass, dark energy, dark matter . . . 14
1.2 The usable forms of energy . . . 15
1.2.1 Scales of energy in matter . . . 15
1.2.2 Classification . . . 17
1.2.3 Some orders of magnitude . . . 18
1.3 Exploitation of energy . . . 19
1.3.1 Energy transformations . . . 19
1.3.2 Exploitation by humans . . . 20
Recap of what must be known for the exam . . . 22
Exercises . . . 24
Ex. 1.1 - Pertes de charges . . . 24

2 Rayonnement **27**
Foreword . . . 27
2.1 Basic thermodynamics of solar energy . . . 28
2.1.1 Blackbody radiation . . . 28
2.1.2 Rough estimate of the solar resource . . . 32
2.1.3 A more accurate approach required for concentration . . . 34
2.2 Solar thermal energy . . . 35
2.2.1 Turning solar radiation into heat . . . 35
2.2.2 Turning solar radiation into work . . . 36
2.3 Solar photovoltaic . . . 37
2.3.1 Short introduction to semi-conductor physics . . . 37
2.3.2 The Trivich-Finn and Shockley-Queisser limits . . . 38
2.3.3 From Shockley-Queisser to real devices . . . 40
2.3.4 Solar cells today . . . 41
Recap of what must be known for the exam . . . 42
Exercises . . . 45
Ex. 2.1 - Panneau solaire thermique . . . 45
Ex. 2.2 - Limite de Shockley-Queisser . . . 46
Ex. 2.3 - Forçage radiatif du CO_2 et réchauffement . . . 49

3 Énergie nucléaire **53**
Foreword . . . 53
3.1 Atomic nuclei . . . 54
3.1.1 Composition . . . 54
3.1.2 Isotopes . . . 54

3.1.3 Energy-mass equivalence . . . 55
3.1.4 Mass excess . . . 55
3.2 Nuclear decay . . . 56
3.2.1 valley of stability . . . 56
3.2.2 Weak interaction and β decay . . . 58
3.2.3 α emission . . . 59
3.2.4 Spontaneous fission . . . 59
3.2.5 Decay chain . . . 60
3.2.6 Gamma radiation . . . 60
3.3 Nuclear reactions . . . 61
3.3.1 Types of nuclear reactions . . . 61
3.3.2 Cross section . . . 62
3.3.3 Radiative capture . . . 63
3.3.4 Fission . . . 64
3.3.5 Elastic scattering . . . 66
3.3.6 Fusion . . . 67
3.4 Applications . . . 67
3.4.1 nuclear reactors . . . 67
3.4.2 Radiation and safety . . . 70
Recap of what must be known for the exam . . . 72
Exercises . . . 75
Ex. 3.1 - Chaîne de décroissance et équation de Bateman . . . 75
Ex. 3.2 - Neutrons et principe de fonctionnement d'un réacteur nucléaire . . . 78
Ex. 3.3 - Modérateur . . . 80

4 Énergie du mouvement des fluides 83
Foreword . . . 83
4.1 Fluid mechanics . . . 84
4.1.1 Continuum description . . . 84
4.1.2 Conservation laws . . . 86
4.1.3 Drag and lift . . . 91
4.1.4 Working principle of an airfoil . . . 97
4.2 Wind energy . . . 99
4.2.1 Wind resource . . . 99
4.2.2 Betz limit . . . 101
4.2.3 Horizontal axis wind turbines . . . 102
4.2.4 Alternative designs . . . 105
4.3 Water-based energy . . . 106
4.3.1 Hydropower . . . 106
4.3.2 Waves . . . 107
4.3.3 Tides and currents . . . 109
4.3.4 Ocean thermal energy . . . 111
Recap of what must be known for the exam . . . 113
Exercises . . . 116
Ex. 4.1 - Effet Magnus . . . 116
Ex. 4.2 - Courant des marées . . . 117
Ex. 4.3 - Usine marémotrice . . . 118
Ex. 4.4 - Limite de Betz . . . 118
Ex. 4.5 - Design de pale éolienne . . . 119

5 Énergie chimique **121**
Foreword . . . 121
5.1 Chemical bonds . . . 121
5.1.1 Electronic orbitals . . . 121
5.1.2 Covalent and ionic bonds . . . 124
5.1.3 Metallic bonds . . . 128
5.1.4 Van der Waals interactions . . . 129
5.2 Chemical reactions . . . 129
5.2.1 What is a reaction ? . . . 129
5.2.2 Basics of thermodynamics . . . 133
5.2.3 Thermodynamics of a reaction . . . 136
5.3 Exploitation of chemical energy . . . 139
5.3.1 Main uses of chemical energy . . . 139
5.3.2 Fossil fuels and carbon cycle . . . 142
Key points . . . 144
Exercises . . . 146
Ex. 5.1 - Haut fourneau et réduction du fer . . . 146
Ex. 5.2 - Batteries Li-ion . . . 148

6 Exercices supplémentaires **153**
6.1 Rayonnement . . . 153
Ex. 6.1 - Bilan radiatif simplifié de la Terre . . . 153
Ex. 6.2 - Rayonnement de la Lune . . . 154
6.2 Énergie nucléaire . . . 155
Ex. 6.3 - Décrément logarithmique d'énergie . . . 155
Ex. 6.4 - Produits de fission . . . 156
Ex. 6.5 - Surgénérateur à neutrons rapides . . . 159
6.3 Énergie du mouvement des fluides . . . 161
Ex. 6.6 - Limite de Betz : prise en compte de la rotation de sillage . . . 161
Ex. 6.7 - Panémone . . . 162

Syllabus

Présentation générale

Ce module est un cours d'introduction à destination des élèves de première année du cursus ingénieur, accessible à toute la promotion, et qui porte sur le socle essentiel des notions de sciences physiques à connaître pour appréhender les enjeux de la transition énergétique.

Le module porte sur les bases nécessaires théoriques permettant d'appréhender la physique des différentes formes d'énergie, et de comprendre les processus en jeu dans les principaux systèmes de transformation d'énergie. Ces bases permettront d'expliquer les enjeux technologiques et environnementaux du secteur de l'énergie, et de maîtriser les principaux ordres de grandeur et indicateurs liés à l'énergie.

Un second module complémentaire de celui-ci (ENRJ2) est proposé au second semestre et porte sur le socle essentiel de sciences humaines permettant d'aborder les enjeux de la transition (coût économique, difficultés d'organisation, incertitudes etc.).

Raison d'être académique

Ce cours vise à fournir aux élèves ingénieurs les bases scientifiques indispensables pour comprendre la physique des différentes formes d'énergie et les processus associés à leur transformation. En première année, il s'agit de poser un socle théorique solide, mobilisant des champs variés de la physique (thermodynamique, mécanique des fluides, physique nucléaire, électromagnétisme...), afin de permettre aux élèves de développer une vision cohérente et rigoureuse des enjeux énergétiques. Ce socle est conçu comme un prérequis transversal et structurant pour les enseignements plus spécialisés de deuxième et troisième années, tout en participant à la culture scientifique générale des futurs ingénieurs.

Raison d'être professionnelle

La transition énergétique constitue un défi majeur du XXIe siècle, auquel les ingénieurs seront confrontés dans de nombreux secteurs d'activité. Pour agir de manière éclairée et responsable dans ce contexte, ils doivent être capables de comprendre les limites physiques et les implications techniques des systèmes énergétiques actuels et futurs. Ce cours leur apporte les outils nécessaires pour évaluer les performances, les impacts environnementaux et les potentialités d'évolution des différentes filières énergétiques. Cette compréhension scientifique est essentielle pour dialoguer efficacement avec les experts techniques, contribuer à la conception de solutions innovantes et participer à la prise de décision dans des contextes professionnels complexes, interdisciplinaires et contraints.

Équipe enseignante

- Laurent Brochard (Navier), responsable du module, laurent.brochard@enpc.fr
- Louis Carillo (CERMICS), louis.carillo@enpc.fr
- Christophe Chevalier (Univ. Gustave Eiffel), christophe.chevalier@univ-eiffel.fr

- Nicolas Daunais (Navier), nicolas.daunais@univ-eiffel.fr
- Caroline Kopecz-Muller (Navier), caroline.kopecz-muller@univ-eiffel.fr
- Mathias Lebihain (Navier), mathias.lebihain@enpc.fr
- Anaël Lemaitre (Navier), anael.lemaitre@enpc.fr
- Bilal Mahmoud Hawchar (Navier), bilal.mahmoud-hawchar@enpc.fr
- Rahima Sidi-Boulenouar (Navier), rahima.sidi-boulenouar@univ-eiffel.fr
- Daniel Suchet (Ecole Polytechnique), daniel.suchet@polytechnique.org

Organisation des séances

Le modules est organisé en 4 thématiques correspondant au 4 premiers chapitres du polycopié. Un $5^{\text{ème}}$ chapitre, consacré à l'énergie chimique, n'est pas au programme. Enfin un $6^{\text{ème}}$ chapitre est dédié à quelques exercices complémentaires, qui ne seront pas traités en séance, mais sont fournis à titre d'entrainement / révision.

- Séance 1 (11/09) - énergie, ressources et climat : la nécessité d'une transition

  Cette séance introductive vise à introduire le concept d'énergie, et son lien avec les problématiques environnementales dont découle la nécessité d'une transition énergétique. On commence par introduire des généralités d'ordre physique : définition de l'énergie comme quantité conservée, les unités d'énergies, les principales formes d'énergies, leurs ordres de grandeur respectifs, et un aperçu rapide des transformations d'énergie qui seront abordées plus en détail dans le reste du module. On s'intéresse ensuite à l'exploitation de l'énergie par l'homme : aperçu historique, état des lieux actuels, impacts environnementaux. Le TD est consacré à deux exercices préliminaires sur le rayonnement (bilan radiatif simplifié de la Terre) et la mécanique des fluides (pertes de charges).

- Séances 2 et 3 (18/09 et 25/09) - rayonnement

  Ces deux séances sont consacrées à l'exploitation de l'énergie du rayonnement. On commence par présenter la loi de Planck prédisant le spectre du corps noir, comme première approximation du rayonnement du soleil, et par évaluer la ressource solaire. On aborde ensuite le solaire thermique (avec et sans concentration), et on discute les rendements associés. A l'issue d'une brève introduction à la physique des semi-conducteurs, on aborde ensuite le photovoltaïque avec notamment le choix du gap et les limites de Trivich-Finn et de Shockley-Queisser. On poursuit par l'état actuel des technologies solaires : les rendements réels, l'exploitation de l'énergie solaire à l'échelle globale, les coûts financiers et matières. On termine enfin par les perspectives futures et notamment le rôle du solaire dans la transition énergétique.

- Séances 4 et 5 (02/10 et 09/10) - énergie nucléaire

  Ces deux séances sont consacrées à l'énergie nucléaire. On commence par introduire quelques fondements de la physique nucléaire : l'énergie de masse, la composition des noyaux atomiques, et les interactions nucléaires. On s'intéresse ensuite à la stabilité des noyaux et aux différentes réactions de desintégration nucléaire et aux chaînes de désintégrations nucléaires. On aborde alors les réactions nucléaires (fission, fusion) et le concept de section efficace. Enfin, le fonctionnement des réacteurs nucléaires est présenté de façon générale, en abordant le cycle des neutrons et les principaux types de réacteurs, ainsi que les enjeux associés aux radiations.

- Séances 6 et 7 (16/10 matin et après-midi) - énergie du mouvement des fluides

  Ces deux séances abordent la mécanique des fluides et son application à l'exploitation de l'énergie du mouvement des fluides, dont les utilisations les plus répandues sont l'énergie éolienne et l'hydroélectricité. La première partie est consacrée aux bases de mécanique des fluides (déjà vues dans certaines filières de prépa): description continue de la matière, lois

de conservation, Bernoulli. La seconde partie porte sur l'application à l'énergie. Nous commençons par l'origine des phénomènes de traînée et de portance à l'origine du principe de fonctionnement des pales d'éolienne. L'énergie éolienne est ensuite discutée en détail : ressources, efficacité (limite de Betz), aérodynamique d'une pale, efficacité réelle des éoliennes à axe horizontal et conceptions alternatives. Enfin, nous nous concentrons sur l'énergie basée sur l'eau en commençant par l'hydroélectricité, l'exploitation la plus courante des flux d'eau. Nous présentons ensuite quelques formes d'énergie hydraulique moins connues : les vagues, les marées, les courants et l'énergie thermique des océans.

Ressources, travail personnel, assiduité et validation

Le polycopié est distribué lors de la première séance et déposé en ligne sur la page Educnet du module : `https://educnet.enpc.fr/course/view.php?id=1761`

Les présentations en amphi et les corrigés des TD seront mis en ligne à chaque séance. Les sujets d'exercice de TD sont dans le polycopié, à la fin des chapitres.

Un travail personnel régulier est attendu entre les séances : lecture du chapitre en lien avec la séance et approfondissement du TD qui ne sera pas toujours intégralement traité en séance. Conformément à la valorisation en ECTS du module, le temps de travail personnel attendu est comparable au temps de travail en présenciel.

La notation est basée sur un examen écrit en fin de module (06/11). Les examens des années précédents seront mis à disposition sur Educnet à titre d'entrainement. Les parties du cours à connaitre pour l'examen sont mis en évidence dans le polycopié avec une ligne rouge dans la marge. Un récapitulatif de ces éléments à connaitre est proposé à la fin des chapitres. Le chapitre sur l'énergie chimique n'est pas au programme de l'examen. La maitrise des TD est indispensable pour la réussite à l'examen.

La présence en amphi et petite classe est obligatoire. Il convient d'informer l'enseignant de petite classe ainsi que l'inspectrice des études de toute absence prévisible. Merci de veiller à la ponctualité, en amphi et en petite classe, gage de bon déroulement du cours.

Chapter 1

Énergie, ressources et climat : la nécessité d'une transition

Contents

Foreword **11**
1.1 What is energy? **11**
1.1.1 Noether's theorem 11
1.1.2 Fundamental forces 13
1.1.3 Energy-mass, dark energy, dark matter 14
1.2 The usable forms of energy **15**
1.2.1 Scales of energy in matter 15
1.2.2 Classification 17
1.2.3 Some orders of magnitude 18
1.3 Exploitation of energy **19**
1.3.1 Energy transformations 19
1.3.2 Exploitation by humans 20
Recap of what must be known for the exam **22**
Exercises **24**
Ex. 1.1 - Pertes de charges 24

Foreword

This introductory lesson aims to introduce the concept of energy, and its link with the environmental issues that motivate the need for an energy transition. We begin by introducing generalities about the physics of energy: the definition of energy as a conserved quantity, the units of energy, the main forms of energy, their respective orders of magnitude, and a quick overview of the energy transformations which will be covered in more detail in this class. We then focus on the exploitation of energy by man: historical overview, current state of affairs, and environmental impacts.

1.1 What is energy?

1.1.1 Noether's theorem

A formal definition of the concept of energy can be established from Noether's theorem. This theorem states that every continuous physical symmetry is associated with a conserved quantity. A physical symmetry is a transformation that leaves the properties and evolution of a physical system invariant. For instance, symmetry by spatial translation is associated with

the conservation of linear momentum, and symmetry by spatial rotation is associated with the conservation of angular momentum. As for energy, it is defined as the conserved quantity associated with symmetry by time translation. Otherwise said, if the physical laws describing the evolution of a system are time-invariant, then there exists a conserved quantity which is called energy. This is a very general definition of the concept of energy, and the corresponding mathematical expression results from the laws describing the time evolution of a system.

Thus, in quantum mechanics, for a single particle, energy is described by the Hamiltonian operator $\widehat{H} = -\frac{\hbar^2}{2m}\Delta + V$ which emerges from the Schrödinger equation, where the first term corresponds to the kinetic energy (Laplacian operator Δ, and mass m) and the second term to the potential energy (V). Likewise, in classical mechanics, for a discrete particle, the expression of energy results from Newton's law of motion: $E = \frac{p^2}{2m} + V$, where, again, the first term corresponds to the kinetic energy (momentum p), and the second term corresponds to the potential energy.

This definition of energy as a conserved quantity considers the particular case of isolated systems. Yet, even if energy is no more conserved for non isolated systems, it is still of great importance to describe the physics as it cannot be created or destroyed but only exchanged between a system and its environment. Exchanges of energy are the basis of any transformation of energy at the heart of human exploitation of energy.

Energy is homogeneous to a mass times a square length divided by a square time. Accordingly, in the International System of Units, energy is expressed in $\mathrm{kg.m^2.s^{-2}}$, also known as a Joule (J). Many other units are often used, depending on the situation:

- For atomic processes, one often considers the eV unit: 1 eV = 1.60218×10^{-19} J. Sometimes the hartree is also used, which refers to the Hartree energy, namely the negative of the electric potential energy of the electron in a hydrogen atom in its ground state (approximately twice the Rydberg energy of 13.6 eV): 1 Ha = 27.21 eV.
- In biology / chemistry, one often refers to a calorie which is the heat needed to raise the temperature of 1 g of liquid water by 1°C: 1 cal = 4.184 J or 1 kcal = 1 Cal = 4 184 J (note that 'cal' and 'Cal' are different units, referred to as small calorie and large calorie, respectively). An equivalent unit in the imperial system is the British thermal unit (Btu) which is the heat needed to raise the temperature of one pound of water by one degree Fahrenheit: 1 Btu = 1 055 J.
- For electrical energy applications, one often uses the kilowatt-hour which is the energy corresponding to a power of 1 kW (where a watt W is a Joule per second J/s) consumed/produced during one hour: 1 kW.h = 3.6 MJ.
- The petroleum industry generally refers to the tonne of oil equivalent (toe) or to the barrel of oil equivalent (boe). A tonne oil equivalent is the thermal energy released by burning a metric ton of crude oil: 1 toe = 41.868 GJ. And a metric ton of oil is equivalent to 7.33 barrel: 1 toe = 7.33 boe.
- Catastrophic events (explosions, earthquakes) are sometimes quantified in ton of TNT, which is the energy released upon the explosion of one ton of TNT: 1 ton of TNT = 4.184 GJ. Note that one gram of TNT actually corresponds to 1 kcal.

In specific fields of physics, one sometimes refers to energy in units that are not units of energy. In such cases, one actually refers to an equivalence between energy and another quantity based on a fundamental relationship. This happens in particular in two domains :

Spectroscopy According to the Planck-Einstein relation, the energy of a photon is related to its frequency ν and wavelength λ : $E = h\nu = \frac{hc}{\lambda}$, with $h = 6.626\,070\,15 \times 10^{-34}$ J.s the Planck constant and $c = 299\,792\,458$ m/s the speed of light. Therefore, when referring

to the energy of a radiation, one sometimes refers to frequency units, e.g., 1 THz = 4.136 meV, or to inverse length units, e.g., 1 cm^{-1} = 0.124 meV or 1 μm^{-1} = 1.24 eV.

Relativity According to special relativity, mass and energy are equivalent for a system at rest: $E = mc^2$, with c the speed of light. Therefore, in nuclear applications, one often refers to changes in energy as to changes in mass, e.g., 1 kg = 8.988×10^{16} J or 1 u = 0.932 GeV with u the atomic mass unit (1/12 of the mass of a carbon 12 atom).

1.1.2 Fundamental forces

The potential energy (V) originates from the interactions between particles. At a fundamental level, there are only 4 fundamental forces at the origin of theses interactions.

Electromagnetism is the interaction due to charges of particles. For two particles of charges q_1 and q_2 at rest separated by a distance r, the interaction potential is given by Coulomb's law $E = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r}$, with $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ $J.m^{-1}.C^{-2}$ the Coulomb constant. Electromagnetism is the interaction that dominates from the scale of the atom to that of the planets. It is responsible for the organization of the electrons within atoms, molecules, and materials, and for all the inter-molecular interactions (except the tiny contribution of gravity, see hereafter). Accordingly, electromagnetism is at the origin of most of the physical phenomena occurring at our macroscopic scale (phase transitions, mechanical properties, chemical reactions, electrical properties...). Photons are the carrier of the electromagnetic interaction, and, electromagnetism is therefore central for all the energy exchanges between radiation (UV, light, IR) and matter. At very large scales ($\geq$ planets) and at very low scales ($\leq$ atom nuclei), other interactions emerges that prevail over electromagnetism, as explained hereafter.

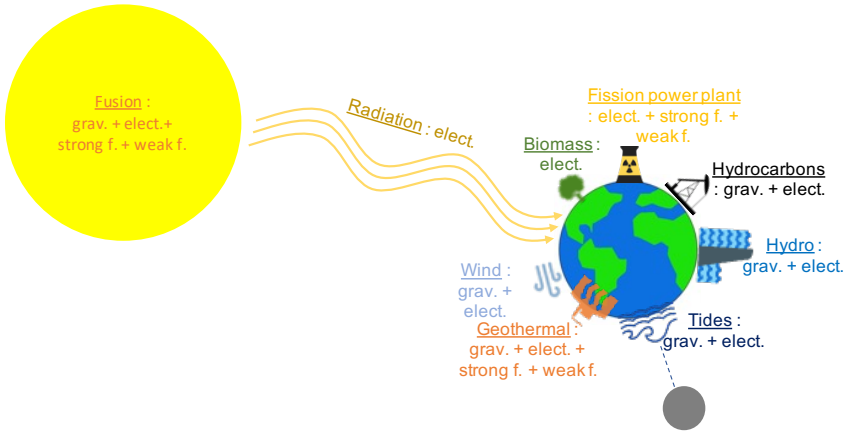
Gravity is a mutual attraction between all particles, and originates from the space-time curvature according to relativity. In a non-relativistic framework, for two particles of mass m_1 and m_2 separated by a distance r, the corresponding potential energy is $E_{grav} = -\frac{Gm_1m_2}{r}$, with $G = 7 \times 10^{-11} J.m^{-1}.kg^{-2}$. This energy is 20 orders of magnitude smaller than electromagnetism for typical macroscopic objects (m = 1 kg, q = 1 C, r = 1 m). But, unlike electromagnetism, gravity is universally attractive since masses are always positive while charges are both positive and negative. As a consequence, screening effects strongly limit the range of electromagnetism (positive and negative charges cancel out so that matter appears approximately neutral), whereas the range of gravity is unlimited. Therefore, gravitational energy dominates over electromagnetic energy for astronomical objects (planets, stars...). Gravity governs the motion of planets, stars, and galaxies. It even determines the motion of light, which is massless but still affected by space-time curvature in relativity.

Strong force At the scale of the atom nucleus, a force stronger than the electromagnetic repulsion emerges and holds the protons and neutrons together. This 'strong force' is about 10 times stronger than the electromagnetic force, but it acts over an extremely short range of about 1 fm (10^{-15} m). The strong force covers actually multiple interactions and has no simple analytic expression. On one hand, it acts between quarks within hadrons (e.g., within protons and neutrons). On the other hand, it acts between hadrons. At the scale of the nuclei, it can be roughly represented by a Yukawa potential $E_{strong} = -\frac{1}{4\pi\varepsilon_0}\frac{(g_{strong})^2}{r}\exp\left(-\frac{r}{b_{strong}}\right)$, with $b_{strong} \approx 1$ fm and $(g_{strong})^2 \approx 10e^2$ (where $e = 1.60218 \times 10^{-19}$ C is the elementary charge). The range of the strong force is at the origin of the size of atomic nuclei and its magnitude controls the energy changes involved in nuclear reactions. It is responsible for the domain of stability of the atom nuclei (size, composition), which explains the processes of nuclear fusion between light nuclei and nuclear fission of heavy nuclei.

Weak force The last fundamental force also acts within atomic nuclei, but is much weaker than the strong force, which is why it is called the 'weak' force. This force acts between all fermions (and also Higgs boson and neutrinos), it is of similar magnitude as the electromagnetic force, i.e., an order of magnitude weaker than the strong force, and its range is about

10^{-3} fm much shorter than the strong force and less than the size of a proton. It remains poorly known and, as for the strong force, it is roughly represented by a Yukawa potential $E_{weak} = -\frac{1}{4\pi\varepsilon_0}\frac{(g_{weak})^2}{r}\exp\left(-\frac{r}{b_{weak}}\right)$, with $b_{weak} \approx 10^{-3}$ fm and $(g_{weak})^2 \approx e^2$. Because of its extremely short range and moderate magnitude, the contribution of the weak force to the potential energy of a nucleus is negligible. However, it is essential to the processes of nuclear fission and fusion as it is the only interaction that allows to convert a neutron into a proton (as in the beta decay) and a proton into a neutron (as in the fusion of hydrogen into helium).

Many of the processes of energy transformation involve the conjunction between several fundamental forces (see figure below). Solar radiation is electromagnetic waves that originate from the nuclear fusion reaction between hydrogen atoms in the sun, and this reaction is possible because of the intense pressure and temperature conditions caused by gravitation. This solar radiation is the main source of energy received by our planet, and is at the origin of all the energy forms that are used by humans except tidal energy (gravitation of the moon), geothermal energy (radioactive decay within the Earth), and nuclear energy (nuclear fission). Solar radiation causes the evaporation of water that fills the dam reservoirs, it triggers the winds that drive the wind turbines, it is converted into chemical energy by plants providing biomass energy and the same process some hundreds of million years ago led to fossil fuels. All these energy forms with no exception involve electromagnetism either through radiation or through the behavior of matter (fluid flow, chemical reactions, phase transitions...).



1.1.3 Energy-mass, dark energy, dark matter

Gravity is apart in the aforementioned fundamental forces because, according to general relativity, it is not really a force but the manifestation of the deformation of space-time. The above expression of gravitation energy is an approximation valid in non-relativistic situations (velocities small compared to the speed of light). As we mentioned at the end of section 1.1.1, special relativity relates mass and energy. More precisely, for a free particle of momentum $\mathbf{p}$ and rest mass m, the energy is given by $E^2 = c^2\mathbf{p}^2 + m^2c^4$. For non-relativistic energies, $\mathbf{p}^2 \ll mc$ so that a first-order expansion gives $E \approx mc^2 + \frac{\mathbf{p}^2}{2m}$ and one recovers the usual expression of kinetic energy shifted by the rest mass energy. In summary, assuming non-relativistic conditions, gravity and kinetic energies can be approximated by their classical expressions. For the applications considered in this class, these approximations will be sufficient.

At astronomical scales, however, relativity is essential to estimate the energy budget of the universe. In particular, two mysterious contributions to the energy of the universe emerged from astronomical observations:

1. The universe is in expansion and this expansion is accelerating. This is in contradiction with the idea that gravity dominates at astronomical scales, as it is universally attractive and, therefore, one would expect that the expansion would slow down. To capture the accelerating expansion, one has to introduce an additional source of curvature in the theory of relativity,

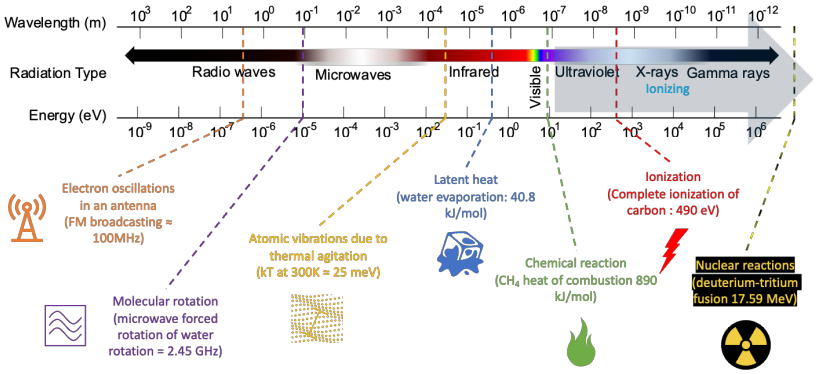
known as the cosmological constant, which acts as a repulsion. This cosmological constant brings an additional energy to the Universe, which is called dark energy.

2. The rotation velocities of the galaxies are independent of the distance to the center of the galaxies. Again, this observation is in contradiction with the estimated distribution of mass within galaxies which would imply a decreasing rotation velocity from the center to the edge. This observation suggests the existence of dark matter, an invisible form of matter that does not interact with ordinary matter or light except through gravity. This dark matter also brings additional energy to the Universe.

Dark energy and dark matter are estimated to represent 69% and 26% of all the energy in the Universe, so that ordinary matter represents only 5% of the total energy in the Universe.

1.2 The usable forms of energy

1.2.1 Scales of energy in matter



Let us now turn to the forms of energy that we actually use. There is a wide variety of ways by which we manipulate energy. Looking at the elementary atomic processes involved in these manipulations, we can classify them by order of scale of energy (or equivalent wavelength of the associated photons), as represented in the Figure above.

The lowest scale of energy we manipulate are radio waves which are generated by the oscillation of electrons in antennas. These radio waves typically have wavelengths of a few meters, corresponding to energies of a few 10^{-7} eV.

Next to radio waves are the microwaves which can again be generated by oscillations of electrons in conductors, but are strong enough to rotate small polar molecules. A well-known application is the microwave oven which consists of rotating water molecules at a frequency of around 2.45 GHz which leads to an optimal dielectric heating of food as these rotations are rapidly converted into disordered kinetic energy of all molecules. The associated scale of energy is about 10^{-5} eV.

After the microwaves, one enters into the infrared domain. The corresponding energies are particularly important as they correspond to the scale of thermal agitation in matter at the temperatures of the Earth's surface. The scale of energy of thermal agitation is $k_B T$ with $k_B = 1.380649 \times 10^{-23}$ J/K the Boltzmann constant and T the temperature (in K). According to statistical physics, $k_B T$ represents the typical energy scale of an atom in thermal equilibrium at the temperature T (up to a multiplicative factor). At 300 K, $k_B T$ is about 25 meV corresponding to a wavelength of 50 µm. All matter at the surface of the Earth contains thermal agitation energy of this magnitude per atom. In solids, for instance, this thermal agitation takes the form of vibrations called phonons. These phonons radiate infrared light to the atmosphere and part of this radiation is absorbed by CO_2 which causes the greenhouse effect (see dedicated exercise in the small class about radiation). Regarding applications, the

k_BT scale of energy corresponds to the magnitude of energy that is needed to change the temperature of matter. This reflects the heat capacity of matter.

Further into the infrared domain, near the visible domain, one reaches energies that are strong enough to break inter-molecular interactions thus causing phase changes. Phase changes are associated with latent heats that are significantly larger than the energies associated with heat capacity. For instance, the latent heat of water evaporation (40.8 kJ/mol) corresponds to 0.4 eV per molecule, which is about 5 times more than the energy needed to heat liquid water from 0°C to 100°C. Latent heat is used in particular in thermal engines as it makes it possible to exchange large amounts of heat over a narrow range of temperatures.

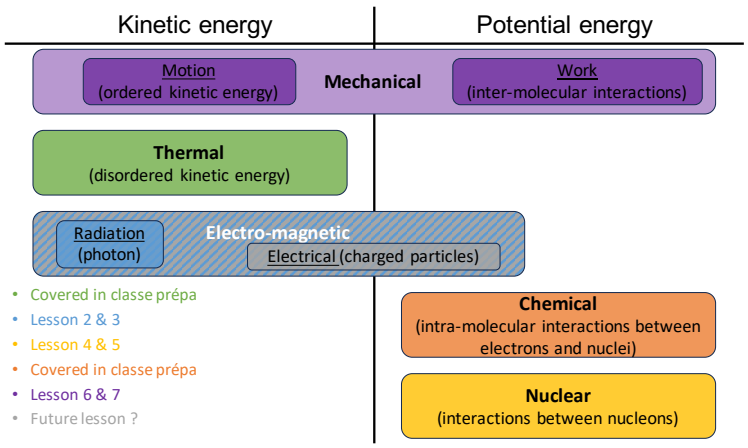
The visible domain is well-known as it corresponds to radiation visible by the naked eye. The solar spectrum is centered on this domain which therefore corresponds to the thermal agitation energy at the temperature of the surface of the sun ($\sim$ 6000K). Precisely, the energy of the visible domain ranges from 1.5 eV (800 nm, red light) to 3 eV (400 nm, blue light). The eV range corresponds to the scale of energy associated with the binding of electrons within atoms. The proper physical theory to understand the behavior of electrons within atoms is no more classical physics, but quantum mechanics. In particular, bound electrons occupy discrete states of energies called orbitals, and the energy is therefore quantified. The typical difference of energy between two orbitals ranges from a fraction of eV to hundreds of eVs (covering the visible and ultraviolet domains). The most well-known orbital energies are the orbitals of the hydrogen atom, which are known analytically: $E_n = \frac{E_{Ryd}}{n^2}$ with $E_{Ryd} = -13.6$ eV the Rydgerg energy and $n \in \mathbb{N}^*$ the principal quantum number. Since an electron can only occupy these states of energy, only discrete energy variations are possible of the form $\Delta E = E_{Ryd}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$, with n_1 and n_2 positive integers. This explains the peculiar structure of the adsorption and emission bands of the hydrogen spectrum in $\lambda^{-1} \propto \frac{1}{n_1^2} - \frac{1}{n_2^2}$ found empirically by Rydberg. Chemical reactions, which involve changes in the orbitals when the electronic clouds of two atoms overlap, involve scales of energies comparable to that in the hydrogen atom. For instance, the heat of combustion of methane (CH_4) is 9.2 eV per molecule (890 kJ/mol). Chemical reactions in the form of fossil fuel combustion are by far the main form of energy used by humans.

Further into the ultraviolet domain and near X-rays, one reaches energies of hundreds to thousands of eVs with are enough to fully ionize atoms, i.e., electrons are no longer bound to nuclei and the state of matter becomes a plasma. For instance, the complete ionization energy of carbon is 490 eV. Plasma is found at extreme temperatures only, such as in the sun or in the path of lightning. Plasma is also made artificially in advanced setups (e.g., tokamak) in the perspective of achieving controlled nuclear fusion.

Beyond X-rays are gamma rays, the most energetic form of radiation, that results only from nuclear reactions. The associated energies are orders of magnitude larger than those involved in chemical reactions or atom ionization. For instance, the fusion between deuterium and tritium, considered for controlled nuclear fusion experiments, releases 17.59 MeV per pair of reacted atoms.

To put these scales into perspective, radiation energies of a few tens of eVs or more are called ionizing as these are strong enough to ionize atoms and are therefore harmful to life. Ultimately, the largest scale of energy is the mass energy $E = mc^2$. For a proton, this gives about 1 GeV, one to two orders of magnitude larger than the energies involved in nuclear reactions. Such a form of energy is not within reach of man exploitation with current technologies, and would involve for instance the annihilation of matter with anti-matter.

1.2.2 Classification



The usable forms of energy are usually classified as shown in the Figure above. The main forms are:

Mechanical energy refers to the kinetic energy associated with the motion of matter on one hand, and to the potential energy associated with the mechanical work of matter (traction-compression) on the other hand. At the atomic scale, motion corresponds to the ordered kinetic energy of the atoms, as opposed to heat energy which corresponds to the disordered kinetic energy of the atoms. As for mechanical work, it originates from the inter-molecular potential energy, that is the part of the interactions between atoms that do not involve any change in the nature of chemistry (electrons) or of atoms (nuclei). For instance, fluid flow through a wind turbine involves a transformation of longitudinal motion into rotational motion, by the action of mechanical work due to pressure differences on the blades.

Thermal energy is the energy associated with the disordered kinetic energy of the atoms. The disordered nature of this energy is quantified by the temperature and entropy, and the second law of thermodynamics limits the amount of thermal energy that can be converted into 'ordered' forms of energy. This gives rise, for instance, to the Carnot efficiency which fundamentally limits the efficiency of thermal engines.

Electro-magnetic energy refers to the radiation energy on one hand, and to the electrical energy on the other hand. Radiation energy is electro-magnetic waves, also viewed as photons, and is the force carrier of the electro-magnetic interactions. Exchanges of energy between radiation and matter are ubiquitous and are used to transform solar radiation into heat (solar thermal energy) or into electricity (photovoltaics). Electrical energy is the energy of charged particles and of the movement of those particles (most often electrons in conductors). It can be potential energy (e.g., energy stored in a capacitor) or kinetic energy (propagation of electromagnetic waves as electrons flow through a wire). Electrical energy is often used as an intermediary form of energy as it is not easy to store in large quantities but it can be efficiently transported and converted into the other forms of energy.

Chemical energy is the energy resulting from chemical reactions. At the atomic scale, it corresponds to the potential energy associated with changes in the orbitals of the electrons within the atoms and molecules. Chemical energy in the form of wood fire is the oldest form of energy controlled by humans, as exothermic combustion is a very easy way to produce thermal energy from chemical energy. Combustion of fossil fuels still dominates the energy use nowadays, and is at the origin of major environmental concerns (see last section).

Nuclear energy refers to the energy associated with nuclear reactions, i.e., potential energy due to the nuclear forces. The use of nuclear energy is rather recent in comparison with the other forms of energy, and involves much larger energy scales. Only a few nuclear reactions can be controlled, which is intimately related to the stability of nuclei. Thus, most current

nuclear fission reactors rely on a particular isotope of uranium (^{235}U, 0.7% of all naturally occurring uranium). Use of alternative, more abundant, nuclei is under development (e.g., breeder reactors using ^{238}U or ^{232}Th).

The lessons of this class are actually structured following this classification of the forms of energy. The lessons 2 and 3 explore radiation, lessons 4 and 5 focus on nuclear energy, and lessons 6 and 7 investigate energy from fluid mechanics. Thermal energy and chemical energy are already covered in classe préparatoire (through thermodynamics and thermochemistry). An additional lesson is planned for future sessions that will consider electrical energy. Of course, this classification is not strict and practical applications often combine different forms of energy, as in most cases one is interested in the transformation of one form of energy into another. For instance, electro-chemistry links chemical energy with electrical energy. Likewise, thermal engines convert thermal energy into mechanical energy.

1.2.3 Some orders of magnitude

So far we have focused on energy scales at the level of the elementary particles involved (electrons, atoms, nuclei). These energies are expressed in eV ($\sim 10^{-19}$ J) which remain insignificant at our macroscopic scale. Hereafter, we provide some macroscopic orders of magnitudes of energy, which are more representative of the magnitudes of energy we experience in practice.

To start with, let us consider the energies representative of our everyday life. A typical order of magnitude of our daily use of energy is the global daily average consumption of primary energy per person: 200 MJ. Accordingly, the MJ (10^6 J) is the appropriate scale when dealing with everyday life use. We provide hereafter a list of typical energies covering the range from a J to a few MJ:

- 1 J: a tennis ball falling from 2 m
- 10 J: heat 1g liquid water by 2°C
- 100 J: heat produced by a person at rest during 1 s
- 1 kJ: energy released by a candle burning for 10 s
- 10 kJ: 4 h of electricity consumption of a smartphone
- 100 kJ: kinetic energy of a car at 50 km/h
- 1 MJ: running for 1 h
- 10 MJ: food energy of a person per day

Our daily use (200 MJ and much more in developed countries) is rather large compared to our basic needs such as food energy. Our energy consumption has increased tremendously with the industrial revolution and can be attributed to modern uses such as transportation or industry. Switching to industrial scales, we provide hereafter some typical orders of magnitudes of energies at a more global scale:

- 1 MJ: 1 ton of water released from a 100 m dam
- 10 MJ: average solar radiation per m^2 per day in France
- 100 MJ: heat of combustion of 2 kg of oil
- 1 GJ: energy of an average gasoline reservoir (50 L)
- 1 TJ (10^{12} J): consumption of a wide-body aircraft for a long-haul flight
- 1 PJ (10^{15} J): 2 weeks of electricity production of a nuclear plant

- 1 EJ (10^{18} J): 6 months of French electricity production
- 600 EJ: global annual consumption
- 50 000 EJ: proven reserves of fossil fuels
- 5 500 000 EJ: global solar radiation on Earth per year

Interestingly, the proven reserves of fossil fuels remain large compared to the global need (about a hundred years of reserves), and the shortage of fossil fuels does not appear as a pressing issue when compared to the climate change caused by their combustion. As for solar radiation on Earth, it is orders of magnitude larger than global needs: 1 hour of solar radiation on Earth corresponds to a year of energy consumption. Otherwise said, energy resources are huge. The real issue is more about how to exploit those resources in a sustainable manner.

1.3 Exploitation of energy

1.3.1 Energy transformations

As energy cannot be created or destroyed but only exchanged, the practical exploitation of energy consists of transforming energy from one form to another. All progress in harvesting energy comes down to new transform processes from the control of fire by early humans (chemical to thermal), to the steam engine (from thermal to mechanical), and to the light bulb (from electrical to radiation). The table below lists examples of transformation processes, either natural or artificial, for all possible transformations between the different forms of energy, except for electrical to nuclear and for chemical to nuclear for which no transformation process exists. This table shows that there is a wide variety of ways to transform energy. In practice, some transformations are widely used in modern societies (e.g., thermal to mechanical in a wide variety of engines) while others are limited to niche applications (e.g., atomic batteries), and some remain beyond the reach of human control (supernova).

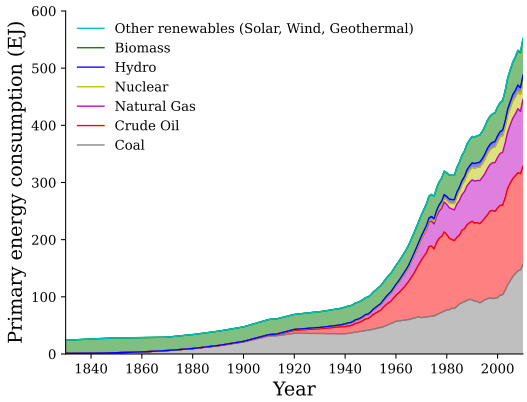
↓from \ to→	Mechanical	Thermal	Radiation	Electrical	Chemical	Nuclear
Mechanical	Gearbox	Friction	Synchrotron radiation	Generator	Metamorphism	Particle accelerator
Thermal	Turbine	Heat exchanger	Black body	Thermocouple (Seebeck effect)	Blast furnace	Supernova
Radiation	Radiometer	Thermal solar collector	Non linear optic	Photovoltaic cell	Photosynthesis	Photodesintegration
Electrical	Electrical engine	Electrical resistance	Lightning, LED	Transformator	Accumulator	-
Chemical	Muscle	Combustion	Luminescence	Fuel cell	Coal gasification	-
Nuclear	Atomic bomb	Nuclear reactor	Gamma radiation	Atomic battery	Radiolysis	Breeder reactor

The efficiency of a transformation of energy is defined by the fraction of the initial form of energy that is effectively converted into the desired form of energy, e.g., the ratio between the mechanical work produced by a thermal engine and the heat it consumes. In most cases, efficiency is limited to a maximum value of 1, i.e., all the initial energy form can be converted into the final energy form. In some particular cases (e.g., heat pumps or refrigerators) it can exceed 1. And in some other cases, it is limited, for physical reasons, to values below 1. Four efficiency limitations are of particular importance for practical applications:

- The Carnot limit for the transformation between thermal energy and mechanical energy
- The maximum recoverable thermodynamic energy (Helmholtz free energy for isotermal systems, enthalpy for chemical reactions producing heat, Gibbs free enthalpy for electro-chemistry etc.)
- The Shockley-Queisser limit for the efficiency of single-layer photovoltaic cells
- The Betz limit for the efficiency of wind turbines

These limits in efficiency are not of the same nature. For instance, the Carnot limit is a hard limit as it arises from the second principle of thermodynamics, and, therefore, it cannot be overcome irrespective of the engine considered. In contrast, the Shockley-Queisser limit is a technical limit of single junction photovoltaic cells, but there is no fundamental principle limiting the transformation of radiation energy into electrical energy like the second principle for the Carnot limit. Accordingly, the Shockley-Queisser limit can be overcome by considering alternative techniques of energy transformation, and this is indeed the case with multi-junction cells which can overcome the Shockley-Queisser limit by 50%.

1.3.2 Exploitation by humans



source: the Oil Drum

The exploitation of energy by humans has evolved tremendously since the industrial revolution (see graphic above). While chemical energy from biomass combustion has been the main source of energy for almost all history, the development of the coal-fired steam engine initiated the use of fossil fuels which are now the predominant source of energy. Alternative energy sources (hydro, nuclear, wind, solar, geothermal) are growing but their total contribution remains small in comparison with that of fossil fuels. The usage of energy has become ubiquitous in all economic sectors and in every aspect of our daily lives (housing, food, transportation...). Out of the 600 EJ/yr of primary energy consumed, one-third only is converted into final energy providing useful energy services. The remaining two-thirds are losses, either due to the efficiency of energy transformation (e.g., Carnot limit) or due to other forms of dissipation (e.g., resistive losses of power lines).

Fossil fuels have become the dominant energy sources because they combine several practical advantages over the other sources of energy:

- Fossil fuels have been rather abundant so far (see table below on the left). Proven reserves of oil, gas, and coal correspond to more than a century of current consumption of primary energy. In comparison, many of the other sources of energy have limited resources (assuming current exploitation techniques), not enough to cover the global needs: biofuels, geothermal, and hydro. Nuclear energy is also very limited (excluding breeder reactors, which would increase reserves by two orders of magnitude). Only wind and solar energy offer scales of energy resources capable of covering the current global needs in energy.

- Fossil fuels are highly concentrated forms of energy (see table below on the right), that make it possible to produce large amounts of usable energy over limited land areas. All alternative sources of energy, except nuclear, require significantly more land areas to produce similar amounts of energy (5 times for wind, 10 times for geothermal, 50 times for solar...).

- Fossil fuels are easy to store and transform, which makes them convenient sources of energy for transport applications. The electricity storage in batteries exhibits energy densities that remain one order of magnitude smaller than the heat of combustion of gasoline, even after taking into account a 30% efficiency of thermal engines. Therefore, fossil fuels are difficult to replace in particular in applications for which weight or volume is critical, such as air transport.

- Fossil fuels offer flexible energy production that can easily adapt to the demand. In contrast, wind and solar energies are intermittent, so sustained electricity production requires either large-scale energy storage or complementary on-demand production which often relies on fossil fuels.

- The exploitation of fossil fuels requires moderate investment costs, but significant operation costs, when compared to alternative energy sources for which the resource is cheap or even free (e.g. wind) but the investment is significant. Thus the development of alternative energies is conditional on low interest rates.

Global need : 600 EJ/yr ≈ 20 TW

Type	Source	Resource	
Renewable	Biofuels	1 TW	Power available (in W = J/s)
	Geothermal	1 TW	
	Hydro	1-2 TW	
	Wind	100 TW	
	Solar	100 000 TW	
Non renewable	Coal	20 000 EJ	Energy reserve (proven)
	Oil	9 000 EJ	
	Gas	36 000 EJ	
	Nuclear	> 2 000 EJ	

Sources :
- Renewables: Jaffe & Taylor (2018) The Physics of Energy. Cambridge University Press
- Fossil fuels: British Petroleum
- Nuclear: IPCC

Electrical power produced per unit area of land

Type	Source	Power density
Renewable	Biofuels	0.23 W/m²
Renewable	Hydro	11 W/m²
Renewable	Solar	11 W/m²
Non renewable	Coal (open cast)	22 W/m²
Renewable	Geothermal	46 W/m²
Renewable	Wind	114 W/m²
Non renewable	Coal (underground)	570 W/m²
Non renewable	Oil & gas	570 W/m²
Non renewable	Nuclear	1 140 W/m²

Source : Uwe R.Fritsche et al. (2017) Energy and Land Use - Global Land Outlook working paper. UNCCD and IRENA

However, the use of fossil fuels causes major environmental impacts, which destabilize ecosystems, degrade health, and compromise the future development of human societies. Let us mention in particular, the following impacts:

- Fossil fuels are the main source of anthropogenic greenhouse gas emissions, at the origin of global warming of the Earth. Climate change is a major concern, and the underlying physical mechanisms are the topic of an exercise in the small class about radiation.

- A much lesser-known consequence of CO_2 emissions is ocean acidification. CO_2 dissolution in the oceans reduces the pH. While pre-industrial pH was about 8.2, the current pH is 8.06 and it could reach 7.74 by the end of the century following a business-as-usual scenario. Such a change in pH is a drastic change in ocean chemistry, in particular regarding solubility. For instance, aragonite, a polymorph of calcium carbonate encountered frequently in biological shells, would become unsaturated (i.e., dissolve) in most of the southern ocean by 2100 in a business-as-usual scenario. Needless to say, this would destabilize ecosystems on a very large scale.

- Fossil fuels are major contributors to many other forms of pollution that impact health or ecosystems: NOx (nitrogen oxides), SOx (sulfur oxides), PM (particular matter), VOCs (volatile organic compounds), etc.

The current use of fossil fuels has reached unsustainable levels, and their predominance in the energy sector calls for a drastic change to sustainable energy, of much reduced environmental impact, in particular regarding carbon emissions, and ensuring long-term supply for future generations. This energy transition is expected to occur within a few decades. The Paris Agreement (COP21) adopted in 2015 by 196 countries has set as an objective to reach net zero carbon emissions by mid-century.

Recap of what must be known for the exam

1. Noether's theorem

Energy is defined as the conserved quantity associated with symmetry by time translation.

As an example, the energy of a particle in classical mechanics is given by $E = \frac{p^2}{2m} + V$ where the first term is the kinetic energy, and the second term is the potential energy.

2. Units of energy

Energy is homogeneous to a mass times a square length divided by a square time. The SI unit of energy is the Joule : 1 J = 1 kg.m^2.s^{-2}

Some alternatives units:

- 1 eV $\approx 1.6 \times 10^{-19}$ J for atomic processes
- 1 kcal $\approx$ 4 kJ used in chemistry and biology
- 1 kW.h = 3.6 MJ for electrical energy

In spectroscopy, one can refer to energy as the corresponding frequency (ν) or wavelength (λ) of the photons having this energy: $E = h\nu = hc/\lambda$. Thus, for instance, 1 THz $\sim$ a few meV (thermal radiation in ambient conditions) and 1 µm^{-1} $\approx$ 1 eV (radiation from the sun).

For nuclear applications, one can refer to energy as the corresponding mass (m) according to the energy-mass equivalence of relativity: $E = mc^2$. The typical scale of mass energy of atoms is the GeV (about 1 GeV for a proton).

3. The usable forms of energy

Energy scales at the level of the elementary particles and processes:

radio waves (a few 10^{-7} eV): oscillation of electrons in antennas, wavelengths of a few meters.

microwaves (about 10^{-5} eV): strong enough to rotate small polar molecules (rotation of water in microwave oven), wavelength of a few cm.

thermal agitation at 300K (25 meV): thermal fluctuation of energy of magnitude k_BT, infrared radiation (wavelength of tens of µm), origin of the heat capacity of matter.

inter-molecular interactions (a fraction of eV): energy strong enough to trigger phase change, near infrared domain (wavelength of a few µm), origin of latent heat (evaporation, solidification).

intra-molecular interactions (a few eV): energy strong enough to break molecule and change the state of electron in atoms, origin of chemical reactions, domain of visible light and solar spectrum (wavelength of hundreds of nm).

ionization energy (hundreds of eV): energy strong enough to separate electron from nucleus, formation of plasma, UV and X-ray domain (wavelengths of a few nm),

nuclear reactions (tens or hundreds of MeV): energy involved in nuclear processes (fission, fusion), emits gamma rays

The ultimate scale of energy is given by the mass energy (GeV) that would result from complete annihilation of matter (non usable).

4. Typical orders of magnitude

Some energies representative of our everyday life:

- 1 MJ - running for 1 h
- 10 MJ - food energy of a person per day
- 200 MJ - global daily average consumption of primary energy per person

Some energies at a more global scale:

- 1 MJ: 1 ton of water released from a 100 m dam
- 10 MJ: average solar radiation per m^2 per day in France
- 1 GJ: energy of an average gasoline reservoir (50 L)
- 2 EJ (10^{18} J): annual French electricity production
- 600 EJ: global annual consumption
- 50 000 EJ: proven reserves of fossil fuels
- 5 500 000 EJ: global solar radiation on Earth per year

Exercises

Ex. 1.1 - Pertes de charges

Cet exercice s'inscrit en préparation des séances consacrées à l'énergie du mouvement des fluides. Il s'agit d'introduire quelques concepts de mécanique des fluides (déjà abordés dans certaines filières de classe préparatoire), en s'intéressant à un problème simple, à savoir la perte de charge dans un tube.

On s'intéresse à l'écoulement d'un fluide newtonien (viscosité μ) incompressible (densité ρ_0). Dans une description continue, l'état du fluide est représenté par le champ de vitesse $\mathbf{v}(\mathbf{x}, t)$ et le champ de pression $p(\mathbf{x}, t)$ en fonction du temps t et de la position $\mathbf{x}$. A partir d'une condition initiale, l'évolution du système est décrite par deux équations de conservation :

- L'équation de conservation de la masse du fluide, dont la forme locale est donnée par l'équation de continuité :

$$\nabla.\mathbf{v} = 0$$

  où $\nabla.$ est l'opérateur divergence.

- La conservation du moment, dont la forme locale est donnée par l'équation de Navier-Stokes (équation vectorielle) :

$$\rho_0 \frac{\partial \mathbf{v}}{dt} + \rho_0 \mathbf{v}.\nabla\mathbf{v} = -\nabla p + \mu\Delta\mathbf{v} + \rho_0\mathbf{g}$$

  où ∇ est l'opérateur gradient, Δ est l'opérateur Laplacien, $\mathbf{g}$ est l'accélération due à la gravité.

La conservation de la masse traduit le fait que la quantité de fluide sortant d'un volume élémentaire est égale à la quantité y entrant (puisque le fluide est incompressible). Quant à la conservation du moment, celle-ci traduit la deuxième loi de Newton à l'échelle du milieu continu : la variation du moment cinétique ($\rho_0 \frac{\partial \mathbf{v}}{dt}$) et le terme de convection ($\rho_0 \mathbf{v}.\nabla\mathbf{v}$) correspondent à "masse $\times$ accélération", tandis que les forces se répartissent entre les forces de volume (∇p), les forces visqueuses ($\mu\Delta\mathbf{v}$), et les forces extérieures ($\rho_0\mathbf{g}$).

1. Verifier que le problème est mathématiquement bien posé.

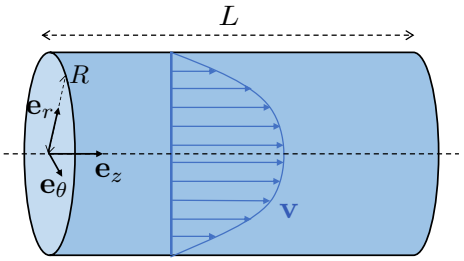
   Rq : Ce problème est particulièrement difficile à résoudre en raison de deux termes en particulier : la convection (terme non linéaire) et force visqueuse (dérivée de second ordre). Il s'agit de l'un des sept problèmes du millénaire.

2. Soit l une longueur caractéristique de variation de la vitesse et v_0 une valeur caractéristique de la vitesse dans le problème considéré. En comparant l'ordre de grandeur des termes de l'équation de Navier-Stokes, que représente la quantité $\mathcal{R}_e = \frac{\rho_0 v_0 l}{\mu}$ (nombre de Reynolds).

   Rq : la transition entre le régime 'laminaire' ($\mathcal{R}_e$ faible) et le régime 'turbulent' ($\mathcal{R}_e$ élevé) a lieu autour de $\mathcal{R}_e \sim 10^3 - 10^4$. La formation de turbulences est observée lorsque $\mathcal{R}_e \gtrsim 10^5$.

3. On considère un écoulement stationnaire d'un fluide newtonien incompressible dans un tube cylindrique de rayon R et de longueur L à $\mathcal{R}_e$ faible (cf. schéma ci-dessous). Stationnaire, signifie qu'il n'y a pas de dépendance en temps des champs de vitesse et de pression ($\mathbf{v}(\mathbf{x}, t) = \mathbf{v}(\mathbf{x})$ et $p(\mathbf{x}, t) = p(\mathbf{x})$). Par ailleurs, on suppose les forces de gravité négligeables ($\|\rho_0\mathbf{g}\| \ll \|-\nabla p + \mu\Delta\mathbf{v}\|$). On se place en coordonnées cylindriques, avec l'axe du tube correspondant à l'axe polaire (Oz) et le fluide s'écoule dans la direction de l'axe : $\mathbf{v}(\mathbf{x}) = v(\mathbf{x})\,\mathbf{e}_z$. A l'interface avec le tube, la vitesse du fluide est nulle (condition de non glissement) : $v(r = R) = 0$. En coordonnées cylindriques, les divers opérateurs impliqués dans les équations du problème prennent les formes suivantes :

Gradient d'un scalaire	Divergence d'un vecteur	Laplacien d'un vecteur
$\nabla A = \begin{pmatrix} \frac{\partial A}{\partial r} \\ \frac{\partial A}{r\partial\theta} \\ \frac{\partial A}{\partial z} \end{pmatrix}$	$\nabla.\mathbf{A} = \frac{1}{r}\frac{\partial(rA_r)}{\partial r} + \frac{\partial A_\theta}{r\partial\theta} + \frac{\partial A_z}{\partial z}$	$\Delta\mathbf{A} = \begin{pmatrix} \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial A_r}{\partial r}\right) + \frac{\partial^2 A_r}{r^2\partial\theta^2} + \frac{\partial^2 A_r}{\partial z^2} - \frac{2}{r^2}\frac{\partial A_\theta}{\partial\theta} - \frac{A_r}{r^2} \\ \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial A_\theta}{\partial r}\right) + \frac{\partial^2 A_\theta}{r^2\partial\theta^2} + \frac{\partial^2 A_\theta}{\partial z^2} + \frac{2}{r^2}\frac{\partial A_r}{\partial\theta} - \frac{A_\theta}{r^2} \\ \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial A_z}{\partial r}\right) + \frac{\partial^2 A_z}{r^2\partial\theta^2} + \frac{\partial^2 A_z}{\partial z^2} \end{pmatrix}$



(a) Pourquoi le champs de vitesse est-il de la forme $\mathbf{v} = v(r,z)\,\mathbf{e}_z$?

(b) Montrer que la vitesse est indépendante de z.

(c) Montrer que l'équation de Navier-Stokes se réduit à $\frac{1}{r}\frac{d}{dr}\left(r\frac{dv}{dr}\right) = -\frac{\Delta p}{\mu L}$ où $\Delta p = p(z=0) - p(z=L)$ est la différence de pression entre l'entrée et la sortie du tube, appelée la perte de charge.

(d) En déduire le profil de vitesse $v(r) = \frac{1}{4}\frac{\Delta p}{\mu L}\left(R^2 - r^2\right)$.

(e) On définit le flux volumique de fluide en intégrant la vitesse à travers la section du tube : $Q = \iint_{\text{section}} v dS$. Établir la relation entre la perte de charge et le flux (loi de Poiseuille).

4. Jusqu'à présent, nous avons considéré la forme locale des équations de conservation. Alternativement, il est également possible d'aborder la mécanique des fluides par une approche intégrale, c'est à dire en considérant des bilans (masse, quantité de mouvement) à l'échelle d'un volume fini. Ainsi, sous forme intégrale pour un volume Ω, la conservation du moment cinétique suivant la direction z s'écrit :

$$\iiint_\Omega \frac{\partial(\rho_0 v_z)}{\partial t} dV + \iint_{\partial\Omega} \rho_0 v_z \mathbf{v}.\mathbf{n} dS = \iint_{\partial\Omega} f_z dS + \iiint_\Omega \rho_0 \mathbf{g}.\mathbf{e}_z dV$$

où :

- Le premier terme représente la variation de moment dans le volume Ω en fonction du temps.
- Le second terme représente le flux de moment à travers le bord du domaine Ω, avec $\rho_0 v_z \mathbf{v}$ le vecteur flux et $\mathbf{n}$ la normale au bord $\partial\Omega$.
- Le troisième terme représente les forces selon z exercées sur le système au bord du domaine Ω, avec f_z est la force par unité de surface suivant z s'exerçant sur le fluide au bord $\partial\Omega$. Pour un fluide newtonien, le vecteur force $\mathbf{f}$ se décompose sous la forme $\mathbf{f} = -p\mathbf{n} + \mu\left(\nabla\mathbf{v} + (\nabla\mathbf{v})^t\right).\mathbf{n}$ avec $\nabla\mathbf{A} = \begin{pmatrix} \frac{\partial A_r}{\partial r} & \frac{\partial A_r}{r\partial\theta} - \frac{A_\theta}{r} & \frac{\partial A_r}{\partial z} \\ \frac{\partial A_\theta}{\partial r} & \frac{\partial A_\theta}{r\partial\theta} + \frac{A_r}{r} & \frac{\partial A_\theta}{\partial z} \\ \frac{\partial A_z}{\partial r} & \frac{\partial A_z}{r\partial\theta} & \frac{\partial A_z}{\partial z} \end{pmatrix}$ l'opérateur gradient d'un vecteur.
- Le dernier terme représente les forces de volume (en l'occurrence la gravité) exercées au sein du volume Ω.

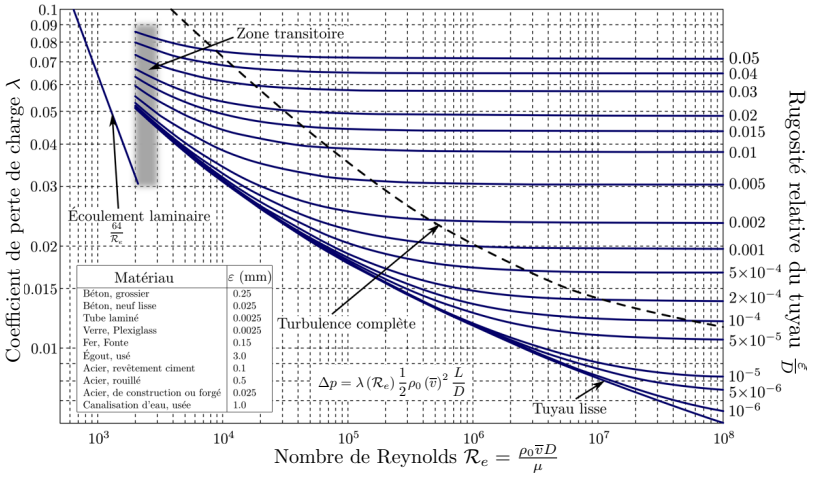
(a) Comme précédemment, on suppose l'écoulement stationnaire et le fluide incompressible. En considérant comme domaine Ω un cylindre d'axe (Oz) et de rayon r compris entre $z = 0$ et $z = L$, montrer que les deux premiers termes sont nuls.

(b) Comme précédemment, on néglige la gravité. Calculer le terme de force au bord du cylindre et en déduire une relation entre la perte de charge Δp et la dérivée de la vitesse $\frac{dv}{dr}$.

(c) Montrer alors que l'on retrouve bien le profil de vitesse de la loi de Poiseuille.

5. Plus généralement, quel que soit la nature de l'écoulement dans un tube (laminaire ou turbulent), on peut montrer par analyse dimensionnelle que la perte de charge est toujours de la forme $\Delta p = \lambda(\mathcal{R}_e)\,\frac{1}{2}\rho_0\,(\overline{v})^2\,\frac{L}{D}$ (équation de Darcy-Weisbach) où $\overline{v} = \frac{Q}{\pi R^2}$ est la vitesse moyenne, $D = 2R$ est le diamètre du tube, et $\lambda(\mathcal{R}_e)$ est un facteur multiplicatif sans dimension qui ne dépend que du nombre de Reynolds $\mathcal{R}_e = \frac{\rho_0 \overline{v} D}{\mu}$. Que vaut $\lambda(\mathcal{R}_e)$ dans le cas de la loi de Poiseuille ? Commenter l'équation de Darcy-Weisbach.

6. Le diagramme de Moody (ci-dessous, Wikimedia commons, CC BY-SA 4.0) représente le coefficient de perte de charge λ en fonction du nombre de Reynolds. Par ailleurs, en régime turbulent, λ dépend également de la rugosité de la surface du tube ε/D (avec ε la taille caractéristique des aspérités). Dans le diagramme de Moody, l'effet de la rugosité est représenté par une série de courbe à différentes valeurs de ε/D. Commenter ce diagramme.



7. Répondre aux cas d'application suivants en s'appuyant sur le diagramme de Moody :

(a) Conduite d'eau urbaine : en ville, la distribution d'eau ($\rho_0 = 1000$ kg/m^3, $\mu = 1$ mPa.s) se fait avec des tuyaux en fonte ($D = 10$ cm, $\varepsilon = 0.15$ mm) à une vitesse moyenne de l'ordre de $\overline{v} = 1$ m/s. La pression typique du réseau est de l'ordre de $p_0 = 5$ bar. Sachant que la pression distribuée à l'usager ne doit pas descendre en dessous de $p_{min} = 3$ bar, estimer la longueur de tuyau telle que la perte de charge reste acceptable.

(b) Oléoduc : le transport de pétrole ($\rho_0 = 850$ kg/m^3, $\mu = 30$ mPa.s) par oléoduc ($D = 1$ m, $\varepsilon = 0.15$ mm, $\overline{v} = 2$ m/s) nécessite des stations de pompage réparties tous les $L = 100$ km environ pour compenser les pertes de charge. Quelle puissance mécanique ces stations doivent-elles être en mesure de fournir ?

(c) Réfrigérant de climatisation : une climatisation doit distribuer un fluide réfrigérant ($\rho_0 = 950$ kg/m^3, $\mu = 0.15$ mPa.s) entre une unité extérieure et une unité intérieure à travers un petit tuyau en cuivre ($D = 6$ mm, $\varepsilon = 0.0025$ mm) de longueur maximum $L = 30$ m. La climatisation doit fournir une puissance thermique de $\mathcal{P}$ par évaporation du fluide réfrigérant (chaleur latente d'évaporation $c = 150$ kJ/kg) tout en limitant la perte de charge à $\Delta p = 1$ bar. Déterminer la puissance thermique maximum de la climatisation.

Chapter 2

Rayonnement

Contents

Foreword **27**
2.1 Basic thermodynamics of solar energy **28**
2.1.1 Blackbody radiation 28
2.1.2 Rough estimate of the solar resource 32
2.1.3 A more accurate approach required for concentration 34
2.2 Solar thermal energy **35**
2.2.1 Turning solar radiation into heat 35
2.2.2 Turning solar radiation into work 36
2.3 Solar photovoltaic **37**
2.3.1 Short introduction to semi-conductor physics 37
2.3.2 The Trivich-Finn and Shockley-Queisser limits 38
2.3.3 From Shockley-Queisser to real devices 40
2.3.4 Solar cells today 41
Recap of what must be known for the exam **42**
Exercises **45**
Ex. 2.1 - Panneau solaire thermique 45
Ex. 2.2 - Limite de Shockley-Queisser 46
Ex. 2.3 - Forçage radiatif du CO_2 et réchauffement 49

Foreword

This chapter is dedicated to the conversion of solar energy. Solar radiation is by far the largest contributor to the energy budget on Earth, exceeding by a factor 10 000 the heat flux provided by radioactive processes inside Earth's crest and core. It generates most of the energy sources at our disposal, either directly (solar energy) or indirectly (wind, hydro, biomass, and subsequent degradation of the biomass into fossil resources).

In this chapter, we first review the basic thermodynamics of radiation (section 2.1). In particular, we introduce Planck's law, which predicts the blackbody spectrum, as a first approximation of solar radiation, and evaluate the solar energy resource. We then consider the direct conversion of solar radiation into heat, with and without concentration, and we provide estimates of the associated efficiencies (section 2.2). After a short introduction to the physics of semiconductors, we then consider the conversion into electricity with photovoltaic cells (section 2.3). We discuss the choice of the gap energy and the Trivich-Finn and Shockley-Queisser limits. We finish with a state of the art of current solar technologies: the actual efficiencies, the development of solar energy on the global scale, the financial costs and material budget, and the future perspectives with respect to the energy transition.

2.1 Basic thermodynamics of solar energy

2.1.1 Blackbody radiation

A bit of history

The discovery of Planck's law describing the blackbody radiation is considered as a landmark in the history of sciences as it constitute the very first step in the construction of the theory of quantum mechanics.

Why incandescent matter glows was a subject of intense debate at the end of the 19^{th} century. Precise measurements showed that the color of incandescent matter, from lava flow to burning wood or steel, does not depend on the matter itself, but primarily on temperature. This is the same phenomenon due to which our own bodies emit infrared light. In 1862, Gustav Kirchhoff introduced the notion of "black body", a somewhat non-intuitive terminology, since a black body refers to an idealized lump of material that, like hot steel, may emit light. The adjective "black" merely indicates here that this lump of material does not reflect incoming light, hence the light it emits may only be attributed to the radiation of internal energy. Light was then already understood to be an electromagnetic radiation governed by Maxwell's equations. Kirchhoff observed that an object receiving electromagnetic radiation (e.g., a coin in the sunlight) does not heat up indefinitely: it eventually reaches a stable temperature; in that state, it must emit as much energy (light) as it receives. He thus considered that a black body should contain energy in the form of electromagnetic waves; and that these waves should be able to escape when reaching its surface from inside. He also showed that since incident light is not reflected, by assumption, outgoing light could not be reflected either as otherwise the body would constitute an energy trap and would heat up indefinitely. Tremendous efforts followed to understand how the glow of hot metals could be explained by radiated electromagnetic waves.

All these descriptions had to deal with the fact that, in a cube of size L^3, the possible standing electromagnetic waves have discrete wavevectors $\vec{k} = \pi\vec{n}/L$ with $\vec{n} \in \mathbb{N}^3$ a 3D vector with integer coordinates. The associated angular frequency is $\omega = ck$ with $k = \|\vec{k}\|$ and $c = 3\times10^8$ m/s the speed of light. In a large enough system, the number N of modes of wavevector amplitude $k < K = \omega/c$ is estimated from the volume of the radius-K sphere divided by the volume π^3/L^3 associated with the discretization of the $\vec{k}$ space, yielding:

$$N \approx 2 \times \left(\frac{1}{8}\frac{4\pi}{3}K^3\right) \times \frac{L^3}{\pi^3} = \frac{1}{3\pi^2}\frac{\omega^3 L^3}{c^3}$$

where the prefactor $1/8$ accounts the fact that the components of $\vec{n}$ (and $\vec{k}$) are positive, and the prefactor 2 accounts for the polarization of electromagnetic waves. Finally, the density of modes per unit volume and per unit angular frequency can be estimated as:

$$D(\omega) = \frac{1}{L^3}\frac{dN}{d\omega} = \frac{\omega^2}{\pi^2 c^3} \tag{2.1}$$

A classical (but erroneous) estimate for the radiated energy presumes that each electromagnetic wave carries a fraction $k_B T$ of the heat, with $k_B = 1.38 \times 10^{-23}$ J/K the Boltzmann's constant and T the temperature[1]. If so, the density of electromagnetic energy per unit volume and angular frequency is $\frac{\omega^2}{\pi^2 c^3} k_B T$, and thus the radiance (i.e., the energy emitted at the surface per unit area, angular frequency and time) is given by:

$$\mathcal{P}_\omega(T) = \frac{c}{4} \times \frac{\omega^2}{\pi^2 c^3} k_B T = \frac{\omega^2}{4\pi^2 c^2} k_B T$$

where the prefactor $c/4$ accounts for the part of the electromagnetic radiation that is emitted at the surface per unit time. This radiance is obviously wrong since it predicts that the total radiated

[1] This result arises from the equipartition theorem of statistical physics, which applies to standing electromagnetic waves whose energy $\int_V \left(\frac{\varepsilon_0 E^2}{2} + \frac{B^2}{2\mu_0}\right) dr^3$ exhibits two quadratic terms (see class of statistical physics). Note that the equipartition theorem assumes that the electric and magnetic fields E and B can take any value (i.e., no quantification), and therefore it is no more valid under Planck's assumption.

energy, which is the integral of this expression over all angular frequencies, is infinite. Moreover, it predicts the density of radiated energy to become infinite at high frequencies, which is called the ultraviolet catastrophe.

In 1900, Max Planck introduced an assumption that would revolutionize Physics. In his own words ("a purely formal assumption [...] actually I did not think much about it..."), it was just meant to formally regularize the ultraviolet catastrophe. He assumed that electromagnetic energy at frequency $\nu = \omega/(2\pi)$ could only be absorbed or emitted as "packets" of an energy E given by what is now known as the Planck-Einstein relation:

$$E = h\nu = \hbar\omega$$

with $h = 6.63 \times 10^{-34}$ a constant called the Planck constant, and $\hbar = h/(2\pi)$ the reduced Planck constant.

In this view, a mode of angular frequency ω may contain any number n of such packets, i.e., present an energy $E = n\hbar\omega$. Using the recently proposed statistical physics, he then calculated the associated partition function[2]:

$$Z = \sum_{n=0}^{\infty} \exp(-\beta n\hbar\omega) = \frac{1}{1 - \exp(-\beta\hbar\omega)}$$

with $\beta = 1/(k_B T)$. This predicts that the average energy contained in modes of angular frequency ω is:

$$E(\omega) = -\left.\frac{\partial \ln Z}{\partial \beta}\right|_{\omega} = \frac{\hbar\omega}{\exp(\beta\hbar\omega) - 1}$$

which vanishes in the $\omega \to \infty$ limit. Using this estimate with the density of modes of Eq. (2.1) it yields, for the radiance:

$$\mathcal{P}_{\omega}(T) = \frac{\hbar\omega^3}{4\pi^2 c^2}\left(\exp\left(\frac{\hbar\omega}{k_B T}\right) - 1\right)^{-1} \tag{2.2}$$

This relation, now known as Planck's law, fits the radiated energy of incandescent objects, and accounts remarkably for the light spectrum of stars!

Another way to examine light-matter interactions is to consider cases when light is absorbed and transformed into a measurable form of energy. In 1839, Edmond Becquerel observed that, when illuminated by light, coated electrodes could generate voltage and electric current. This is the photovoltaic effect, which lies at the basis of solar cells. A related, but different, situation is the photoelectric effect, discovered by Heinrich Hertz in 1887, in which light absorption leads to the instantaneous ejection of electrons outside the electrode. This phenomenon garnered a tremendous amount of interest at the turn of the twentieth century, leading to the identification of several laws governing it. It was thus recognized that: (a) the photoelectric current is proportional to the intensity of incident light; (b) emission occurred only if the light frequency was above a certain, material-dependent, threshold ν_0; (c) above this threshold, the maximum kinetic energy K^{max} of the ejected electrons is a function of the light frequency, and more precisely of the form $K^{max} \propto \nu - \nu_0$. While the first law is consistent with electromagnetism, the last two are not. Within a classical interpretation of the phenomenon, indeed, the energy of the incoming light does not depend on its frequency, but on its intensity. The latter quantity only should hence determine the properties of the photoelectric current.

In 1905, a few years after Planck's theory of radiation, Albert Einstein explained the photoelectric effect by proposing that Planck's assumption reflected a physical reality: light is emitted or

[2]Here we refer to the canonical partition function, defined as $Z = \sum_i \exp(\beta E_i)$, where the sum is over all possible microstates i and E_i is the energy of the microstate. Here, the microstates are all the possible number of quanta n and the associated energy is $n\hbar\omega$. The partition function is a central concept of statistical physics, which is related to usual thermodynamics thanks to its relationship with the Helmholtz free energy: $F = -k_B T \ln(Z)$. Here, we use in particular the relation $E = -\left.\frac{\partial \ln Z}{\partial \beta}\right|_{\omega}$ relating the partition function to the average energy of a mode of angular frequency ω. See the class of statistical physics for more details.

absorbed as particles of energy $h\nu$, which are now called "photons". Einstein reasoned that electrons would have a binding energy E_0 in the electrode and that this energy would have to be paid to extract them so that they could be produced with the maximum kinetic energy $K^{max} = h\nu - E_0$. Identifying $E_0 = h\nu_0$ explains that electrons cannot be emitted for light of frequency $\nu < \nu_0$ and yields $K^{max} = h(\nu - \nu_0)$. It not only explains the linear empirical relationship but also identifies its prefactor to be Planck's constant.

It cannot be sufficiently stressed how remarkable it is that the coefficient h that fits radiative spectra (Planck's law) and the coefficient h that fits the linear $K^{\max}$ vs $\nu - \nu_0$ relationship in photoelectric emission are one and the same! This identifies a universal constant and establishes the physical reality of photons.

Planck's law

The above formulation of Planck's law (Eq. 2.2) provides the power density emitted at the surface of a blackbody in all directions per unit area and per unit angular frequency. In what follows, we will consider a formulation per unit energy $\hbar\omega$ rather than per unit angular frequency ω, which introduces a multiplicative factor $\hbar^3$; and we will need to consider the spectral density $\mathcal{B}(T)$, i.e., number of photons, rather than the power density $\mathcal{P}(T)$, which introduces an additional multiplicative factor $\hbar\omega$[4]. Moreover, we will also need to consider the radiation emitted in a particular direction and not in all directions, which requires introducing the concept of solid angle Ω characterizing the direction in which the radiation is emitted (Ω can be represented by the angles $\theta \in \left[0; \frac{\pi}{2}\right]$ and $\varphi \in [0; 2\pi]$ in spherical coordinates with the polar direction normal to the surface as depicted in Figure 2.1).

The formulation of Planck's law for the spectral density taking into account the direction of radiation is given by:

$$\mathcal{B}_{\hbar\omega,\Omega}(T) = \frac{\cos(\theta)}{\pi} B_{\hbar\omega}(T) \text{ with } B_{\hbar\omega}(T) = \frac{(\hbar\omega)^2}{4\pi^2\hbar^3c^2}\left(\exp\left(\frac{\hbar\omega}{k_BT}\right) - 1\right)^{-1}$$

$\mathcal{B}_{\hbar\omega,\Omega}(T)\, d\Omega d(\hbar\omega)$ represents the number of photons with energy between $\hbar\omega$ and $\hbar\omega + d(\hbar\omega)$ emitted per unit time and per unit surface in the directions between Ω and $\Omega + d\Omega$ (using the spherical coordinates $d\Omega = \sin(\theta)\, d\theta d\varphi$). And $B_{\hbar\omega}(T)$ represents the spectral density emitted in all directions, obtained by integration over all the possible orientations ($\mathcal{B}_{\hbar\omega}(T) = \int_0^{2\pi}\int_0^{\frac{\pi}{2}} \mathcal{B}_{\hbar\omega,\Omega}(T)\sin(\theta)\, d\theta d\varphi$).

The power density is directly related to the spectral density by multiplying by the energy of a photon $\hbar\omega$: $\mathcal{P}_{\hbar\omega,\Omega}(T) = \hbar\omega\mathcal{B}_{\hbar\omega,\Omega}(T)$. Integrating over all directions gives the power density emitted in all directions: $\mathcal{P}_{\hbar\omega}(T) = \hbar\omega\mathcal{B}_{\hbar\omega}(T)$ (one readily recovers the expression of Eq. 2.2 with $\mathcal{P}_\omega(T) = \hbar\mathcal{P}_{\hbar\omega}(T)$). And further integrating over all energies, one obtains the total power emitted, also known as the Stefan-Boltzmann law:

$$\begin{aligned}\mathcal{P}(T) &= \int_0^{+\infty} \hbar\omega\mathcal{B}_{\hbar\omega}(T)\, d(\hbar\omega) = \int_0^{+\infty} \frac{(\hbar\omega)^3}{4\pi^2\hbar^3c^2}\left(\exp\left(\frac{\hbar\omega}{k_BT}\right) - 1\right)^{-1} d(\hbar\omega) \\ &\underset{X=\frac{\hbar\omega}{k_BT}}{=} \frac{(k_BT)^4}{4\pi^2\hbar^3c^2}\underbrace{\int_0^{+\infty}\frac{X^3}{\exp(X)-1}dX}_{\pi^4/15} = \sigma T^4\end{aligned} \tag{2.3}$$

where $\sigma = \frac{\pi^4}{15}\frac{k_B^4}{4\pi^2\hbar^3c^2} = 5.67 \times 10^{-8}$ W.m^{-2}.K^{-4} is the Stefan-Boltzmann constant. And the total power emitted in a given direction is therefore $\mathcal{P}_\Omega(T) = \frac{\cos(\theta)}{\pi}\sigma T^4$.

[3] $\mathcal{P}_\omega(T)\, d\omega = \mathcal{P}_{\hbar\omega}(T)\, d(\hbar\omega) \Rightarrow \mathcal{P}_{\hbar\omega}(T) = \frac{\mathcal{P}_\omega(T)}{\hbar}$

[4] $\mathcal{P}(T) = \hbar\omega\mathcal{B}(T)$

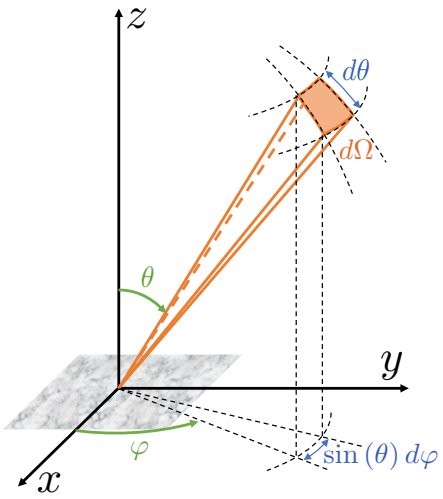


Figure 2.1: Representation of a solid angle in spherical coordinates.

Note that, in the above expressions of Plank's law, the spectrum is expressed in terms of energy $\hbar\omega$. One can also express Planck's law in terms of wavelength λ. The respective spectral densities must satisfy $|\mathcal{B}_{\hbar\omega,\Omega} d(\hbar\omega)| = |\mathcal{B}_{\lambda,\Omega}(T)\, d\lambda|$. Introducing the relationship between the wavelength and the energy $\hbar\omega = \frac{hc}{\lambda}$, one gets:

$$\mathcal{B}_{\lambda,\Omega}(T) = \frac{\cos(\theta)}{\pi}\mathcal{B}_{\lambda}(T) \text{ with } \mathcal{B}_{\lambda}(T) = \frac{2\pi c}{\lambda^4}\left(\exp\left(\frac{hc}{\lambda k_B T}\right) - 1\right)^{-1}$$

$$\mathcal{P}_{\lambda,\Omega}(T) = \frac{\cos(\theta)}{\pi}\mathcal{P}_{\lambda}(T) \text{ with } \mathcal{P}_{\lambda}(T) = \frac{2\pi hc^2}{\lambda^5}\left(\exp\left(\frac{hc}{\lambda k_B T}\right) - 1\right)^{-1}$$

The two representations of Planck's law are strictly equivalent, and both are often used. Be careful however to note that the units are not the same: $\mathcal{B}_{\hbar\omega}(T)$ is a quantity per unit area, time and energy ($\mathrm{m}^{-2}.\mathrm{s}^{-1}.\mathrm{J}^{-1}$ in SI units), whereas $\mathcal{B}_{\lambda}(T)$ is a quantity per unit area, time and wavelength ($\mathrm{m}^{-3}.\mathrm{s}^{-1}$ in SI units).

Kirchhoff's law

While a blackbody is defined as the best possible absorber (and emitter), it is necessary to also consider gray bodies, with non-unit absorptivity. The absorptivity $\alpha(\hbar\omega, \Omega)$ is defined as the fraction of incident photons successfully absorbed by the system when illuminated with a monochromatic radiation at frequency ω from an incident direction Ω:

$$\alpha(\hbar\omega, \Omega) = \frac{\text{absorbed photons}}{\text{incident photons with energy } \hbar\omega \text{ coming from direction } \Omega}$$

For a blackbody, $\alpha(\hbar\omega, \Omega) = 1$ for all wavelengths and from all directions. On the contrary, a greybody is defined as a body that does not absorb all the photons, i.e., $\alpha(\hbar\omega, \Omega) < 1$. Conversely, a greybody emits less radiation than a blackbody at the same temperature. The emissivity $\varepsilon(\hbar\omega, \Omega)$ represents the ability of the body to emit a radiation at frequency ω in the direction Ω compared to that of a blackbody:

$$\mathcal{B}_{\text{greybody}} = \varepsilon(\hbar\omega, \Omega)\, \mathcal{B}_{\hbar\omega,\Omega}(T)$$

Kirchhoff's law of radiation stipulates that the ability of a body to absorb any wavelength

in any direction is equal to its ability to emit the same wavelength in the same direction:

$$\alpha(\hbar\omega, \Omega) = \varepsilon(\hbar\omega, \Omega)$$

The proof of this theorem is actually not straightforward, and the angular part requires additional assumptions on the problem. But the Kirchhoff law averaged over all directions ($\alpha(\hbar\omega) = \varepsilon(\hbar\omega)$) is not too complicated to derive and can be obtained from a careful energy balance. Let us consider a body with emissivity $\varepsilon_1(\hbar\omega)$ and absorptivity $\alpha_1(\hbar\omega)$. The energy balance for an absorber in equilibrium at temperature T_{abs} exposed to a thermal radiation at temperature T_{rad} imposes:

$$\int_0^{+\infty} \varepsilon_1(\hbar\omega)\,\mathcal{P}_{\hbar\omega}(T_{abs})\,d(\hbar\omega) = \int_0^{+\infty} \alpha_1(\hbar\omega)\,\mathcal{P}_{\hbar\omega}(T_{rad})\,d(\hbar\omega)$$

which is not enough to conclude the equality between emissivity and absorptivity at all energies. Let us now put the body inside a cavity, the radiative behavior of which is characterized by $\varepsilon_2(\hbar\omega)$ and $\alpha_2(\hbar\omega)$. We assume thermal equilibrium at temperature T between the body and the cavity.

The spectral irradiance arriving on the body is $\mathcal{P}^{in}_{\hbar\omega} = \varepsilon_2(\hbar\omega)\,\mathcal{P}_{\hbar\omega}(T) + (1-\alpha_2(\hbar\omega))\,\mathcal{P}^{out}_{\hbar\omega}$ where the spectral irradiance reaching the cavity is expressed as $\mathcal{P}^{out}_{\hbar\omega} = \varepsilon_1(\hbar\omega)\,\mathcal{P}_{\hbar\omega}(T) + (1-\alpha_1(\hbar\omega))\,\mathcal{P}^{in}_{\hbar\omega}$. Solving this simple system leads to:

$$\begin{cases} \mathcal{P}^{in}_{\hbar\omega} = \frac{\varepsilon_2(\hbar\omega)+\varepsilon_1(\hbar\omega)(1-\alpha_2(\hbar\omega))}{1-(1-\alpha_1(\hbar\omega))(1-\alpha_2(\hbar\omega))}\mathcal{P}_{\hbar\omega}(T) \\ \mathcal{P}^{out}_{\hbar\omega} = \frac{\varepsilon_1(\hbar\omega)+\varepsilon_2(\hbar\omega)(1-\alpha_1(\hbar\omega))}{1-(1-\alpha_1(\hbar\omega))(1-\alpha_2(\hbar\omega))}\mathcal{P}_{\hbar\omega}(T) \end{cases}$$

and the energy balance $\int_0^{+\infty} \mathcal{P}^{in}_{\hbar\omega} d(\hbar\omega) = \int_0^{+\infty} \mathcal{P}^{out}_{\hbar\omega} d(\hbar\omega)$ requires:

$$\int_0^{+\infty} \frac{\alpha_1(\hbar\omega)\,\alpha_2(\hbar\omega)}{1-(1-\alpha_1(\hbar\omega))(1-\alpha_2(\hbar\omega))}\left(\frac{\varepsilon_1(\hbar\omega)}{\alpha_1(\hbar\omega)} - \frac{\varepsilon_2(\hbar\omega)}{\alpha_2(\hbar\omega)}\right)\mathcal{P}_{\hbar\omega}(T)\,d(\hbar\omega) = 0$$

The trick is that this relation holds for whatever cavity material, i.e.; whatever $\varepsilon_2(\hbar\omega)$ and $\alpha_2(\hbar\omega)$. This is possible if and only if $\varepsilon \propto \alpha$, and the first equation imposes $\varepsilon(\hbar\omega) = \alpha(\hbar\omega)$ for each energy, hence the direction-averaged Kirchhoff law of radiation.

2.1.2 Rough estimate of the solar resource

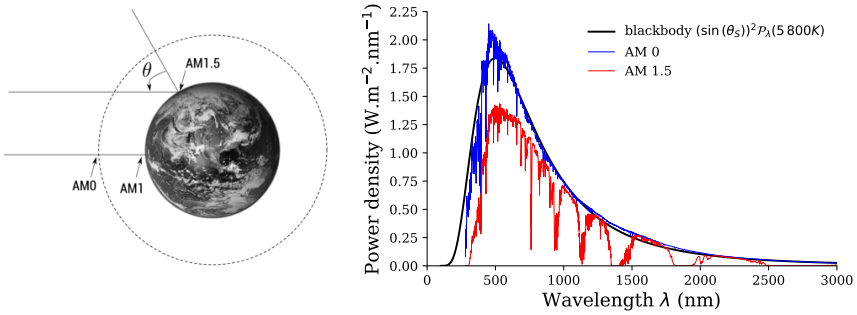


Figure 2.2: Solar spectrum reaching the Earth

Using the Stefan-Boltzmann law (Eq. 2.3), we can reach a first estimate of the solar resource. The Sun has a radius of $R_\odot = 700\,000$ km, and the temperature of the photo-sphere reaches $T_\odot = 5\,800$K (see Fig. 2.2). The total power radiated by the Sun is thus:

$$\mathcal{P}_\odot = 4\pi R_\odot^2 \times \sigma T_\odot^4 \approx 4 \times 10^{26}\ \text{W}$$

The Earth is located $D_{T\odot} = 150 \times 10^6$ km away from the Sun, and receives a solar flux:

Location	Average illumination (W/m^2)
At the top of the atmosphere facing the Sun (AM 0)	1360
At ground level facing the Sun (AM 1.5)	1000
Average value in Arizona, US (with day/night cycle, weather and seasons)	200
Average value in France (idem)	150
Average value in Germany (idem)	100
Average value in Sichuan, China (idem)	50

Table 2.1: Average direct illumination at different locations on Earth (data from https://globalsolaratlas.info/)

$$\phi_{\text{AM 0}} = \frac{\mathcal{P}_\odot}{4\pi \left(D_{T-\odot}\right)^2} = 1\,360 \text{ W/m}^2$$

This value is known as the solar constant. It corresponds to the power per unit area received at the top of the atmosphere when facing the Sun ('AM 0' spectrum). The total solar power reaching the Earth is thus:

$$\mathcal{P}_{\odot\to T} = \pi R_T^2 \times \phi_{\text{AM0}} \approx 10^{17} \text{ W}$$

This by far the main energy input to Earth. Recalling that the average power consumption of humanity is currently $\sim 10^{13}$ W, the order of magnitude of the solar input is roughly 10 000 times larger than our current needs.

When propagating through the atmosphere, sunlight is partially absorbed and only a fraction of the incident power actually reaches the surface. This fraction depends on the considered wavelength and on the optical distance traveled by the light. The minimal distance, reached at the equator when facing the Sun, corresponds simply to the atmosphere thickness $w_{atm} \approx 9$ km, and results in the 'AM 1' spectrum. Further away from the equator, sunlight has to travel through an increasing thickness of the atmosphere, resulting in a stronger absorption. The corresponding spectra are labeled as a function of the Air Mass (AM), i.e., the ratio between the effective atmosphere thickness crossed by light, and the actual thickness of the atmosphere. This value depends on the latitude θ as:

$$\text{AM} = \sqrt{\left(\frac{w_{atm}}{R_T}\cos\left(\theta\right)\right)^2 + 2\frac{w_{atm}}{R_T} + 1} - \frac{w_{atm}}{R_T}\cos\left(\theta\right)$$

For solar application, the convention is usually to consider the 'AM 1.5' spectrum (see Fig. 2.2), which corresponds to an integrated irradiance of:

$$\phi_{\text{AM 1.5}} = 1\,000 \text{ W/m}^2$$

This value corresponds to irradiance perceived at a latitude of 48° when facing the Sun.
At a latitude θ, a surface lying on the ground when the Sun is at the zenith receives an irradiance:

$$\phi_{horizontal}^{max} = \cos\left(\theta\right)\phi_{\text{AM 1.5}}$$

Due to the rotation of the Earth, the Sun is not always at the zenith. The average irradiance is:

$$\phi_{horizontal}^{average} = \frac{1}{2\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos\left(\varphi\right)\phi_{horizontal}^{max} d\varphi = \frac{\cos\left(\theta\right)}{\pi}\phi_{horizontal}^{max}$$

On top of these geometrical effects, weather conditions will decrease the average illumination by 30-50%. Numerical values for different locations on Earth are presented in Table 2.1. The difference between the maximal irradiance (obtained at noon in mid-June when facing the Sun with

no clouds) and the average irradiance (taking clouds, day/night cycles, Sun's position...) explains the difference between the installed capacity and the actual energy production of an installation. The installed capacity is defined as the maximal power that the installation is able to produce in the best possible conditions - which occurs once a year, or less. The ratio between the installed capacity and the actual average power is called the load factor, or capacity factor:

$$LF = \frac{\text{average power}}{\text{installed capacity}} = \frac{\text{yearly energy output}}{\text{installed capacity } \times 24 \times 365[\text{h/yr}]}$$

and the typical capacity factor for solar is 10-20%.

As a simple application, we can use these figures to calculate the surface of solar panels required to provide as much electricity throughout the year as the French domestic consumption. France uses around 530 TWh of electricity per year, corresponding to an average power of 60 GW. Assuming a power conversion efficiency of about 20% and an average illumination of 150 W/m^2 (or equivalently assuming an installed capacity of 200 W/m^2 with a load factor of 15%), this corresponds to 2 000 km^2 of solar panels, to be compared to the 20 000 km^2 covered with roads. Note however that this value corresponds only to a yearly average balance, and does not mean that covering 2 000 km^2 with solar panels would be sufficient to power the electrical system.

2.1.3 A more accurate approach required for concentration

The previous calculation evaluates the fraction of the solar power reaching the Earth as the ratio between the surface of the Earth and that of a sphere with a radius equal to the distance between the Earth and the Sun. The approximation holds as long as the Sun occupies only a small solid angle in the sky. With an angular radius $\sin(\theta_S) = \frac{R_\odot}{D_{T-\odot}} \approx 0.26°$, this assumption is valid in the absence of concentrating optics. However, a more careful estimation is required when concentrators are involved.

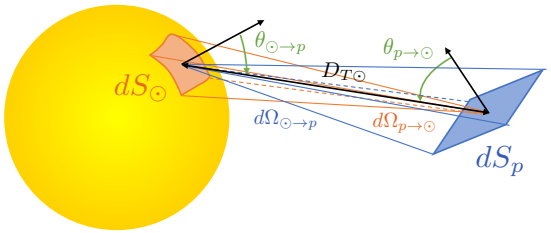


Figure 2.3: Schematic representation of a surface element of a solar panel receiving sunlight from a surface element of the Sun.

To do so, let us evaluate the power received by a small surface element of a solar panel dS_p from a small surface element of the Sun $dS_\odot$ separated by a distance $D_{T-\odot}$. The Sun's surface emits radiation in all directions, and only a fraction is emitted towards the direction Ω of the panel element (see Fig. 2.3).Following Planck's law (Eq. 2.3), the radiation emitted by the surface element of the Sun in the direction of the surface element of the panel is:

$$d\mathcal{P}^{dS_\odot\to dS_p} = \mathcal{P}_\Omega(T_\odot)\, d\Omega_{\odot\to p} dS_\odot = \frac{\cos(\theta_{\odot\to p})}{\pi}\sigma T_\odot^4 d\Omega_{\odot\to p} dS_\odot$$

A solid angle is defined by the area dS of the surface it intercepts at a distance r: $d\Omega = \frac{\vec{r}\cdot\vec{n}}{r^3}dS$, with $\vec{r}$ the vector of the source at the surface and $\vec{n}$ the normal to the surface. Accordingly, the solid angle $d\Omega_{\odot\to p}$ is related to the surface element of the panel by $d\Omega_{\odot\to p} = \frac{\cos(\theta_{p\to\odot})dS_p}{(D_{T\odot})^2}$. Conversely, the solid angle from which the panel sees the surface element of the Sun is $d\Omega_{p\to\odot} = \frac{\cos(\theta_{\odot\to p})dS_\odot}{(D_{T\odot})^2}$. Thus, we have $\cos(\theta_{\odot\to p})\, d\Omega_{\odot\to p} dS_\odot = \cos(\theta_{p\to\odot})\, d\Omega_{p\to\odot} dS_p$,

so one can rewrite the power transmitted from the point of view of the panel: $d\mathcal{P}^{dS_\odot \to dS_p} = \frac{\cos(\theta_{p\to\odot})}{\pi}\sigma T_\odot^4 d\Omega_{p\to\odot} dS_p$

In the ideal case of a panel oriented normal to the direction of the Sun, we have $\theta_{p\to\odot} \in [0;\theta_S]$, where θ_S is the angle occupied by the radius of the Sun in the sky, and $\varphi_{p\to\odot} \in [0;2\pi]$. Integrating over all the possible values of $\theta_{p\to\odot}$ and $\varphi_{p\to\odot}$, one obtains the power received per unit surface of the panel:

$$\begin{aligned}\phi_{1\odot} = \frac{d\mathcal{P}^{\odot\to dS_p}}{dS_p} &= \int_0^{2\pi}\int_0^{\theta_S} \frac{\cos(\theta_{p\to\odot})}{\pi}\sigma T_\odot^4 \sin(\theta_{p\to\odot})\, d\theta_{p\to\odot} d\varphi_{p\to\odot} \\ &= \frac{\sigma T_\odot^4}{\pi}\int_0^{2\pi} d\varphi_{p\to\odot}\int_0^{\theta_S}\frac{\sin(2\theta_{p\to\odot})}{2} d\theta_{p\to\odot} = \frac{\sigma T_\odot^4}{\pi}(2\pi)\underbrace{\frac{1-\cos(2\theta_S)}{4}}_{\frac{1}{2}(\sin(\theta_S))^2} \\ &= (\sin(\theta_S))^2 \sigma T_\odot^4\end{aligned} \tag{2.4}$$

This expression gives an upper bound for the ability of an optical system to concentrate sunlight. In the absence of concentration, the solar flux corresponds to $\phi_{1\odot} = (\sin(\theta_S))^2 \sigma T_\odot^4 = 1\,360$ W/m^2. Under maximal concentration, the Sun seems to occupy the full sky and $\theta_{p\to\odot} \in \left[0;\frac{\pi}{2}\right]$, leading to a 46 000 times increase of power density: $\phi_{46\,000\odot} = \sigma T_\odot^4 = 6\times 10^7$ W/m^2. And it is not thermodynamically possible to increase solar power above this value with simple optics. In between these two values, the concentration factor for a system such that the Sun appears with an angular radius θ_C in the sky is defined as:

$$1 \leq C = \left(\frac{\sin(\theta_C)}{\sin(\theta_S)}\right)^2 \leq \frac{1}{(\sin(\theta_S))^2} = 46\,000$$

2.2 Solar thermal energy

2.2.1 Turning solar radiation into heat

Solar thermal energy, or solar heat, aims at collecting the incident radiation into usable heat. Let us consider a solar panel as a greybody of area S_p, ideally oriented orthogonal to the sunlight, with a concentrator of concentration factor C. Using the results of the previous section, we can estimate the power absorbed by the panel as:

$$\mathcal{P}_{received} = \alpha_p C (\sin(\theta_S))^2 \sigma T_\odot^4 S_p$$

where α_p is the absorptivity of the panel, assumed to be constant over all wavelengths. Because of its finite temperature T_p, the panel also radiates and the corresponding power loss is:

$$\mathcal{P}_{loss} = \varepsilon_p \sigma T_p^4 S_p$$

where $\varepsilon_p = \alpha_p$ is the emissivity of the panel.

These powers allow us to estimate the equilibrium temperature of a panel with no concentration ($C = 1$) simply put under sunlight. Indeed, assuming no other energy exchanges:

$$T_p = \sqrt{\sin(\theta_S)}T_\odot = 122°\text{C}$$

Under the Sun, the equilibrium temperature is 122°C, which means that it is impossible to reach a higher temperature simply with unconcentrated sunlight. Under full concentration ($C(\sin(\theta_S))^2 = 1$), the temperature tends toward $T_\odot$. It is of course impossible to reach a temperature higher than that of the Sun using even concentrated sunlight!

Let us now consider that some energy is extracted from the panel, providing a heat flux $\dot{Q}$. This heat can then be used to provide hot water or cook food in a solar oven... The energy balance is now:

$$\mathcal{P}_{received} = \mathcal{P}_{loss} + \dot{Q} \Rightarrow \dot{Q} = \mathcal{P}_{received} \underbrace{\left(1 - \frac{1}{C\left(\sin\left(\theta_S\right)\right)^2}\left(\frac{T_p}{T_\odot}\right)^4\right)}_{\eta_Q}$$

The corresponding efficiency η_Q is shown in Figure 2.4. Obviously, the colder the panel, the better the efficiency as less heat is wasted as radiation. Note that under 1 Sun, the efficiency is limited to 66% even if the absorber remains at the ambient 300K. Note also that increasing the concentration increases not only the incident power but also the efficiency. It seems that concentration is nothing but beneficial to the system and that any installation should aim at the largest possible concentration. However, this thermodynamic argument neglects practical considerations. In particular:

- Only direct light, not diffuse light, can be effectively concentrated. When a cloud passes in front of the Sun, a concentrated system is more affected than an unconcentrated one.
- The concentrating system focuses the solar disk into a spot. As the Sun moves through the sky, this spot will move away from the absorber unless a tracking system re-orientates the concentrating optics. This requires an additional mechanical system, which can be a weak point for the installation.
- Optics require maintenance to be kept clean and avoid optical losses.

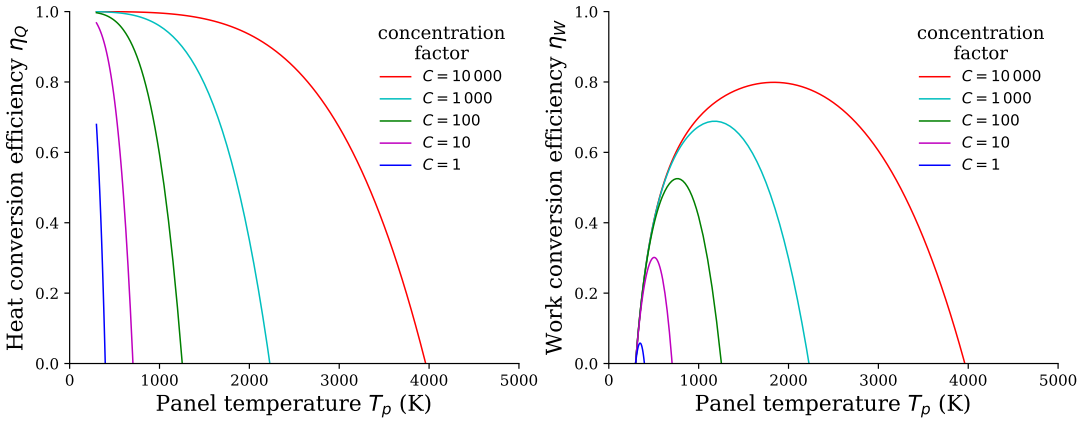


Figure 2.4: Conversion efficiency of solar radiation into heat η_Q and work η_W as a function of the temperature of a blackbody panel at different optical concentrations.

2.2.2 Turning solar radiation into work

The heat flux $\dot{Q}$ can also be used to power a heat engine and to produce some work. The maximal efficiency for this conversion is provided by a Carnot engine:

$$\eta_{Carnot} = 1 - \frac{T_0}{T_p}$$

where Tp is the temperature of the panel and T_0 is the ambient temperature, acting as a cold

source. The efficiency of this "Muzer engine", composed of an ideal absorber and an ideal converter is:

$$\dot{W} = \mathcal{P}_{received} \times \underbrace{\eta_Q \times \eta_{Carnot}}_{\eta_W}$$

and the numerical values are plotted in Figure 2.4.

As compared to the previous situation, where we were happy with just a heat flux $\dot{Q}$, getting work out of solar radiation is much more difficult, and requires a trade-off: if the temperature of the panel is too high, too much energy is lost as radiation and the efficiency η_Q is bad. But if the panel is too cold, the temperature difference between the hot and the cold sources for the Carnot engine is too reduced, and the efficiency η_{Carnot} is poor. In the absence of concentration, this trade-off limits the efficiency below 6%, and concentration appears absolutely compulsory to reach a satisfying efficiency. Under maximum concentration, the maximal efficiency reaches 86%.

2.3 Solar photovoltaic

2.3.1 Short introduction to semi-conductor physics

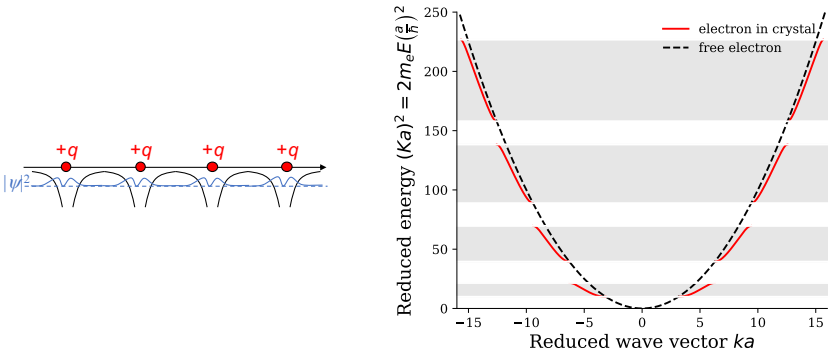


Figure 2.5: (left) Schematic representation of an electron in a periodic lattice. (right) Band and gap structure as a function of the wave vector ka for the Dirac comb problem (spinless particle of mass m_e in 1D submitted to an infinite series of Dirac wells periodically distributed with period a).

As discussed in the lesson about chemical energy, the behavior of electrons in matter is governed by the laws of quantum mechanics. In particular, the electrons can be found only in discrete states of well-defined energies (electronic orbitals). A particular situation is that of crystalline material, which exhibits a periodic lattice and, therefore, the Hamiltonian is periodic in space (see section dedicated to metallic bonds in the lesson about chemical energy). As a consequence, we have seen that the electronic orbitals are delocalized in space, and the energy values of those orbitals are grouped in bands separated by gaps (Fig. 2.5). The differences of energy within a band are very small compared to the thermal agitation energy (k_BT) so that an electron can easily move from one orbital to the other if permitted by the Pauli principle. In contrast, the gaps of energy can be large compared to k_BT so that the transition from one band to another strongly depends on the width of the gap in between. In the ground state (0K limit), the electrons occupy only the states of smallest energy (two electrons by orbitals, following the Pauli principle). The highest energy level occupied in this 0K limit defines the Fermi level. This peculiar electronic structure has some important implications regarding electrical conductivity:

- If the Fermi level lies within a band (i.e., a band is not fully occupied), the material is a conductor, since electrons can easily move by small changes in energy.

- If the Fermi level lies within a gap, the band just below the Fermi level (valence band) is fully occupied, and the band just above (conduction band) is fully unoccupied. In that case, the electron cannot easily move, and the material is an insulator.
- In the particular case of a 'small' gap, i.e., a gap width comparable to $k_B T$, the probability for the electrons to cross the gap becomes non-negligible. In such cases, when the Fermi level lies within a gap, the material can still have a small fraction of electrons moving from the valence band to the conduction providing a small conductivity. And this conductivity can be dramatically enhanced by a small external 'help' such as a photon or an electric field. Such materials are called 'semiconductors'.

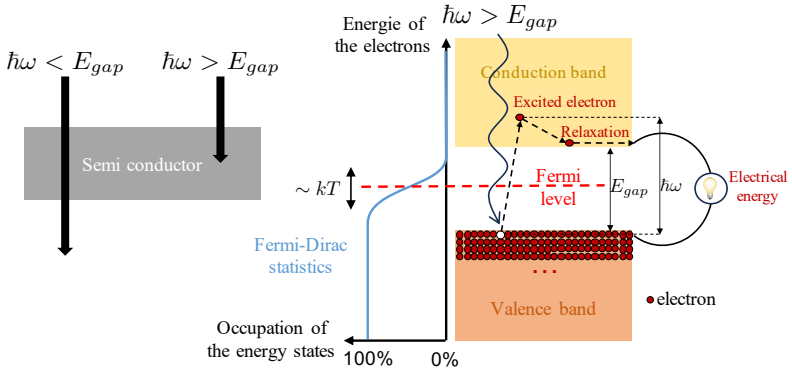


Figure 2.6: Schematic representation of the working principle of a photovoltaic solar cell.

Semiconductors and their bandgap are the heart of photovoltaic energy, converting solar radiation into electricity. These materials have peculiar absorptivity/emissivity spectra: the absorption of radiation by electrons is possible only if the energy of the radiation is larger than the energy gap E_{gap} separating the valence and conduction bands (see Fig. 2.6). Accordingly, the absorptivity and emissivity are of the form: $\alpha(\hbar\omega) = \varepsilon(\hbar\omega) = \begin{cases} 1 & \text{if } \hbar\omega > E_{gap} \\ 0 & \text{otherwise} \end{cases}$. Accordingly, semiconductors are peculiar greybody, which absorb/emit radiation only for frequency above a threshold $\hbar\omega > E_{gap}$ that depends on the material. One can take advantage of this property in the perspective of capturing sunlight and transforming it into electrical work.

As discussed in the previous section, turning solar radiation into work with a blackbody is challenging because the absorber collects energy only from the Sun's direction, but radiates in all directions. A possible approach to improve this trade-off is to take advantage of the spectral separation between the absorbed radiation (blackbody radiation at 5800 K from the Sun) and the emitted radiation (blackbody radiation at 300 K from the cell) as shown in Figure 2.7. Since the majority of the Sun spectrum is at higher frequencies (lower wavelength) than most of the 300 K spectrum, using a semi-conductor with a well-chosen bandgap makes it possible to absorb most of the sunlight while avoiding most emissions at ambient temperature.

2.3.2 The Trivich-Finn and Shockley-Queisser limits

Let us consider a semi-conductor of bandgap E_{gap} receiving sunlight on Earth in ideal conditions, and maintained at ambient temperature. As this greybody absorbs and emits only the energies $\hbar\omega > E_{gap}$, the powers absorbed and emitted are:

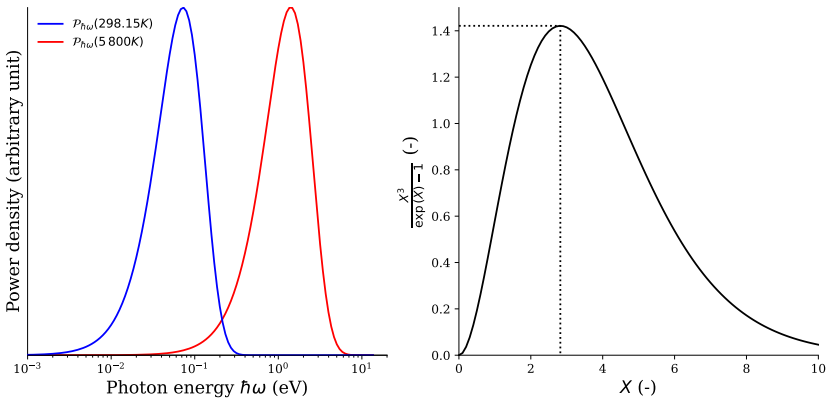


Figure 2.7: (left) The blackbody radiation at the Sun's temperature (5800 K) and at room temperature (300 K) are well separated spectrally. (right) Introducing a gap can reduce significantly the radiated power without affecting the absorbed power too much. This is achieved by choosing a gap energy such that most of the panel spectrum is excluded from emission ($\frac{E_{gap}}{k_B T_p} \gg X_{max}$) and most of the Sun spectrum is absorbed ($\frac{E_{gap}}{k_B T_\odot} \ll X_{max}$).

$$\begin{aligned}
\mathcal{P}_{received} &= (\sin(\theta_S))^2 \int_{E_{gap}}^{+\infty} \mathcal{P}_{\hbar\omega}(T_\odot)\, d(\hbar\omega) \\
&= (\sin(\theta_S))^2 \left(\int_{E_{gap}}^{+\infty} \frac{(\hbar\omega)^3}{4\pi^2\hbar^3c^2} \left(\exp\left(\frac{\hbar\omega}{k_B T_\odot}\right) - 1 \right)^{-1} d(\hbar\omega) \right) \\
&\underset{X=\frac{\hbar\omega}{k_B T_\odot}}{=} (\sin(\theta_S))^2 \frac{k^4 T_\odot^4}{4\pi^2\hbar^3c^2} \int_{\frac{E_{gap}}{k_B T_\odot}}^{+\infty} \frac{X^3}{\exp(X)-1} dX \\
\mathcal{P}_{loss} &= \int_{E_{gap}}^{+\infty} \mathcal{P}_{\hbar\omega}(T_p)\, d(\hbar\omega) \\
&= \int_{E_{gap}}^{+\infty} \frac{(\hbar\omega)^3}{4\pi^2\hbar^3c^2} \left(\exp\left(\frac{\hbar\omega}{k_B T_p}\right) - 1 \right)^{-1} d(\hbar\omega) \\
&\underset{X=\frac{\hbar\omega}{k_B T_p}}{=} \frac{k^4 T_p^4}{4\pi^2\hbar^3c^2} \int_{\frac{E_{gap}}{k_B T_p}}^{+\infty} \frac{X^3}{\exp(X)-1} dX
\end{aligned}$$

Introducing a gap reduces the absorbed power (which is bad) but also reduces the emitted power (which is good). The function $f(X) = \frac{X^3}{\exp(X)-1}$ reaches a maximum for $X_{max} = 2.82$ (Fig. 2.7). By choosing an energy gap E_{gap} such that $\frac{E_{gap}}{k_B T_p} \gg X_{max}$ and $\frac{E_{gap}}{k_B T_\odot} \ll X_{max}$, most of the spectrum at the temperature of the panel is excluded from the integration, whereas most of the spectrum at the temperature of the Sun is included. By doing so, one can optimize the capture of solar radiation. These two conditions correspond to 0.072 eV $\ll E_{gap} \ll$ 1.41 eV (assuming a panel at ambient temperature). Thus, introducing a gap is a good way to reach a significant power conversion efficiency without resorting to concentration.

Let us now turn to the conversion of radiation to electricity. What is the optimal gap to convert solar radiation into electricity, and what is the corresponding efficiency? A qualitative way to estimate these quantities is to consider the following assumptions (see Fig. 2.6):

- Each photon with energy above the bandgap is absorbed and generates one electron

which will contribute to the electrical current of the cell. The corresponding current is thus:

$$\mathcal{I}_0 = e\left(\sin\left(\theta_S\right)\right)^2 \int_{E_{gap}}^{+\infty} B_{\hbar\omega}\left(T_\odot\right) d\left(\hbar\omega\right) = e\left(\sin\left(\theta_S\right)\right)^2 \int_{E_{gap}}^{+\infty} \frac{\left(\hbar\omega\right)^2}{4\pi^2\hbar^3c^2} \left(\exp\left(\frac{\hbar\omega}{k_B T_\odot}\right) - 1\right)^{-1} d\left(\hbar\omega\right)$$

with $e = 1.6 \times 10^{-19}$ C the elementary charge.

- Each electron, regardless of the energy of the photo-generating photon, relaxes toward the bottom of the conduction band. The energy provided by each electron is thus E_{gap} and the corresponding voltage

$$V_0 = \frac{E_{gap}}{e} \tag{2.5}$$

The output power $\mathcal{P}_{elect} = \mathcal{I}_0 \times V_0$ is limited by voltage if the gap is too small and by current if the gap is too large. The optimum can be found numerically and corresponds to the Trivich-Finn limit around 43% for a gap at 1.1 eV (Figure 2.8).

This naive model can be refined in the so-called detailed balance approach with two additional assumptions to take into account the operating point of the device:

- While each electron provides an energy E_{gap}, the voltage actually covers only a fraction of this energy corresponding to the work provided by the electron. Equation (2.5) should be replaced by:

$$eV = \Delta\mu$$

where $\Delta\mu$ is the difference in Fermi levels between the two contacts of the cell.

- All photogenerated electrons are not extracted, some will recombine and re-emit a photon, producing the radiative losses for the cell. The number of electrons recombining per second is increased by the voltage as:

$$\begin{aligned}\mathcal{R} &= \exp\left(\frac{eV}{k_B T_p}\right) \times \left(\int_{E_{gap}}^{+\infty} B_{\hbar\omega}\left(T_p\right) d\left(\hbar\omega\right)\right) \\ &= \exp\left(\frac{eV}{k_B T_p}\right) \times \left(\int_{E_{gap}}^{+\infty} \frac{\left(\hbar\omega\right)^2}{4\pi^2\hbar^3c^2} \left(\exp\left(\frac{\hbar\omega}{k_B T_p}\right) - 1\right)^{-1} d\left(\hbar\omega\right)\right)\end{aligned}$$

and the net current is thus $\mathcal{I} = \mathcal{I}_0 - e\mathcal{R}\left(V\right)$

The overall power $\mathcal{P}_{elect} = \mathcal{I} \times V$ leads to another trade-off: considering a given material with a given gap, working at low voltage will degrade the produced power because V is small, but working at high voltage the increased recombination reduce the current, and thus the power. For each gap, we can therefore find a maximum power point V_{opt}, corresponding to the maximal power achievable (see exercise in small class). Considering this value for all possible gaps leads to the celebrated Shockley-Queisser limit around 30% for a gap of 1.25 eV, as shown in Figure 2.8.

2.3.3 From Shockley-Queisser to real devices

The Shockley-Queisser model makes very bold assumptions:

- Absorption is ideal. In reality, there could be many optical losses: reflection at the surface of the cell, parasitic absorption by non-active regions, limited absorption even above the band gap ($\alpha < 1$), etc.
- Electrical losses are only due to radiative recombinations. In reality, electrons have many other channels to return to the valence band without contributing to the electrical current (trap-assisted recombination, carriers-carriers interactions...).

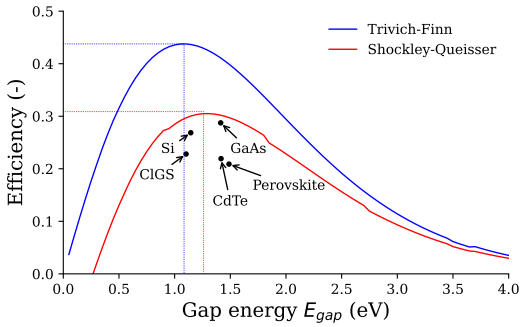


Figure 2.8: Trivich-Finn and Shockley-Queisser limits as a function of the band gap energy.

- Collection is ideal. In reality, transporting carriers from the photogeneration depth to the electrical contact can be challenging because of their finite mobility.
- Selectivity is perfect. In reality, the fraction of electrons going in the "wrong" direction will reduce the net current output.

The Shockley-Queisser limit is thus an upper bound for solar power conversion efficiency. Yet, despite these simplifications, the Shockley-Queisser limit offers an excellent key to understanding materials issues for solar photovoltaics (see Fig. 2.8). The model provides a good insight into the optimal value of the gap and explains why all technologies (and in particular silicon) have similar band gaps, around 1 eV. Record systems show efficiencies very close to the Shockley-Queisser limit: 27% for Silicon at 1.1 eV (a detailed calculation gives a limit at 29%), 28% for GaAs at 1.4 eV (for a limit at 30%)... Alternative approaches can be considered to overcome this 30% maximum efficiency (e.g., multi-junction cells, see exercise in small class). The only strict limit is the 86% thermodynamic efficiency already encountered for thermal solar energy with maximum concentration and ideal conversion. Note that this discussion highlights the 4 main functionalities required from a material to provide photovoltaic energy: absorbing the light (absorptivity), preventing recombinations (lifetime), transporting carriers (mobility), and breaking a symmetry (selectivity). There are several ways of implementing these functionalities, but every PV technology has to produce them one way or the other.

2.3.4 Solar cells today

While many technologies offer efficiencies exceeding 10% (see Fig. 2.9), silicon solar represents more than 95% of the market as of 2020. Silicon cells are formed on Silicon wafers (typical size: 156 mm × 156 mm × 0.18 mm) doped with Boron and Phosphorus, coated with Silicon oxide and Silicon nitride, and contacted with Silver and Aluminium. These cells are connected in series, encapsulated, and framed to form a module, the "solar panel" you can buy on the market (see Fig. 2.10). As of 2020, commercial modules have nominal efficiencies of about 20%, and cost about 0.23 € per Watt-peak (Wp). The spectacular decrease of costs goes together with the fast growth of the sector, which contributed to about 2% of the world electricity production in 2019 (see Fig. 2.11). Note however that this cost represents only the production cost, not the life cycle analysis (Levelized Cost Of Electricity, LCOE) nor the cost of grid integration.

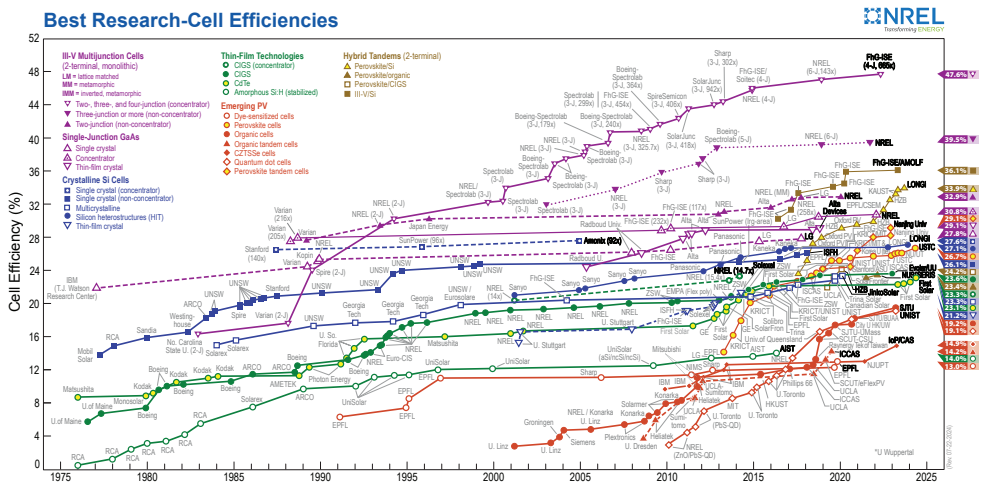


Figure 2.9: Best Research-Cell Efficiency Chart (from National Renewable Energy Laboratory NREL, see interactive version online: `https://www.nrel.gov/pv/cell-efficiency.html`)

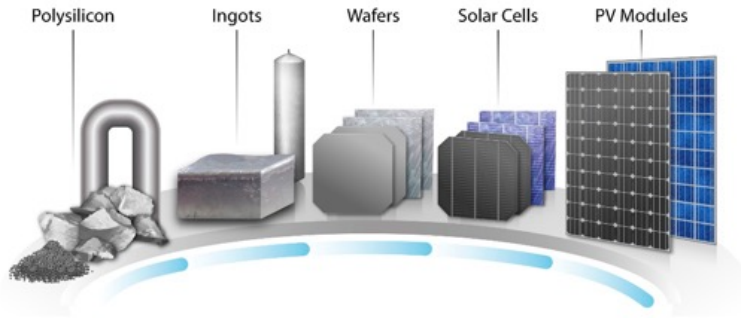


Figure 2.10: Schematic of c-Si PV module supply chain (from Woodhouse et al. (2020) Crystalline Silicon Photovoltaic Module Manufacturing Costs and Sustainable Pricing, National Renewable Energy Laboratory NREL, `https://www.nrel.gov/docs/fy19osti/72134.pdf`)

Recap of what must be known for the exam

1. Planck's law

The spectral density of a blackbody at temperature T is :

$$\mathcal{B}_{\hbar\omega,\Omega}(T) = \frac{\cos(\theta)}{\pi} B_{\hbar\omega}(T) \text{ with } B_{\hbar\omega}(T) = \frac{(\hbar\omega)^2}{4\pi^2\hbar^3c^2}\left(\exp\left(\frac{\hbar\omega}{k_BT}\right)-1\right)^{-1}$$

$\mathcal{B}_{\hbar\omega,\Omega}(T)\, d(\hbar\omega)\, d\Omega d(\hbar\omega)$ represents the number of photons with energy between $\hbar\omega$ and $\hbar\omega + d(\hbar\omega)$ emitted per unit time and per unit surface in the directions between Ω and $\Omega + d\Omega$. $B_{\hbar\omega}(T)$ is the spectral density emitted in all direction.

The power density $\mathcal{P}_{\hbar\omega,\Omega}(T) = \hbar\omega\mathcal{B}_{\hbar\omega,\Omega}(T)$ represents the power emitted (since $\hbar\omega$ is the energy of a photon). Integration of $\mathcal{P}_{\hbar\omega,\Omega}$ over all energies and all directions give the total power emitted per unit surface of a blackbody (Stefan-Boltzmann law) :

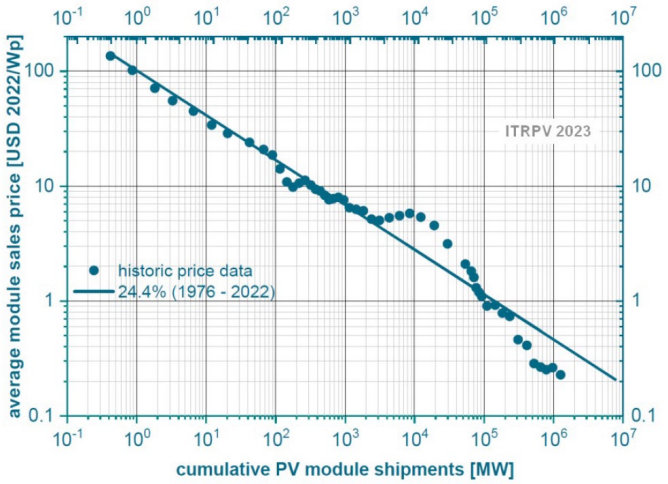


Figure 2.11: Evolution of the module cost as a function of the world installed capacity. (from the 2023 International Technology Roadmap for Photovoltaics, Machinery and Equipment Manufacturers Association VDMA, `https://www.vdma.org/international-technology-roadmap-photovoltaic`).

$$\mathcal{P}(T) = \int_0^{+\infty} \hbar\omega \mathcal{B}_{\hbar\omega}(T)\, d(\hbar\omega) = \sigma T^4 \text{ with } \sigma = \frac{\pi^4}{15}\frac{k_B^4}{4\pi^2\hbar^3c^2} \text{ the Stefan-Boltzmann constant}$$

Photons can be described equivalently in terms of their energy $\hbar\omega$, frequency $\nu = \frac{\omega}{2\pi}$, or wavelength $\lambda = \frac{c}{\nu}$. Therefore, Planck's law can also be expressed in terms of ν or λ. The respective spectral densities have different units and satisfy:

$$|\mathcal{B}_{\hbar\omega,\Omega} d(\hbar\omega)| = |\mathcal{B}_{\nu,\Omega}(T)\, d\nu| = |\mathcal{B}_{\lambda,\Omega}(T)\, d\lambda|$$

2. Kirchhoff's law

The absorptivity $\alpha(\hbar\omega, \Omega)$ is the fraction of incident photons absorbed at frequency ω from an incident direction Ω). The emissivity $\varepsilon(\hbar\omega, \Omega)$ is the ability of the body to emit a radiation at frequency ω in the direction Ω relative to the emission of a blackbody.

$$\text{Kirchhoff's law: } \alpha(\hbar\omega, \Omega) = \varepsilon(\hbar\omega, \Omega)$$

For a blackbody, absorptivity and emissivity are equal to 1 at all frequencies. A material with absorptivity and emissivity < 1 is called a grey body.

3. Solar resource

Applying the Stefan-Boltzmann law, the total radiation power emitted by the Sun is $\mathcal{P}_\odot = 4\pi R_\odot^2 \times \sigma T_\odot^4$, with $R_\odot$ the radius of the Sun, and $T_\odot = 5\,800$K the surface temperature of the Sun. The power per unit surface received on Earth at a distance $D_{T-\odot}$ in ideal conditions (surface perfectly oriented, before the atmosphere) is therefore $\phi_{\text{AM }0} = \frac{\mathcal{P}_\odot}{4\pi(D_{T-\odot})^2} = 1\,360$ W/m^2 (called solar constant). The total power received from the Sun on Earth is 10 000 times larger than humanity's energy needs.

Taking into account the atmosphere, the seasonal variations and the day/night cycles, the actual power received on average at the surface of the Earth is about 10 times less than the above estimate, e.g., 150 W/m^2 in France.

A more accurate calculation of the solar power received per unit surface is:

$$\phi = (\sin(\theta))^2 \sigma T_\odot^4$$

where θ is the angle occupied by the radius of the Sun in the sky. In absence of concentration $\sin(\theta) = \sin(\theta_S) = R_\odot / D_{T-\odot}$ and one recovers the AM 0 estimate (1 360 W/m^2). With a concentration system, θ can be increased : $(\sin(\theta))^2 = C(\sin(\theta_S))^2$, with $C \geq 1$ the concentration factor. For the maximum concentration ($\sin(\theta) = 1$) the solar power received can be increased by a factor $C = 46\,000$.

3. Solar thermal energy

A thermal solar panel absorbs radiation from the Sun, remits a radiation that depends on its temperature because of its emissivity, and provides heat for subsequent use (e.g., domestic heating). In stationary conditions, the efficiency is obtained from the energy balance and is displayed in Figure 2.4. This efficiency falls to zero as the temperature of the panel approaches the maximum temperature $T_p = \sqrt{\sin(\theta_S)} T_\odot = 122°$C. Turning the heat into work proves very inefficient (efficiency $< 6\%$), even with an ideal Carnot engine, because the temperature of the panel remains small. Solar concentration is necessary to reach a reasonable efficiency.

4. Solar photovoltaic

Photovoltaic pannels are made of semiconductors that are greybody, which absorb/emit radiation only for frequencies above a given threshold $\hbar\omega > E_{gap}$ (energy gap). Accordingly, $\alpha(\hbar\omega) = \varepsilon(\hbar\omega) = 1$ if $\hbar\omega > E_{gap}$ and $\alpha(\hbar\omega) = \varepsilon(\hbar\omega) = 0$ otherwise. The semiconductor is chosen so that the energy gap E_{gap} allows most of most of the solar spectrum to be absorbed, while most of blackbody emission at the temperature of the panel is avoided.

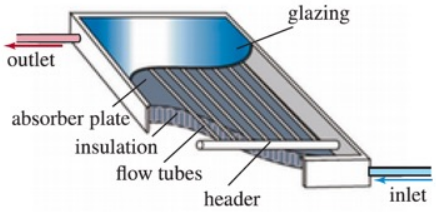
One can estimate an upper bound of the efficiency of a photovoltaic pannel by assuming that all electrons excited by a photon of energy $> E_{gap}$ contribute to the current collected (no loss of current), and that the electrical energy provided by each electron is exactly E_{gap} (no loss of voltage). One obtains the so-called Trivich-Finn limit, which reaches a maximum of 43% for a energy gap of 1.1 eV (Fig. 2.8) :

$$\eta = E_{gap} (\sin(\theta_S))^2 \int_{E_{gap}}^{+\infty} \frac{(\hbar\omega)^2}{4\pi^2\hbar^3c^2} \left(\exp\left(\frac{\hbar\omega}{k_B T_\odot} \right) - 1 \right)^{-1} d(\hbar\omega)$$

A more accurate estimation of the efficiency is given by the Shockley-Queisser limit, which takes into account the loss of current to the electrons recombining before being collected, and the loss of voltage required to collect the electrons. The Shockley-Queisser limit reaches a maximum of about 30% for a gap of 1.25 eV (Fig. 2.8). Most solar cells are made of silicon, a widely available resource, with an energy gap of 1.1 eV close to the optimal gap .

Exercises

Ex. 2.1 - Panneau solaire thermique



On s'intéresse dans cet exercice à la performance d'un panneau solaire thermique (cf. image ci-contre[5]). Le panneau considéré est un panneau plan de longueur $L = 2$ m et de largeur $l = 1$ m sans dispositif de concentration, que l'on supposera dans des conditions d'ensoleillement idéales. En bonne approximation, le soleil peut être vu comme un corps noir à une température $T_\odot = 6\,000$K, émettant un rayonnement dont la puissance par unité de surface est donnée par la loi de Stefan-Boltzmann : $\mathcal{P}_{\text{émise}} = \sigma T_\odot^4$, avec $\sigma = 5.67 \times 10^{-8}$ W.m^{-2}.K^{-4} la constante de Stefan-Boltzmann. Lorsque le rayonnement solaire traverse l'atmosphère terrestre, il subit une atténuation si bien que la puissance du rayonnement après la traversée de l'atmosphère est réduite d'un facteur $f = 0.73$ par rapport à la puissance avant la traversée de l'atmosphère. On supposera que le panneau lui-même est constitué d'un collecteur parfaitement absorbant dans toutes les longueurs d'onde (corps noir).

1. Déterminer la puissance par unité de surface $\mathcal{P}_{\text{reçue}}$ du rayonnement que le panneau reçoit du Soleil, en fonction de l'angle $2\theta_s = 0.52°$ occupé par le Soleil dans le ciel.

2. En négligeant toute autre forme d'échange d'énergie que le rayonnement reçu du soleil et le rayonnement émis par le collecteur, déterminer l'expression de la température d'équilibre T_{max} du panneau (on fera appel régulièrement à cette température T_{max} dans la suite de l'exercice). Faire l'application numérique (on donne : $\left(f\left(\sin\left(\theta_s\right)\right)^2\right)^{1/4} \approx 0.06$).

3. La température du panneau est maintenue à $T_p < T_{max}$, de sorte que l'on récupère l'excès d'énergie sous forme de chaleur. Toujours en négligeant toute autre forme d'échange d'énergie, déterminer le rendement théorique du panneau η_0 en fonction de T_p et de T_{max}. Faire l'application numérique pour un panneau à température ambiante $T_p = 300$K (on donne $(1.2)^4 \approx 2$).

4. La chaleur fournie par le panneau est mise à profit pour produire du travail avec une machine thermique utilisant la température ambiante ($T_{atm} = 280$K) comme source froide. Quelle est la fraction de l'énergie du rayonnement solaire reçu qui peut être convertie en travail ? Montrer que ce rendement est maximum pour une température du panneau intermédiaire entre T_{atm} et T_{max}. En donner une estimation numérique pour $T_p = \frac{T_{max}+T_{atm}}{2}$.

5. Le panneau échange de l'énergie avec l'atmosphère, d'une part sous forme de rayonnement, d'autre part sous forme de conduction/convection. Afin de limiter ces échanges, on introduit un vitrage séparant le collecteur de l'atmosphère. On suppose ce vitrage sans réflexion, transparent à la lumière visible (soleil), mais opaque pour les infra-rouges (rayonnement de l'atmosphère et du panneau). Le vitrage étant en contact avec l'atmosphère, il est supposé à température fixe $T_{atm} = 285$ K. L'espace entre le vitrage et le collecteur est considéré vide d'air (ni convection, ni conduction).

   Expliquer pourquoi la puissance rayonnée par unité de surface du vitrage peut être raisonnablement approximée par σT_{atm}^4. Établir le bilan d'énergie à l'équilibre du collecteur et en déduire le rendement avec vitrage η_1. Faire l'application numérique pour une utilisation sous forme de chauffage individuel avec $T_p = 330$K (on donne : $\frac{11^4-9.5^4}{12^4} \approx 0.31$).

6. Comme pour les vitrages de bâtiments, on peut envisager un double vitrage au dessus du collecteur. En supposant le même type de vitrage (pas de réflexion, transparent au visible, opaque à l'infra-rouge) et le vide entre les vitrages, comment le rendement (η_2) est-il modifié par rapport au simple vitrage (η_1) ? Faire l'application numérique.

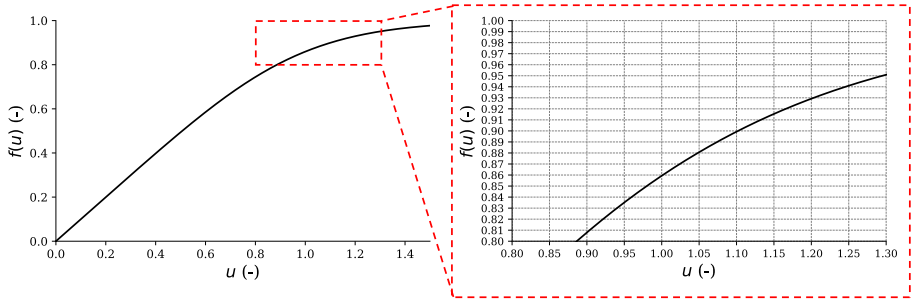
[5] Jaffe & Taylor (2018) The Physics of Energy. Cambridge University Press

7. Les panneaux double vitrages restent onéreux et les panneaux simple vitrage restent la norme pour des installations domestiques. On s'intéresse à présent plus en détail au fonctionnement d'un panneau simple vitrage. Un liquide caloporteur (mélange eau propylène glycol à basse température de fusion) circule dans la direction de la longueur L à travers une série de tubes parallèles (cf. figure en début d'exercice). La température du liquide augmente au fur et à mesure qu'il progresse le long des tubes. Dès lors, la température du panneau n'est pas uniforme mais évolue en fonction de la position z le long du panneau : $T_p(z)$. L'augmentation de la température du liquide caloporteur est reliée à la chaleur reçue et à la capacité calorifique à pression constante du liquide : $c_p = 3.6$ kJ.kg^{-1}.K^{-1}. On considère un débit de liquide constant $\dot{m} = 15$ g/s pour l'ensemble du panneau. Par simplicité ici, on néglige le rayonnement reçu du vitrage. En régime stationnaire, établir le bilan d'énergie d'une tranche de panneau situé entre z et $z + dz$. En déduire que le profil de température du panneau vérifie l'équation différentielle :

$$\frac{dT_p(z)}{dz} = \frac{T_{max}}{z_{cr}} \left(1 - \left(\frac{T_p(z)}{T_{max}}\right)^4\right)$$

où z_{cr} est une longueur caractéristique dont on déterminera l'expression et on montrera qu'elle vaut $z_{cr} \approx 20$ m (on donne $\frac{150}{5.67 \times 3.6^2} \approx 2$).

8. La figure ci-dessous représente la fonction $f(u)$ solution du problème sans dimension $\frac{df}{du} = 1 - (f(u))^4$ avec $f(0) = 0$. Faire le lien avec le problème de la question précédente. En supposant que la température à l'entrée du panneau est $T_p(0) = T_{in} = 320$K, exploiter le graphique, en faisant bien attention à adapter la condition initiale, pour déterminer l'augmentation de température en sortie $\Delta T_p = T_p(L) - T_p(0)$ (on donne $\frac{8}{9} \approx 0.89$).



9. En déduire le rendement global du panneau et faire l'application numérique.

Ex. 2.2 - Limite de Shockley-Queisser

On s'intéresse dans cet exercice au rendement des panneaux solaires photovoltaïques, et plus particulièrement à la limite dite de Shockley-Queisser. Un panneau photovoltaïque vise à convertir de l'énergie sous forme de rayonnement en énergie électrique. Le rayonnement reçu est celui du Soleil, qui peut être vu comme un corps noir sphérique de rayon $R_\odot$, de température de surface $T_\odot$, et situé à une distance $D_{T\odot}$ de la Terre (données numériques ci-dessous). Le spectre du rayonnement émis par un corps noir est donné par la loi de Planck. Un corps noir à température T émet un rayonnement dont la densité spectrale énergétique des photons émis par unité de surface vaut :

$$\mathcal{B}_{\hbar\omega}(T) = \frac{(\hbar\omega)^2}{4\pi^2\hbar^3c^2} \left(\exp\left(\frac{\hbar\omega}{k_B T}\right) - 1\right)^{-1}$$

où $\hbar\omega$ est l'énergie d'un photon. Ainsi $\mathcal{B}_{\hbar\omega}(T)\, d(\hbar\omega)$ représente le nombre de photons d'énergie comprise entre $\hbar\omega$ et $\hbar\omega + d(\hbar\omega)$ émis par unité de temps et par unité de surface (toutes directions confondues).

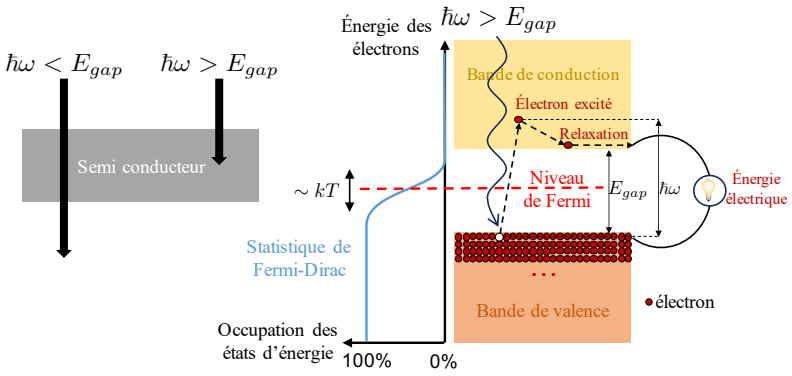
Données numériques :

- Rayon du Soleil : $R_\odot = 700 \times 10^3$ km
- Température de surface du Soleil : $T_\odot = 5800$K
- Distance Terre-Soleil $D_{T\odot} = 150 \times 10^6$ km
- Charge élémentaire : $e = 1.6 \times 10^{-19}$ C
- Constante de Boltzmann : $k_B = 1.38 \times 10^{-23}$ J.K^{-1}
- Constante de Stefan-Boltzmann : $\sigma = \frac{\pi^4}{15}\frac{k_B^4}{4\pi^2\hbar^3c^2} = 5.67 \times 10^{-8}$ W.m^{-2}.K^{-4}

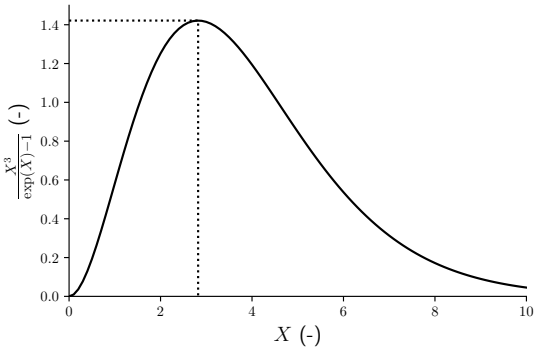
1. On considère un panneau plan sur Terre et on cherche à déterminer la densité spectrale énergétique $\mathcal{B}_{\hbar\omega}^{\odot\to p}$ qu'il reçoit du Soleil (toutes directions confondues). En supposant le panneau parfaitement orienté, le rayonnement reçu par le panneau est le rayonnement solaire moyen à une distance $D_{T\odot}$ du soleil. Montrer alors que la densité spectrale reçue par unité de surface du panneau vaut $\mathcal{B}_{\hbar\omega}^{\odot\to p} = (\sin(\theta_S))^2 \mathcal{B}_{\hbar\omega}(T_\odot)$ avec $\sin(\theta_S) = \frac{R_\odot}{D_{T\odot}}$ où θ_S est l'angle occupé par le rayon du Soleil dans le ciel.

2. En déduire la puissance reçue $\mathcal{P}_{\text{reçue}}$ par unité de surface du panneau sachant que $\int_0^{+\infty} \frac{X^3}{\exp(X)-1}dX = \frac{\pi^4}{15}$ et faire l'application numérique. Commenter vis-à-vis de la puissance moyenne effectivement mesurée (150 W.m^{-2} en France).

3. Un panneau photovoltaïque est un corps 'gris' qui n'absorbe le rayonnement qu'en partie. En effet, les matériaux utilisés pour les panneaux photovoltaïques sont des semi-conducteurs pour lesquels l'absorption et l'émission de rayonnement n'est possible que si l'énergie des photons reçus ou émis est supérieure à un seuil correspondant au gap d'énergie E_{gap} séparant les bandes de valence et de conduction (cf. schéma). Dès lors l'absorptivité et l'émissivité sont de la forme : $\alpha(\hbar\omega) = \varepsilon(\hbar\omega) = \begin{cases} 1 & \text{si } \hbar\omega > E_{gap} \\ 0 & \text{sinon} \end{cases}$.



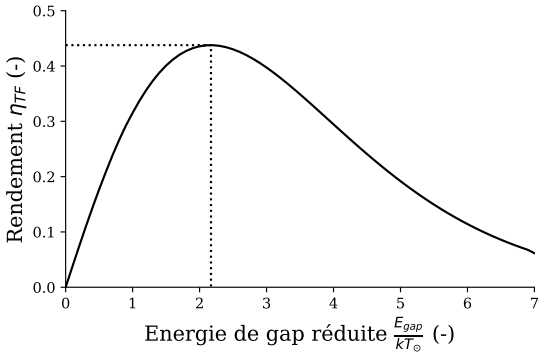
   (a) Quelles sont les expressions des puissances absorbée $\mathcal{P}_{\text{absorbée}}$ et émise $\mathcal{P}_{\text{émise}}$ par unité de surface du panneau ? Quel gap d'énergie E_{gap} est intéressant pour capturer l'énergie du rayonnement solaire ? Qu'en est-il du silicium ($E_{gap} = 1.124$ eV) ? On donne l'allure de la fonction $f(X) = \frac{X^3}{\exp(X)-1}$ ci-dessous. Celle-ci présente un maximum pour $X_{max} = 2.82$.



(b) Les électrons qui sont excités au delà de l'énergie minimum de la bande de conduction, relaxent rapidement jusqu'à cette énergie minimum, les états d'énergie au sein de la bande de valence étant très rapprochés. Cette relaxation dissipe l'énergie sous forme de chaleur et n'est pas récupérable sous forme de courant électrique. Ainsi, seule l'énergie du gap E_{gap} peut être convertie en énergie électrique. Déterminer l'expression de la puissance électrique $\mathcal{P}_{\text{élect}}$ pouvant être produite par le panneau, et en déduire le rendement théorique (limite de Trivich-Flinn) :

$$\eta_{TF} = \frac{\mathcal{P}_{\text{élect}}}{\mathcal{P}_{\text{reçue}}} = \frac{15}{\pi^4} \frac{E_{gap}}{k_B T_\odot} \int_{\frac{E_{gap}}{k_B T_\odot}}^{+\infty} \frac{X^2}{\exp(X) - 1} dX$$

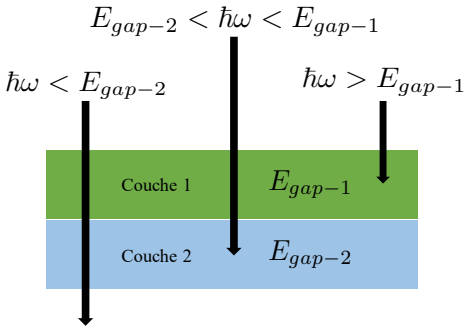
(c) Le graphique ci-dessous représente le rendement théorique en fonction du rapport $\frac{E_{gap}}{k_B T_\odot}$. Pourquoi le silicium est-il un choix intéressant ?



(d) Une amélioration consiste à utiliser des panneaux à plusieurs couches, telles que $E_{gap-1} > E_{gap-2} > ... > E_{gap-n}$ (cf. illustration ci-dessous pour deux couches). Pour un panneau double couche, montrer que le rendement théorique est alors :

$$\eta_{TF} = \frac{\mathcal{P}_{\text{élect}}}{\mathcal{P}_{\text{reçue}}} = \frac{15}{\pi^4} \left(\frac{E_{gap-1}}{k_B T_\odot} \int_{\frac{E_{gap-1}}{k_B T_\odot}}^{+\infty} \frac{X^2}{\exp(X) - 1} dX + \frac{E_{gap-2}}{k_B T_\odot} \int_{\frac{E_{gap-2}}{k_B T_\odot}}^{\frac{E_{gap-1}}{k_B T_\odot}} \frac{X^2}{\exp(X) - 1} dX \right)$$

et proposer une représentation graphique illustrant l'amélioration de rendement.



4. La limite de Trivich-Flinn néglige deux choses. D'une part la loi de Kirchhoff impose que le matériau émette du rayonnement dans les mêmes longueurs d'onde qu'il absorbe. Ainsi, une partie des électrons excités se recombine (i.e., reviennent dans la bande de valence). D'autre part, l'énergie du gap n'est pas convertie en totalité en travail électrique. Il y a deux conséquences importantes concernant le rendement :

   - L'énergie électrique récupérée n'est pas le gap d'énergie mais la différence de potentiel aux contacts avec le circuit électrique induite par le rayonnement : $\Delta\mu = eV < E_{gap} = eV_0$, avec V_0 le potentiel électrique si l'énergie récupérée était E_{gap} et V le potentiel réel.
   - Les électrons qui se recombinent ne contribuent pas au courant électrique. En l'absence de potentiel ($V = 0$, pas de jonction), le taux de recombinaison correspond à l'équilibre avec le rayonnement émis à température T_p soit un courant $\mathcal{I}_0^{\text{recomb}} = e\int_{E_{gap}}^{+\infty} \mathcal{B}_{\hbar\omega}(T_p)\, d(\hbar\omega)$ (courant d'obscurité). Lorsque qu'un potentiel V est appliqué, les recombinaisons deviennent plus probables (car les populations d'électrons et de trous augmentent) et le taux de recombinaison augmente selon : $\mathcal{I}^{\text{recomb}} = \mathcal{I}_0^{\text{recomb}} \exp\left(\frac{eV}{k_B T_p}\right)$.

   (a) Quel est le courant $\mathcal{I}^{\text{photo}}$ produit par unité de surface du panneau en l'absence du phénomène de recombinaison ? En déduire le courant effectivement délivré en tenant compte du phénomène de recombinaison.

   (b) En déduire que la puissance électrique fournie peut s'exprimer sous la forme :

$$\mathcal{P}_{\text{élect}} = V\left(1 - \exp\left(\frac{e\left(V - V_{max}\left(E_{gap}\right)\right)}{k_B T_p}\right)\right)\mathcal{I}^{\text{photo}}\left(E_{gap}\right)$$

   où V_{max} est un potentiel maximum qui ne dépend que du gap d'énergie E_{gap} et de la température du panneau T_p (donner son expression). Commenter l'évolution de la puissance en fonction du potentiel V d'une part, et en fonction du gap d'énergie E_{gap} d'autre part.

   (c) En considérant une maximisation de la puissance électrique à V constant, déterminer une relation que doivent satisfaire E_{gap} et V. On exprimera cette relation sous la forme $V = V_{opt}\left(E_{gap}\right)$ de sorte que l'on puisse se ramener à un problème à un seul paramètre : E_{gap}.

   (d) En conclure l'expression du rendement η_{SQ} du panneau en identifiant deux contributions supplémentaires par rapport au résultat de la question 3b (limite de Shockley-Queisser).

Ex. 2.3 - Forçage radiatif du CO_2 et réchauffement

L'effet de serre dû à l'atmosphère retient une partie de la puissance radiative rayonnée par la Terre. Le forçage radiatif de l'atmosphère est défini comme la différence entre la puissance rayonnée par la Terre en l'absence d'atmosphère et celle rayonnée en présence d'atmosphère. On s'intéresse dans cet exercice à estimer quantitativement la part du forçage radiatif due au CO_2 et comment celle-ci

augmente avec la concentration atmosphérique, principal phénomène à l'origine du réchauffement climatique[6].

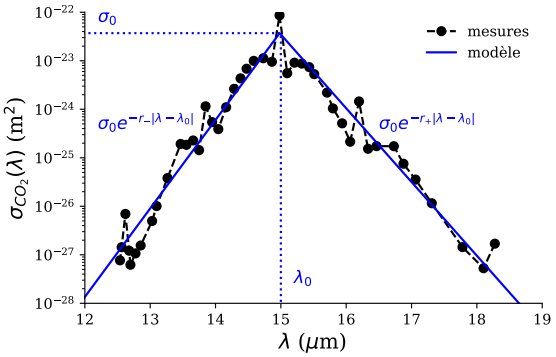
On adopte une représentation continue des échanges radiatifs dans l'atmosphère, paramétrés par l'altitude z. On considèrera que la composition de l'atmosphère est homogène, avec une fraction de CO_2 $x_{CO_2} = 400$ ppm (1 ppm $= 10^{-6}$) constante, et on supposera que la densité numérique des molécules de l'atmosphère suit la loi barométrique $\rho = \rho_0 \exp\left(-\frac{z}{H}\right)$ avec $\rho_0 = 2.51 \times 10^{25}$ m^{-3} et $H = 8260$ m.

On s'intéresse au rayonnement émis par la surface de la Terre dans la direction des z croissants. La Terre est supposée être un corps noir parfait à $T_0 = 288$K, si bien que la puissance rayonnée par unité de surface est distribuée en fonction de la longueur d'onde λ selon la loi de Planck : $\mathcal{P}_\lambda(T) = \frac{2\pi hc^2}{\lambda^5}\left(\exp\left(\frac{hc}{\lambda k_B T}\right) - 1\right)^{-1}$, avec $h = 6.63 \times 10^{-34}$ J.s la constante de Planck, $c = 3 \times 10^8$ m/s la vitesse de la lumière, et $k_B = 1.38 \times 10^{-23}$ J/K la constante de Boltzmann. $\mathcal{P}_\lambda$ a la dimension d'une puissance par unité de volume (W.m^{-3}) de sorte que $\mathcal{P}_\lambda d\lambda$ représente la puissance rayonnée par unité de surface du corps noir pour les longueurs d'onde comprises entre λ et $\lambda + d\lambda$. Une fois émis, ce rayonnement traverse l'atmosphère et est soumis à l'absorption par le milieu. La puissance par unité de surface et de longueur d'onde I_λ (également en W.m^{-3}) du rayonnement incident est absorbée avec un taux d'absorption κ_λ (dimension inverse d'une longueur) qui dépend de la longueur d'onde : $\frac{dI_\lambda}{dz}\big|_{\text{absorption}} = -\kappa_\lambda I_\lambda$. Par ailleurs, le milieu émet, dans la direction des z croissants un rayonnement distribué selon la loi de Planck : $\frac{dI_\lambda}{dz}\big|_{\text{émission}} = \kappa_\lambda \mathcal{P}_\lambda(T)$, où la température de l'atmosphère change avec l'altitude ($T = T(z)$). L'émission est proportionnelle au coefficient d'absorption κ_λ pour garantir l'équilibre thermique (loi du rayonnement de Kirchhoff).

1. Formuler l'équation différentielle que vérifie la puissance rayonnée I_λ (équation de Schwarzschild), ainsi que la condition aux limites en $z = 0$.

2. On s'intéresse ici au spectre compris entre $\lambda = 10$ μm et $\lambda = 20$ μm (infrarouge moyen), pour lequel la grande majorité de l'absorption du rayonnement est attribuable au CO_2 de l'atmosphère. Le coefficient d'absorption se réduit alors à $\kappa_\lambda = \rho_{CO_2}\sigma_{CO_2}(\lambda)$ avec ρ_{CO_2} la densité numérique du CO_2 et $\sigma_{CO_2}(\lambda)$ la section efficace du CO_2 (unité d'aire). En ne retenant que le phénomène d'absorption, quelle est la solution à l'équation de Schwarzschild ? Que représente la longueur caractéristique $H_0^\lambda = \left(x_{CO_2}\rho_0\sigma_{CO_2}(\lambda)\right)^{-1}$?

3. Le CO_2 présente une bande d'absorption autour de $\lambda = 15$ μm (cf. courbe de $\sigma_{CO_2}(\lambda)$ ci-dessous). Quelle est la probabilité que le rayonnement émis par la Terre parvienne jusqu'à l'espace pour une longueur d'onde au centre de la bande d'absorption ? et pour une longueur d'onde au bord de la bande ? Tracer approximativement la probabilité de transmission en fonction de la longueur d'onde.



4. En considérant à présent absorption et émission, procéder au changement de variable $\xi = 1 - \exp\left(-\frac{z}{H}\right)$ dans l'équation de Schwarzschild. En multipliant cette nouvelle équation par

[6] Cet exercice s'inspire de l'approche proposée par Wilson & Gea-Banacloche (2012). Am. J. Phys., 80(4), 306-315.

$\exp\left(\frac{H}{H_0^\lambda}\xi\right)$ et en remarquant que $\frac{dI_\lambda}{d\xi}\exp\left(\frac{H}{H_0^\lambda}\xi\right) + \frac{H}{H_0^\lambda}I_\lambda \exp\left(\frac{H}{H_0^\lambda}\xi\right) = \frac{d}{d\xi}\left[I_\lambda \exp\left(\frac{H}{H_0^\lambda}\xi\right)\right]$, montrer que l'expression suivante est solution :

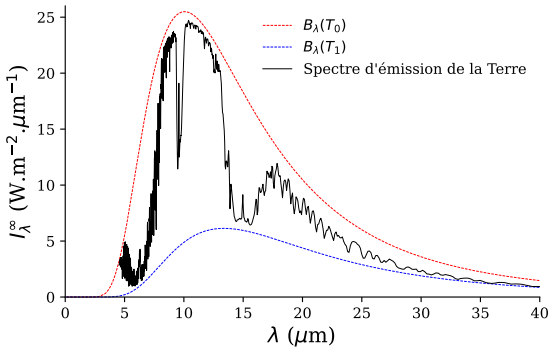
$$I_\lambda = \exp\left(-\frac{H}{H_0^\lambda}\xi\right)\mathcal{P}_\lambda\left(T_0\right) + \frac{H}{H_0^\lambda}\int_0^\xi \exp\left(-\frac{H}{H_0^\lambda}\left(\xi-\xi'\right)\right)\mathcal{P}_\lambda\left(T\left(\xi'\right)\right)d\xi'$$

Donner une interprétation à chacun des deux termes.

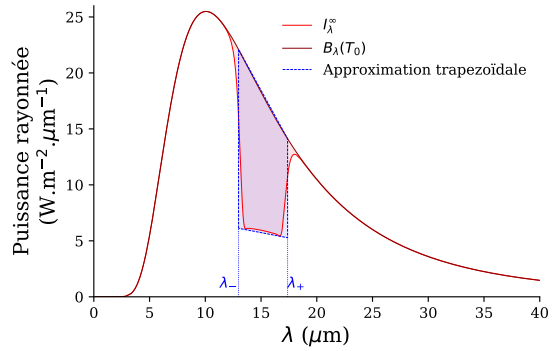
5. Quel serait le rayonnement transmis vers l'espace I_λ^∞ si l'atmosphère était isotherme $T = T_0$?

6. En moyenne, l'atmosphère présente un gradient thermique approximativement constant $\nabla_z T \approx -6.5\text{K/km}$ pour les altitudes inférieures à $z_1 = 11$ km (troposphère), puis une température constante $T_1 = T_0 + \nabla_z T.z_1 = 217\text{K}$ au delà (stratosphère). Déterminer l'expression du rayonnement émis vers l'espace. Identifier les contributions i) du rayonnement initial émis par la Terre, ii) du rayonnement émis entre $z = 0$ et $z = z_1$, et iii) du rayonnement émis au delà de $z = z_1$. Discuter les contributions relatives de ces trois contributions en fonction de la longueur d'onde.

7. Interpréter alors le spectre ci-dessous correspondant au rayonnement effectivement émis par la planète[7].



8. Le forçage radiatif ΔI^∞ est défini comme la différence entre la puissance totale rayonnée par la Terre en l'absence d'atmosphère et celle rayonnée en présence d'atmosphère : $\Delta I^\infty = \int_0^{+\infty}\left(\mathcal{P}_\lambda\left(T_0\right) - I_\lambda^\infty\right)d\lambda$. Cela quantifie l'effet de serre en évaluant la puissance retenue par l'atmosphère par comparaison au cas sans atmosphère. On estime le forçage radiatif du au CO_2 en l'approximant par un trapèze délimité par les longueurs d'onde $\lambda_\pm$ telles que 50% du rayonnement de la terre est absorbé par l'atmosphère (cf. figure ci-dessous).



[7]MODTRAN Infrared Light in the Atmosphere (case 1976 U.S. Standard Atmosphere), The University of Chicago, https://climatemodels.uchicago.edu/modtran

(a) La section efficace du CO_2 peut-être modélisée raisonnablement par $\sigma_{CO_2}(\lambda) = \sigma_0 \exp(-r_\pm |\lambda - \lambda_0|)$, avec $\sigma_0 = 3.71 \times 10^{-23}$ m^2, $\lambda_0 = 15$ μm, $r_+ = 3.5$ μm^{-1}, $r_- = 4.2$ μm^{-1} (cf. figure de la question 3) . Déterminer les expressions de $\lambda_\pm$.

(b) En déduire que le forçage radiatif du CO_2 peut être estimé par :

$$\Delta I^\infty_{CO_2} \approx \left(\frac{1}{r_+} + \frac{1}{r_-}\right) \ln\left(\frac{x_{CO_2}\sigma_0\rho_0 H}{\ln(2)}\right) \left(\mathcal{P}_{\lambda_0}(T_0) - \mathcal{P}_{\lambda_0}(T_1)\right)$$

(c) Estimer le forçage radiatif pour la concentration actuelle de CO_2 ($x_{CO_2} = 400$ ppm) et pour une concentration deux fois supérieure ($x_{CO_2} = 800$ ppm). L'augmentation de forçage radiatif $\Delta I^\infty_{CO_2}\big|_{2\times x_{CO_2}} - \Delta I^\infty_{CO_2}\big|_{\text{actuel}}$ quantifie le pouvoir de réchauffement global du CO_2 et est estimé par le GIEC à 3.75 W/m^2. Comparer à la valeur obtenue. Quelles parties du spectre contribuent à ce réchauffement ?

Pour aller plus loin :

- Vidéo de vulgarisation sur le sujet : `https://youtu.be/ewc8FBtEKPs?si=HCdaxZ3oanaUS1V1`
- Un simulateur en ligne du spectre émis par la planète vers l'espace (MODTRAN Infrared Light in the Atmosphere (case 1976 U.S. Standard Atmosphere), The University of Chicago) : `https://climatemodels.uchicago.edu/modtran`

Chapter 3

Énergie nucléaire

Contents

Foreword **53**
3.1 Atomic nuclei **54**
3.1.1 Composition 54
3.1.2 Isotopes 54
3.1.3 Energy-mass equivalence 55
3.1.4 Mass excess 55
3.2 Nuclear decay **56**
3.2.1 valley of stability 56
3.2.2 Weak interaction and β decay 58
3.2.3 α emission 59
3.2.4 Spontaneous fission 59
3.2.5 Decay chain 60
3.2.6 Gamma radiation 60
3.3 Nuclear reactions **61**
3.3.1 Types of nuclear reactions 61
3.3.2 Cross section 62
3.3.3 Radiative capture 63
3.3.4 Fission 64
3.3.5 Elastic scattering 66
3.3.6 Fusion 67
3.4 Applications **67**
3.4.1 nuclear reactors 67
3.4.2 Radiation and safety 70
Recap of what must be known for the exam **72**
Exercises **75**
Ex. 3.1 - Chaîne de décroissance et équation de Bateman 75
Ex. 3.2 - Neutrons et principe de fonctionnement d'un réacteur nucléaire 78
Ex. 3.3 - Modérateur 80

Foreword

Nuclear energy is the main source of electricity in France (75%) and represents 10% of the global electricity production. In 2020, 441 nuclear reactors were in operation worldwide, and 54 were under construction. Nuclear energy is an interesting source of electricity as it is controllable with a low carbon footprint. Yet, spent nuclear fuels emit hazardous ionizing radiations for up to a

million years, and, therefore, their disposal must be handled cautiously. Also, the Chernobyl and Fukushima accidents remind us that the exploitation of nuclear energy raises safety issues, which requires great care in the design of nuclear reactors. Nuclear energy is a nonrenewable energy, and the vast majority of current nuclear reactors use $^{235}_{92}$U as nuclear fuel, the reserves of which are rather limited (about 130 yrs of proven reserves). $^{235}_{92}$U is the only naturally occurring fissile nucleus, but it is possible to produce fissile nuclei from non-fissile nuclei (e.g., $^{238}_{92}$U) in 'breeder' reactors. The development of such reactors would allow for a much longer prospect regarding the reserves of nuclear fuels ($\sim$ 10 000 yrs) as more than 99% of uranium on Earth is $^{238}_{92}$U. While all current nuclear energy is based on the fission reaction, another prospect is the exploitation of nuclear fusion. Yet, the extreme conditions needed to initiate the fusion reaction make it a huge technological challenge to sustain this reaction and exploit the energy it releases. Thus, the development of fusion reactors remains a distant prospect, and this technology is not considered a viable solution for the energy transition before the second half of the century.

Hereafter, we begin by introducing some of the fundamentals of atomic nuclei: composition, isotopes, and mass-energy (section 3.1). We then look at the stability of nuclei and the nuclear decay reactions (section 3.2). We then discuss nuclear reactions, the concept of cross-section to characterize the probabilities of these reactions, and we have a more precise look at the key processes for energy applications, namely fission, fusion, and moderation (section 3.3). Finally, the operation of nuclear reactors is presented in a general way, addressing the neutron cycle and the main types of reactors, as well as the issues associated with radiation (section 3.4).

3.1 Atomic nuclei

3.1.1 Composition

Any atom is made of protons and neutrons packed together in a positively charged nucleus, around which electrons are distributed in orbitals. Protons and neutrons, with a mass of $m = 1.672 \times 10^{-27}$ kg each, are 2 000 times heavier than electrons. Following quantum mechanics, the mass of a particle is intimately related to the spatial extent of its wave function (think for instance of the uncertainty principle). As a consequence, the spatial extent of a nucleus is about 10 fm = 10^{-14} m, which is 10 000 smaller than the size of the electronic orbitals (about $1\mathring{A} = 10^{-10}$ m).

The protons bear a charge $q = +e = 1.60217663 \times 10^{-19}$ C, the electrons have the opposite charge ($q = -e$), and the neutrons are neutral ($q = 0$). The attractive electromagnetic interaction between the protons and the electrons is what determines the structures of the electronic orbitals and how electrons are organized around the nucleus. The same electromagnetic interaction acts repulsively between protons, with a much larger intensity since the distance between protons is much smaller than between protons and electrons. Yet, the protons are tightly held together within the nucleus thanks to the strong force. The strong force is the force that acts between quarks within protons and neutrons and holds them together. The fluctuations of the strong force also act between protons and neutrons (similarly to the London dispersion forces between atoms that result from the fluctuations of the electrostatic interactions). Thus, the strong force provides an attraction between protons and neutrons within the nucleus, and the magnitude of this attractive energy exceeds by one order of magnitude the repulsive energy due to electromagnetism. Yet, the strong force acts over extremely short ranges (a few fm) so that if a nucleus is too big, it becomes unstable.

3.1.2 Isotopes

The number of charge Z and the number of mass A fully characterize the composition of a nucleus. Z represents the number of protons, and A the total number of protons and neutrons. As a consequence, the number of neutrons is given by $N = A - Z$. An atom with a number of charge Z and a number of mass A is noted $^{A}_{Z}$X, where X is the usual symbol of the element (e.g., $^{1}_{1}$H and $^{12}_{6}$C for the usual isotopes of hydrogen and carbon). Since electrons interact with the protons via electromagnetism and neutrons have no charge, the electronic orbitals are

insensitive to the number of neutrons. Therefore, the behavior of electrons around a nucleus depends only on the number of charges Z, which implies that most of the properties of atoms depend on Z only, with the exception of their mass and the nuclear reactions. Two atoms with the same number of charge Z but different numbers of mass A are called isotopes. A well-known example is ${}^{14}_{6}\mathrm{C}$ which is a rare isotope of ${}^{12}_{6}\mathrm{C}$. Unlike ${}^{12}_{6}\mathrm{C}$, ${}^{14}_{6}\mathrm{C}$ is unstable and exhibits a nuclear decay with a characteristic half-life time of 5 700, which makes it particularly suitable for determining the age of ancient objects. Since isotopes are chemically identical, the only way to separate them is by taking advantage of the difference in mass (hence the need for centrifugation to enrich nuclear fuel in ${}^{235}_{92}\mathrm{U}$).

3.1.3 Energy-mass equivalence

In the field of nuclear energy, energy is often referred to as a mass. Indeed, following the theory of relativity, an object of mass m at rest has an energy $E = mc^2$, with $c = 299\,792\,458$ m/s the speed of light. Accordingly, one can convert a mass to an energy by simply introducing a multiplicative factor c^2. For instance, the mass of a proton $M_p = 1.672\,62 \times 10^{-27}$ kg corresponds to an energy $E_p = 938.27$ MeV. Mass and energy can be used interchangeably, and, by extension, one often writes $M_p = 938.27$ MeV, which should be understood as $M_p = \frac{938.27\ \mathrm{MeV}}{c^2}$. Note that the scale of energy (hundreds of MeV) is huge compared to all other forms of energy discussed in this class (all other processes involve energies less than a few hundred eV). The conversion factor $1\ \mathrm{MeV} = 1.782\,662 \times 10^{-30}$ kg illustrates how even a very small mass (10^{-30} kg) corresponds to huge energies (MeV).

A direct consequence of the energy-mass equivalence is that when a system changes energy, it changes mass. Therefore, when an electron binds to a proton forming the hydrogen atom ${}^{1}_{1}\mathrm{H}$, the resulting atoms have a smaller mass than the sum of the mass of a free proton (M_p) and a free electron ($M_e = 9.109 \times 10^{-31}$ kg), and the difference of mass corresponds to the binding energy $B = 13.6$ eV between the electron and the proton:

$$M_{\mathrm{H}} = M_p + M_e - \frac{B}{c^2}$$

The binding energy B is so small compared to the mass-energy of a proton that its contribution to the total mass is imperceptible (10 parts per billion). This remains true for all the binding energies between electrons and nuclei. However, when it comes to the binding energies within the nucleus between protons and neutrons, the corresponding scale of energies corresponds to about 1% of the total mass energy (i.e., tens of MeV), and the associated difference of mass is significant.

All the binding energy of an atom is captured by the difference in mass between that of the free protons, neutrons, and electrons, and that of the atom. Yet, since a proton always comes with an electron, it is more convenient to compare the energy of the atom to that of Z hydrogen ${}^{1}_{1}\mathrm{H}$ atoms and N neutrons. One thus defines the binding energy of an atom as:

$$B\left(Z, A\right) = Z \times M_{\mathrm{H}} + N \times M_n - M_{\mathrm{atom}}\left(Z, A\right)$$

Note that following this definition, the binding energy is positive, and the stronger the binding energy, the lower the energy of the atoms compared to that of its constituents. Note also, that the above definition does not strictly reflect the binding energy of the electrons since the energy of the electronic orbitals of an atom with Z charges is not the same as Z times the energy of the electronic orbital in hydrogen. Yet, as the electronic binding energy is extremely small compared to the other energies involved, this approximation is of little importance.

3.1.4 Mass excess

The mass of atoms increases linearly with the number of mass A, apart from the minor contribution of the binding energy and the mass of the electrons. Therefore, one often refers to

Particle	Mass			Excess Mass		
	(kg)	(MeV)	(u)	(kg)	(MeV)	(10^-3 u)
proton	1.67262×10^{-27}	938.27	1.00728	1.20811×10^{-29}	6.7770	7.2754
neutron	1.67493×10^{-27}	939.57	1.00867	1.43911×10^{-29}	8.0728	8.6665
electron	9.10938×10^{-31}	0.510999	0.00054858	9.10938×10^{-31}	0.510999	0.548580
$^1_1$H	1.67353×10^{-27}	938.78	1.00783	1.29938×10^{-29}	7.289	7.825
$^{12}_6$C	1.99265×10^{-26}	11 177.93	12	0	0	0
$^4_2$He	6.64648×10^{-27}	3728.40	4.00260	4.3230×10^{-30}	2.425	2.603

Table 3.1: A few examples of total and excess masses.

the concept of excess mass Δ rather than total mass M, where the excess mass is the difference between the total mass and A times a reference mass called the atomic mass unit u. By convention, this reference mass is chosen as $1/12^{th}$ of the mass of $^{12}_6$C, and also serves as the definition of a unit of mass called 'dalton':

$$1u = \frac{1}{12} M_{^{12}_6\mathrm{C}} = 931.494 \text{ MeV} = 1 \text{ dalton}$$

$$\Delta(Z, A) = M_{\text{atom}}(Z, A) - Au$$

A few examples of total and excess masses are shown in the Table 3.1.

Nuclear reactions are reactions that involve changes in the composition of one or several nuclei (e.g., capturing a neutron, releasing a proton, fusion of two nuclei etc.). However, a fundamental aspect of all the nuclear reactions we will consider hereafter is that they conserve the number of mass between the reactants and the products. As a consequence, the balance of energy of a nuclear reaction is the same when expressed in terms of mass or of excess mass. The energy Q released by a nuclear reaction is given by:

$$\begin{aligned} Q_{\text{reaction}} &= \sum_{\text{reactants}} M_i - \sum_{\text{products}} M_i \\ &= \sum_{\text{reactants}} (\Delta_i + A_i u) - \sum_{\text{products}} (\Delta_i + A_i u) \\ &= \sum_{\text{reactants}} \Delta_i - \sum_{\text{products}} \Delta_i \text{ since } \sum_{\text{reactants}} A_i = \sum_{\text{products}} A_i \end{aligned}$$

3.2 Nuclear decay

3.2.1 valley of stability

When looking at the existing atoms, one immediately understands that only a quite restricted number of atoms and isotopes can be found in nature. The heaviest atom is uranium 238 ($^{238}_{92}$U), which suggests that heavier atoms are all unstable. Moreover, it is striking to observe that there are only a few naturally occurring isotopes of each atom, with a number of neutrons often a little larger than the number of protons. All this can be understood by looking at the profile of binding energy per nucleon $b(Z, A) = \frac{B(Z,A)}{A}$, which exhibits a 'valley of stability' (see Fig. 3.1). Indeed, as we shall see, nuclear reactions make it possible to vary the composition of the nuclei in terms of protons and neutrons under the constraint that the number of mass A is conserved. So, one can expect that nuclear reactions favor nuclei compositions with strong binding energy per nucleon (low value of $-b$). In Figure 3.1, one can observe that the strongest binding energy is obtained for A near 60 and Z near 25. Precisely, $^{56}_{26}$Fe is the

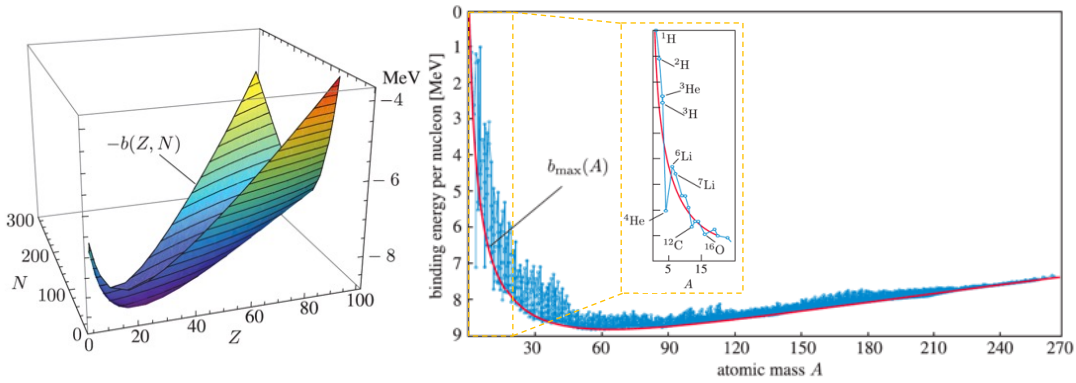


Figure 3.1: (left) Binding energy per nucleon $-b\left(Z,A\right)=-\frac{B(Z,A)}{A}$ as a function of the number of protons Z and of the number of neutrons $N=A-Z$ (approximate profile, based on the semi-empirical mass formula of the liquid drop model). All known nuclei (either naturally occurring or synthetic) are found in the 'valley of stability' (blue domain). (right) Actually measured binding energies per nucleon as a function of the atomic mass A. The strongest binding $b_{max}\left(A\right)$ for each atomic mass following the liquid drop model is displayed in red. A few nuclei exhibit stronger binding than predicted by the model (blue points below the red curve), such as ${}_{2}^{4}\mathrm{He}$, ${}_{6}^{12}\mathrm{C}$, or ${}_{8}^{16}\mathrm{O}$. (from Jaffe & Taylor (2018) The Physics of Energy. Cambridge University Press)

most tightly bound nucleus. Only the nuclei with binding energy not too large compared to ${}_{26}^{56}\mathrm{Fe}$ are expected to be potentially stable. The binding energy profile appears quite narrow, and the existing stable nuclei are restricted domain of energies in blue in Fig 3.1 (left), which allows only a few isotopes for each atom. The binding energy profile also explains why it is possible to fusion nuclei lighter than ${}_{26}^{56}\mathrm{Fe}$ to produce a heavier nucleus, whereas it is possible to fission a nucleus heavier than ${}_{26}^{56}\mathrm{Fe}$ to produce lighter nuclei. The valley of stability is not perfectly aligned with the bisector $Z=N$, but tends to favor neutron-rich nuclei. A more subtle observation is that nuclei with even numbers of protons and neutrons tend to be more tightly bound than nuclei with odd numbers of protons or neutrons ('pairing effect').

Nuclear decays are nuclear reactions by which a nucleus can spontaneously change Z and/or decrease A[a]. Nuclei that exhibit no decay are said to be stable. For a nucleus that exhibits decay, the decay process is a random Poisson process, which implies that the amount of this nuclide in a system decreases exponentially with time:

$$N\left(t\right)=N\left(0\right)\exp\left(-\frac{\ln\left(2\right)}{\tau_{1/2}}t\right)$$

where $\tau_{1/2}$ is the half-life time (time after which half of the nuclei have decayed: $N\left(\tau_{1/2}\right)=N\left(0\right)/2$). Figure 3.2 displays all the known nuclei with their respective half-life time (stable nuclei are displayed in black). One clearly observes the neutron-rich effect, especially for heavy nuclei. The pairing effect manifests itself with the 'steps' in the chain of stable nuclei. The half-life time changes very rapidly away from the valley of stability. Naturally occurring nuclei must have a half-life time comparable to the age of the Earth (4.54×10^{9} yrs), or must result from the decay of such nuclei. Uranium ($Z=92$) is the heaviest naturally occurring nucleus on Earth.

[a] a change of A requires the emission of another nucleus to ensure the overall conservation of A

So far, we have discussed the energies of the nuclei, but we have not explained how unstable nuclei can decay to transform into stable nuclei. Nuclear decay actually occurs in one of three ways,

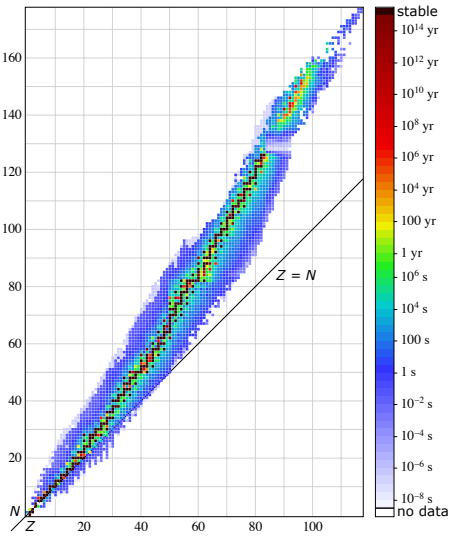


Figure 3.2: Half-life time of all known isotopes as a function of the number of protons Z and neutrons $N = A - Z$. (from wikimedia commons, public domain)

which are described in the following sections.

3.2.2 Weak interaction and β decay

In addition to the strong force, protons and neutrons also interact via another fundamental force called the weak force. The strength and range of the weak force are very small compared to that of the strong force, and the structure of a nucleus is essentially dictated by the strong force. The weak force manifests itself only on very rare occasions but exhibits a unique feature as it can change the flavor of quarks, the elementary particles forming protons and neutrons. Therefore, the weak force interaction can convert a neutron into a proton and vice-versa, while conserving[1] the charge by releasing or absorbing an electron or a positron (antiparticle of the electron).

The process by which a neutron is converted into a proton is called the β^- decay, and takes the form:

	for a free neutron	for an atom
β^- decay	$n \rightarrow p + e^- + \bar{\nu}_e$	${}^A[\mathrm{Z}] \rightarrow {}^A[\mathrm{Z}+1] + e^- + \bar{\nu}_e$

where $\bar{\nu}_e$ is an electron antineutrino.

The process by which a proton is converted into a neutron is called the β^+ decay, and takes the form:

	for a free proton	for an atom
β^+ decay	$p \rightarrow n + e^+ + \nu_e$	${}^A[\mathrm{Z}] \rightarrow {}^A[\mathrm{Z}-1] + e^+ + \nu_e$

where ν_e is an electron neutrino.

A last conversion is when a proton combines with an electron to form a neutron, and is called electron capture[a]:

	for a free proton	for an atom
e^- capture	$p + e^- \rightarrow n + \nu_e$	${}^A[\mathrm{Z}] + e^- \rightarrow {}^A[\mathrm{Z}-1] + \nu_e$

[1] In addition to the charge, the conversion between a neutron and a proton must also conserve a quantum number called lepton number, which imposes the emission of a neutrino or anti-neutrino during the conversion process

β^+ decay and e^- capture are endothermic for a free proton and never occur. In contrast, β^- decay is exothermic for a free neutron and β^- decay of a free neutron occurs within 10 minutes with an energy release of about $Q \approx 0.78$ MeV.

For proton and neutron in nuclei, all three processes can be observed. β^- decay is expected in nuclei with more neutrons than the most stable nucleus with the same A. Conversely, β^+ and e^- capture can occur in proton-rich nuclei. Yet, β^+ releases less energy than e^- capture while connecting the same nuclei and is therefore less frequent. Thus, one often refers to β^- decay as to β decay.

[a]The capture of a positron by a neutron to form a proton is not possible because, unlike electron, positron are not found bound around a nucleus.

3.2.3 α emission

Another way for a nucleus to decay is to break into two pieces. Breaking into two large pieces, a process known as spontaneous fission, is rather infrequent, and is the object of the next section. Here we consider the case where a small nucleus ${}^{a}[\mathrm{z}]$ is emitted from a larger nucleus ${}^{A}[\mathrm{Z}]$, a process called particle emission:

$$ {}^{A}[\mathrm{Z}] \rightarrow {}^{A-a}[\mathrm{Z}-\mathrm{z}] + {}^{a}[\mathrm{z}] $$

For instance, a proton and a neutron emissions take the form:

$$ {}^{A}[\mathrm{Z}] \rightarrow {}^{A-1}[\mathrm{Z}-1] + {}^{1}_{1}\mathrm{H} $$

$$ {}^{A}[\mathrm{Z}] \rightarrow {}^{A-1}[\mathrm{Z}] + {}^{1}_{0}\mathrm{n} $$

In practice, a proton or a neutron emission is endothermic and is very unfrequent. Emissions of ${}^{3}_{1}\mathrm{H}$ and ${}^{3}_{2}\mathrm{He}$ are also not expected because of unfavorable energy release. In contrast, the emission of ${}^{4}_{2}\mathrm{He}$, also known as α emission, is energetically favorable for nuclei ${}^{A}[\mathrm{Z}]$ with atomic mass $A \gtrsim 155$. α emission is actually a very frequent decay process for heavy nuclei. In contrast, α emission is not energetically favorable for lighter nuclei. The lightest naturally occurring nucleus that exhibits α emission is an isotope of neodymium (${}^{144}_{60}\mathrm{Nd}$). α emission takes the form:

$$ {}^{A}[\mathrm{Z}] \rightarrow {}^{A-4}[\mathrm{Z}-2] + {}^{4}_{2}\mathrm{He} $$

Emission of heavier nuclei can be energetically favorable but is rarely observed, and α emission represents the vast majority of all particle emission processes.

Half-time life associated with α emission exhibits widely different values from one nucleus to another (from nanoseconds to billions of years). Indeed, there is a high energy barrier preventing α emission, but the barrier is overcome by quantum tunneling. Yet, the probability of quantum tunneling is highly sensitive to the width and height of the barrier, which causes a wide dispersion in the probability of α emission.

3.2.4 Spontaneous fission

The last decay process is spontaneous fission. Spontaneous fission is the process by which a heavy nucleus breaks into two lighter nuclei of similar size. When the number of mass A is very large, the size of the nucleus approaches the range of the strong force interactions. On some rare occasions, a fluctuation of the spherical shape of the nucleus can lead to an irreversible deformation in which two parts of the nucleus separate (Fig. 3.3). For a nucleus with number of mass $A \gtrsim 100$, the energy of two nuclei of number of mass $A/2$ is generally lower than the energy of the original nucleus, so that spontaneous fission becomes energetically favorable. Yet, the energy barrier against fission is usually very large (Fig. 3.3a) and the process of spontaneous fission is extremely unlikely. As the number of mass A approaches 230,

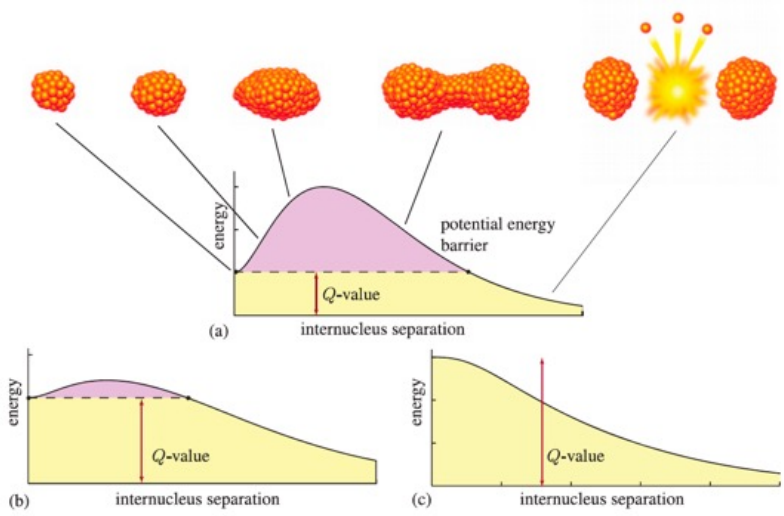


Figure 3.3: Schematic representation of the spontaneous fission process. (a) For nuclei with $A \gtrsim 100$, the fission is favorable energetically (i.e., the energy of the resulting nuclei is lower than the energy of the initial nucleus), but the large energy barrier makes the spontaneous fission extremely unlikely. (b) For nuclei with $A \sim 230$ the energy barrier becomes small enough for the spontaneous fission to occur by quantum tunneling. (c) For very heavy nuclei ($A \gtrsim 290$), there is no more energy barrier and fission becomes instantaneous. (from Jaffe & Taylor (2018) The Physics of Energy. Cambridge University Press)

however, the energy barrier becomes small enough for spontaneous fission to become possible by quantum tunneling (Fig. 3.3a). This is the case for instance of $^{235}_{92}$U, which usually decays by α emission but can occasionally exhibit spontaneous fission (once every 10^{10} decays). For very heavy nuclei ($A \gtrsim 290$), the energy profile resembles that of Fig. 3.3c, where there is no more energy barrier, and fission is instantaneous. Therefore nuclei with number of mass $A \gtrsim 290$ cannot be observed.

3.2.5 Decay chain

Starting from a given heavy unstable element, the sequence of α emissions and β decays that follows the nucleus to reach stability is called a decay chain. A common decay chain is that of the long-lived naturally occurring $^{238}_{92}$U, shown in Figure 3.4 (see exercise in small class). There are only four primary decay chains that can be observed in nature starting from $^{232}_{90}$Th, $^{237}_{93}$Np[a], $^{238}_{92}$U, and $^{235}_{92}$U. By convention, these four decay chains are called $4n$, $4n+1$, $4n+2$, and $4n+3$ for $^{232}_{90}$Th, $^{237}_{93}$Np, $^{238}_{92}$U, and $^{235}_{92}$U, respectively. Note that sometimes a nucleus may decay following different possible processes. In such cases, the decay chain contains different branches.

[a]Because of the short half-life of $^{237}_{93}$Np ($\tau_{1/2} = 2.14 \times 10^6$ yrs) compared to the age of the Earth, the primary decay chain of $^{237}_{93}$Np is actually extinct in nature. However, this decay chain is active in the decay of radioactive waste resulting from nuclear reactor operations.

3.2.6 Gamma radiation

As nuclear decay processes release large amounts of energy, the nuclei resulting from a nuclear decay may be found in excited states of energy, which are metastable. Such excited states are called nuclear isomers. As nuclear isomers are metastable, they ultimately relax to the non-excited state of the respective nuclei, sometimes going through intermediary excited states.

$$^{238}_{92}\mathrm{U} \xrightarrow[4.51\times10^{9}\ \mathrm{yrs}]{\alpha} {}^{234}_{90}\mathrm{Th} \xrightarrow[24.5\ \mathrm{days}]{\beta^-} {}^{234}_{91}\mathrm{Pa} \xrightarrow[6.7\ \mathrm{h}]{\beta^-} {}^{234}_{92}\mathrm{U} \xrightarrow[2.33\times10^{5}\ \mathrm{yrs}]{\alpha} {}^{230}_{90}\mathrm{Th} \xrightarrow[8.3\times10^{4}\ \mathrm{yrs}]{\alpha} {}^{226}_{88}\mathrm{Ra}$$

$$^{226}_{88}\mathrm{Ra} \xrightarrow[1\,622\ \mathrm{yrs}]{\alpha} {}^{222}_{86}\mathrm{Rn} \xrightarrow[3.825\ \mathrm{days}]{\alpha} {}^{218}_{84}\mathrm{Po} \xrightarrow[3.05\ \mathrm{min}]{\alpha} {}^{214}_{82}\mathrm{Pb} \xrightarrow[26.8\ \mathrm{min}]{\beta^-} {}^{214}_{83}\mathrm{Bi}$$

$$^{214}_{83}\mathrm{Bi} \xrightarrow[29.7\ \mathrm{min}]{\beta^-} {}^{214}_{84}\mathrm{Po} \xrightarrow[1.5\times10^{-4}\ \mathrm{s}]{\alpha} {}^{210}_{82}\mathrm{Pb} \xrightarrow[22\ \mathrm{days}]{\beta^-} {}^{210}_{83}\mathrm{Bi} \xrightarrow[5\ \mathrm{days}]{\beta^-} {}^{210}_{84}\mathrm{Po} \xrightarrow[140\ \mathrm{days}]{\alpha} {}^{206}_{82}\mathrm{Pb}$$

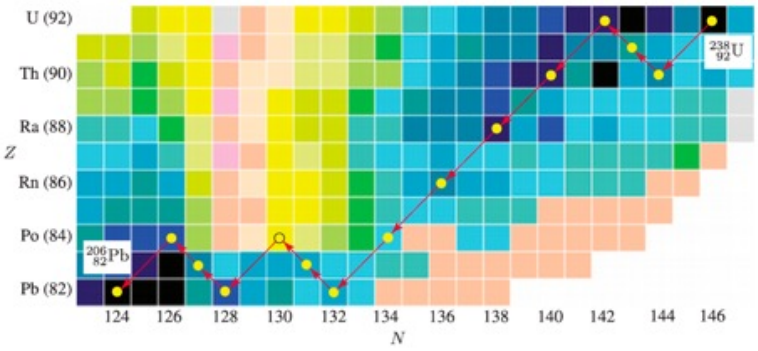


Figure 3.4: Decay chain of $^{238}_{92}\mathrm{U}$ and its representation in the N-Z plane with the color representing the half-lives of the nuclei (light for short-lived nuclei, and dark for long-lived nuclei). (from Jaffe & Taylor (2018) The Physics of Energy. Cambridge University Press)

This relaxation emits photons in the γ-ray domain corresponding to energies in the range of MeV. This process of relaxation is called γ decay. The example of the γ decay of $^{60}_{28}\mathrm{Ni}$ following the β decay of $^{60}_{27}\mathrm{Co}$ is illustrated in Figure 3.5. In this example, two successive excited states of $^{60}_{28}\mathrm{Ni}$ are encountered, leading to the emission of two types of γ-rays. γ-rays are the most energetic radiation and are particularly harmful for living organisms as they can penetrate deep into matter and ionize any atom. Highly radioactive substances must therefore be handled with great precaution to avoid any exposition to γ-rays.

3.3 Nuclear reactions

3.3.1 Types of nuclear reactions

$$^{60}_{27}\mathrm{Co} \longrightarrow {}^{60}_{28}\mathrm{Ni}^{**} + e^- \,(0.31\ \mathrm{MeV}) + \bar{\nu}_e$$
$$^{60}_{28}\mathrm{Ni}^{**} \longrightarrow {}^{60}_{28}\mathrm{Ni}^{*} + \gamma\,(1.17\ \mathrm{MeV})$$
$$^{60}_{28}\mathrm{Ni}^{*} \longrightarrow {}^{60}_{28}\mathrm{Ni} + \gamma\,(1.33\ \mathrm{MeV})$$

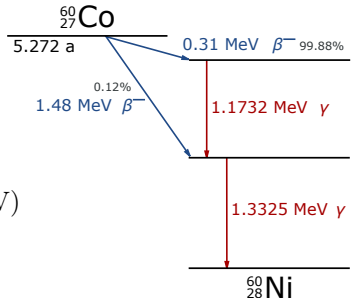


Figure 3.5: Nuclear decay scheme of Cobalt 60 ($^{60}_{27}\mathrm{Co}$). The majority of nuclei exhibits a β^- decay to the second excited state $^{60}_{28}\mathrm{Ni}^{**}$ of Nickel 60 followed by two successive γ decays (chain of reactions shown on the left). A small proportion of $^{60}_{27}\mathrm{Co}$ exhibits a β^- decay directly to the first excited state $^{60}_{28}\mathrm{Ni}^{*}$ followed by a single *gamma* decay. (chart from wikimedia commons, public domain)

A nuclear reaction is a process during which two nuclei interact to produce one or several new nuclei:

$$^{A_1}[\mathrm{Z}_1] + ^{A_2}[\mathrm{Z}_2] \rightarrow ^{A_3}[\mathrm{Z}_3] + ...$$

There is a wide variety of nuclear reactions. The most important with respect to energy applications are:

Neutron capture: Since the neutron has no charge, the most likely nuclear reactions involve the capture of a neutron:

$$^{A}[\mathrm{Z}] + {}^{1}_{0}\mathrm{n} \rightarrow ^{A+1}[\mathrm{Z}]^*$$

The exponent '*' means that the resulting nucleus is in an excited state. This excited nucleus may result in further decays (γ emission, α emission, proton emission).

Induced fission: After a neutron capture, a decay of special interest is fission. Such fission is referred to as induced fission, as it is triggered by the capture of a neutron, in contrast with spontaneous fission, discussed earlier, and which occurs spontaneously. The most widely used nuclear reaction is the induced fission of ${}^{235}_{92}\mathrm{U}$ which writes:

$${}^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow \text{fission}$$

Fusion: Two light nuclei with enough kinetic energy can come close enough to trigger a fusion. The fusion reaction occurs in the heart of the stars, which makes it the most prevalent reaction in the universe. Fusion is extremely hard to reproduce on Earth as it requires extremely high temperatures to provide enough kinetic energy to the nuclei. A fusion reaction that is actively studied is the reaction between deuterium and tritium:

$${}^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n}$$

A last process that is important for energy applications, but is not a nuclear reaction per se, is scattering. When a neutron interacts with a nucleus but is not captured it can be scattered thereby exchanging energy with the nucleus.

3.3.2 Cross section

Let us consider a nucleus immersed in an incoming flux ϕ of particles (neutron, α etc.). ϕ is a number of particles per unit area and times. Some of the incoming particles interact with the nucleus, which leads to a number of events N (induced fission, scattering, capture etc.). Since nuclear processes are probabilistic, one expects the probability of an event to occur per unit time $\frac{dN}{dt}$ to be proportional to the amount of particles that cross the nuclei per unit time, that is $\frac{dN}{dt} = \sigma\phi$, with σ a quantity with unit of area. This quantity σ is called the cross-section and can be interpreted as the equivalent surface area of the nucleus with respect to the type of event considered. The cross-section is a property that depends on:

- the type of nucleus (both A and Z)
- the type of event (induced fission, neutron capture, neutron scattering etc.)
- the kinetic energy of the incoming particles

As nuclei have a spatial extent of about 10 fm, a common unit for cross section is the barn: 1 barn = 100 fm^2. Note, however, that actual cross-sections can vary widely, and can be much larger than the spatial extent of a nucleus. For instance, ${}^{135}_{54}\mathrm{Xe}$ is well-known for its very large cross section for neutron capture ($\sigma_c = 2.5 \times 10^6$ barns) which makes it a fission poison for nuclear reactors (see exercise on the working principle of a nuclear reactor).

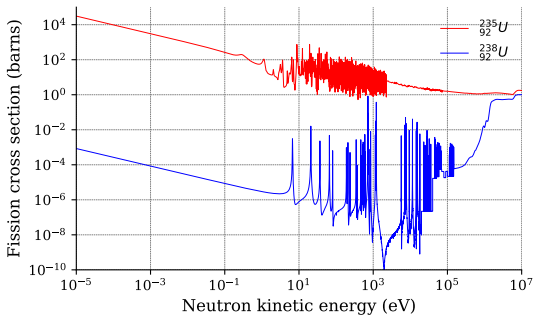


Figure 3.6: Induced fission cross section of $^{235}_{92}$U and $^{238}_{92}$U as a function of the incident energy of the neutrons. (data from the National Nuclear Data Center, Brookhaven National Lab)

A cross-section is highly sensitive to the kinetic energy of the incident particles, as is illustrated in Figure 3.6 regarding the fission cross-sections of $^{235}_{92}$U and $^{238}_{92}$U. The cross-section of $^{238}_{92}$U is rather small at all energies of the incident neutrons, and $^{238}_{92}$U is unlikely to exhibit induced fission. In contrast, the cross-section of $^{235}_{92}$U reaches very high values, up to 10^4 barns. The cross sections of both $^{235}_{92}$U and $^{238}_{92}$U vary by many orders of magnitude between low-energy neutrons and high-energy neutrons. Thus, the probability of nuclear processes varies widely with the incident velocity of the neutrons, and each nucleus can exhibit multiple types of possible processes. Cross sections are the appropriate concept to estimate, for a given nucleus, what nuclear processes can occur and with what probability.

To illustrate this, we consider the case of $^{235}_{92}$U. Three nuclear processes have significant probabilities: induced fission, elastic scattering, and radiative capture (neutron capture followed by γ emissions). Figure 3.7 displays the cross-sections of $^{235}_{92}$U for these three processes. From these curves, one immediately sees that the response to a flux of incident neutrons is very different for fast neutrons (energy $\sim$ MeV) and for thermal neutrons (ambient kinetic energy $\approx$ 25 meV). For thermal neutrons, the most likely process is neutron capture followed by induced fission (86% probability) or by radiative capture (14% probability). Elastic scattering is quite unlikely (2%). For fast neutrons, however, the cross sections are much smaller, corresponding to 100 times less nuclear events than for thermal neutrons. And the most likely process becomes elastic scattering, i.e., most neutrons simply pass through the nucleus, a few bounce off the nucleus, and very few are captured. Accordingly, sustaining an induced fission of $^{235}_{92}$U is much more likely for thermal neutrons. Yet, free neutrons originating from a previous fission are produced with high kinetic energy (typically a few MeV) and therefore they must be slowed down in order to trigger a new fission of $^{235}_{92}$U. This is the reason why it is necessary to use a 'moderator' in a nuclear reactor.

3.3.3 Radiative capture

Radiative capture designates the process by which a neutron is captured by a nucleus, followed by γ emissions. The Figure 3.8 shows the example of gold 197 ($^{197}_{79}$Au) which becomes $^{198}_{79}$Au* after radiative capture of thermal neutron. The excited nucleus $^{198}_{79}$Au* rapidly decays in the ground state of $^{198}_{79}$Au. This capture process followed by the γ emission releases 0.411 MeV of radiation and is formally noted $^{197}_{79}\mathrm{Au}\,(n,\gamma)\,^{198}_{79}\mathrm{Au}$. The resulting nucleus $^{198}_{79}$Au is an unstable isotope and exhibits β^- decay in $^{198}_{80}$Hg with a half-life of 2.7 days, yielding additional γ emissions.

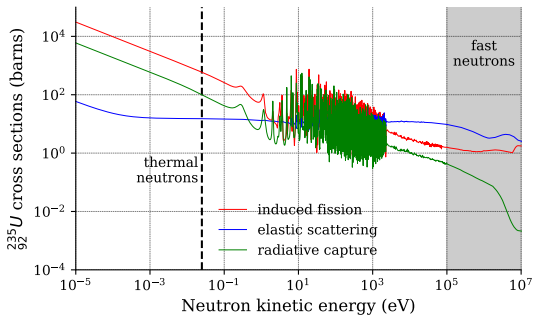


Figure 3.7: Cross sections of $^{235}_{92}$U regarding induced fission, elastic scattering of neutrons, and radiative capture. (data from the National Nuclear Data Center, Brookhaven National Lab)

$$\underbrace{^{197}_{79}\mathrm{Au} + {}^{1}_{0}\mathrm{n}\,(\mathrm{thermal}) \rightarrow {}^{198}_{79}\mathrm{Au} + \gamma\,(0.411\ \mathrm{MeV})}_{^{197}_{79}\mathrm{Au}(n,\gamma)^{198}_{79}\mathrm{Au}}$$

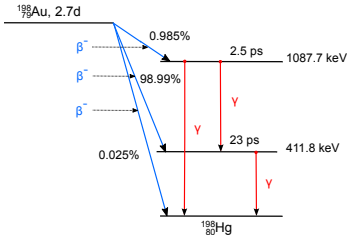


Figure 3.8: Radiative capture of thermal neutron by $^{197}_{79}$Au followed by β decay of $^{198}_{79}$Au. (chart from wikimedia commons, public domain)

Radiative capture is an important process with respect to nuclear reactor operations as it consumes neutrons and thus reduces the amount of neutrons available for induced fission. Some nuclei with large radiative capture cross-sections are used as neutron absorbers in control rods to regulate the chain reaction. Material frequently considered are boron-rich materials with boron 10 ($^{10}_{5}$B, $\sigma_c = 3835$ barns) or silver-cadmium-indium alloys. Radiative capture can also be an issue for nuclear reactor operations when a highly absorbing nucleus is produced by the nuclear reactions themselves. If such a nucleus is present in too large quantities, it can hinder the chain reaction, acting as a fission poison. An example is $^{135}_{54}$Xe (see exercise in small class), which exhibits a huge radiative capture cross section ($\sigma_c = 2.65 \times 10^6$ barns).

3.3.4 Fission

Only a few heavy nuclei are 'fissile', i.e., exhibit a significant induced fission cross-section, the most commonly encountered being $^{233}_{92}$U, $^{235}_{92}$U, $^{239}_{94}$Pu, and$^{241}_{94}$Pu. Among those, $^{235}_{92}$U is the only naturally occurring fissile nuclei. Let us describe in more detail the induced fission process of $^{235}_{92}$U. As we have discussed before, $^{235}_{92}$U fission requires slow neutrons. When a $^{235}_{92}$U nucleus captures a slow neutron, it is converted into an unstable excited $^{236}_{92}$U* which either radiates γ rays (14% probability) or exhibits fission and breaks into two fission fragments (on very rare occasions it can also break into three fragments). As the fission fragments are much smaller than the initial uranium nucleus, they tend to favor a number of neutrons closer to the number of protons in order to follow the valley of stability. As a consequence, fission also yields a few lone neutrons along with the two fission fragments. The overall process can be written:

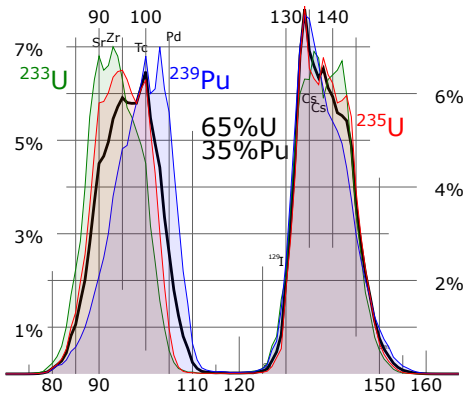


Figure 3.9: Distribution of fission products with respect to the number of mass (adapted from wikimedia commons, public domain)

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n}\,(\text{thermal}) \rightarrow {}^{236}_{92}\mathrm{U}^{*}$$

$$\text{or} \begin{cases} {}^{236}_{92}\mathrm{U}^{*} \rightarrow {}^{236}_{92}\mathrm{U} + \gamma \text{ with } 14\% \text{ probability} \\ {}^{236}_{92}\mathrm{U}^{*} \rightarrow \text{fission products} + 2.4\,{}^{1}_{0}\mathrm{n} + 192.9 \text{ MeV with } 86\% \text{ probability} \end{cases}$$

The fission reaction is not deterministic but is a stochastic process, and the fission products are not well defined. Instead, there is a wide variety of possible fission products, with a notable asymmetry in number of mass (Fig. 3.9), i.e., one of the two products is heavier than the other. Likewise, the number of neutrons produced by fission is not fixed but varies from one fission event to another, hence the 2.4 neutrons released on average. Likewise, the energy released is not fixed and 192.9 MeV is an average value. A constraint though, is that the total numbers of protons and neutrons must be conserved during the fission process. A few examples of actual fission products are:

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{90}_{36}\mathrm{Kr} + {}^{144}_{56}\mathrm{Ba} + 2{}^{1}_{0}\mathrm{n}$$
$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{92}_{36}\mathrm{Kr} + {}^{143}_{56}\mathrm{Ba} + 3{}^{1}_{0}\mathrm{n}$$
$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{94}_{38}\mathrm{Sr} + {}^{140}_{54}\mathrm{Xe} + 2{}^{1}_{0}\mathrm{n}$$
$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{94}_{40}\mathrm{Zr} + {}^{139}_{52}\mathrm{Te} + 3{}^{1}_{0}\mathrm{n}$$

The fission products themselves are often unstable and follow decay chains after the fission. For instance, for the first of the above reactions, both $^{90}_{36}\mathrm{Kr}$ and $^{144}_{56}\mathrm{Ba}$ are unstable, and their respective decay chains are:

$$^{90}_{36}\mathrm{Kr} \xrightarrow[\beta-]{32.3\text{ s}} {}^{90}_{37}\mathrm{Rb} \xrightarrow[\beta-]{158\text{ s}} {}^{90}_{38}\mathrm{Sr} \xrightarrow[\beta-]{28.9\text{ y}} {}^{90}_{39}\mathrm{Y} \xrightarrow[\beta-]{64.1\text{ h}} {}^{90}_{40}\mathrm{Zr}$$
$$^{144}_{56}\mathrm{Ba} \xrightarrow[\beta-]{11.5\text{ s}} {}^{144}_{57}\mathrm{La} \xrightarrow[\beta-]{40.8\text{ s}} {}^{144}_{58}\mathrm{Ce} \xrightarrow[\beta-]{285\text{ d}} {}^{144}_{59}\mathrm{Pr} \xrightarrow[\beta-]{17.3\text{ m}} {}^{144}_{60}\mathrm{Nd}$$

A majority of the energy is released during the fission reaction (for instance, 181.747 MeV is released by the fission of $^{235}_{92}\mathrm{U}$ into $^{90}_{36}\mathrm{Kr}$ and $^{144}_{56}\mathrm{Ba}$). But, because of the subsequent decays, part of the energy is released after some delay (10.99 MeV within a few minutes when reaching $^{90}_{38}\mathrm{Sr}$ and $^{144}_{58}\mathrm{Ce}$, and an additional 6.14 MeV over the complete decay chains). For this reason, nuclear fuels continue to release heat and harmful radiation long after their service life in a reactor. The fission products with long lives in particular (e.g., $^{90}_{38}\mathrm{Sr}$ in the above decay chains) are an issue as one must ensure that these wastes remain confined for long times (see radiation

		Energy (MeV)
Prompt release	Fission fragments	169.1
	Fission neutrons	4.8
	γ (and associated electrons)	7.0
Delayed release	β decay electrons	6.5
	β decay neutrinos (lost)	8.8
	γ emission	6.3
Subtotal		202.5
Neutron capture		8.8
Total		211.3

Table 3.2: Average energy budget for $^{235}_{92}$U fission.

and safety section). The detailed average energy budget of $^{235}_{92}$U fission is given in Table 3.2.

The fact that each induced fission event produces more than one neutron makes it possible to establish a chain reaction, i.e., each fission event can induce further fission events. The key to the stable operation of a nuclear reactor is to ensure a multiplication factor of one, that is, for each fission, on average one of the produced neutrons will induce a new fission, while the others don't (e.g., radiative capture). While the majority of neutrons, called prompt neutrons, are emitted at the time of the fission event, a small fraction of the neutrons are emitted during the subsequent decay chains as direct neutron emission from an excited nucleus. These neutrons are called delayed neutrons, and, in the case of $^{235}_{92}$U fission, they represent 0.65% of the neutrons produced with an average delay of 12 s. Even though delayed neutrons are a small minority, these are important in the control of a nuclear reactor: they provide a time scale that is long enough to permit engineering control of the multiplication factor.

Some non-fissile nuclei can produce fissile nuclei after a radiative capture. Such nuclei are called 'fertile'. $^{238}_{92}$U is a fertile nucleus as radiative capture with this nucleus and the subsequent decays lead to the fissile $^{239}_{94}$Pu:

$$^{238}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{239}_{92}\mathrm{U} \xrightarrow[\beta^-,1.265\ \mathrm{MeV}]{23.5\ \mathrm{mn}} {}^{239}_{93}\mathrm{Np} \xrightarrow[\beta^-,0.722\ \mathrm{MeV}]{2.365\ \mathrm{d}} {}^{239}_{94}\mathrm{Pu}$$

$^{235}_{92}$U represents only 0.72% of naturally occurring uranium, and the proven reserves correspond to about 130 years of current consumption, which is somehow comparable to fossil fuels (oil, gas, coal). In contrast, 99.275% of uranium is $^{238}_{92}$U, and, therefore, the possibility of converting it into a fissile nucleus would dramatically change the perspective regarding available reserves. Breeder reactors are experimented to do so. These work with fast neutrons instead of slow neutrons to favor radiative capture by $^{238}_{92}$U over other possible processes.

3.3.5 Elastic scattering

Most nuclear reactors are based on fission induced by slow neutrons, which requires a way to moderate the speed of the neutrons produced by fission. Moderation is achieved by elastic scattering on appropriate nuclei ('moderator'). This scattering process involves collisions between neutrons and nuclei, such that the total angular momentum and total kinetic energy are conserved. When a neutron of incident energy E is scattered by a nucleus, it exchanges energy with this nucleus so that the resulting energy E' of the neutron is decreased after the scattering event. The decrease of kinetic energy of the neutron is characterized by the so called 'logarithmic energy decrement', defined as $\xi = \left\langle -\ln\left(\frac{E'}{E}\right)\right\rangle$, where the average is over all possible direction of scattering. One can show (see supplementary exercise in chapter 6) that the logarithmic decrement depends only on the number of mass of the nucleus as follows:

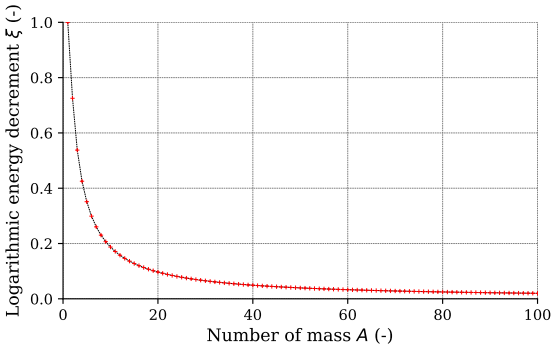


Figure 3.10: Logarithmic energy decrement after an elastic scattering event.

$$\xi(A) = 1 + \frac{(A-1)^2}{2A} \ln\left(\frac{A-1}{A+1}\right)$$

This relation between the logarithmic decrement and the number of mass is shown in Figure 3.10. The larger the logarithmic decrement is, the faster the speed of the neutron is reduced. After each scattering event, the energy is reduced by a factor $\frac{E'}{E} = \exp(-\xi)$. Therefore, after n events, $\frac{E_n}{E_0} = \exp(-n\xi)$. From Figure 3.10, we see that light nuclei are much more efficient at moderating the velocity of neutrons than heavy nuclei. With the lightest nucleus ${}^{1}_{1}\mathrm{H}$, a fast neutron of initial energy $E_0 = 2$ MeV is thermalized ($E_n = 25$ meV) in 18 scattering events. In contrast, for ${}^{238}_{92}\mathrm{U}$, thermalization requires about 2000 events.

Being a light nucleus with a large enough scattering cross-section is not enough to be a good moderator. Indeed, as we discussed in the previous section, the capture of neutrons is detrimental to maintaining a chain reaction in a nuclear reactor as it consumes part of the neutrons released by fission. Accordingly, a good moderator should combine the following properties:

- Have a high logarithmic decrement, i.e., be a light nucleus
- Have a significant scattering cross-section
- Have a low capture cross-section

A frequently used and widely available moderator is water ${}^{1}_{1}\mathrm{H}_2{}^{16}_{8}\mathrm{O}$ where the hydrogens moderate the neutrons while both the hydrogens and oxygens have rather small capture cross sections.

3.3.6 Fusion

under preparation

3.4 Applications

3.4.1 nuclear reactors

The working principle of a nuclear reactor consists in maintaining a chain of fission reactions so that the neutrons produced by each fission event induce a new fission event. Thus, the heart of the matter is to understand in detail the neutron budget during a cycle from one fission event to another as schematically represented in Figure 3.11. Let us follow a neutron throughout this cycle:

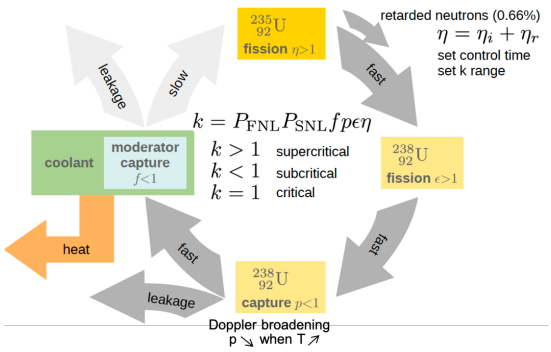


Figure 3.11: Schematic representation of the neutron cycle in a nuclear reactor with the different contributions to the multiplication factor.

1. We start from a thermal neutron that is captured by a uranium nucleus present in the nuclear fuel (top box). Considering that the nuclear fuel contains a fraction x of $^{235}_{92}$U and $(1-x)$ of $^{238}_{92}$U, the probability that this neutron capture induces a fission of $^{235}_{92}$U is given by the ratio of cross-section $\frac{x\sigma_f\left(^{235}_{92}\mathrm{U}\right)}{x\sigma_a\left(^{235}_{92}\mathrm{U}\right)+(1-x)\sigma_a\left(^{238}_{92}\mathrm{U}\right)}$, where the subscript $_f$ refers to fission, and the subscript $_a$ to absorption (i.e., fission and radiative capture). As we have seen previously, this fission produces 2.42 fast neutrons on average. Therefore, the number of fast neutrons produced on average for each thermal neutron capture is given by the reproduction factor:

$$\eta = 2.42 \times \frac{x\sigma_f\left(^{235}_{92}\mathrm{U}\right)}{x\sigma_a\left(^{235}_{92}\mathrm{U}\right) + (1-x)\,\sigma_a\left(^{238}_{92}\mathrm{U}\right)}$$

2. On rare occasions, a fast neutron induces the fission of a $^{238}_{92}$U. Indeed, although $^{238}_{92}$U exhibits a much smaller fission cross-section than $^{235}_{92}$U, this cross-section increases significantly at high neutron energies and approaches that of $^{235}_{92}$U (Fig. 3.6). This fast neutron cross section is too small for $^{238}_{92}$U to sustain a chain reaction alone, but it has a non-negligible contribution to the neutron budget of a $^{235}_{92}$U chain reaction. The fission of $^{238}_{92}$U consumes 1 fast neutron and then produces 1.7 fast neutrons, thus increasing the number of fast neutrons in the cycle. This increase is measured by the fast fission factor $\epsilon > 1$ so that after the $^{238}_{92}$U fission a total of $\eta \times \epsilon$ neutrons are available.

3. As we discussed in the previous sections, fast neutrons must be slowed down by a moderator to initiate new fission of $^{235}_{92}$U. When slowing down, a neutron can be captured by $^{238}_{92}$U whose capture cross section exhibits numerous resonance peaks between 10 eV and 10 keV (caused by the bound state of energies within the nucleus). As a result, a fraction p of the fast neutrons are captured by $^{238}_{92}$U before being thermalized, resulting in a total of $\eta \times \epsilon \times p$ neutrons. The resonance peaks get wider as temperature increases, a phenomenon called 'Doppel broadening'. Therefore, the fraction p decreases with temperature.

4. Once the neutrons have been thermalized, they can still be captured by the moderator rather than by the nuclear fuel. Therefore, the fraction f of the neutrons that are not captured by the moderator is called the thermal utilization factor. Eventually, the number of neutrons that close the neutron cycle is given by:

$$k_\infty = \eta \times \epsilon \times p \times f$$

k_∞ is the neutron-multiplication factor for a virtually 'infinite' reactor.

5. In practice, reactors are necessarily finite so that one cannot avoid some neutron leakage at the boundary of the reactor. Fast neutron leakage reduces the number of available neutrons by a factor P_{FNL}. Likewise, slow neutron leakage reduces the number of neutrons by a factor P_{SNL}. Ultimately, the actual neutron multiplication factor is given by:
$$k = P_{FNL}P_{SNL}k_\infty = P_{FNL}P_{SNL}\eta\epsilon pf$$

For a constant multiplication factor k, the time evolution of the number of neutrons during the chain reaction is given by:

$$N(t) = N_0 \exp\left((k-1)\frac{t}{t_0}\right)$$

where t_0 is the time it takes for each generation of neutrons to produce the next generation. A reactor with $k > 1$ has a growing population of neutrons and is said supercritical. Conversely, a reactor with $k < 1$ has a decreasing population of neutrons and is said subcritical. The particular case $k = 1$ corresponds to a stable population of neutrons and is said critical. Nuclear reactors are designed to keep the multiplication factor very close to $k = 1$. To do so, one plays on the different factors affecting k to adjust it. Active control is achieved by using control rods containing neutron absorbers. In addition, intrinsic retroaction loops are implemented by design so that if k departs too much from 1 it is brought back to 1. For instance, in water-pressurized reactors, an increase of k makes the water boil, which reduces dramatically the efficiency of moderation, since ${}^{1}_{1}\mathrm{H}$ density in a vapor is much lower than in a liquid, and thus reduces k. This negative feedback between water moderation and temperature is also known as the negative void coefficient of reactivity. Another intrinsic retroaction is the Doppler broadening phenomenon that increases the radiative capture of neutrons by ${}^{238}_{92}\mathrm{U}$ as temperature increases.

Another key feature in the control of the neutron cycle is the existence of delayed neutrons. Indeed, in the above description, we assumed a single time scale t_0 for the neutron cycle, as if all neutrons were prompt neutrons. Prompt neutrons are emitted within $t_{emission} \approx 10^{-15}$ s, they are then slowed down in about $t_{moderation} \approx 5.6 \times 10^{-6}$ s (for a water-moderated thermal reactor), and, once thermalized, neutrons diffuse for about $t_{capture} \approx 2.1 \times 10^{-4}$ s before being captured by the nuclear fuel. Therefore, the characteristic time is of the order of $t_0^{prompt} \approx 2 \times 10^{-4}$ s for prompt neutrons, which is way too fast to allow proper engineering control with control rods: even for a multiplication factor very close to 1, for instance $k = 1.001$, the number of neutrons would increase by a factor $\exp(5) = 150$ within a second. The existence of delayed neutrons introduces an additional time scale. Indeed, as mentioned in the previous section, a small fraction of the neutrons produced by fission are emitted with a delay during the decay of the fission products. For ${}^{235}_{92}\mathrm{U}$ fission, a fraction $d = 0.65$ % of neutrons are delayed neutrons, with an average delay of 12 s, much longer than the characteristic time scale t_0^{prompt} for prompt neutrons. So there is a second characteristic time scale $t_0^{delayed} \approx 12$ s governing the neutron cycle of a small fraction d of the neutrons. On the short time scale t_0^{prompt}, the effective multiplicative factor is $k - d$. Therefore, if $k < 1 + d$ there is no exponential growth on the short time scale. It is only after the long time scale $t_0^{delayed}$ that critical condition is reached. Thus, as long as the multiplication factor does not exceed $1 + d$, the time scale for the growth of the neutron population is that of the delayed neutrons, i.e., ~ 10 s, which is now long enough to allow for an engineering control. Accordingly, delayed neutrons are essential for the control of nuclear fission.

Natural uranium contains only $x = 0.72\%$ of ${}^{235}_{92}\mathrm{U}$, which corresponds to a rather small reproduction factor $\eta = 1.33$. Reaching criticality with such a reproduction factor requires very little capture during thermalization and diffusion, which is achieved in practice in only a few types of reactors such as heavy water ${}^{2}_{1}\mathrm{H}_2{}^{16}_{8}\mathrm{O}$ reactors taking advantage of the very low capture cross-section of deuterium ${}^{2}_{1}\mathrm{H}$ for moderation. Instead, the majority of fission reactors

Typology	Nuclear fuel	Moderator	Coolant	Comment
pressurized water	enriched	high-pressure water		80% of reactors
boiling water	enriched	low-pressure water		
pressurized heavy water	natural	high-pressure heavy water		refuelable in service
gas-cooled	graphite	CO_2	high T	
pebbled-bed	pebbles	graphite	helium	first gen. IV
molten salt	dissolved in salt	graphite	molten salt mixture	
liquid metal fast breeder	breeder	none	liquid metal lead/sodium	pb. sodium/water
aqueous homogeneous	nuclear salts dissolved in coolant/moderator	high-pressure light or heavy water		

Table 3.3: Main typologies of nuclear reactors.

use enriched nuclear fuel in which the $^{235}_{92}$U content is significantly larger in the nuclear fuel, typically $x = 3.5\%$ to 5%. Table 3.3 compares the main typologies of nuclear reactors.

3.4.2 Radiation and safety

As we have seen, nuclear reactions yield a wide variety of unstable nuclei, that continue to decay long after the reactions. There are 4 main types of radiation resulting from those decay with different levels of hazard for living organisms:

- α emission is the least hazardous radiation. $^{4}_{2}$He is a complete atom whose electronic cloud is subjected to steric repulsion in any material. As a consequence, α particles are easily stopped by a simple sheet of paper or by the skin.
- β decay emits an electron/positron that is a bit more penetrating. It can be stopped by a thin aluminum foil.
- γ rays resulting from γ emission are the most energetic photons. They are highly penetrating, and their energy is so high that they can ionize any bound electron they encounter. A material's half-value layer (HVL) is the thickness of this material for which the radiation intensity is reduced by half. For γ rays, the HVL is 1 cm for lead, 4 cm for granite, and 6 cm for concrete.
- Neutrons emitted by fission are also highly penetrating. As we have seen, neutron moderation and absorption are strongly nucleus-dependent. Good materials to stop neutron radiation are water, concrete, graphite, etc.

Table 3.4 shows the main nuclei involved in radioactive wastes. The table on the left shows the typical composition of the spent nuclear fuel collected after operating a nuclear reactor. One distinguishes:

- The $^{235}_{92}$U and $^{238}_{92}$U residues that correspond to the fraction of the original nuclear fuel that has not been involved in the nuclear reactions and has not decayed.
- The fission products, which cover the fission fragments produced by the fission reaction, and the nuclei resulting from the decay of these fragments.
- The heavy nuclei that result from $^{235}_{92}$U/$^{238}_{92}$U subjected the (possibly multiple) radiative captures and decays. These heavy nuclei are called actinides. Apart from uranium, plutonium isotopes are by far the most frequent actinides. Therefore, uranium and plutonium isotopes are called major actinides, while all other actinides are referred to as minor actinides.

Category	Nucleus	Mass	Half-life	Decay
Residues of original fuel	$^{238}_{92}$U	20 t	4.5 Gy	α
	$^{235}_{92}$U	162 kg	700 My	α
Plutonium isotopes	$^{239}_{94}$Pu	120 kg	24 ky	α
	$^{240}_{94}$Pu	50 kg	6.6 ky	α
	$^{241}_{94}$Pu	18 kg	14 y	β^-
	$^{242}_{94}$Pu	10 kg	375 ky	α
	$^{238}_{94}$Pu	2 kg	88 y	α
Minor actinides	$^{237}_{93}$Np	10 kg	2.2 My	α
	$^{241}_{95}$Am		432 y	α
	$^{243}_{95}$Am		7.4 ky	α
	$^{244}_{96}$Cm		18 y	α
	$^{245}_{96}$Cm	1 kg	8.5 ky	α
Fission products	$^{99}_{43}$Tc	18 kg	211 ky	β^-
	$^{93}_{40}$Zr	19 kg	1.5 My	β^-
	$^{135}_{55}$Cs	9 kg	2.3 My	β^-
	$^{107}_{46}$Pd	5 kg	6.5 My	β^-
	$^{129}_{53}$I	3 kg	15 My	β^-
	total	760 kg	-	-

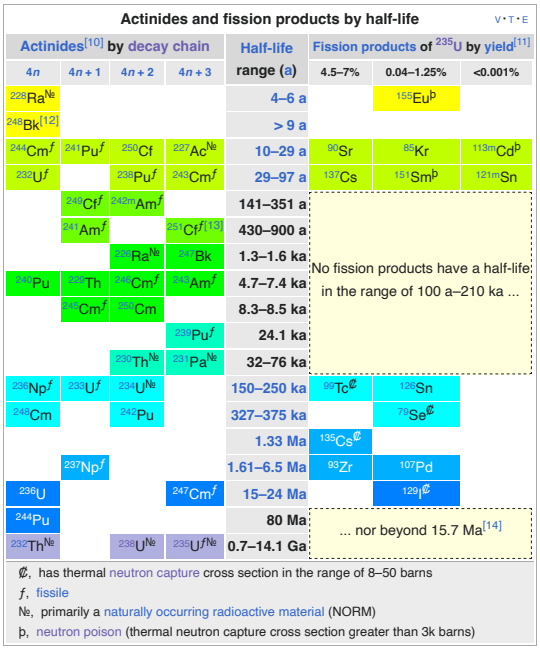


Table 3.4: (left) Approximate composition of 21 t of spent nuclear fuel from a light-water reactor fueled with low-enriched uranium. (right) List of the main radioactive wastes ranked by half-life. The nuclei on the right part are the main fission products, the nuclei on the left are actinides from nuclear fuel resulting from radiative capture and subsequent decays. (right table from `https://en.wikipedia.org/wiki/Radioactive_waste`)

The nuclear wastes present in spent nuclear fuel exhibit very different half-life times (see Table 3.4 right), and are classified into three main categories. Firstly, short-lived wastes with half-lives of days or less (e.g., $^{131}_{53}$I) almost disappear after a few years and are not an issue for long-term waste management. Secondly, medium-lived nuclei with half-lives ranging from a few years to a few hundred years (e.g., $^{90}_{38}$Sr) disappear after a few thousand years. These wastes exhibit high radioactivity and must be confined for long times compared to a human life. Thirdly, long-lived nuclei with half-lives exceeding a thousand years and up to a few million years (e.g., $^{135}_{55}$Cs) are less radioactive but require to be confined for extremely long times. Medium-lived and long-lived wastes cannot be confined in a conventional manner and must be stored in a way that does not rely on human action to guarantee long-term confinement. In this respect, nuclear waste storage in underground facilities located in low permeability rocks is planned in many countries, including France.

Some important nuclear wastes are:

- $^{90}_{38}$Sr and $^{137}_{55}$Cs which are the most frequent medium-lived fission products.
- $^{99}_{43}$Tc, $^{135}_{55}$Cs, and $^{93}_{40}$Zr which are the most frequent long-lived fission products.
- $^{237}_{93}$Np, $^{241}_{95}$Am, $^{243}_{95}$Am, $^{244}_{96}$Cm, and $^{245}_{96}$Cm which are the most frequent medium and long-lived minor actinides.
- $^{233}_{92}$U, $^{234}_{92}$U, $^{235}_{92}$U, $^{238}_{92}$U, $^{239}_{94}$Pu, $^{240}_{94}$Pu, and $^{242}_{94}$Pu which are the long-lived major actinides.

Note that many of the actinides are fissile and can be used as nuclear nuclear fuel. In particular $^{235}_{92}$U and $^{239}_{94}$Pu, which are present in significant amounts in spent nuclear fuel, can be separated from the other wastes to be recycled as new nuclear fuel.

Recap of what must be known for the exam

1. Nuclei

A nucleus, noted ${}^{A}_{Z}\mathrm{X}$ with X the symbol of the element, is characterized by its **number of charge** Z (number of protons) and its **number of mass** A (number of protons and neutrons). The number of charges determines how the electrons arrange around the nucleus. Thus, two nuclei with same Z but different A, called **isotopes**, have identical chemical properties, and differ only by their mass.

Because of the energy-mass equivalence of relativity ($E = mc^2$ for a particle of mass m at rest), any change of mass characterizes a change of energy. Therefore, the difference of mass between an atom and the same number of free protons and free neutrons characterizes the binding energy of an atom (the mass of the electrons is negligible):

$$B\left(Z,A\right) = Z \times M_{\mathrm{H}} + \left(A - Z\right) \times M_n - M_{\mathrm{atom}}\left(Z,A\right)$$

Binding energies are usually small compared to the total mass, so one often refers to the concept of **mass excess** instead of the total mass. The mass excess is the difference between the total mass and A times a reference mass (atomic mass unit u or 'dalton', corresponding to $1/12^{th}$ of the mass of ${}^{12}_{6}\mathrm{C}$:

$$\Delta\left(Z,A\right) = M_{\mathrm{atom}}\left(Z,A\right) - Au \text{ with } 1u = \frac{1}{12}M_{{}^{12}_{6}\mathrm{C}} = 1 \text{ dalton}$$

Since all nuclear reactions of interest for energy applications conserve the number of mass between the reactants and the products, the balance of energy of a nuclear reaction is the same when expressed in terms of mass or of mass excess:

$$Q_{\mathrm{reaction}} = \sum_{\mathrm{reactants}} M_i - \sum_{\mathrm{products}} M_i = \sum_{\mathrm{reactants}} \Delta_i - \sum_{\mathrm{products}} \Delta_i$$

2. Nuclear decay

Nuclear reactions can vary the composition of the nuclei in terms of protons and neutrons provided that the number of mass A is conserved. Accordingly, the binding energy per nucleon $b\left(Z,A\right) = \frac{B(Z,A)}{A}$ can be used to compare the level of energy of the nuclei. $b\left(Z,A\right)$ exhibits a '**valley of stability**' (see Fig. 3.1) and ${}^{56}_{26}\mathrm{Fe}$ is the most tightly bound nucleus. Nuclei lighter than ${}^{56}_{26}\mathrm{Fe}$ can fusion to produce a heavier nucleus, whereas a nucleus heavier than ${}^{56}_{26}\mathrm{Fe}$ can fission to produce lighter nuclei. The 'valley of stability' is quite narrow, only a few isotopes for each atom can exist.

Nuclei can reduce their binding energy by nuclear decays, i.e., nuclear reactions by which a nucleus can spontaneously emit a small particle to change Z and/or decrease A. **Nuclear decays** are random processes and the probability that a nucleus has not decayed decreases exponentially with time $\exp\left(-\frac{\ln(2)}{\tau_{1/2}}t\right)$. The characteristic time $\tau_{1/2}$ represents the **half-life time**. The most frequent nuclear decays are :

β^- **decay:** a neutron is converted into a proton

$$^{A}[\mathrm{Z}] \rightarrow {}^{A}[\mathrm{Z}+1] + e^- + \bar{\nu}_e$$

β^+ **decay:** a proton is converted into a neutron

$$^{A}[\mathrm{Z}] \rightarrow {}^{A}[\mathrm{Z}-1] + e^+ + \nu_e$$

electron capture: a proton combines with an electron to form a neutron

$$^{A}[\mathrm{Z}] + e^{-} \rightarrow {}^{A}[\mathrm{Z}-1] + \nu_e$$

α **emission:** emission of an $^{4}_{2}\mathrm{He}$

$$^{A}[\mathrm{Z}] \rightarrow {}^{A-4}[\mathrm{Z}-2] + {}^{4}_{2}\mathrm{He}$$

β^{+} is less frequent than e^{-} (releases less energy) while connecting the same nuclei. Thus, one often refers to β^{-} decay as to β decay.

A few heavy nuclei also exhibit **spontaneous fission**, a process by which the nucleus breaks into two lighter nuclei of similar size. Spontaneous fission is a rare event compared to the other decays, but it produces free neutrons, which can serve to initiate nuclear chain reactions.

A **decay chain** is a sequence of decay reactions that follows a unstable nucleus to reach stability. There are only four primary decay chains that can be observed in nature starting from $^{232}_{90}\mathrm{Th}$, $^{237}_{93}\mathrm{Np}$, $^{238}_{92}\mathrm{U}$, and $^{235}_{92}\mathrm{U}$.

Nuclei resulting from a nuclear reaction can be found in excited states of energy, called nuclear isomers. Nuclear isomers are metastable and they ultimately relax to the non-excited state of the respective nuclei by emitting high energy photons (MeV) in the γ-ray domain. These γ **emissions** are particularly harmful for living organisms as they can penetrate deep into matter and ionize any atom.

3. nuclear reactions

A nuclear reaction is a process during which two nuclei interact to produce one or several new nuclei. Of particular interest are:

Neutron capture: a neutron is captured by a nucleus leading to a heavier excited nucleus ('*') that may further decay. If the decay is a gamma emission, the reaction reaction is called a **radiative capture**

$$^{A}[\mathrm{Z}] + {}^{1}_{0}\mathrm{n} \rightarrow {}^{A+1}[\mathrm{Z}]^{*}$$

Induced fission: after a neutron capture, the nucleus breaks into lighter nuclei.

$$^{A}[\mathrm{Z}] + {}^{1}_{0}\mathrm{n} \rightarrow {}^{a}[\mathrm{z}] + {}^{A-a-x}[\mathrm{Z}-\mathrm{z}] + (x+1)\,{}^{1}_{0}\mathrm{n}$$

Induced fission is possible only for a few heavy nuclei that are said 'fissile'. $^{235}_{92}\mathrm{U}$ is the only naturally occurring fissile nuclei. Some non-fissile nuclei are said 'fertile' because they can produce fissile nuclei after a radiative capture. Fission is a stochastic process, and there is a wide variety of possible fission products. Induced fission also produces a few neutrons of high kinetic energy (fast neutrons).

Fusion: Two light nuclei with enough kinetic energy can come close enough to form a single heavier nucleus. Fusion requires extreme temperatures and takes place in particular in stars. Fusion between deuterium and tritium is actively studied for energy applications:

$$^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n}$$

Neutron scattering is the process by which a neutron interacts with a nucleus but is not captured. Neutron scattering is not a nuclear reaction but leads to an exchange of energy with the nucleus.

For a nucleus submitted to an incoming flux ϕ of particles (number of particles per unit area and time), the probability that a nuclear process occurs per unit time is proprotional to the flux: $\frac{dp_{\text{reaction}}}{dt} = \sigma\phi$. The proportionality factor σ has a unit of area and is called the cross-section.

4. fission reactors

The vast majority of nuclear reactors use the induced fission of $^{235}_{92}$U by thermal neutrons to produce heat which is then converted to electricity. Key physical principles are:

- This fission reaction produces 2.42 fast neutrons on average. So a chain reaction is possible if at least 1 one those 2.42 fast neutrons triggers another fission.
- The cross section of $^{235}_{92}$U induced fission is much smaller for fast neutrons than for thermal neutrons. So the fast neutrons produced by the fission must be slowed down to allow a chain reaction. This is done by using a 'moderator', i.e., a medium made of light nuclei with high scattering cross section (to easily transfer the kinetic energy of the fast neutrons to the medium) and low capture cross section (to avoid consuming the neutrons). Water is frequently used as a moderator.
- During the thermalization process, part of the neutrons are inevitably lost because of 1) radiative capture by $^{238}_{92}$U, by the moderator or by some fission products with high capture cross-section (e.g., $^{135}_{54}$Xe), 2) leakage at the boundary of the reactor.

The reproduction factor (number of fast neutrons produced on average for each thermal neutron capture) of natural uranium (0.72% of $^{235}_{92}$U) is only $\eta = 1.33$. Unless a very efficient moderator is used (e.g., heavy water), a nuclear reactor is unable to maintain a chain reaction with natural uranium. Therefore, most nuclear reactors use enriched uranium with 3.5% to 5% of $^{235}_{92}$U.

Exercises

Ex. 3.1 - Chaîne de décroissance et équation de Bateman

Dans cet exercice, nous nous intéressons au processus de décroissance radioactive. Le tableau ci-après fournit les données nécessaires à l'exercice (données issues du National Nuclear Data Center, https://www.nndc.bnl.gov/nudat3/). Chaque ligne correspond à un type d'atomes ${}_{Z}\mathrm{X}$ (caractérisé par son nombre de charge Z) et liste différents isotopes ${}^{A}_{Z}\mathrm{X}$ de cet atome qui diffèrent par leurs nombre de masse A. Lorsque l'isotope existe dans la nature, l'abondance naturelle est précisée entre parenthèse après le symbole de l'isotope. En rouge est donné l'excès de masse Δ, en vert le(s) mode(s) de décroissance, et en bleu le temps de demi-vie τ (sauf si le nucléide est stable). Les modes de décroissance possibles sont les décroissances β^- et β^+, la décroissance α, la capture électronique ε. Quand plusieurs modes de décroissance sont possible pour un nucléide donné, la probabilité de chaque mode est indiquée à la suite du mode. Parfois la décroissance peut s'accompagner d'une émission de neutron notée (+ n X %), où X % représente la probabilité qu'une décroissance émette un neutron. Par railleurs, dans de rares cas, certains nucléides peuvent décroitre par fission spontanée (notée FS dans le tableau). Au début de l'exercice, on négligera la fission spontanée, qui sera abordée explicitement à la question 11. On précise enfin l'excès de masse de l'${}^{4}_{2}\mathrm{He}$ (particule alpha) : $\Delta\left(4He\right) = 2.42$ MeV. On négligera l'énergie des électrons et des neutrinos impliqués dans les réactions de décroissance.

Atome	Isotopes									
$_{81}$Tl	$^{203}_{81}$Tl (29.5%) $\Delta = -25.76$ MeV stable - -	$^{204}_{81}$Tl $\Delta = -24.35$ MeV β^- 97.08 % & $\varepsilon + \beta^+$ 2.92 % $\tau = 3.78$ ans -	$^{205}_{81}$Tl (70.5%) $\Delta = -23.82$ MeV stable - -	$^{206}_{81}$Tl $\Delta = -22.25$ MeV β^- $\tau = 4.202$ min -	$^{207}_{81}$Tl $\Delta = -21.03$ MeV β^- $\tau = 4.77$ min -	$^{208}_{81}$Tl $\Delta = -16.75$ MeV β^- $\tau = 3.053$ min -	$^{209}_{81}$Tl $\Delta = -13.65$ MeV β^- $\tau = 2.162$ min -	$^{210}_{81}$Tl $\Delta = -9.25$ MeV β^- (+ n 0.007 %) $\tau = 1.30$ min -	$^{211}_{81}$Tl $\Delta = -6.08$ MeV β^- (+ n 2.2 %) $\tau = 79.6$ s -	$^{212}_{81}$Tl $\Delta = -1.55$ MeV β^- (+ n 1.8 %) $\tau = 30.9$ s -
$_{82}$Pb	$^{205}_{82}$Pb $\Delta = -23.77$ MeV ε $\tau = 1.70 \times 10^7$ ans -	$^{206}_{82}$Pb (24.1%) $\Delta = -23.79$ MeV stable - -	$^{207}_{82}$Pb (22.1%) $\Delta = -22.45$ MeV stable - -	$^{208}_{82}$Pb (52.4%) $\Delta = -21.75$ MeV stable - -	$^{209}_{82}$Pb $\Delta = -17.61$ MeV β^- $\tau = 3.235$ h -	$^{210}_{82}$Pb $\Delta = -14.73$ MeV β^- $\tau = 22.20$ ans -	$^{211}_{82}$Pb $\Delta = -10.49$ MeV β^- $\tau = 36.165$ min -	$^{212}_{82}$Pb $\Delta = -7.55$ MeV β^- $\tau = 10.628$ h -	$^{213}_{82}$Pb $\Delta = -3.20$ MeV β^- $\tau = 10.2$ min -	$^{214}_{82}$Pb $\Delta = -0.18$ MeV β^- $\tau = 27.06$ min -
$_{83}$Bi	$^{207}_{83}$Bi $\Delta = -20.05$ MeV $\varepsilon + \beta^+$ $\tau = 31.22$ ans -	$^{208}_{83}$Bi $\Delta = -18.87$ MeV $\varepsilon + \beta^+$ $\tau = 3.68 \times 10^5$ ans -	$^{209}_{83}$Bi (100%) $\Delta = -18.26$ MeV α $\tau = 2.05 \times 10^{19}$ ans -	$^{210}_{83}$Bi $\Delta = -14.79$ MeV β^- $\tau = 5.012$ jours -	$^{211}_{83}$Bi $\Delta = -11.86$ MeV α 99.724 % & β^- 0.276 % $\tau = 2.14$ min -	$^{212}_{83}$Bi $\Delta = -8.12$ MeV β^- 64.07 % & α 35.93 % $\tau = 60.551$ min -	$^{213}_{83}$Bi $\Delta = -5.23$ MeV β^- 97.872 % & α 2.128 % $\tau = 45.607$ min -	$^{214}_{83}$Bi $\Delta = -1.20$ MeV β^- 99.979 % & α 0.021 % $\tau = 19.71$ min -	$^{215}_{83}$Bi $\Delta = 1.63$ MeV β^- $\tau = 7.6$ min -	$^{216}_{83}$Bi $\Delta = 5.87$ MeV β^- $\tau = 2.25$ min -
$_{84}$Po	$^{209}_{84}$Po $\Delta = -16.37$ MeV α 99.545 % & $\varepsilon + \beta^+$ 0.455 % $\tau = 124$ ans -	$^{210}_{84}$Po $\Delta = -15.95$ MeV α $\tau = 138.378$ jours -	$^{211}_{84}$Po $\Delta = -12.43$ MeV α $\tau = 0.516$ s -	$^{212}_{84}$Po $\Delta = -10.37$ MeV α $\tau = 295.0$ ns -	$^{213}_{84}$Po $\Delta = -6.65$ MeV α $\tau = 3.706$ µs -	$^{214}_{84}$Po $\Delta = -4.47$ MeV α $\tau = 163.68$ µs -	$^{215}_{84}$Po $\Delta = -0.54$ MeV α $\tau = 1.781$ ms -	$^{216}_{84}$Po $\Delta = 1.78$ MeV α $\tau = 144.0$ ms -	$^{217}_{84}$Po $\Delta = 5.88$ MeV α 97.5 % & β^- 2.5 % $\tau = 1.53$ s -	$^{218}_{84}$Po $\Delta = 8.36$ MeV α 99.978 % & β^- 0.022 % $\tau = 3.097$ min -
$_{85}$At	$^{211}_{85}$At $\Delta = -11.65$ MeV ε 58.22 % & α 41.78 % $\tau = 7.216$ h -	$^{212}_{85}$At $\Delta = -8.63$ MeV α $\tau = 314.0$ ms -	$^{213}_{85}$At $\Delta = -6.58$ MeV α $\tau = 124.0$ ns -	$^{214}_{85}$At $\Delta = -3.38$ MeV α $\tau = 558.0$ ns -	$^{215}_{85}$At $\Delta = -1.26$ MeV α $\tau = 37.0$ µs -	$^{216}_{85}$At $\Delta = 2.26$ MeV α $\tau = 0.30$ ms -	$^{217}_{85}$At $\Delta = 4.40$ MeV α 99.993 % & β^- 0.007 % $\tau = 32.6$ ms -	$^{218}_{85}$At $\Delta = 8.10$ MeV α 99.95 % & β^- 0.05 % $\tau = 1.28$ s -	$^{219}_{85}$At $\Delta = 10.40$ MeV α 93.6 % & β^- 6.4 % $\tau = 56.0$ s -	$^{220}_{85}$At $\Delta = 14.38$ MeV β^- 92 % & α 8 % $\tau = 3.71$ min -
$_{86}$Rn	$^{214}_{86}$Rn $\Delta = -4.32$ MeV α $\tau = 259.0$ ns -	$^{215}_{86}$Rn $\Delta = -1.17$ MeV α $\tau = 2.3$ µs -	$^{216}_{86}$Rn $\Delta = 0.25$ MeV α $\tau = 29.0$ µs -	$^{217}_{86}$Rn $\Delta = 3.66$ MeV α $\tau = 0.59$ ms -	$^{218}_{86}$Rn $\Delta = 5.22$ MeV α $\tau = 33.75$ ms -	$^{219}_{86}$Rn $\Delta = 8.83$ MeV α $\tau = 3.96$ s -	$^{220}_{86}$Rn $\Delta = 10.61$ MeV α $\tau = 55.6$ s -	$^{221}_{86}$Rn $\Delta = 14.47$ MeV β^- 80 % & α 20 % $\tau = 25.7$ min -	$^{222}_{86}$Rn $\Delta = 16.37$ MeV α $\tau = 3.821$ jours -	$^{223}_{86}$Rn $\Delta = 20.39$ MeV β^- $\tau = 24.3$ min -
$_{87}$Fr	$^{217}_{87}$Fr $\Delta = 4.32$ MeV α $\tau = 22.0$ µs -	$^{218}_{87}$Fr $\Delta = 7.06$ MeV α $\tau = 1.1$ ms -	$^{219}_{87}$Fr $\Delta = 8.62$ MeV α $\tau = 22.0$ ms -	$^{220}_{87}$Fr $\Delta = 11.48$ MeV α 99.65 % & β^- 0.35 % $\tau = 27.4$ s -	$^{221}_{87}$Fr $\Delta = 13.28$ MeV α 99.995 % & β^- 0.005 % $\tau = 4.801$ min -	$^{222}_{87}$Fr $\Delta = 16.38$ MeV β^- $\tau = 14.2$ min -	$^{223}_{87}$Fr $\Delta = 18.38$ MeV β^- 99.994 % & α 0.006 % $\tau = 21.99$ min -	$^{224}_{87}$Fr $\Delta = 21.75$ MeV β^- $\tau = 3.33$ min -	$^{225}_{87}$Fr $\Delta = 23.82$ MeV β^- $\tau = 4.0$ min -	$^{226}_{87}$Fr $\Delta = 27.52$ MeV β^- $\tau = 49.0$ s -
$_{88}$Ra	$^{220}_{88}$Ra $\Delta = 10.27$ MeV α $\tau = 18.0$ ms -	$^{221}_{88}$Ra $\Delta = 12.96$ MeV α $\tau = 16.0$ s -	$^{222}_{88}$Ra $\Delta = 14.32$ MeV α $\tau = 33.6$ s -	$^{223}_{88}$Ra $\Delta = 17.23$ MeV α $\tau = 11.435$ jours -	$^{224}_{88}$Ra $\Delta = 18.83$ MeV α $\tau = 3.631$ jours -	$^{225}_{88}$Ra $\Delta = 21.99$ MeV β^- $\tau = 14.8$ jours -	$^{226}_{88}$Ra $\Delta = 23.67$ MeV α $\tau = 1600$ ans -	$^{227}_{88}$Ra $\Delta = 27.18$ MeV β^- $\tau = 42.2$ min -	$^{228}_{88}$Ra $\Delta = 28.94$ MeV β^- $\tau = 5.75$ ans -	$^{229}_{88}$Ra $\Delta = 32.56$ MeV β^- $\tau = 4.0$ min -
$_{89}$Ac	$^{223}_{89}$Ac $\Delta = 17.83$ MeV α 99 % & ε 1 % $\tau = 2.12$ min -	$^{224}_{89}$Ac $\Delta = 20.23$ MeV $\varepsilon + \beta^+$ 90.9 % & α 9.1 % $\tau = 2.78$ h -	$^{225}_{89}$Ac $\Delta = 21.64$ MeV α $\tau = 9.919$ jours -	$^{226}_{89}$Ac $\Delta = 24.31$ MeV β^- 83 % & ε 17 % $\tau = 29.37$ h -	$^{227}_{89}$Ac $\Delta = 25.85$ MeV β^- 98.62 % & α 1.38 % $\tau = 21.773$ jours -	$^{228}_{89}$Ac $\Delta = 28.89$ MeV β^- $\tau = 6.15$ h -	$^{229}_{89}$Ac $\Delta = 30.69$ MeV β^- $\tau = 62.7$ min -	$^{230}_{89}$Ac $\Delta = 33.84$ MeV β^- $\tau = 122.0$ s -	$^{231}_{89}$Ac $\Delta = 35.76$ MeV β^- $\tau = 7.5$ min -	$^{232}_{89}$Ac $\Delta = 39.15$ MeV β^- $\tau = 119.0$ s -
$_{90}$Th	$^{226}_{90}$Th $\Delta = 23.20$ MeV α $\tau = 30.69$ min -	$^{227}_{90}$Th $\Delta = 25.80$ MeV α $\tau = 18.694$ jours -	$^{228}_{90}$Th $\Delta = 26.77$ MeV α $\tau = 1.912$ ans -	$^{229}_{90}$Th $\Delta = 29.59$ MeV α $\tau = 7907.5$ ans -	$^{230}_{90}$Th (0.02%) $\Delta = 30.86$ MeV α $\tau = 75400$ ans -	$^{231}_{90}$Th $\Delta = 33.82$ MeV β^- $\tau = 25.52$ h -	$^{232}_{90}$Th (99.98%) $\Delta = 35.45$ MeV α & FS 1.1×10^{-9}% $\tau = 1.407 \times 10^{10}$ ans -	$^{233}_{90}$Th $\Delta = 38.73$ MeV β^- $\tau = 21.83$ min -	$^{234}_{90}$Th $\Delta = 40.61$ MeV β^- $\tau = 24.107$ jours -	$^{235}_{90}$Th $\Delta = 44.02$ MeV β^- $\tau = 7.2$ min -
$_{91}$Pa	$^{229}_{91}$Pa $\Delta = 29.90$ MeV ε 99.49 % & α 0.51 % $\tau = 1.55$ jours -	$^{230}_{91}$Pa $\Delta = 32.17$ MeV $\varepsilon + \beta^+$ 92.2 % & β^- 7.8 % $\tau = 17.4$ jours -	$^{231}_{91}$Pa (100%) $\Delta = 33.42$ MeV α $\tau = 32740$ ans -	$^{232}_{91}$Pa $\Delta = 35.95$ MeV β^- $\tau = 1.31$ jours -	$^{233}_{91}$Pa $\Delta = 37.49$ MeV β^- $\tau = 26.975$ jours -	$^{234}_{91}$Pa $\Delta = 40.34$ MeV β^- $\tau = 6.671$ h -	$^{235}_{91}$Pa $\Delta = 42.29$ MeV β^- $\tau = 24.4$ min -	$^{236}_{91}$Pa $\Delta = 45.33$ MeV β^- $\tau = 9.1$ min -	$^{237}_{91}$Pa $\Delta = 47.53$ MeV β^- $\tau = 8.7$ min -	$^{238}_{91}$Pa $\Delta = 50.89$ MeV β^- $\tau = 2.3$ min -
$_{92}$U	$^{231}_{92}$U $\Delta = 33.81$ MeV ε 99.996 % & α 0.004 % $\tau = 4.2$ jours -	$^{232}_{92}$U $\Delta = 34.61$ MeV α & FS 2.7×10^{-12}% $\tau = 68.90$ ans -	$^{233}_{92}$U $\Delta = 36.92$ MeV α $\tau = 1.5919 \times 10^5$ ans -	$^{234}_{92}$U (0.0054%) $\Delta = 38.15$ MeV α & FS 1.64×10^{-9}% $\tau = 2.455 \times 10^5$ ans -	$^{235}_{92}$U (0.7204%) $\Delta = 40.92$ MeV α & FS 7×10^{-9}% $\tau = 7.040 \times 10^8$ ans -	$^{236}_{92}$U $\Delta = 42.44$ MeV α & FS 9.4×10^{-8}% $\tau = 2.343 \times 10^7$ ans -	$^{237}_{92}$U $\Delta = 45.39$ MeV β^- $\tau = 6.752$ jours -	$^{238}_{92}$U (99.2742%) $\Delta = 47.31$ MeV α & FS 5.44×10^{-5}% $\tau = 4.463 \times 10^9$ ans -	$^{239}_{92}$U $\Delta = 50.57$ MeV β^- $\tau = 23.45$ min -	$^{240}_{92}$U $\Delta = 52.72$ MeV β^- $\tau = 14.1$ h -
$_{93}$Np	$^{233}_{93}$Np $\Delta = 37.95$ MeV ε $\tau = 36.2$ min -	$^{234}_{93}$Np $\Delta = 39.96$ MeV $\varepsilon + \beta^+$ $\tau = 4.4$ jours -	$^{235}_{93}$Np $\Delta = 41.04$ MeV ε 99.9974 % & α 0.0026 % $\tau = 396.1$ jours -	$^{236}_{93}$Np $\Delta = 43.38$ MeV ε 87.84 % & β^- 12 % & α 0.16 % $\tau = 1.550 \times 10^5$ ans -	$^{237}_{93}$Np $\Delta = 44.87$ MeV α $\tau = 2.144 \times 10^6$ ans -	$^{238}_{93}$Np $\Delta = 47.45$ MeV β^- $\tau = 2.0993$ jours -	$^{239}_{93}$Np $\Delta = 49.31$ MeV β^- $\tau = 2.3558$ jours -	$^{240}_{93}$Np $\Delta = 52.32$ MeV β^- $\tau = 61.9$ min -	$^{241}_{93}$Np $\Delta = 54.32$ MeV β^- $\tau = 13.9$ min -	$^{242}_{93}$Np $\Delta = 57.42$ MeV β^- $\tau = 2.2$ min -
$_{94}$Pu	$^{235}_{94}$Pu $\Delta = 42.18$ MeV $\varepsilon + \beta^+$ 99.9972 % & α 0.0028 % $\tau = 25.3$ min -	$^{236}_{94}$Pu $\Delta = 42.90$ MeV α & FS 1.9×10^{-7}% $\tau = 2.858$ ans -	$^{237}_{94}$Pu $\Delta = 45.09$ MeV ε 99.9968 % & α 0.0032 % $\tau = 45.64$ jours -	$^{238}_{94}$Pu $\Delta = 46.16$ MeV α & FS 1.9×10^{-7}% $\tau = 87.7$ ans -	$^{239}_{94}$Pu $\Delta = 48.59$ MeV α & FS 3.01×10^{-10}% $\tau = 24109$ ans -	$^{240}_{94}$Pu $\Delta = 50.13$ MeV α & FS 5.796×10^{-6}% $\tau = 6562.2$ ans -	$^{241}_{94}$Pu $\Delta = 52.96$ MeV β^- 99.99755 % & α 0.00245 % $\tau = 14.329$ ans -	$^{242}_{94}$Pu $\Delta = 54.72$ MeV α & FS 5.5×10^{-4}% $\tau = 3.750 \times 10^5$ ans -	$^{243}_{94}$Pu $\Delta = 57.76$ MeV β^- $\tau = 4.955$ h -	$^{244}_{94}$Pu $\Delta = 59.81$ MeV α 99.877 % & FS 0.123 % $\tau = 8.13 \times 10^7$ ans
$_{95}$Am	$^{237}_{95}$Am $\Delta = 46.57$ MeV $\varepsilon + \beta^+$ 99.975 % & α 0.025 % $\tau = 73.0$ min -	$^{238}_{95}$Am $\Delta = 48.42$ MeV $\varepsilon + \beta^+$ $\tau = 97.4$ min -	$^{239}_{95}$Am $\Delta = 49.39$ MeV ε 99.99 % & α 0.01 % $\tau = 11.9$ h -	$^{240}_{95}$Am $\Delta = 51.51$ MeV $\varepsilon + \beta^+$ $\tau = 50.9$ h -	$^{241}_{95}$Am $\Delta = 52.93$ MeV α & FS 3.6×10^{-10}% $\tau = 432.6$ ans -	$^{242}_{95}$Am $\Delta = 55.47$ MeV β^- 83 % & ε 17 % $\tau = 16.02$ h -	$^{243}_{95}$Am $\Delta = 57.18$ MeV α & FS 3.8×10^{-9}% $\tau = 7345.0$ ans -	$^{244}_{95}$Am $\Delta = 59.88$ MeV β^- $\tau = 10.02$ h -	$^{245}_{95}$Am $\Delta = 61.90$ MeV β^- $\tau = 2.047$ h -	$^{246}_{95}$Am $\Delta = 64.99$ MeV β^- $\tau = 39.0$ min -

1. On s'intéresse à un bloc d'uranium dont on suppose qu'il était constitué d'uranium-238 pur dans un passé lointain (à qualifier). Quelle équation régit le nombre $N_1(t)$ d'atomes d'uranium-238? L'écrire en introduisant un taux de décroissance λ_1 (probabilité qu'un atome subisse une décroissance radioactive par unité de temps). Résoudre cette équation, et écrire λ_1 en fonction du temps de demie-vie, τ_1 (temps au bout duquel la moitié des atomes ont subi une décroissance radioactive).

2. La radioactivité (A) d'un objet est définie comme le nombre de transformations radioactives par unité de temps. Son unité est le Becquerel, avec 1 Bq = 1 événement par seconde, soit $1\,\mathrm{Bq}{=}1\,\mathrm{s}^{-1}$. La radioactivité spécifique, usuellement notée a, est l'activité par unité de masse. Ici, pour l'uranium-238, on notera ces quantités A_1 et a_1. Les calculer. On rappelle le nombre d'Avogadro $N_A = 6.022 \times 10^{23}\ \mathrm{mol}^{-1}$.

3. Calculer la puissance spécifique, i.e., la puissance par unité de masse, émise par de l'uranium-238 pur. On rappelle que 1 eV $= 1.6022 \times 10^{-19}$ J.

4. À partir des données du tableau, identifier la chaîne de décroissance de l'uranium-238.

5. En indexant par $i \in \{1, \ldots, N\}$ les atomes impliqués dans la chaîne de décroissance de l'uranium-238, écrire l'équation qui gouverne leurs nombres $N_i(t)$ (qui est appelée l'équation de Bateman).

6. Résoudre l'équation vérifiée par le nombre d'atomes de $^{234}_{90}$Th (indication : chercher la solution sous la forme $N_2(t) = y(t)\,\exp(-\lambda_2 t)$). Tracer la solution en fonction du temps. Calculer aussi et tracer le ratio des nombres d'atomes de $^{234}_{90}$Th et d'$^{238}_{92}$U en fonction du temps.

7. Montrer qu'un des termes de la solution est très dominant à partir d'un temps fini à qualifier. En déduire une expression approchée pour $N_2(t)$, et donner une estimation pour le ratio des nombres de $^{234}_{90}$Th et d'$^{238}_{92}$U. Est-ce que ça vous semble significatif?

8. En déduire itérativement une expression approchée pour les nombres de chaque nucléide intermédiaire le long de la chaîne.

   Rq: On observe que la solution approchée vérifie $\mathrm{d}N_i/\mathrm{d}t = 0$ *pour tous les nucléides intermédiaires. Cela provient du fait que l'échelle de temps de décroissance de chaque* $N_i(t)$ *est beaucoup plus rapide que celle de* $N_1(t)$*, de sorte que l'évolution des nucléides le long de la chaîne est pilotée par la désintégration de* $^{238}_{92}$U*. Tout ce passe comme si tous les états intermédiaires atteignait une sorte de quasi-équilibre—on parle d'équilibre dynamique. Quand, de plus, le temps d'évolution global (soit* τ_1*) est très grand par rapport à des échelles de temps humaines, toutes les concentrations apparaissent stationnaires, et on parle d'équilibre séculaire.*

9. Quelle est la radioactivité spécifique d'un échantillon d'uranium-238 ayant atteint l'équilibre dynamique (on normalisera par la masse d'uranium-238)? Quelles sont les particules émises? Quelle est la puissance émise? Comparer avec les résultats des questions 2 & 3.

10. Rappeler quelle est la concentration relative du $^{234}_{90}$Th dans l'état d'équilibre séculaire (Question 7). Quelle est sa contribution à la radioactivité totale? Commenter.

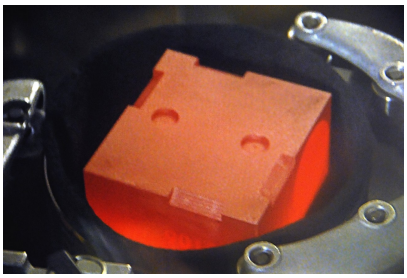
11. Quelques nucléides, dont en particulier certains isotopes de l'uranium, sont susceptibles de subir une fission spontanée avec une faible probabilité par rapport au mode de décroissance principal (cf. tableau) La fission spontanée scinde le noyau en deux produits de fissions, et émet également quelques neutrons. Le tableau ci-dessous précise le nombre de neutrons émis (ν) lors de la fission spontanée des isotopes de l'uranium (National Nuclear Data Center, https://www.nndc.bnl.gov/endf/). Déterminer le nombre de neutrons émis par seconde par un bloc d'uranium naturel de 1kg ? Quel isotope en est principalement responsable ?

$^{232}_{92}$U	$^{234}_{92}$U	$^{235}_{92}$U	$^{236}_{92}$U	$^{238}_{92}$U
$\nu = 1.71$	$\nu = 1.80$	$\nu = 1.87$	$\nu = 1.90$	$\nu = 2.00$

Rq : La radioactivité de l'uranium naturel est principalement due aux émissions α et β qui sont énergétiques, mais peuvent être arrêtées par une feuille d'aluminium. Ainsi, un bloc d'uranium-238 ne nécessite donc en principe qu'une protection radiologique modérée. Il y a cependant un faible taux de fission spontanée. Dans la chaîne de décroissance, les radionucléides les plus dangereux sont le radium et ses descendants de courtes durées de vie, jusqu'au polonium-214. Les produits les plus dangereux du point de vue radiologique sont $^{214}_{82}$Pb et $^{214}_{83}$Bi qui peuvent émettre du rayonnement γ de haute énergie. De plus, $^{226}_{88}$Ra et $^{210}_{82}$Pb sont très assimilables par l'organisme; enfin, $^{222}_{86}$Rn est un gaz: il peut être facilement inhalé, et ses produits de décroissance peuvent s'attacher sur des particules fines, ou s'accumuler sur des surfaces, ce qui facilite ensuite leur assimilation dans l'organisme. Il est donc essentiel de prendre en compte la toxicité chimique de ces atomes et de leurs composés.

12. Par comparaison, on cherche à calculer la radioactivité et la puissance spécifique émise par du plutonium-238. Écrire les équations gouvernant la décroissance de $^{238}_{94}$Pu et de son premier descendant, comme à la question 6 et en tracer les solutions. Quelle approximation est maintenant pertinente pour calculer l'activité d'un bloc de $^{238}_{94}$Pu sur un temps de plusieurs fois sa demie-vie?

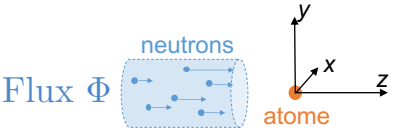
Rq : Des blocs de plutonium sont installés sur les explorateurs spatiaux (type Curiosity) afin de fournir de l'électricité et la chaleur nécessaire au maintien des composants—voir image ci-dessous. Le plutonium a aussi été utilisé pour alimenter des pacemakers (quelques uns sont encore en fonction dans le monde).



Ex. 3.2 - Neutrons et principe de fonctionnement d'un réacteur nucléaire

Rappel sur la section efficace

Considérons un noyau d'atome placé à l'origine du référentiel du laboratoire, dans un faisceau de neutrons se propageant le long de l'axe z à vitesse $v > 0$. On suppose que le faisceau incident est homogène, de flux ϕ, sur une aire A circulaire très grande par rapport à l'atome. Le flux ϕ mesure le nombre de particules par unité d'aire et de temps (dimension $\mathrm{T}^{-1}.\mathrm{L}^{-2}$).



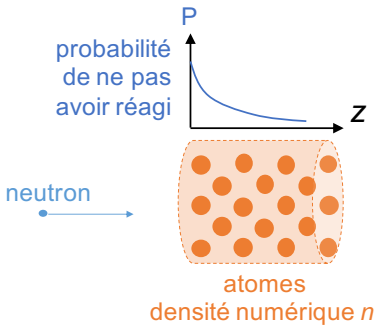
Certains des neutrons interagissent avec le noyau, ce qui produit un nombre d'évènements N (fission induite, diffusion élastique, capture radiative etc.). Les réactions nucléaire sont probabilistes et on s'attend à ce que le nombre d'évènements par unité de temps $\frac{dN}{dt}$ (dimension T^{-1}) soit proportionnel au flux ϕ, c'est à dire :

$$\frac{dN}{dt} = \sigma\phi$$

où la constante multiplicative σ a une dimension d'aire (L^2) et est appelée section efficace. σ peut être interprété (au sens classique) comme l'aire équivalente du noyau projeté sur le plan orthogonal au flux de neutron pour la réaction nucléaire considérée. Comme les noyaux ont une taille de l'ordre de 10 fm (distance entre les neutrons et protons), l'unité usuelle de section efficace est le barn: 1 barn = 100 fm^2. La section efficace dépend du nucléide, de la réaction nucléaire, et de l'énergie cinétique du neutron incident. Le tableau ci-dessous fournit les sections efficaces pour les réactions de fission induite (σ_f) et de capture radiative (σ_c) avec des neutrons thermiques pour différents noyaux ainsi que le nombre de noyaux produits par fission (ν) dont nous auront besoin dans la suite de l'exercice.

Nucléide	σ_c (barns)	σ_f (barns)	ν (-)
$^{16}_{8}$O	1.698×10^{-4}	0	-
$^{235}_{92}$U	98.79	586.7	2.43
$^{238}_{92}$U	2.684	1.687×10^{-5}	2.44

Si l'on considère maintenant un neutron unique qui se propage à vitesse $\vec{v} = v\vec{e}_z$ dans un matériau formé d'atomes de densité numérique n (dimension L^{-3}), alors on se retrouve avec un problème analogue avec une particule (neutron) soumise à un flux $\phi = -n\vec{v}$ de l'autre type de particules (noyaux). Dès lors, la probabilité qu'un réaction se produise par unité de temps, aussi appelé taux de réaction, est donné par :



$$\frac{dp_{\text{réaction}}}{dt} = \sigma\,|\phi| = \underbrace{\sigma n}_{\Sigma} v = \Sigma v$$

où Σ (dimension L^{-1}) est appelé section efficace macroscopique, et s'apparente à l'inverse du libre parcours moyen du neutron à l'intérieur du matériau : $l = 1/\Sigma$ (distance caractéristique au bout de laquelle le neutron a réagit).

Si plusieurs réactions réactions sont possible pour le neutron (fission ou capture avec différents noyau) alors les probabilité de réaction s'additionnent et la section efficace macroscopique devient :

$$\frac{dp_{\text{réaction}}}{dt} = \left(\sum_i \sigma_i n_i\right) v = \Sigma v \Rightarrow \Sigma = \sum_i \sigma_i n_i$$

Fonctionnement d'un réacteur

Le combustible des réacteurs nucléaires peut être vu en première approximation comme composé d'oxyde d'uranium UO_2, les isotopes $^{235}_{92}\mathrm{U}$ et $^{238}_{92}\mathrm{U}$ étant en proportions relatives x et $1-x$. On notera n la densité numérique des atomes d'uranium dans le matériau, et on considérera que la densité massique $\rho = 10.96\,\mathrm{g/cm^3}$ pour $x = 0$.

1. Expliquer quel effet isotopique on anticipe sur la densité numérique de l'oxyde et calculer n.

2. Identifier les 5 réactions qui consomment des neutrons dans le combustible neuf. Expliquer pourquoi deux d'entre elles peuvent être négligées. Puis écrire en fonction de x le libre parcours moyen d'un neutron lent dans ce matériau avant tout événement d'absorption. Même chose pour les événements de fission induite. Donner des évaluations numériques de ces quantités dans le cas où $x = 0.0072$ (uranium naturel) ou $x = 0.035$ (uranium enrichi).

3. À la lumière de ce résultat, commenter sur ce qu'il est indispensable de faire pour que le combustible nucléaire soit utilisé de la façon la plus efficace? En quoi le résultat ci-dessus affecte-t-il la conception d'un réacteur?

   Rq: On peut ainsi s'attendre à ce que la structure globale d'un réacteur dépende sensiblement du choix de combustible (uranium naturel ou enrichi). On voit aussi que prédire et optimiser le fonctionnement d'un réacteur nécessite la prise en compte des variations spatiales des flux de neutrons. On doit aussi s'attendre à ce que les gradients de neutrons et donc de taux de réactions introduisent des gradients thermiques, chimiques et isotopiques à l'intérieur des éléments de combustible avec de multiples conséquences (contraintes, gonflement, modification de taux de réactions, ...). La neutronique vise à étudier l'ensemble de ces phénomènes à travers, en particulier, la résolution d'équations différentielles décrivant ces flux et leurs interactions avec les éléments du réacteur.

4. Exprimer en fonction de x le nombre η de neutrons rapides émis en moyenne pour chaque neutron lent absorbé dans le combustible neuf, puis fournir les valeur pour $x = 0.0072$ et $x = 0.035$.

5. On s'intéresse à l'évolution du nombre $N\left(^{235}_{92}\mathrm{U}\right)$ d'atomes d'uranium-235 pendant le fonctionnement du réacteur. On supposera pour cela que le flux ϕ est homogène et constant. En s'appuyant sur la définition de la section efficace, écrire l'équation d'évolution de $N\left(^{235}_{92}\mathrm{U}\right)$ et la résoudre. Identifier un taux de décroissance et un temps de demie-vie, et commenter par comparaison avec le phénomène de décroissance analysé dans l'exercice précédent.

6. Par la suite, on caractérisera la concentration résiduelle en uranium-235 par la quantité:

$$x(t) = \frac{N\left({}^{235}_{92}\mathrm{U}, t\right)}{N\left({}^{235}_{92}\mathrm{U}, 0\right) + N\left({}^{238}_{92}\mathrm{U}, 0\right)} \quad .$$

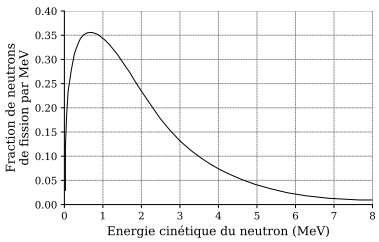
Un réacteur à uranium enrichi ($x_0 = 0.035$) est conçu de telle façon que l'on remplace 1/3 du combustible tous les ans alors que la concentration en ${}^{235}_{92}\mathrm{U}$, par rapport à la quantité d'uranium initiale, y a atteint $x_f = 0.01$. On suppose que le combustible est irradié uniformément et à flux constant. Quelle est la valeur de ϕ? Commenter sur les paramètres qui déterminent cette quantité. Calculer le temps de demie-vie de l'uranium-235 dans ces conditions.

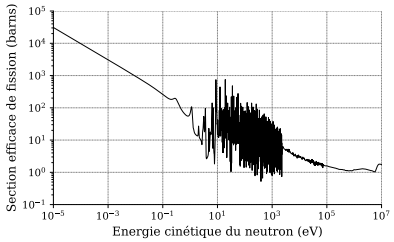
7. On cherche à dimensionner un réacteur pour produire 1000 MW thermique en moyenne. On supposera que l'énergie est essentiellement produite par la fission de l'uranium-235 et que celle-ci libère en moyenne 202.5 MeV par événement en comptant les fissions retardées. Quelle quantité de combustible doit contenir le réacteur?

Ex. 3.3 - Modérateur

On s'intéresse dans cet exercice à la modération des neutrons dans les réacteurs nucléaires.

1. La fission de ${}^{235}_{92}\mathrm{U}$ émet des neutrons dont l'énergie cinétique est distribuée selon la densité de probabilité donnée dans le premier graphique ci-dessous (gauche). Par ailleurs, la section efficace de fission induite de ${}^{235}_{92}\mathrm{U}$ dépend de l'énergie du neutron incident comme présenté dans le second graphique ci-dessous (droite). Dans un réacteur nucléaire, la modération consiste à 'thermaliser' les neutrons issus de la fission, c'est à dire ramener leur énergie cinétique à celle correspondant à la température ambiante (~ 0.025 eV). Pourquoi doit on modérer les neutrons ?





2. Le mécanisme de modération principal est la diffusion élastique des neutrons par les noyaux du milieu qu'ils traversent. Ce processus de diffusion correspond à des collisions entre neutrons et noyaux, de sorte que l'énergie cinétique des neutrons est progressivement transmise aux noyaux. Le rapport entre l'énergie cinétique du neutron avant collision (E) et après collision (E'), est caractérisé par le décrément logarithmique d'énergie $\xi = \left\langle -\ln\left(\frac{E'}{E}\right)\right\rangle$, où $\langle\cdot\rangle$ représente la moyenne sur toutes les directions possibles de diffusion. Le décrément ξ ne dépend que du nombre de masse A des noyaux (cf. démonstration dans le polycopié) :

$$\xi(A) = 1 + \frac{(A-1)^2}{2A}\ln\left(\frac{A-1}{A+1}\right)$$

Faire un tracé approximatif du décrément logarithmique en fonction du nombre de masse A. Commenter.

3. On considère un neutron rapide issue d'une fission d'énergie 2 MeV.

   (a) Estimer le nombre de collisions n avec des atomes de nombre de masse A qui sont nécessaires pour thermaliser ce neutron.

(b) Faire l'application numérique approximative pour une diffusion par l'hydrogène ${}^{1}_{1}\mathrm{H}$ et l'uranium ${}^{238}_{92}\mathrm{U}$. On donne $\ln(10) \approx 2.3$ et $\xi(238) \approx 0.01$.

(c) Le tableau ci-dessous liste le décrément logarithmique d'énergie pour une diffusion par divers noyaux, ainsi que les sections efficaces de diffusion (σ_d) et d'absorption (σ_a). Un bon modérateur doit permettre de thermaliser les neutrons tout en évitant de les consommer. Pourquoi l'eau et le graphite sont-ils de bons modérateurs ?

noyau	${}^{1}_{1}\mathrm{H}$	${}^{2}_{1}\mathrm{H}$	${}^{3}_{1}\mathrm{H}$	${}^{3}_{2}\mathrm{He}$	${}^{7}_{3}\mathrm{Li}$	${}^{9}_{4}\mathrm{Be}$	${}^{11}_{5}\mathrm{B}$	${}^{12}_{6}\mathrm{C}$	${}^{14}_{7}\mathrm{N}$	${}^{16}_{8}\mathrm{O}$	${}^{238}_{92}\mathrm{U}$
ξ	1	0.725	0.538	0.538	0.260	0.207	0.171	0.158	0.136	0.120	0.0084
σ_d (barns)	82.03	7.64	3.03	6	1.4	7.63	5.77	5.559	11.53	4.232	8.871
σ_a (barns)	0.332	0.000519	0	5333	0.0454	0.0076	0.0055	0.00353	1.91	0.0001	2.68

(d) Le deutérium ${}^{2}_{1}\mathrm{H}$, aussi noté D, est un isotope rare de l'hydrogène (0.015% de l'hydrogène naturel). Les réacteurs de type CANDU (Canada) fonctionnent à l'uranium naturel avec de l'eau lourde $\mathrm{D_2O}$ comme modérateur. Commenter. Les réacteurs de ce type ont la particularité de produire du tritium ${}^{3}_{1}\mathrm{H}$ qui est un noyau instable se désintégrant par radioactivité β^- avec un temps de demi-vie de 12.3 ans. Quelles difficultés cela pose ?

Chapter 4

Énergie du mouvement des fluides

Contents

Foreword **83**
4.1 Fluid mechanics **84**
4.1.1 Continuum description 84
4.1.2 Conservation laws 86
4.1.3 Drag and lift 91
4.1.4 Working principle of an airfoil 97
4.2 Wind energy **99**
4.2.1 Wind resource 99
4.2.2 Betz limit 101
4.2.3 Horizontal axis wind turbines 102
4.2.4 Alternative designs 105
4.3 Water-based energy **106**
4.3.1 Hydropower 106
4.3.2 Waves 107
4.3.3 Tides and currents 109
4.3.4 Ocean thermal energy 111
Recap of what must be known for the exam **113**
Exercises **116**
Ex. 4.1 - Effet Magnus 116
Ex. 4.2 - Courant des marées 117
Ex. 4.3 - Usine marémotrice 118
Ex. 4.4 - Limite de Betz 118
Ex. 4.5 - Design de pale éolienne 119

Foreword

This lesson addresses the energy of fluid motion, the most widespread use of which are wind energy and hydropower. These are among the oldest forms of energy used by humans, with the first known windmills dating back to Babylonian antiquity. These energies are still widely used today. Hydropower and wind turbines are currently the two most developed renewable electric power sources, with 15% and 7.3% of the global electricity production (2022). From a physical point of view, the elementary energies involved are the smallest in comparison with the other forms of energy (thermal, radiation, chemical, nuclear). The gravitational energy of a water molecule falling from a 100 m dam is 1.8×10^{-4} eV, and the average kinetic energy of an air molecule associated with the collective motion of wind during a hurricane (50 m/s) is 3.9×10^{-4} eV. These energies are

orders of magnitude smaller than the thermal agitation energy ($k_BT \sim 0.025$ eV), solar radiation ($\sim$ eV), chemical reaction energies ($\sim$ 10eV), and nuclear reaction energies ($\sim$ MeV). Accordingly, the energy densities involved in wind, hydro, and ocean powers are quite small. But this form of energy is easily converted into useful mechanical work, which is why it has been used for so long. Windmills and watermills have been the only sources of artificial mechanical work for most of the history until the development of thermal engines.

In this lesson, we focus on the physics of fluid motion and its conversion into work. We begin with some fundamentals of fluid mechanics (conservation laws, Navier-Stokes, Bernoulli) and its application to the drag and lift phenomenon at the origin of the working principle of airfoils (section 4.1). Wind energy is then discussed in detail (section 4.2): resources, efficiency (Betz limit), aerodynamics of a blade, actual efficiency of horizontal axis wind turbines, and alternative designs. Finally, we focus on water-based energy (section 4.3) starting with hydropower, the most common exploitation of water flows. We then present some less-known forms of water-based energy: waves, tides, currents, and thermal energy of the oceans.

4.1 Fluid mechanics

4.1.1 Continuum description

Fluid mechanics is the core theory to understand the physics of wind turbines and other fluid-based energies. This theory relies on a continuum thermodynamic description of matter, well-suited to describe systems at scales larger than a micrometer. Indeed, according to statistical physics, a detailed description of matter involving atoms and molecules is unnecessary when considering macroscopic systems as molecular fluctuations become negligible and a thermodynamic description based on a few state parameters (e.g., density) and the maximization/minimization of a state function (e.g., entropy) is sufficient to fully describe a system and its evolution. Relative molecular fluctuations typically decrease in $N^{-1/2}$, with N the number of atoms, so that, one can reasonably assume non-fluctuating thermodynamic averages even at the micrometer scale ($N \sim 10^{10}$).

Before turning to the continuum description, however, let us clarify the nature of fluid motion at the molecular scale. Even for a fluid at rest, temperature originates from a permanent thermal agitation of the atoms and molecules. More precisely, statistical physics relates temperature to the average kinetic energy of atoms: $\langle K \rangle = \sum_{i \in \text{atoms}} \frac{1}{2} m_i \left\langle \mathbf{v}_i^2 \right\rangle = \frac{3}{2} N k_B T$. At $T = 300$K, the average velocity of a water molecule is therefore $\sqrt{\langle v^2 \rangle} = \frac{3 k_B T}{m_{\text{H}_2\text{O}}} \approx 650$ m/s, much larger than any wind speed. The motion of the fluid corresponds to a collective movement of all the molecules so that their center of mass moves. In such cases, the molecules are still under thermal agitation but there is a small bias in the velocities that produces a collective motion. One can then distinguish a 'macroscopic' kinetic energy corresponding to the velocity of the center of mass $K_{macro} = \frac{1}{2} M \left(\mathbf{v}_{\text{c.o.m.}}\right)^2$ (with $M = \sum_{i \in \text{atoms}} m_i$), and the temperature is then defined from the remaining 'microscopic' kinetic energy: $\frac{3}{2} N k_B T = \langle K_{micro} \rangle = \sum_{i \in \text{atoms}} \frac{1}{2} m_i \left\langle \mathbf{v}_i^2 \right\rangle - K_{macro}$. Such macroscopic kinetic energy is not accounted for in the internal energy $U = K_{micro} + V$ (with V the potential energy). Accordingly, the total energy is:

$$E = K_{macro} + U = K_{macro} + K_{micro} + V$$

So far, for the study of thermal energy conversion or thermo-chemistry, we have never considered macroscopic motion when describing the thermodynamics of a system, so that the internal energy and total energy were equal. For what follows, we do take into account a macroscopic motion and the two quantities are no more equal. As the velocities associated with thermal agitation are much larger than the center-of-mass velocity, the magnitude of the two kinetic energies are very different, the macroscopic one being much smaller (at ambient temperature): $K_{macro} \ll K_{micro}$. Thermal agitation energy is a 'disordered' kinetic energy and, as a consequence, its conversion into work is limited by the second principle of thermodynamics (e.g., Carnot efficiency for a thermal engine operating between two reservoirs at fixed temperature). In contrast, the macroscopic kinetic energy is an 'ordered' kinetic energy, and there is no thermodynamic limitation on its conversion into mechanical work.

Let us now turn to the continuum description of matter. Such description relies on local intensive fields describing the thermodynamic state at any position $\mathbf{x}$ in an infinitely small volume V around $\mathbf{x}$ (but large enough for the continuum description to be valid). Mass, volume, and internal energy are the usual extensive state parameters in statistical physics and their field representation in a continuum description are:

- The mass density $\rho(\mathbf{x},t) = \frac{M}{V} = \frac{1}{V}\sum_{i\in V} m_i$, with m_i the masses of the atoms contained in V.
- The internal energy density : $\rho(\mathbf{x},t)\, u(\mathbf{x},t) = \frac{U}{V}$.

The intensive quantities conjugated to mass, volume, and internal energy are the chemical potential, pressure, and temperature, respectively. But, here we disregard chemical potential and chemical composition as we are not interested in osmotic equilibrium (exchange of molecules) for the study of wind turbines or hydropower. A more complete continuum description would distinguish between the different chemical species in the system, by defining a mass density for each species and the associated chemical potentials. We are thus left with two intensive conjugated quantities, namely pressure and temperature. Although fluids cannot sustain shear stress at equilibrium, their time-dependent response can exhibit some shear stresses because of viscosity. Accordingly, a continuum description should account for deviatoric stresses. The fields of conjugated quantities are:

- The stress tensor $\underline{\underline{\sigma}}(\mathbf{x},t)$, which can be decomposed into spherical (pressure p) and deviatoric ($\underline{\underline{\tau}}$) contributions $\underline{\underline{\sigma}} = -p\underline{\underline{I}} + \underline{\underline{\tau}}$, with $\underline{\underline{I}}$ the identity tensor.
- The temperature $T(\mathbf{x},t)$.

As explained above, one should also account for the collective motion of atoms and molecules, which is described by the center-of-mass velocity. Therefore, we complete the continuum description by defining:

- The velocity $\mathbf{v}(\mathbf{x},t) = \mathbf{v}_{\text{c.o.m.}}(V)$, which refers to the center of mass velocity of the atoms contained in V.
- The macroscopic kinetic energy density: $\rho(\mathbf{x},t)\, e_{kin}(\mathbf{x},t) = \frac{1}{2}\rho(\mathbf{x},t)\,|\mathbf{v}(\mathbf{x},t)|^2 = \frac{1}{2V}\sum_{i\in V} m_i\,|\mathbf{v}_{\text{c.o.m.}}|^2 = \frac{K_{macro}}{V}$.

Note that $e_{kin}(\mathbf{x},t)$ represents the macroscopic kinetic energy per unit mass, and $u(\mathbf{x},t)$ the internal energy per unit mass. The sum of the two $e(\mathbf{x},t) = u(\mathbf{x},t) + e_{kin}(\mathbf{x},t)$ is the total energy per unit mass, and $\rho(\mathbf{x},t)\, e(\mathbf{x},t)$ represents the total energy density. Since the matter has a macroscopic motion, energy and mass are transported over time, and we define the associated flux as follows:

- The mass flux $\mathbf{J}_\rho(\mathbf{x},t) = \rho(\mathbf{x},t)\,\mathbf{v}(\mathbf{x},t)$
- The kinetic energy flux $\mathbf{J}_{e_{kin}}(\mathbf{x},t) = \rho(\mathbf{x},t)\, e_{kin}(\mathbf{x},t)\,\mathbf{v}(\mathbf{x},t)$, internal energy flux $\mathbf{J}_u(\mathbf{x},t) = \rho(\mathbf{x},t)\, u(\mathbf{x},t)\,\mathbf{v}(\mathbf{x},t)$, and total energy flux $\mathbf{J}_e(\mathbf{x},t) = \rho(\mathbf{x},t)\, e(\mathbf{x},t)\,\mathbf{v}(\mathbf{x},t)$.

The magnitude of the kinetic energy flux defines the power density which is the power that one tries to harvest with a wind turbine:

$$\mathcal{P} = |\mathbf{J}_{e_{kin}}(\mathbf{x},t)| = \frac{1}{2}\rho(\mathbf{x},t)\,|\mathbf{v}(\mathbf{x},t)|^3$$

The power density is the energy of motion crossing a surface orthogonal to the flow per unit time and area. As it is proportional to the density, it strongly depends on the phase of the fluid: the power density of liquid water is three orders of magnitude larger than the

power density of air at the same velocity. One can estimate the energy resource based on the power density. For instance, a 30 km/h wind ($\rho = 1.3$ kg/m^3) corresponds to a power density of $\mathcal{P} = 1\,740$ W/m^2, which is approximately equivalent to a 5 km/h water stream. This is a magnitude similar to the solar radiation before entering the atmosphere (1 380 W/m^2). So windy regions represent substantial energy resources.

The above fluxes are convective fluxes, and one can also have diffusive fluxes. Diffusion is involved in the transport of chemical species, but we disregard chemical composition here. Diffusion is also involved in the transport of energy in the form of heat. So we define the heat flux $\mathbf{q}(\mathbf{x}, t)$ which corresponds to exchanges of entropy within the system without exchange of matter.

4.1.2 Conservation laws

During its evolution, an isolated continuum system conserves mass, energy, linear and angular momentum. These conservation laws originate from the fundamental physical symmetries in the laws of physics. Energy conservation arises from the time translation symmetry (see introductory lesson). Linear and angular momentum arise from the space translation and rotation symmetries. Mass, however, is not strictly conserved as it can be converted to energy (see lesson on nuclear energy). Yet, for non-relativistic systems that do not involve nuclear reactions, mass changes are negligible and we can consider that mass is conserved. There are other fundamental quantities that are conserved (e.g., charge) but accounting for these conservation laws is unnecessary here (e.g., matter is assumed electrically neutral).

Conservation law in the continuum description

Let us first explain how a conservation principle is formulated in a continuum description. We consider the field of a scalar quantity $A(\mathbf{x}, t)$ that is conserved for an isolated system. In non-isolated conditions, the amount of A can change because of an external volumetric source term $S_A(\mathbf{x}, t)$ (amount of A that is produced per unit volume and per unit time) and because of exchanges at the boundaries of the system due to a non-convective flux $\mathbf{q}_A$. We consider a given amount of matter that occupies the volume Ω at time t. The conservation of A for this amount of matter writes:

$$\frac{D}{Dt}\left(\int_\Omega A(\mathbf{x}, t)\, dV\right) = \int_{\partial\Omega} \mathbf{q}_A(\mathbf{x}, t).\mathbf{n} dA + \int_\Omega S_A(\mathbf{x}, t)\, dV \tag{4.1}$$

where $\frac{DA}{Dt} = \lim_{\delta t \to 0} \frac{A(\mathbf{x}+\mathbf{v}\delta t, t+\delta t) - A(\mathbf{x},t)}{\delta t} = \frac{\partial A}{\partial t} + \nabla A.\mathbf{v}$ represents the material derivative, that is the derivative attached to the matter.

For the field of position, the material derivative is trivially $\frac{D\mathbf{x}}{Dt} = \frac{\partial \mathbf{x}}{\partial t} = \mathbf{v}$. Applying the material derivative to a space integral requires to derive the volume increment $dV = dxdydz$. The material derivative of a position increment is $\frac{D(dx)}{Dt} = \frac{D(x+dx)}{Dt} - \frac{D(x)}{Dt} = v_x(x + dx) - v_x(x) = \frac{\partial v_x}{\partial x} dx$. Therefore, the material derivative of a volume increment is $\frac{D(dV)}{Dt} = \frac{D(dx)}{Dt} dydz + dx \frac{D(dy)}{Dt} dz + dxdy \frac{D(dz)}{Dt} = \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}\right) dxdydz = (\nabla.\mathbf{v})\, dV$. Applying the material derivative to the integral of A in Ω gives:

$$\frac{D}{Dt}\left(\int_\Omega A dV\right) = \int_\Omega \left[\underbrace{\left(\frac{\partial A}{\partial t} + \nabla A.\mathbf{v}\right)}_{\frac{DA}{Dt}} dV + A\underbrace{(\nabla.\mathbf{v})\, dV}_{\frac{D(dV)}{Dt}}\right] = \int_\Omega \left(\frac{\partial A}{\partial t} + \nabla.(A\mathbf{v})\right) dV$$

Using this expression, and applying the divergence theorem to the boundary integral, the integral formulation of the conservation law (Eq. 4.1) can be reformulated into a local formulation:

$$\frac{\partial A}{\partial t} + \nabla.\mathbf{J}_A = \nabla.\mathbf{q}_A + S_A \tag{4.2}$$

with $\mathbf{J}_A = A\mathbf{v}$, the convective flux of A.

In summary, the conservation of a quantity A can be written both in an integral form, and in a local form :

$$\int_\Omega \frac{\partial A}{\partial t} dV + \int_{\partial\Omega} \mathbf{J}_A.\mathbf{n} dA = \int_{\partial\Omega} \mathbf{q}_A.\mathbf{n} dA + \int_\Omega S_A dV$$

$$\frac{\partial A}{\partial t} + \nabla.\mathbf{J}_A = \nabla.\mathbf{q}_A + S_A$$

Conservation of mass

We consider the particular case $A(\mathbf{x},t) = \rho(\mathbf{x},t)$ with $\mathbf{q}_A = 0$ and $S_A(\mathbf{x},t) = 0$. The conservation law is:

$$\frac{\partial \rho}{\partial t} + \nabla.\mathbf{J}_\rho = 0 \tag{4.3}$$

This conservation of mass equation is also known as the *continuity equation*.

Conservation of linear momentum

Considering the case of the linear moment in x, we have $A(\mathbf{x},t) = \rho(\mathbf{x},t) v_x$ with $\mathbf{q}_A = \sigma_{xx}.\mathbf{e}_x + \sigma_{xy}.\mathbf{e}_y + \sigma_{xz}.\mathbf{e}_z$ (force in the x direction) and $S_A(\mathbf{x},t) = \rho g_x$ (gravity). The conservation law is:

$$\frac{\partial(\rho v_x)}{\partial t} + \nabla.(\rho v_x \mathbf{v}) = \nabla.(\sigma_{xx}.\mathbf{e}_x + \sigma_{xy}.\mathbf{e}_y + \sigma_{xz}.\mathbf{e}_z) + \rho g_x$$

Developing the divergence operators, we have:

$$\frac{\partial(\rho v_x)}{\partial t} + \left(\frac{\partial(\rho v_x v_x)}{\partial x} + \frac{\partial(\rho v_x v_y)}{\partial y} + \frac{\partial(\rho v_x v_z)}{\partial z}\right) = \left(\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial \sigma_{xz}}{\partial z}\right) + \rho g_x$$

Similar equations hold for the linear momentum in y and z. Arranging the three conservation equations together in a vector form with the x momentum equation along $\mathbf{e}_x$, y momentum along $\mathbf{e}_y$, and z momentum along $\mathbf{e}_z$, we obtain:

$$\frac{\partial}{\partial t}\left(\rho \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}\right) + \begin{bmatrix} \frac{\partial(\rho v_x v_x)}{\partial x} + \frac{\partial(\rho v_x v_y)}{\partial y} + \frac{\partial(\rho v_x v_z)}{\partial z} \\ \frac{\partial(\rho v_y v_x)}{\partial x} + \frac{\partial(\rho v_y v_y)}{\partial y} + \frac{\partial(\rho v_y v_z)}{\partial z} \\ \frac{\partial(\rho v_z v_x)}{\partial x} + \frac{\partial(\rho v_z v_y)}{\partial y} + \frac{\partial(\rho v_z v_z)}{\partial z} \end{bmatrix} = \begin{bmatrix} \frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial \sigma_{xz}}{\partial z} \\ \frac{\partial \sigma_{yx}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{yz}}{\partial z} \\ \frac{\partial \sigma_{zx}}{\partial x} + \frac{\partial \sigma_{zy}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} \end{bmatrix} + \rho \begin{bmatrix} g_x \\ g_y \\ g_z \end{bmatrix}$$

This can be written in the following more synthetic form:

$$\frac{\partial(\rho \mathbf{v})}{\partial t} + \nabla.(\rho \mathbf{v} \otimes \mathbf{v}) = \nabla.\underline{\underline{\sigma}} + \rho \mathbf{g} \tag{4.4}$$

where $\mathbf{u} \otimes \mathbf{v} = \mathbf{u}.\mathbf{v}^T \Rightarrow (\mathbf{u} \otimes \mathbf{v})_{ij} = u_i v_j$ is the outer product. The left-hand part of the equation can be rearranged as follows:

$$\begin{aligned}
\nabla.\left(\rho\mathbf{v}\otimes\mathbf{v}\right) &= \begin{bmatrix} \frac{\partial(\rho v_x v_x)}{\partial x}+\frac{\partial(\rho v_x v_y)}{\partial y}+\frac{\partial(\rho v_x v_z)}{\partial z} \\ \frac{\partial(\rho v_y v_x)}{\partial x}+\frac{\partial(\rho v_y v_y)}{\partial y}+\frac{\partial(\rho v_y v_z)}{\partial z} \\ \frac{\partial(\rho v_z v_x)}{\partial x}+\frac{\partial(\rho v_z v_y)}{\partial y}+\frac{\partial(\rho v_z v_z)}{\partial z} \end{bmatrix} \\
&= \begin{bmatrix} \rho v_x\frac{\partial v_x}{\partial x}+\rho v_y\frac{\partial v_x}{\partial y}+\rho v_z\frac{\partial v_x}{\partial z} \\ \rho v_x\frac{\partial v_y}{\partial x}+\rho v_y\frac{\partial v_y}{\partial y}+\rho v_z\frac{\partial v_y}{\partial z} \\ \rho v_x\frac{\partial v_z}{\partial x}+\rho v_y\frac{\partial v_z}{\partial y}+\rho v_z\frac{\partial v_z}{\partial z} \end{bmatrix} + \begin{bmatrix} v_x\frac{\partial(\rho v_x)}{\partial x}+v_x\frac{\partial(\rho v_y)}{\partial y}+v_x\frac{\partial(\rho v_z)}{\partial z} \\ v_y\frac{\partial(\rho v_x)}{\partial x}+v_y\frac{\partial(\rho v_y)}{\partial y}+v_y\frac{\partial(\rho v_z)}{\partial z} \\ v_z\frac{\partial(\rho v_x)}{\partial x}+v_z\frac{\partial(\rho v_y)}{\partial y}+v_z\frac{\partial(\rho v_z)}{\partial z} \end{bmatrix} \\
&= \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{bmatrix}.\begin{bmatrix} \rho v_x \\ \rho v_y \\ \rho v_z \end{bmatrix} + \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}\left(\frac{\partial(\rho v_x)}{\partial x}+\frac{\partial(\rho v_y)}{\partial y}+\frac{\partial(\rho v_z)}{\partial z}\right) \\
&= \rho\mathbf{v}.\nabla\mathbf{v}+\mathbf{v}\nabla.\left(\rho\mathbf{v}\right)
\end{aligned}$$

$$\frac{\partial(\rho\mathbf{v})}{\partial t}+\nabla.\left(\rho\mathbf{v}\otimes\mathbf{v}\right)=\mathbf{v}\frac{\partial\rho}{\partial t}+\rho\frac{\partial\mathbf{v}}{\partial t}+\rho\mathbf{v}.\nabla\mathbf{v}+\mathbf{v}\nabla.\left(\rho\mathbf{v}\right)=\mathbf{v}\underbrace{\left(\frac{\partial\rho}{\partial t}+\nabla.\left(\rho\mathbf{v}\right)\right)}_{0\text{ (conservation of mass)}}+\rho\frac{\partial\mathbf{v}}{\partial t}+\rho\mathbf{v}.\nabla\mathbf{v}=\rho\left(\frac{\partial\mathbf{v}}{\partial t}+\mathbf{v}.\nabla\mathbf{v}\right)$$

Therefore, an alternative form of the conservation of linear momentum (Eq. 4.4), is:

$$\frac{\partial\mathbf{v}}{\partial t}+\mathbf{v}.\nabla\mathbf{v}=\frac{\nabla.\underline{\underline{\sigma}}}{\rho}+\mathbf{g} \tag{4.5}$$

This vectorial equation summarizes three scalar equations which are also known as the *Navier-Stokes equations*.

Conservation of angular momentum

The derivation of the conservation law for the angular momentum is quite similar to that for the linear momentum. One eventually obtains:

$$\mathbf{x}\times\left(\frac{\partial(\rho\mathbf{v})}{\partial t}+\nabla.\left(\rho\mathbf{v}\otimes\mathbf{v}\right)-\nabla.\underline{\underline{\sigma}}-\rho\mathbf{g}\right)+\begin{bmatrix}\sigma_{yz}-\sigma_{zy}\\ \sigma_{xz}-\sigma_{zx}\\ \sigma_{xy}-\sigma_{yx}\end{bmatrix}=0$$

One readily identifies the Navier-Stokes equation in the first term, so that the conservation of angular momentum reduces to $\sigma_{yz}-\sigma_{zy}=\sigma_{xz}-\sigma_{zx}=\sigma_{xy}-\sigma_{yx}=0$, that is the stress tensor must be symmetrical.

Conservation of energy

For the conservation of energy, $A\left(\mathbf{x},t\right)=\rho\left(\mathbf{x},t\right)e\left(\mathbf{x},t\right)$ with $\mathbf{q}_A=\mathbf{q}+\underline{\underline{\sigma}}.\mathbf{v}$ (heat and work exchanges) and $S_A\left(\mathbf{x},t\right)=\rho\mathbf{g}.\mathbf{v}$ (work of gravity). The conservation of energy is:

$$\frac{\partial\left(\rho e\right)}{\partial t}+\nabla.\left(\mathbf{J}_e\right)=\nabla.\left(\underline{\underline{\sigma}}.\mathbf{v}\right)+\rho\mathbf{g}.\mathbf{v}+\nabla.\mathbf{q} \tag{4.6}$$

This equation is also known as the *energy balance* equation.

General formulation of fluid mechanics

The mass, momentum and energy conservation laws (Eqs. 4.3, 4.5, 4.6) are not enough to fully describe the evolution of a system. One must also provide:

- A behavior law relating the internal energy and stress tensor to the state parameters of the system (density, temperature, velocity): $\left(u, \underline{\underline{\sigma}}\right) = f\left(\rho, T, \mathbf{v}\right)$. For instance, for a diatomic ideal gas: $u = \frac{5}{2}\frac{RT}{M_{gas}}$, $p = \frac{\rho RT}{M_{gas}}$, and $\underline{\underline{\tau}} = 0$, with R the ideal gas constant and M_{gas} the molar mass of the gas.

- A heat flux law relating the heat flux $\mathbf{q}$ to the state parameters: $\mathbf{q} = g\left(\rho, T, \mathbf{v}\right)$. One often considers Fourier's law: $\mathbf{q} = -\kappa \nabla T$, with κ the thermal conductivity.

In fine, the complete formulation is:

$$\begin{cases} \frac{\partial \rho}{\partial t} + \nabla.\left(\rho \mathbf{v}\right) = 0 & \text{Continuity} \\ \frac{\partial \mathbf{v}}{\partial t} + \mathbf{v}.\nabla \mathbf{v} = \frac{\nabla.\underline{\underline{\sigma}}}{\rho} + \mathbf{g} & \text{Navier-Stokes} \\ \frac{\partial (\rho e)}{\partial t} + \nabla.\left(\rho e \mathbf{v}\right) = \nabla.\left(\underline{\underline{\sigma}}.\mathbf{v}\right) + \rho \mathbf{g}.\mathbf{v} + \nabla.\mathbf{q} & \text{Energy balance} \end{cases} \quad \text{with} \quad \begin{cases} e = u + \frac{\mathbf{v}^2}{2} \\ \left(u, \underline{\underline{\sigma}}\right) = f\left(\rho, T, \mathbf{v}\right) \\ \mathbf{q} = g\left(\rho, T, \mathbf{v}\right) \end{cases}$$

There are 5 unknowns ($\rho, T, \mathbf{v} = v_x \mathbf{e}_x + v_y \mathbf{e}_y + v_z \mathbf{e}_z$) and 5 conservation equations (continuity : 1, Navier-Stokes : 3, energy balance : 1), so the formulation is complete.

Some simplifying assumptions

When applying these equations to air or water flows, some simplifying assumptions can be made.

- The fluid can be considered incompressible, i.e., its density is constant $\rho = \rho_0$. Liquid water is indeed poorly compressible (bulk modulus of 2.2 GPa). Air is a compressible fluid, but changes in density near the blades of a wind turbine can be estimated to be less than 1% (see application of Bernoulli theorem below). The continuity equation reduces to:

$$\nabla.\mathbf{v} = 0$$

- The fluid follows adiabatic evolutions, i.e., $\mathbf{q} = 0$. This assumption is valid because of the small magnitude of the macroscopic kinetic energy compared to the microscopic one due to thermal agitation ($e_{kin} \ll u \sim k_B T$). Even if all the macroscopic kinetic energy was dissipated, this would induce a negligible temperature change so that there is almost no temperature gradient and no heat flux.

- The fluid is a Newtonian fluid, i.e., the deviatoric stress is proportional to the rate of strain. As mentioned before, one can decompose the stress tensor in two contributions: pressure and deviatoric stress ($\underline{\underline{\sigma}} = -p\underline{\underline{I}} + \underline{\underline{\tau}}$). For an incompressible Newtonian fluid, $\underline{\underline{\tau}} = \mu\left(\nabla \mathbf{v} + \nabla \mathbf{v}^T\right)$ with μ the dynamic viscosity.

Under these simplifying assumptions, the divergence of the deviatoric stress in the Navier-Stokes equation can be simplified:

$$
\begin{aligned}
\nabla.\underline{\underline{\tau}} &= \mu\left(\nabla.\begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{bmatrix} + \nabla.\begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_y}{\partial x} & \frac{\partial v_z}{\partial x} \\ \frac{\partial v_x}{\partial y} & \frac{\partial v_y}{\partial y} & \frac{\partial v_z}{\partial y} \\ \frac{\partial v_x}{\partial z} & \frac{\partial v_y}{\partial z} & \frac{\partial v_z}{\partial z} \end{bmatrix}\right) \\
&= \mu\left(\nabla.\begin{bmatrix} 2\frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y}+\frac{\partial v_y}{\partial x} & \frac{\partial v_x}{\partial z}+\frac{\partial v_z}{\partial x} \\ \frac{\partial v_x}{\partial y}+\frac{\partial v_y}{\partial x} & 2\frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z}+\frac{\partial v_z}{\partial y} \\ \frac{\partial v_x}{\partial z}+\frac{\partial v_z}{\partial x} & \frac{\partial v_y}{\partial z}+\frac{\partial v_z}{\partial y} & 2\frac{\partial v_z}{\partial z} \end{bmatrix}\right) \\
&= \mu\begin{bmatrix} \frac{\partial}{\partial x}\left(2\frac{\partial v_x}{\partial x}\right)+\frac{\partial}{\partial y}\left(\frac{\partial v_x}{\partial y}+\frac{\partial v_y}{\partial x}\right)+\frac{\partial}{\partial z}\left(\frac{\partial v_x}{\partial z}+\frac{\partial v_z}{\partial x}\right) \\ \frac{\partial}{\partial x}\left(\frac{\partial v_x}{\partial y}+\frac{\partial v_y}{\partial x}\right)+\frac{\partial}{\partial y}\left(2\frac{\partial v_y}{\partial y}\right)+\frac{\partial}{\partial z}\left(\frac{\partial v_y}{\partial z}+\frac{\partial v_z}{\partial y}\right) \\ \frac{\partial}{\partial x}\left(\frac{\partial v_x}{\partial z}+\frac{\partial v_z}{\partial x}\right)+\frac{\partial}{\partial y}\left(\frac{\partial v_y}{\partial z}+\frac{\partial v_z}{\partial y}\right)+\frac{\partial}{\partial z}\left(2\frac{\partial v_z}{\partial z}\right) \end{bmatrix} \\
&= \mu\left(\begin{bmatrix} \frac{\partial^2 v_x}{\partial x^2}+\frac{\partial^2 v_x}{\partial y^2}+\frac{\partial^2 v_x}{\partial z^2} \\ \frac{\partial^2 v_y}{\partial x^2}+\frac{\partial^2 v_y}{\partial y^2}+\frac{\partial^2 v_y}{\partial z^2} \\ \frac{\partial^2 v_z}{\partial x^2}+\frac{\partial^2 v_z}{\partial y^2}+\frac{\partial^2 v_z}{\partial z^2} \end{bmatrix} + \begin{bmatrix} \frac{\partial}{\partial x}\left(\frac{\partial v_x}{\partial x}+\frac{\partial v_y}{\partial y}+\frac{\partial v_z}{\partial z}\right) \\ \frac{\partial}{\partial y}\left(\frac{\partial v_x}{\partial x}+\frac{\partial v_y}{\partial y}+\frac{\partial v_z}{\partial z}\right) \\ \frac{\partial}{\partial z}\left(\frac{\partial v_x}{\partial x}+\frac{\partial v_y}{\partial y}+\frac{\partial v_z}{\partial z}\right) \end{bmatrix}\right) \\
&= \mu\left(\Delta\mathbf{v} + \nabla\underbrace{(\nabla.\mathbf{v})}_{0\text{ (incompressible)}}\right) = \mu\Delta\mathbf{v}
\end{aligned}
$$

$$
\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v}.\nabla\mathbf{v} = -\frac{\nabla p}{\rho_0} + \frac{\mu}{\rho_0}\Delta\mathbf{v} + \mathbf{g} \tag{4.7}
$$

Likewise, the energy balance can be simplified :

$$
\nabla.(e\mathbf{v}) = e\underbrace{\nabla.\mathbf{v}}_{0\text{ (incompressible)}} + \mathbf{v}.\nabla e = \mathbf{v}.\nabla e
$$

$$
\nabla.\left(\underline{\underline{\sigma}}.\mathbf{v}\right) = -\nabla.(p\mathbf{v}) + \nabla.\left(\underline{\underline{\tau}}.\mathbf{v}\right) = \mathbf{v}.\nabla p + p\underbrace{\nabla.\mathbf{v}}_{0\text{ (incompressible)}} + \nabla.\left(\underline{\underline{\tau}}.\mathbf{v}\right) = \mathbf{v}.\nabla p + \nabla.\left(\underline{\underline{\tau}}.\mathbf{v}\right)
$$

$$
\frac{\partial e}{\partial t} + \mathbf{v}.\nabla e = -\frac{\mathbf{v}.\nabla p}{\rho_0} + \frac{\nabla.\left(\underline{\underline{\tau}}.\mathbf{v}\right)}{\rho_0} + \mathbf{g}.\mathbf{v}
$$

Additional assumptions are sometimes made, the validity of which may depend on the situation.

- The fluid flow is sometimes assumed stationary, i.e., all local properties are time-independent ($\frac{\partial X}{\partial t} = 0$). This assumption excludes any turbulent flows.
- The fluid is sometimes assumed ideal, i.e., non-viscous ($\mu = 0$). This assumption is valid if the velocity gradient is small, which cannot be true close to surfaces where the velocity goes from 0 at the interface to some given value far from the surface (boundary layer).

Stationary and ideal fluid assumptions are of course not valid in general for air and water flows, but will be useful to derive some key fundamental results (Bernoulli, Kutta-Zhukovskii) that help understand the physical origin of drag and lift.

4.1.3 Drag and lift

Bernoulli's principle

A fundamental result to understand the origin of lift is Bernoulli's principle which relates the changes in velocity and pressure in the stationary flow of an incompressible ideal fluid. Let us consider an ideal fluid (possibly compressible) in stationary adiabatic conditions. As the flow is steady the material derivative reduces to: $\frac{DA}{Dt} = \cancel{\frac{\partial A}{\partial t}} + \nabla A.\mathbf{v} = \nabla A.\mathbf{v}$. The continuity equation can therefore be written as:

$$\cancel{\frac{\partial \rho}{\partial t}} + \nabla.(\rho\mathbf{v}) = 0 \Rightarrow \nabla\rho.\mathbf{v} + \rho\nabla.\mathbf{v} = 0 \Rightarrow \nabla.\mathbf{v} = -\frac{1}{\rho}\frac{D\rho}{Dt}$$

Likewise, the energy balance can be written

$$\underbrace{\cancel{\frac{\partial(\rho e)}{\partial t}}}_{\text{stationary}} + \underbrace{\nabla.(\rho e\mathbf{v})}_{\nabla(\rho e).\mathbf{v}+\rho e\nabla.\mathbf{v}} = -\underbrace{\nabla.(p\mathbf{v})}_{\nabla p.\mathbf{v}+p\nabla.\mathbf{v}} + \underbrace{\cancel{\nabla.(\underline{\underline{\tau}}.\mathbf{v})}}_{\text{no viscosity}} + \rho\underbrace{\mathbf{g}}_{\nabla(-gz)}.\mathbf{v} + \underbrace{\cancel{\nabla.\mathbf{q}}}_{\text{adiabatic}}$$

Using the continuity equation $\nabla.\mathbf{v} = -\frac{1}{\rho}\frac{D\rho}{Dt}$ and introducing $\frac{D(\rho e)}{Dt} = \nabla(\rho e).\mathbf{v}$, $\frac{Dp}{Dt} = \nabla p.\mathbf{v}$, and $\frac{D(gz)}{Dt} = \nabla(gz).\mathbf{v}$, one can reformulate the energy balance in terms of material derivatives:

$$\underbrace{\frac{D(\rho e)}{Dt} - e\frac{D\rho}{Dt}}_{\rho\frac{De}{Dt}} = \underbrace{-\frac{Dp}{Dt} + \frac{p}{\rho}\frac{D\rho}{Dt}}_{-\rho\frac{D\left(\frac{p}{\rho}\right)}{Dt}} - \rho\frac{D(gz)}{Dt} \Rightarrow \rho\frac{D\left(e + \frac{p}{\rho} + gz\right)}{Dt} = 0$$

Therefore the material derivative of the quantity $e + \frac{p}{\rho} + gz = u + \frac{\mathbf{v}^2}{2} + \frac{p}{\rho} + gz$ is zero, which means that this quantity is constant during the flow of the fluid:

$$u + \frac{\mathbf{v}^2}{2} + \frac{p}{\rho} + gz = \text{constant}$$

This is Bernoulli's principle for a compressible fluid. In the case of an incompressible fluid, $\rho = \rho_0$ and $u = u_0$ (no heat and no work exchange), and Bernoulli's principle reduces to:

$$\frac{\mathbf{v}^2}{2} + \frac{p}{\rho_0} + gz = \text{constant}$$

According to Bernoulli's principle, the pressure of a fluid drops as its velocity (or its altitude) increases. The quantity $p_{dyn} = \frac{1}{2}\rho_0\mathbf{v}^2$ is called the dynamical pressure and represents the maximum increase of pressure as the fluid velocity is reduced to zero. For a 50 km/h wind, $p_{dyn} = 125$ Pa which is about 0.1% of the atmospheric pressure. Thus, pressure changes induced by velocity changes remain modest in comparison with the atmospheric pressure. Following the ideal gas law, the same is true for the density, which is why one can reasonably assume an incompressible fluid even for air.

Bernoulli's principle explains why a fluid flow around an object can experience a lift. An object surrounded by a steady flow perturbates the nearby velocity field. If the perturbation of the velocity is asymmetrical with respect to the object, the associated pressure changes are asymmetrical as well, resulting in a net force on the object orthogonal to the fluid flow. This 'asymmetry' of the fluid flow is measured by the concept of circulation, which is the object of the next paragraph.

circulation and vorticity

The circulation Γ is defined as a line integral of the velocity around a closed curve $\mathcal{C}$:

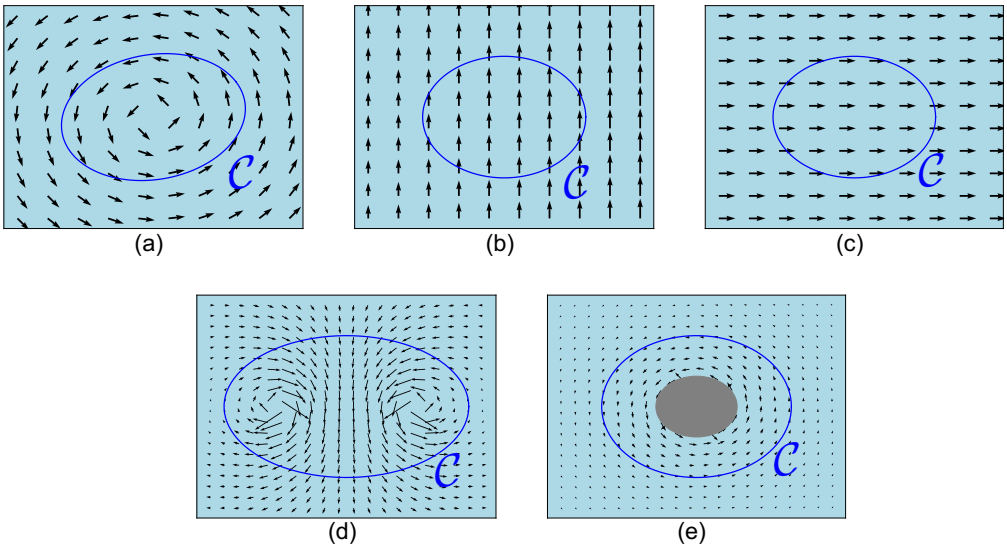


Figure 4.1: Circulation and vorticity illustrated on an few examples. (a) A fluid flow with circulation and vorticity. (b) A uniform velocity gradient is a flow with vorticity and circulation. (c) A uniform velocity field has no circulation and no vorticity. (d) A fluid flow can have local vorticity but no net circulation on a particular closed curve. (e) A fluid flow can have a non-zero circulation around an object while the fluid is irrotational everywhere.

$$\Gamma(\mathcal{C}) = \oint_{\mathcal{C}} \mathbf{v}.d\mathbf{x}$$

Applying the Stokes theorem, the circulation is also the flow of $\boldsymbol{\omega} = \nabla \times \mathbf{v}$, called vorticity, through the section delimited by $\mathcal{C}$:

$$\Gamma(\mathcal{C}) = \int_S (\nabla \times \mathbf{v}).d\mathbf{S} = \int_S \boldsymbol{\omega}.d\mathbf{S}$$

The vorticity measures the magnitude of the local circulation of the fluid. Circulation and vorticity are associated with the rotational flow of the fluid (Fig. 4.1a). But non-zero circulation also emerges from gradients in the velocity field with no apparent 'rotation' (Fig. 4.1b). Obviously a stationary uniform velocity field (e.g., undisturbed wind flow) has no vorticity and no circulation (Fig. 4.1c). A fluid with no vorticity is said irrotational. Yet vorticity and circulation should not be confounded as one is purely local and the other one global for a given closed curve. In particular, it is obvious from the above relation that a system can have zero circulation globally but non-zero vorticity locally (Fig. 4.1d). The inverse is also possible but less obvious: a fluid can have zero vorticity but a non-zero circulation. Such a situation is possible if the fluid circulates around an object (Fig. 4.1e): considering a velocity field $\mathbf{v} = \frac{\gamma}{2\pi r}\mathbf{e}_\theta$, one easily checks[a] that $\boldsymbol{\omega} = \nabla \times \mathbf{v} = 0$ whereas $\Gamma(\mathcal{C}) = \gamma$ for any circle larger than the object. In such a case, the circulation is the same whatever the closed curve around the object and equal to the circulation at the interface with the object.

[a] In polar coordinates, the curl operator of a vector field $X(r)\,\mathbf{e}_\theta$ is $\nabla \times (X(r)\,\mathbf{e}_\theta) = \frac{1}{r}\frac{d(rX)}{dr}\mathbf{e}_z$.

A key result regarding circulation and vorticity is Kelvin's circulation theorem: for an ideal fluid, the vorticity of a packet of fluid is constant as the packet moves through the flow. In

particular, an ideal fluid that is irrotational will remain irrotational. Of course, ideal fluids are a limit case, and in reality, all fluids exhibit some viscosity except very peculiar cases (e.g., superfluid helium). And viscosity eventually dissipates the vorticity so that a real fluid loses its vorticity over time: a vortex stirred up in a glass of water eventually comes to rest. Yet Kelvin's theorem remains a useful result to analyze fluid flow over short time scales (in comparison with the time scale of viscous dissipation).

Let us analyze the case of a rotating ball moving through the air (Figure 4.2). Assuming the steady flow of an ideal incompressible fluid, the equations of fluid mechanics are linear and one can therefore superpose the fluid flow resulting from a translation and that resulting from a rotation. In both cases, the flows can be calculated analytically, and we just discuss here the results. The velocity field of a non-rotating ball in translation is a non-circulating air flow (Figure 4.2a), and that of a rotating ball with no translation is a circulating air flow (Figure 4.2b). The fluid far away from the ball has no vorticity, and, according to Kelvin's circulation theorem, it remains irrotational near the ball. Therefore, we are in the particular case of an irrotational fluid circulating around an object. According to Bernoulli's principle, the pressure changes along the interface with the ball (Figure 4.2d). As the velocity is larger along the upper part than along the bottom part (Figure 4.2c), the ball experiences higher pressures below than above, i.e., it is lifted. This lift results directly from the circulating part of the velocity field due to the rotation of the ball. Indeed, the translational part (Figure 4.2a) exhibits symmetric velocities above and below the ball and the corresponding pressure changes would therefore compensate and generate no lift. Therefore a lift appears only if the ball is rotating. This illustrates the well-known Magnus effect. Two special points, called stagnation points, can be identified where the fluid velocity is exactly zero. These are also the points where the fluid pressure is maximum equal to the dynamic pressure. As we shall see, the location of stagnation points is central to explaining lift on airfoils.

Kutta-Zhukovskii theorem

A key result relating circulation and lift is the Kutta-Zhukovskii theorem. Let us consider a steady flow of an incompressible ideal fluid around an elongated object in the $\mathbf{e}_z$ direction, with $v_\infty \mathbf{e}_x$ and p_∞ the free-stream velocity and pressure (velocity and pressure at large distances). Figure 4.3 shows a cross-section of the system in the $(\mathbf{e}_x, \mathbf{e}_y)$ plane.

As the far field velocity is irrotational, the fluid flow remains irrotational everywhere (Kelvin's theorem). Since there is an obstructing object, the circulation around the object can be non-zero, and is the same irrespective of the closed curve considered. We call $\mathcal{C}_0$ the closed curve delimiting the object. The total force per unit length along $\mathbf{e}_z$ exerted by the fluid on the object is related to the pressure field along $\mathcal{C}_0$:

$$\mathbf{f} = -\oint_{\mathcal{C}_0} p\mathbf{n}dl$$

where $\mathbf{n}$ is the outward normal unit vector to the curve $\mathcal{C}_0$. Applying Bernoulli's principle assuming negligible effect of gravity, the pressure p and velocity $\mathbf{v}$ at any point on the surface of the object are related to the remote pressure p_∞ and velocity v_∞ as follows: $\frac{1}{2}\rho_0 (v_\infty)^2 + p_\infty = \frac{1}{2}\rho_0 \mathbf{v}^2 + p$. Therefore, the expression of the force on the object can be related to the fluid velocity at the surface of the object:

$$\begin{aligned}
\mathbf{f} &= -\oint_{\mathcal{C}_0} \left(\frac{1}{2}\rho_0 (v_\infty)^2 + p_\infty - \frac{1}{2}\rho_0 \mathbf{v}^2\right) \mathbf{n}dl \\
&= -\left(\frac{1}{2}\rho_0 (v_\infty)^2 + p_\infty\right) \cancel{\oint_{\mathcal{C}_0} \mathbf{n}dx} + \frac{1}{2}\rho_0 \oint_{\mathcal{C}_0} \mathbf{v}^2 \mathbf{n}dx \\
&= \frac{1}{2}\rho_0 \oint_{\mathcal{C}_0} \mathbf{v}^2 \mathbf{n}dx
\end{aligned}$$

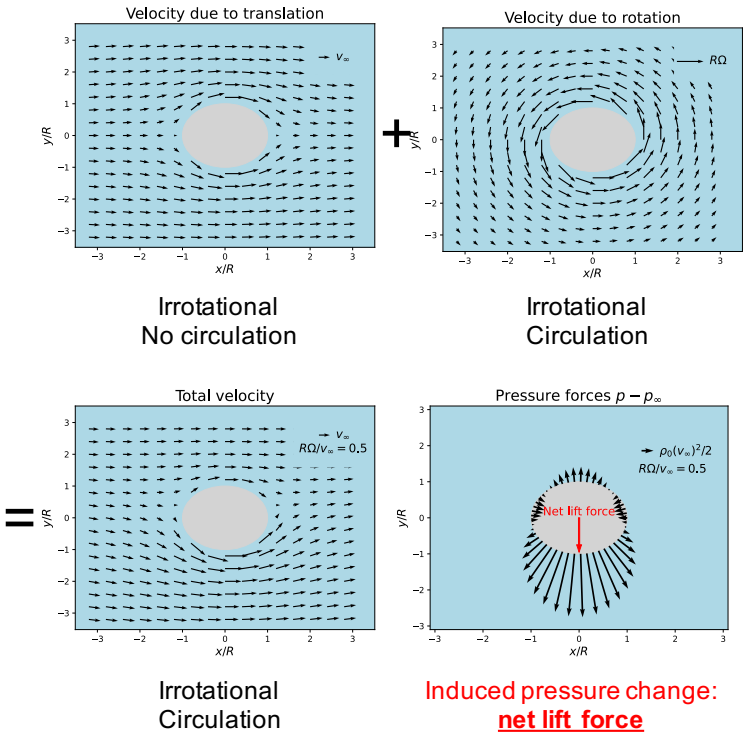


Figure 4.2: A flow with both translational and rotational contributions around an object generates a net lift.

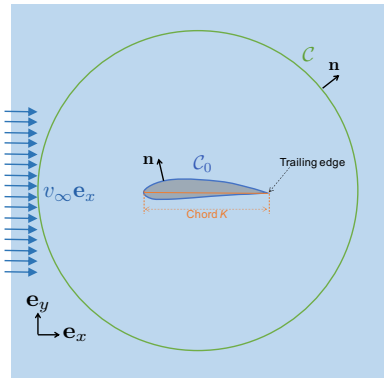


Figure 4.3: The Kutta-Zhukovskii theorem applies to a circulating irrotational steady flow of an incompressible ideal fluid around an object. We consider a 2D problem assuming the section of the object remains identical along its axis. $\mathcal{C}_0$ represents the boundary of the object, and $\mathcal{C}$ represent a circular closed curve far from the object.

Let us now consider a circular closed curve $\mathcal{C}$ far from the object (Fig. 4.3). Applying the divergence theorem in two dimensions[1], we have:

$$\begin{cases} f_x = \frac{1}{2}\rho_0 \oint_{\mathcal{C}} \mathbf{v}^2 n_x dl - \frac{1}{2}\rho_0 \iint_\Omega \frac{\partial(\mathbf{v}^2)}{\partial x} dS \\ f_y = \frac{1}{2}\rho_0 \oint_{\mathcal{C}} \mathbf{v}^2 n_y dl - \frac{1}{2}\rho_0 \iint_\Omega \frac{\partial(\mathbf{v}^2)}{\partial y} dS \end{cases}$$

where Ω is the domain delimited by $\mathcal{C}_0$ and $\mathcal{C}$. Since the fluid is incompressible and irrotational, we have $\nabla.\mathbf{v} = 0 \Rightarrow \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0$ and $\nabla \times \mathbf{v} = 0 \Rightarrow \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} = 0$. Using these relations, we can rewrite $\frac{\partial(\mathbf{v}^2)}{\partial x}$ and $\frac{\partial(\mathbf{v}^2)}{\partial y}$ in the surface integrals as follows:

$$\begin{cases} \frac{1}{2}\frac{\partial(\mathbf{v}^2)}{\partial x} = v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_y}{\partial x} = \left(v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y}\right) + v_x \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y}\right) \\ \qquad = 2v_x \frac{\partial v_x}{\partial x} + \left(v_x \frac{\partial v_y}{\partial y} + v_y \frac{\partial v_x}{\partial y}\right) = \frac{\partial(v_x^2)}{\partial x} + \frac{\partial(v_x v_y)}{\partial y} = \nabla. \begin{bmatrix} v_x^2 \\ v_x v_y \end{bmatrix} \\ \frac{1}{2}\frac{\partial(\mathbf{v}^2)}{\partial y} = v_x \frac{\partial v_x}{\partial y} + v_y \frac{\partial v_y}{\partial y} = \left(v_x \frac{\partial v_y}{\partial x} + v_y \frac{\partial v_y}{\partial y}\right) + v_y \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y}\right) \\ \qquad = \left(v_x \frac{\partial v_y}{\partial x} + v_y \frac{\partial v_x}{\partial x}\right) + 2v_y \frac{\partial v_y}{\partial y} = \frac{\partial(v_x v_y)}{\partial x} + \frac{\partial(v_y^2)}{\partial y} = \nabla. \begin{bmatrix} v_x v_y \\ v_y^2 \end{bmatrix} \end{cases}$$

Therefore, we can apply again the divergence theorem:

$$\begin{cases} f_x = \frac{1}{2}\rho_0 \oint_{\mathcal{C}} \mathbf{v}^2 n_x dl - \left(\rho_0 \oint_{\mathcal{C}} \left(v_x^2 n_x + v_x v_y n_y\right) dl - \rho_0 \oint_{\mathcal{C}_0} \left(v_x^2 n_x + v_x v_y n_y\right) dl\right) \\ f_y = \frac{1}{2}\rho_0 \oint_{\mathcal{C}} \mathbf{v}^2 n_x dl - \left(\rho_0 \oint_{\mathcal{C}} \left(v_x v_y n_x + v_y^2 n_y\right) dl - \rho_0 \oint_{\mathcal{C}_0} \left(v_x v_y n_x + v_y^2 n_y\right) dl\right) \end{cases}$$

The integrals on $\mathcal{C}_0$ cancel out since $v_x^2 n_x + v_x v_y n_y = v_x \mathbf{v}.\mathbf{n}$ and $v_x v_y n_x + v_y^2 n_y = v_y \mathbf{v}.\mathbf{n}$ while $\mathbf{v}.\mathbf{n} = 0$ all along the surface of the object. The components of the force are thus:

$$\begin{cases} f_x = \frac{1}{2}\rho_0 \oint_{\mathcal{C}} \left(\mathbf{v}^2 n_x - 2v_x v_x n_x - 2v_x v_y n_y\right) dl \\ f_y = \frac{1}{2}\rho_0 \oint_{\mathcal{C}} \left(\mathbf{v}^2 n_y - 2v_x v_y n_x - 2v_y v_y n_y\right) dl \end{cases}$$

We decompose the velocity field as follows: $\mathbf{v}(x,y) = v_\infty \mathbf{e}_x + \mathbf{w}(x,y)$, where $\mathbf{w}(x,y)$ is the distortion of the flow. $\mathbf{w}$ vanishes at large distances from the object. As the curve $\mathcal{C}$ can be taken arbitrarily far from the object, we can therefore consider $|\mathbf{w}|$ small compared to v_∞, and limit ourselves to a first-order expansion in the components of $\mathbf{w}$:

$$\begin{cases} f_x = \frac{1}{2}\rho_0 \oint_{\mathcal{C}} \left(\left(v_\infty + w_x\right)^2 n_x + w_y^2 n_x - 2\left(v_\infty + w_x\right)^2 n_x - 2\left(v_\infty + w_x\right) w_y n_y\right) dl \\ \quad \approx \frac{1}{2}\rho_0 \oint_{\mathcal{C}} \left(-\left(v_\infty\right)^2 n_x - 2v_\infty w_x n_x - 2v_\infty w_y n_y\right) dl \\ \quad \approx -\frac{1}{2}\rho_0 \left(v_\infty\right)^2 \cancel{\oint_{\mathcal{C}} n_x dl} - \rho_0 v_\infty \oint_{\mathcal{C}} \left(w_x n_x + w_y n_y\right) dl \\ \quad \approx -\rho_0 v_\infty \oint_{\mathcal{C}} \left(\mathbf{w}.\mathbf{n}\right) dx \\ f_y = \frac{1}{2}\rho_0 \oint_{\mathcal{C}} \left(\left(v_\infty + w_x\right)^2 n_y + w_y^2 n_y - 2\left(v_\infty + w_x\right) w_y n_x - 2w_y^2 n_y\right) dl \\ \quad \approx \frac{1}{2}\rho_0 \oint_{\mathcal{C}} \left(\left(v_\infty\right)^2 n_y + 2v_\infty w_x n_y - 2v_\infty w_y n_x\right) dl \\ \quad \approx \frac{1}{2}\rho_0 \left(v_\infty\right)^2 \cancel{\oint_{\mathcal{C}} n_y dl} + \rho_0 v_\infty \oint_{\mathcal{C}} \left(w_x n_y - w_y n_x\right) dl \end{cases}$$

[1] $\oint_{\partial\Omega} \mathbf{X}.\mathbf{n} dl = \iint_\Omega \nabla.\mathbf{X} dS$ with $\partial\Omega$ the contour of Ω and $\mathbf{n}$ the outward normal with respect to the interior of Ω.

For the component of the force f_x in the direction of the remote flow, we can apply again the divergence theorem:

$$f_x = -\rho_0 v_\infty \oint_{\mathcal{C}_0} (\mathbf{w}.\mathbf{n})\, dl - \rho_0 v_\infty \iint_\Omega \nabla.\mathbf{w} dS$$

Using the decomposition of the velocity field $\mathbf{w} = \mathbf{v} - v_\infty \mathbf{e}_x$, the integral on $\mathcal{C}_0$ is: $\oint_{\mathcal{C}_0} (\mathbf{w}.\mathbf{n})\, dx = \oint_{\mathcal{C}_0} (\mathbf{v}.\mathbf{n})\, dx - v_\infty \cancel{\oint_{\mathcal{C}_0} n_x dx}$. And since $\mathbf{v}.\mathbf{n} = 0$ all along $\mathcal{C}_0$, we have $\oint_{\mathcal{C}_0} (\mathbf{w}.\mathbf{n})\, dx = 0$. Moreover, since $\mathbf{w}$ and $\mathbf{v}$ differ only by a constant vector, $\nabla.\mathbf{w} = \nabla.\mathbf{v}$. As the fluid is incompressible, the conservation of mass reduces to $\nabla.\mathbf{v} = 0 \Rightarrow \nabla.\mathbf{w} = 0$. Therefore, the component of the force in the direction of the remote flow is zero, and there is no drag:

$$f_x = 0$$

Regarding the component f_y orthogonal to the direction of the remote flow, one finds that the integral involves the dot product $\mathbf{w}.\mathbf{t}$, where $\mathbf{t} = \begin{bmatrix} n_y \\ -n_x \end{bmatrix}$ is the unit vector tangent to the curve $\mathcal{C}$ in the clockwise direction:

$$f_y = \rho_0 v_\infty \oint_{\mathcal{C}} (w_x t_x + w_y t_y)\, dl = \rho_0 v_\infty \oint_{\mathcal{C}} (\mathbf{w}.\mathbf{t})\, dl$$

Introducing the decomposition the the velocity $\mathbf{w} = \mathbf{v} - v_\infty \mathbf{e}_x$, the contribution of the remote velocity v_∞ cancels out:

$$f_y = \rho_0 v_\infty \oint_{\mathcal{C}} (\mathbf{v}.\mathbf{t})\, dl - \rho_0 (v_\infty)^2 \cancel{\oint_{\mathcal{C}} t_x dl}$$

The integral $\oint_{\mathcal{C}} (\mathbf{v}.\mathbf{t})\, dl = \oint_{\mathcal{C}} \mathbf{v}.d\mathbf{x} = \Gamma(\mathcal{C})$ is the circulation of the fluid around the object counted in the clockwise direction. Therefore, the component f_y orthogonal to the direction of the remote flow is proportional to the circulation:

$$f_y = \rho_0 v_\infty \Gamma(\mathcal{C})$$

In summary, the force exerted by the steady flow of an incompressible ideal fluid on an object exhibits no drag and a lift proportional to the circulation of the fluid around the object:

$$\begin{cases} f_{\text{lift}} = \rho_0 v_\infty \Gamma(\mathcal{C}) \\ f_{\text{drag}} = 0 \end{cases}$$

This expression of the lift is known as the Kutta-Zhukovskii theorem. The fact that there is no drag is known as d'Alembert's paradox. The absence of drag originates from the absence of viscosity of the fluid. Real fluids have non-zero viscosity and there is always a non-zero drag force.

Role of viscosity

As we have seen with d'Alembert's paradox, ignoring viscosity is a critical assumption regarding drag. Whether a fluid mechanics problem is dominated by viscosity or not can be estimated from an analysis of the different dimensions of the problem. To do so, one has to reformulate the conservation of momentum in a dimensionless form. This is done by identifying the characteristic dimensions of the problem (e.g., far-field values) : a velocity v_∞, a density ρ_0, and a characteristic length L. From these reference dimensions, one can reduce every term of the conservation of momentum to reach a dimensionless version of the Navier-Stokes equations (Eq. 4.7):

$$\mathbf{x}^* = \frac{\mathbf{x}}{L} \quad \mathbf{v}^* = \frac{\mathbf{v}}{v_\infty} \quad p^* = \frac{p}{\rho_0 v_\infty^2} \quad t^* = \frac{v_\infty}{L} t$$

$$\Rightarrow \frac{\partial \mathbf{v}^*}{\partial t^*} + \mathbf{v}^*.\nabla_{\mathbf{x}^*}\mathbf{v}^* = -\nabla_{\mathbf{x}^*} p^* + \frac{\mu}{\rho_0 v_\infty L} \Delta_{\mathbf{x}^*}\mathbf{v}^* + \frac{gL}{v_\infty^2}\mathbf{e}_z$$

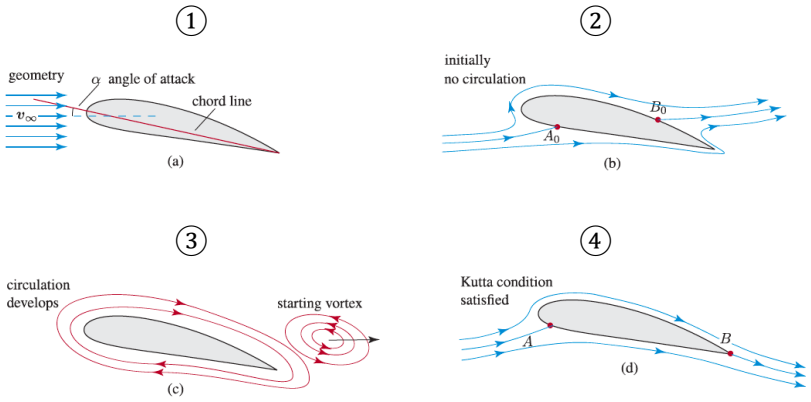


Figure 4.4: How a circulation develops around an airfoil (from Jaffe & Taylor (2018) The Physics of Energy. Cambridge University Press).

where $\nabla_{\mathbf{x}^*}$ and $\Delta_{\mathbf{x}^*}$ are the gradient and laplacian operators with respect to the dimensionless position $\mathbf{x}^*$. One identifies in this formulation of the Navier-Stokes equations two dimensionless numbers: the Reynolds number $\mathcal{R}e = \frac{\rho_0 v_\infty L}{\mu}$ that provides the magnitude of the viscous term, and the Froude number $\mathcal{F}r = \frac{v_\infty}{\sqrt{gL}}$ that provides the magnitude of the gravity term.

$$\frac{\partial \mathbf{v}^*}{\partial t^*} + \mathbf{v}^*.\nabla_{\mathbf{x}^*}\mathbf{v}^* = -\nabla_{\mathbf{x}^*}p^* + \frac{1}{\mathcal{R}e}\Delta_{\mathbf{x}^*}\mathbf{v}^* + \frac{1}{\mathcal{F}r^2}\mathbf{e}_z \tag{4.8}$$

A large value of the Reynolds number $\mathcal{R}e$ means that the viscous term is negligible, and a small value means that it dominates. The same remark applies to the Froude number $\mathcal{F}r$ with respect to the importance of the gravity term. Therefore the Reynolds number is the quantity to consider to evaluate the importance of viscosity. It can be viewed as the ratio between the dynamic pressure and the typical viscous force: $\mathcal{R}e = \frac{\rho_0 v_\infty^2}{\mu v_\infty / L}$. The transition from dominant viscous forces to negligible viscous forces typically occurs for $\mathcal{R}e \sim 10^3 - 10^4$. For instance, for pipes when considering L as the diameter, the flow is systematically laminar and dominated by viscosity for $\mathcal{R}e < 2\,300$, and it is always turbulent for $\mathcal{R}e > 4\,000$. The definition of the Reynolds number is somehow ambiguous as it relies on the choice of a characteristic length L. There is always a length small enough for which the viscous force becomes dominant. For a typical airfoil of chord length $L = 1$ m in a 50 km/h wind, the Reynolds number is large ($\sim 10^6$) which suggests that viscous forces are negligible at the scale of the airfoil. But zooming at the scale of the boundary layer ($L = 1$ mm) the Reynolds number is much smaller ($\sim 10^3$) and falls in the domain for which viscous forces dominate. Therefore, viscosity is unimportant for the aerodynamics of an airfoil, except for a very thin layer close to the surface of the airfoil. Since drag originates from the shear forces within the boundary layer, viscosity appears critical for this force. Conversely, lift originates from the global circulation around the object, for which the detail of the boundary layer is unimportant. For application to water flows, the Reynolds number is also very large: $\mathcal{R}e \sim 10^5$ for a hydrofoil with 20 cm thick blades in a 1.5 km/h water stream.

4.1.4 Working principle of an airfoil

As we have seen, lift originates from the circulation of a fluid around an object. But this does not explain how a circulation develops, starting from an initially non-circulating flow. As we shall explain below, this is where the shape of an airfoil is important, and more specifically the sharp trailing edge. Let us describe step by step the emergence of the circulation as illustrated in Figure 4.4:

1. The airfoil begins to move with an angle of attack α. The flow is initially irrotational, and there is no circulation. As the flow develops and the velocity gradients are small, viscous

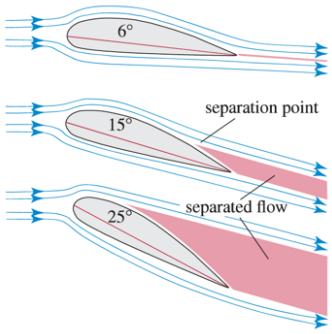


Figure 4.5: As the angle of attack increases, the steady flow separates from the top of the airfoil, and a region of turbulence develops, until the airfoil stalls (from Jaffe & Taylor (2018) The Physics of Energy. Cambridge University Press).

effects are negligible and, following Kelvin's theorem, the flow remains irrotational and with no circulation. So, at this early stage, there is no lift.

2. Two stagnation points appear (A_0 and B_0), that are positioned so that there is no circulation around the airfoil. This situation would persist indefinitely for an ideal fluid. But, since the trailing edge is sharp, the velocities around the trailing edge are very large and one cannot neglect viscous effects around this point.

3. The high velocities that develop between B_0 and the trailing edge induce non-negligible viscous forces in the boundary layer. Ideal fluid assumption breaks down, and Kelvin's theorem does not apply anymore. Some vorticity initiates near the trailing edge, which causes the formation of a clockwise circulation around the airfoil while and counter-clockwise vortex forms in the wake of the airfoil.

4. The vortex moves away in the wake leaving an irrotational circulation around the airfoil, with the rear stagnation point (B) shifted to the sharp trailing edge. The air flow now exerts a net lift on the airfoil.

The final stage is known as the Kutta condition: when an airfoil with a sharp trailing edge is immersed in a uniform fluid flow, viscous forces in the boundary layer establish a circulation of exactly the magnitude necessary to keep the rear stagnation point at the trailing edge of the airfoil. In this respect, the most important feature of an airfoil is the sharp trailing edge. Of course, the other aspects of the shape of the airfoil can be optimized to improve the lift, but the sharp trailing edge is essential to the emergence of circulation and lift.

The Kutta condition relies on a steady flow around the airfoil, with no turbulence except in a very narrow region in the wake of the trailing edge. This is valid for small angles of attack α, but this analysis fails as the angle of attack increases (Fig. 4.5): at larger α, the steady flow separates from the top of the airfoil leaving behind an enlarged region of turbulence. The lift starts to decrease as the turbulent region becomes too large, and the airfoil stalls.

In practice, analytical calculations are limited to very simplified cases, and estimating the lift and drag of actual airfoil requires computational fluid mechanics or wind tunnel experimental tests. As discussed before, the assumptions of incompressible fluid, adiabatic evolution, and Newtonian fluid are reasonably valid. But one cannot neglect viscosity (ideal fluid) and transient evolutions (turbulence). So the general problem to solve is that of a viscous incompressible Newtonian fluid in adiabatic transient evolution. This problem can expressed in dimensionless form as follows (same approach as for Eq. 4.8):

$$\begin{cases} \nabla_{\mathbf{x}^*}.\mathbf{v}^* = 0 \\ \frac{\partial \mathbf{v}^*}{\partial t^*} + \mathbf{v}^*.\nabla_{\mathbf{x}^*}\mathbf{v}^* = -\nabla_{\mathbf{x}^*}p^* + \frac{1}{\mathcal{R}e}\Delta_{\mathbf{x}^*}\mathbf{v}^* \\ \frac{\partial e^*}{\partial t^*} + \mathbf{v}^*.\nabla_{\mathbf{x}^*}e^* = -\mathbf{v}^*.\nabla_{\mathbf{x}^*}p^* + \frac{1}{\mathcal{R}e}\nabla_{\mathbf{x}^*}.\left(\left(\nabla_{\mathbf{x}^*}\mathbf{v}^* + \left(\nabla_{\mathbf{x}^*}\mathbf{v}^*\right)^T\right).\mathbf{v}^*\right) \end{cases}$$

where we have considered the reduced position $\mathbf{x}^* = \frac{\mathbf{x}}{L}$, velocity $\mathbf{v}^* = \frac{\mathbf{v}}{v_\infty}$, pressure $p^* = \frac{p}{\rho_0 v_\infty^2}$, and energy $e^* = \frac{u}{v_\infty^2} + \frac{1}{2}\frac{v^2}{v_\infty^2}$. The boundary conditions of the above problem are $\mathbf{v}\left(\mathbf{x} \in \mathcal{C}_0\right) = 0$ (stick condition at the interface with the airfoil), and $\mathbf{v}\left(|\mathbf{x}| \to \infty\right) = v_\infty \mathbf{e}_x$ (far field undisturbed flow). These boundary conditions can be turned dimensionless: $\mathbf{v}^*\left(\mathbf{x}^* \in \mathcal{C}_0^*\right) = 0$ and $\mathbf{v}^*\left(|\mathbf{x}^*| \to \infty\right) = \mathbf{e}_x$, where $\mathcal{C}_0^*$ is the reduced shape of the airfoil (reduced with a characteristic length L which we take here equal to the longest dimension of the airfoil, called the chord length K). The solution of this dimensionless problem leads to a dimensionless stress field $\underline{\underline{\sigma}}^* = \frac{1}{\rho_0 v_\infty^2}\underline{\underline{\sigma}} = -p^*\underline{\underline{1}} + \frac{1}{\mathcal{R}e}\left(\nabla_{\mathbf{x}^*}\mathbf{v}^* + \left(\nabla_{\mathbf{x}^*}\mathbf{v}^*\right)^T\right)$ of the general form: $\underline{\underline{\sigma}}^* = f\left(\mathbf{x}^*, t^*, \mathcal{R}e, \mathcal{C}_0^*\right)$. The total force on the airfoil is the integration of the stresses over the airfoil boundary $\mathcal{C}_0$:

$$\mathbf{f} = \oint_{\mathcal{C}_0} \underline{\underline{\sigma}}.\mathbf{n}dl$$

Introducing the dimensionless solution to the problem, we have:

$$\mathbf{f} = \frac{\rho_0 v_\infty^2}{2} K \underbrace{\left(2\oint_{\mathcal{C}_0^*} \underline{\underline{\sigma}}^*\left(\mathbf{x}^*, \mathbf{t}^*, \mathcal{R}e, \mathcal{C}_0^*\right)\mathbf{n}dl^*\right)}_{c_d\mathbf{e}_x + c_l\mathbf{e}_y} = c_d\frac{\rho_0 v_\infty^2}{2}K\mathbf{e}_x + c_l\frac{\rho_0 v_\infty^2}{2}K\mathbf{e}_y$$

where we have introduced the drag and lift coefficients c_d and c_l. These coefficients are related to the dimensionless problem only and depend on $\mathbf{t}^*$, $\mathcal{R}e$, and $\mathcal{C}_0^*$. One can therefore estimate these coefficients by testing a mockup with the same dimensionless shape $\mathcal{C}_0^*$ and in a fluid flow with the same Reynolds number $\mathcal{R}e$. Restricting ourselves to stationary solutions, at a fixed Reynolds number, and for a reduced shape that is only allowed to rotate along the airfoil axis, then the drag and lift coefficients are functions of the angle of attack α only:

$$\begin{cases} f_{drag} = c_d\left(\alpha\right)\frac{\rho_0 v_\infty^2}{2}K \\ f_{\text{lift}} = c_l\left(\alpha\right)\frac{\rho_0 v_\infty^2}{2}K \end{cases}$$

The profile of the drag and lift coefficients as a function of the angle of attack for a typical airfoil is shown in Figure 4.6. The domain of interest is when the lift is large while the drag is small. This is the case at low angles of attack of about 5°-10° for which the ratio between drag and lift coefficients is about 1:100. This provides optimal conditions that will constrain the design of a wind turbine blade (see next section).

4.2 Wind energy

4.2.1 Wind resource

Before looking at the conversion of wind into useful mechanical work, we discuss here the wind resource.

A first important aspect of the wind resource, well-known from our everyday experience, is that wind is a highly variable resource (Fig. 4.7). Wind speed and orientation fluctuates on multiple time scales, which presents challenges for wind energy production. On very short timescales, wind can vary within minutes due to turbulence. Over a 24-hour period, wind patterns shift between day and night, while weather changes can cause variability over the course of a few days. Additionally, wind speeds are influenced by seasonal changes, leading to variations over months or even years. A significant portion of these fluctuations, especially on shorter time scales (day or less), is poorly predictable, making it difficult to forecast wind energy output with precision. Because of this

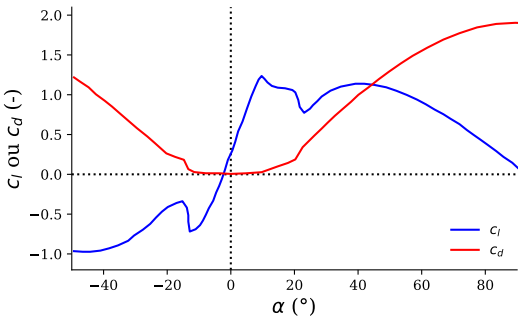


Figure 4.6: Example of drag and lift coefficients as a function of the angle of attack for a typical airfoil shape. The maximum lift coefficient is usually around 1, and the maximum drag coefficient around 2. For applications to wind turbines, one is usually interested in a high lift coefficient and a low drag coefficient, which is achieved for angles of attack around 5°-10°.

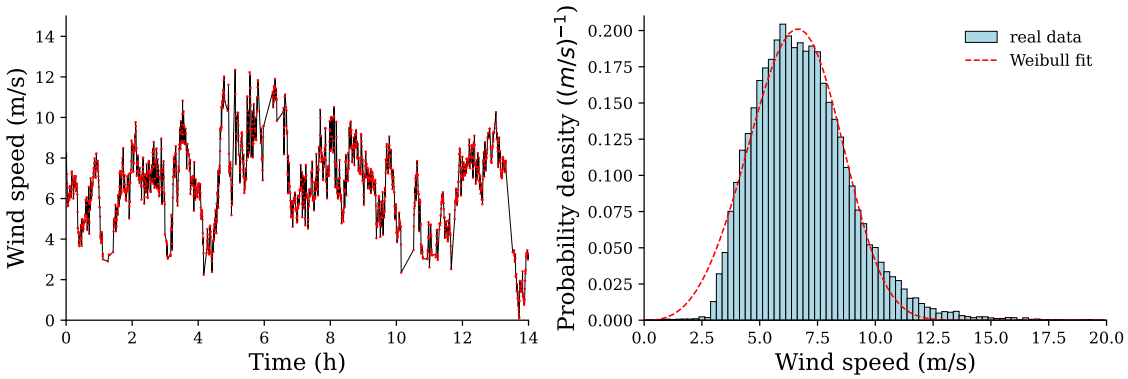
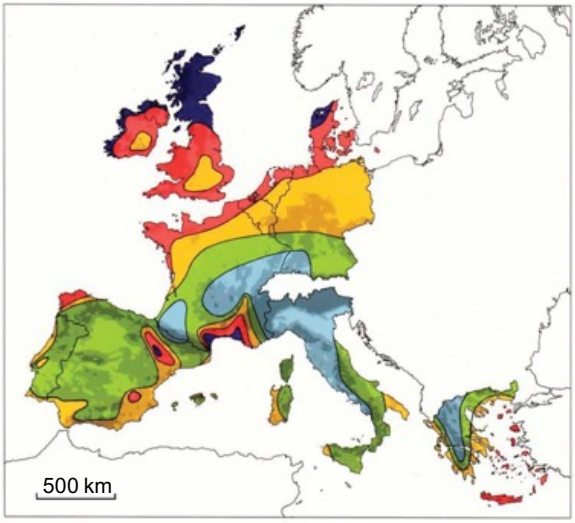


Figure 4.7: (left) Two weeks wind speed times series measured at the location of a turbine on an inland wind farm. Each red point is the average wind speed over 10 min. (right) Probability density of the wind speed at the same location calculated from the times series over a year; and fit of the probability density with a Weibull law. (Data of the time series from Ding, Y. (2021). Wind Time Series Dataset [Data set]. Zenodo. https://doi.org/10.5281/zenodo.5516539)

unpredictability, wind energy production requires careful management and integration with other energy sources or storage systems to ensure a reliable power supply.

Despite this variability, wind is a widely available resource, with global estimates of wind energy potential ranging from 1 TW (conservative) to 100 TW (optimistic), which could cover all current global energy demand of around 20 TW. The best wind resources are found in specific geographic locations such as coastal areas, open oceans, hills, and ridges, where wind speeds are consistently higher (see for instance the wind power atlas of Europe in Figure 4.8). However, these ideal locations are often far from major population centers, posing challenges for energy transmission. Offshore wind farms, for example, can harness powerful winds over the ocean, but they require extensive infrastructure to transport the electricity to consumers on land.

Currently, existing wind turbines represent a total of 0.8 TW of installed capacity. Because of intermittency, these turbines are running at an average capacity factor between 20% and 40% (ratio between the actual energy output and the installed capacity). Despite the challenge of intermittency, the abundance of wind resources offers immense potential for sustainable energy production worldwide.



wind resources at 50m above ground level for five different topographic conditions

	sheltered terrain		open plain		at a sea coast		open sea		hills and ridges	
	m.s^{-1}	W.m^{-2}	m.s^{-1}	W.m^{-2}	m.s^{-1}	W.m^{-2}	m.s^{-1}	W.m^{-2}	m.s^{-1}	W.m^{-2}
	>6.0	>250	>7.5	>500	>8.5	>700	>9.0	>800	>11.5	>1800
	5.0-6.0	150-250	6.5-7.5	300-500	7.0-8.5	400-700	8.0-9.0	600–800	10.0-11.5	1200-1800
	4.5-5.0	100-150	5.5-6.5	200-300	6.0-7.0	250-400	7.0-8.0	400–600	8.5-10.0	700-1200
	<3.5	50-100	4.5-5.5	100-200	5.0-6.0	150-250	5.5-7.0	200–400	7.0-8.5	400-700
	3.5-4.5	<50	<4.5	<100	<5.0	<150	<5.5	<200	<7.0	<400

Figure 4.8: Annual average wind power atlas for western Europe. Wind power is expressed in W/m^2 at 50 m over the ground. Powers greater than 250 W/m^2 are the most suitable for for utility scale wind power production (from Leithead (2007) Philos. Trans. R. Soc. A, 365(1853), 957-970)

4.2.2 Betz limit

Let us now investigate the conversion of wind into useful work. A first fundamental result is the Betz limit which states that no wind turbine can capture more than $\frac{16}{27} \approx 59.3\%$ of the available wind energy. This limit holds irrespective of the nature of the wind power system, and results from the fact that, as wind passes through the turbine, some energy must remain in the wind to allow it to continue moving (see Figure 4.9 left). Reducing too much the wind velocity would force the flow to bypass the turbine. The derivation of the Betz limit (see exercise in small class) assumes a stationary flow of an ideal incompressible fluid, and involves the conservation equations of mass, axial momentum (in the direction of the wind) and energy (Bernoulli). This theoretical limit highlights the inherent limitations in wind energy extraction, and modern wind turbines can reach around 70-80% of the Betz limit.

Note that the nature of the Betz limit is rather different from that of other limits discussed in this class. It is not a fundamental limit like the increase of entropy behind the Carnot efficiency, since the wind velocity in the wake of the wind power system is not lost can be further captured by additional turbines. And it is not related to a particular system of material like the Shockley-Queisser limit for photovoltaics. It applies to any type of harvesting device that tries to capture the wind kinetic power in open air.

The Betz limit can be refined to account for the specificity of the way wind energy is captured. In particular, for horizontal axis wind turbines, wind energy is captured by rotating blades taking advantage of the lift force due to the wind flow around the blades. Conservation of momentum implies that not only the wind velocity is reduced after the blade, but it is deflected so that the wind flow in the wake of the turbine exhibits a rotation (Figure 4.9 right). To address the effect of this rotation, the derivation of the Betz limit can be extended by accounting for the conservation of angular momentum (see supplementary exercise in chapter 6).

As a result, this extension of the Betz limit is found to depend on the ratio $\lambda = \frac{\Omega R}{v_{-\infty}}$

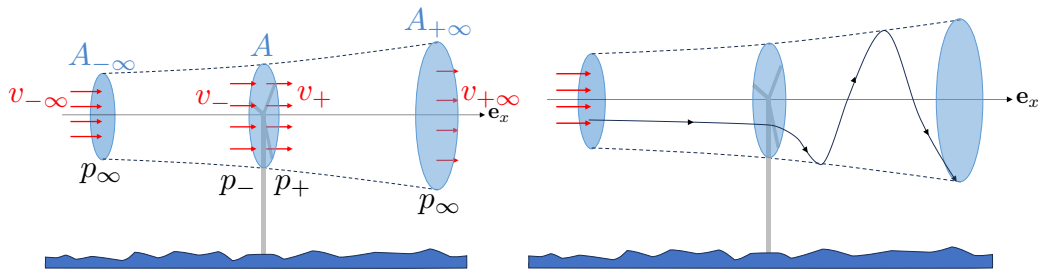


Figure 4.9: (left) As a wind power system captures the kinetic power of the wind, the wind speed is smaller in the wake of the system than upstream of the system. This gives rise to the Betz limit. (right) A refined formulation of the Betz limit takes into account the wake rotation: for a rotating wind power system, the wind acquires an angular momentum as it interacts with the turbine. The associated rotational kinetic energy reduces the power that can be harvested by the wind turbine.

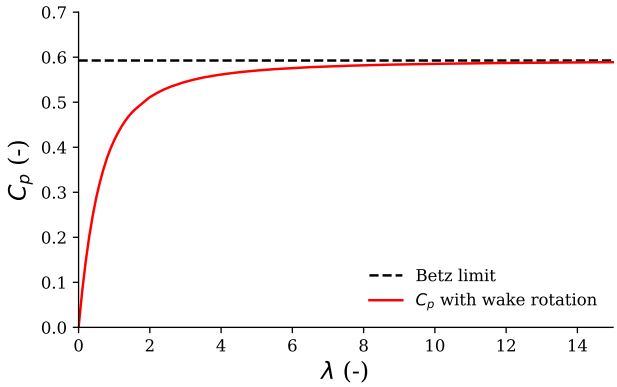


Figure 4.10: Maximum power coefficient C_p as given by the Betz limit (dashed line) and by the refined Betz limit taking into account the wake rotation (plain line).

between the rotation speed at the tip of a blade (ΩR) and the far field velocity of the incoming wind ($v_{-\infty}$). At large λ (fast rotation), the wake rotation is small and one recovers the general Betz limit; but as λ decreases the maximum efficiency decreases and falls to zero when the rotation stops. This effect is due to the fact that, for a given power, a higher torque is needed for a slow rotation than for a fast rotation. More torque means higher lift forces, hence a larger deflection of the air flow and a larger angular momentum in the wake of the turbine. So a slowly rotating turbine looses more energy in the wake rotation than a rapidly rotating turbine.

4.2.3 Horizontal axis wind turbines

Let us now turn to the design of the most common type of wind turbines, namely horizontal axis wind turbines. These wind turbines consist of blades attached to the horizontal axis at one end and rotating at an angular velocity Ω around this axis. Placing ourselves in the reference frame of a blade in the plane orthogonal to the direction of the blade and at a distance r from the axis of the wind turbine, we can schematize the cross-section of the blade as shown in Figure 4.11. Because of the rotation, the apparent air velocity in the frame attached to the

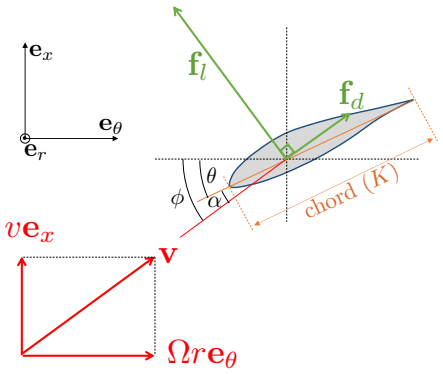


Figure 4.11: Schematic representation of the geometry of the section of a blade of an horizontal axis wind turbine.

blade is the sum of the wind velocity $\mathbf{e}_x$ and of the rotation velocity $\Omega r\mathbf{e}_\theta$ at distance r from the axis. The direction of this apparent speed forms an angle ϕ with the direction of rotation $\mathbf{e}_\theta$.

The wind turbine is designed to capture the energy of the wind with the lift force exerted on the blades. Therefore, as we have seen before, in order to maximize the lift while avoiding drag as much as possible, the blade should be oriented so that the angle of attack is $\alpha \approx 5° - 10°$. Yet, the direction of the apparent velocity varies significantly along the blade since the rotational component Ωr goes from 0 at near the axis to ΩR at the end of the blade (often much larger than the wind velocity). Therefore, to maintain an optimal lift force all along the axis, the orientation of the blade is varied with the distance to the axis, according to (see exercise in small class):

$$\phi = \tan^{-1}\left(\frac{2}{3}\frac{R}{\lambda r}\right) \tag{4.9}$$

It is important to note that the above expression of the orientation of the blade involves the tip-speed ratio $\lambda = \frac{\Omega R}{v_{-\infty}}$. Accordingly, the geometric optimization of blade orientation is only valid for a specific tip-speed ratio. If the tip-speed ratio were to deviate too much from the value for which the blade is optimized, the angle of attack would no longer be optimal, at least over part of the blade, resulting in significant drag and a net loss of efficiency. Thus, the actual efficiency of a wind turbine is optimal only a specific portion of tip-speed ratio, as shown in Figure 4.12.

Beyond the optimization of the orientation of the Blade, one can also optimize the chord length K and the number of blades N to reach the maximum efficiency (see exercise in small class). To do so one should ensure that the following relation is true all along the blade:

$$\frac{K}{R} = \frac{8\pi}{3c_l N\lambda}\sin(\phi) \tag{4.10}$$

where c_l is the lift coefficient. Since the optimal orientation ϕ varies all along the blade, so does the optimal chord length (see Figure 4.13). Provided the two geometric conditions (4.9) and (4.10) are satisfied, the coefficient of performance of the wind turbine reaches the Betz limit (see exercise in small class). Yet, close to the axis, the condition (4.10) for the optimal chord length cannot be satisfied since the rotation velocity Ωr is small (so ϕ is close to $\pi/2$) requiring an optimal chord length larger than the distance between successive blades.

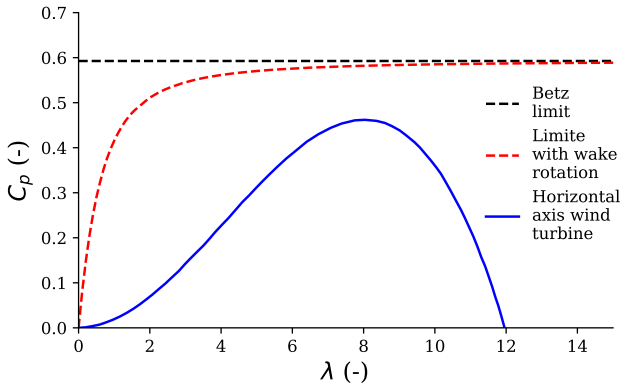


Figure 4.12: Comparison between the maximum wind power efficiency as giver by the Betz limit (dashed curves) and the actual power coefficient of and horizontal axis wind turbine optimized for $\lambda = 8$ (plain line).

Regarding the number of blades N, one can see from equation (4.10) that it is inversely proportional to the tip-seep ratio, all other things being equal. This is a rather intuitive results: as long as successive blades do not interfere, the torque Q is directly proportional to the number of blades. So for a given power ($\mathcal{P} = Q\Omega$), the angular velocity is inversely proportional to the number of blade (and thus so is the tip-speed ratio $\lambda = \Omega R/v_{-\infty}$). In most case, $N = 3$ blades are used, because one is more interested in the power output than in the torque, and the smallest number of blades is preferred to minimize costs. $N = 1$ is impractical as it would correspond to a high eccentricity, and $N = 2$ is also avoided because the moment of inertia of the blades about any axis in the plane of the turbine would oscillate during the rotation. In both case this would cause unwanted effort on the axis. $N = 3$ has none of these defects, and is thus the most optimal choice when considering power output and cost. Yet, in practice, for some applications, torque is more important than power, and therefore a higher number of blades is preferred. This is the case for instance for the well-known American farm windmill used to pump water: for this application, one wants to pump water frequently and not particularly large amounts, so one wants enough torque to activate the pumping even for low winds whereas power is unimportant.

Obviously, an horizontal axis wind turbine is designed to capture the energy of a wind orthogonal to the plane of the turbine. In order to control the orientation of the turbine and adapt to changes in the wind direction, motors are installed enabling active yaw control. Also, to avoid damage due to excessive wind and turbine rotation speeds, the power output of a wind turbine is deliberately caped when reaching a maximum. This can be done by actively adjusting the orientation of the blades (pitch-regulated) to moderate the lift and increase the drag. This can also be done passively by using blades shaped to stall progressively under excessive wind speeds.

Modern wind turbines can reach efficiencies of about 50%, not far from the Betz limit (59%). As we have seen, some source of inefficiency are due to the wake rotation, and the shape of the blade that cannot be optimal near the axis. Another major source of inefficiency are the blade tip vorticies: at the end of the blade vorticies form because of the difference of pressures between the two sides of the blade that induces a turbulent air flow. On a larger scale, a wind turbine disturbs the wind in its surrounding and thus there is a wind 'shadow' in the wake of a turbine that limits the efficiency of other nearby turbines if these are too close. As a rule of thumb, the wind turbines in a wind farm should be spaced by $\sim 10R$, i.e., the larger the turbines, the longer the distance between two turbines. Interestingly, since each turbine captures an area of wind of πR^2, the total area captured by a wind farm is independent of R. Thus, the power density of a wind farm is about $2-5$ W/m^2 irrespective of R. Based on this result, it is somehow surprising to see that wind

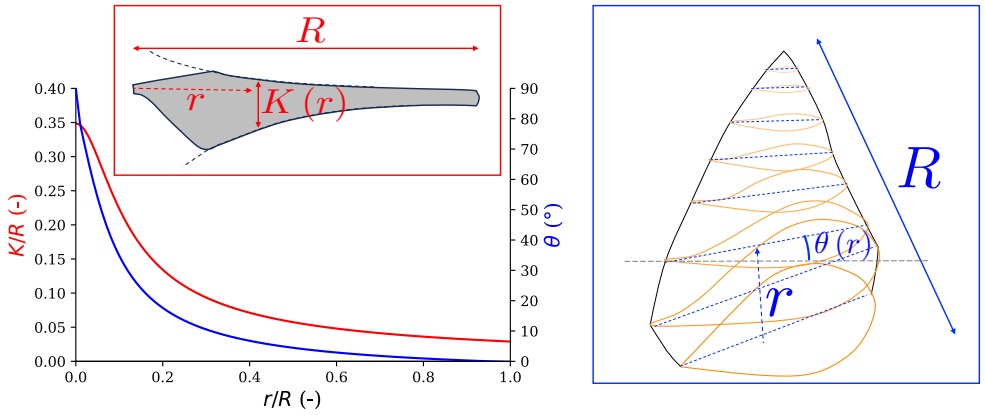


Figure 4.13: Optimal chord length (K), in red, and twist (θ), in blue, of a blade as a function of the distance r from the axis of the turbine.

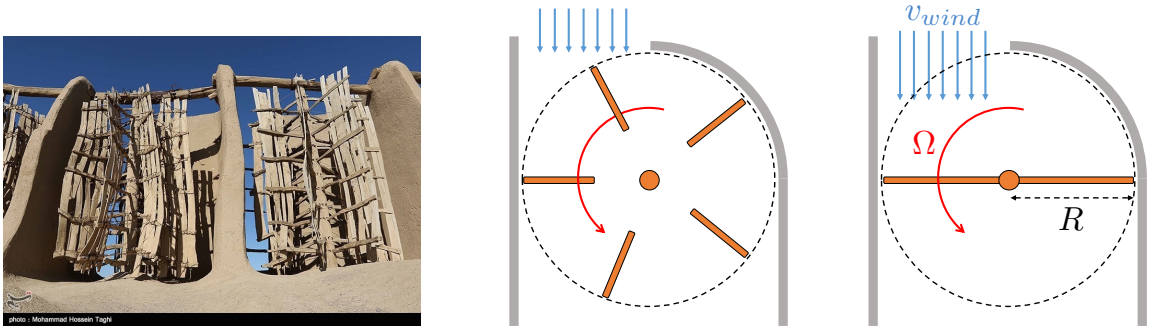


Figure 4.14: A panemone is an ancestral vertical axis wind turbine using the drag force instead of the lift force (photography from wikimedia commons, CC BY 4.0). The working principle is schematically represented (center): the blades in rotation around the vertical axis are facing the wind when rotating downstream, but are protected from the wind when rotating upstream. A simplified version of this working principle is proposed (right).

turbines are becoming bigger and bigger. The reason is that, at the second order, the size of the wind turbines does improve the power output because the winds are stronger and more stable at higher altitudes. Also, using fewer bigger turbines turns out to be more interesting regarding the material efficiency (i.e., less materials are needed to build the wind farm).

4.2.4 Alternative designs

In addition to the conventional horizontal-axis wind turbines, there are alternative designs that offer different advantages, particularly for specific conditions or environments.

The panemone, for instance, is an ancestral vertical axis wind turbine that uses the drag force instead of the lift force (Fig. 4.14): the blades in rotation around the vertical axis are facing the wind when rotating downstream, but are protected from the wind when rotating upstream. The efficiency of a panemone is rather poor, about 10%, in comparison with the Betz limit (59%) but these systems were operational 3 000 years ago, at a time when the only other way to provide useful mechanical work was was through human or animal power.

Nowadays, two well-known alternatives of the horizontal-axis wind turbines are the Savonius and Darrieus wind turbines, both examples of vertical-axis wind turbines:

- The Savonius turbine is a kind of modern panemone that uses both drag and lift. It is typically

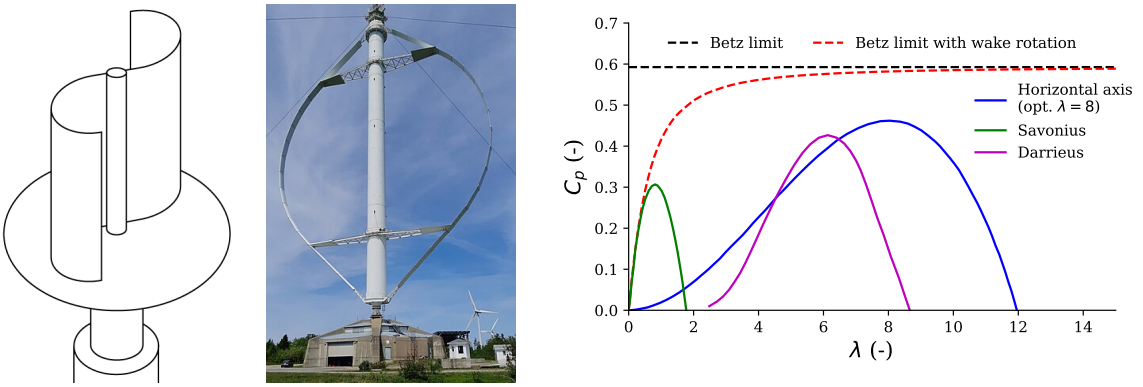


Figure 4.15: Alternative designs of wind turbines. (left) Inspired by the panemone, a Savonius wind turbine exploits both the drag and the lift forces and is well-adapted to low wind speed. (center) Darrieus wind turbine of the Éole project in Québec (from wikimedia commons, CC BY-SA 4.0). (right) Power coefficient of different types of wind turbines, compared to the Betz limit.

composed of two or more curved blades that resemble an "S" shape when viewed from above (Figure 4.15 left). It works by harnessing the drag forces when the "S" is orthogonal to the wind, and the lift forces when the "S" is parallel to the wind, making it highly efficient at low wind speeds (Figure 4.15 right) and offering good self-starting capabilities. However, since its optimal regime is at low angular velocity, it tends to be less efficient in terms of energy conversion compared to lift-based turbines with much larger angular velocities. It is thus better suited for smaller-scale applications, such as rooftop installations or in urban environments.

- The Darrieus turbine, in contrast, is lift-based and features curved, vertical blades connected to a central rotor (Figure 4.15 center). It relies on aerodynamic lift to generate rotational motion, and it can achieve higher efficiency than the Savonius design (Figure 4.15 right). However, Darrieus turbines typically require external power or complex mechanisms to start, as they don't self-start easily. Their main advantage is that they can capture wind from any direction, making them ideal for locations with variable wind patterns.

Vertical-axis wind turbines are generally better suited for locations with turbulent or lower-speed winds where conventional horizontal turbines might not perform optimally.

4.3 Water-based energy

In this section we present other forms of energy based on fluid mechanics, but this time the fluid is water rather than wind.

4.3.1 Hydropower

The most important form of water-based energy is hydropower. Hydropower is the most established form of renewable electric power generation, producing an average of 430 GW globally. This accounts for approximately one-sixth of the world's electricity. Despite its strong presence, the global potential for hydropower is estimated to be about 1 600 GW, a relatively modest figure when compared to the massive potential of wind power (up to 100 000 GW). And considering the global energy demand (20 000 GW), the potential of hydropower appears limited.

One of hydropower's key advantages is its energy density. It is about 1 000 times more concentrated than wind energy, making it easier to harness efficiently. Therefore, hydropower has a long history, having been exploited for over 2 000 years, making it one of the oldest forms of energy generation.

Figure 4.16: (left) Watermill of Braine-le-Château in Belgium as an example of ancestral hydropower system from the 12$^{\text{th}}$ century (from wikimedia commons, CC BY 2.0). (center) Hoover dam in Arizona as an example of reservoir dam (from wikimedia commons, CC BY-SA 3.0). (right) Chief Joseph dam on the Columbia River in Washington state as an example of run-of-river plant (from wikimedia commons, public domain).

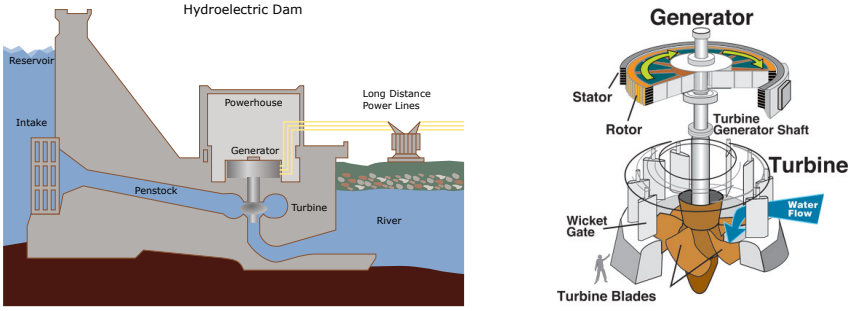


Figure 4.17: (left) Schematic diagram of hydroelectric power plant (from wikimedia commons, CC BY-SA 3.0). (right) Schematic diagram of a modern hydro turbine. (from top-alternative-energy-sources.com).

There are two main types of hydropower plants: reservoir dams and run-of-river plants (Fig. 4.16). Reservoir dams are particularly versatile as they can store water, allowing them to produce electricity on demand. Run-of-river plants, in contrast, do not store water but generate electricity continuously based on the river flow. As a renewable energy source, hydropower is highly flexible. It offers the ability to respond quickly to electricity demand, and by pumping water back in the reservoir it can also be used to store large amounts of energy. Reservoir dams are the primary energy storage systems in power grids and play a crucial role in balancing electricity surpluses and shortages.

The working principle of a hydropower dam involves converting the potential energy of stored water ($\rho g z$) into mechanical energy, which is then transformed into electrical energy. A schematic representation is shown in Figure 4.17. Water is accumulated in a reservoir behind the dam, creating a height difference (head) between the water level and the turbine below. When electricity is needed, water is released from the reservoir through intake gates and directed down through penstocks (large pipes) to the turbines. The flowing water spins the turbines, which are connected to generators that convert the mechanical energy of the spinning turbines into electrical energy. After passing through the turbines, the water is discharged back into the river downstream. The process is extremely efficient reaching almost 100% efficiency. It is important to note that the Betz limit does not apply to the turbine in the dam. Indeed, in contrast with a wind turbine, the flow of the fluid is constrained to go through the turbine and no by-pass is possible.

4.3.2 Waves

Wave energy has a significant potential as a renewable energy source but remains almost unexploited so far. When wave forms, the peak and trough of the waves store a significant potential energy,

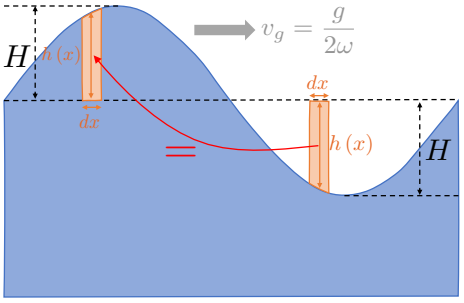


Figure 4.18: The potential energy associated with a wave can be estimated by imagining lifting water in an interval dx from the trough to the corresponding point in the peak. And this energy is transported at the group velocity of the wave.

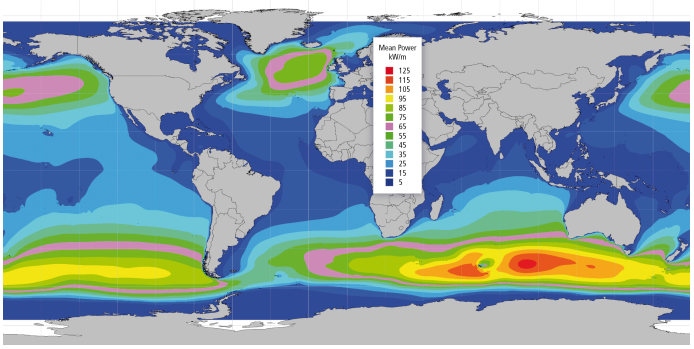


Figure 4.19: Global offshore annual wave power level distribution (from Lewis et al. (2011) Ocean Energy. In IPCC Special Report on Renewable Energy Sources and Climate Change Mitigation, Cambridge University Press).

while the movement of the wave brings kinetic energy (Fig. 4.18). The energy density of a traveling surface wave is $\frac{E}{A} = \frac{1}{2}\rho g h^2$, with h the wave amplitude (height). And the traveling velocity is given by $v_g = \frac{g}{2\omega}$. Therefore the linear power density brought with a traveling wave is:

$$\frac{\mathcal{P}}{L} = \frac{E}{A} v_g = \frac{\rho g^2 h^2}{4\omega}$$

For instance, a typical ocean wave with a height of 1 meter and a period ($T = 2\pi/\omega$) of 10 seconds can generate about 40 kW per meter of wavefront. This is a substantial power density compared to other renewable energy sources, such as solar radiation, which averages around 0.15 kW per square meter (at the ground surface). This high concentration of energy in wave fronts offers a compelling opportunity to harness renewable energy for coastal regions.

However, the practical utilization of wave energy faces several challenges. One of the main issues is that wave energy is dispersed across vast areas of the oceans, far from consumers and population centers (Fig. 4.19). Limiting ourselves to coastlines, only about 2 to 3 terawatts (TW) of wave energy are available, which is far below the global energy demand of approximately 20 TW. Another significant challenge is the highly variable nature of wave energy, which fluctuates based on weather conditions and ocean behavior. This variability requires highly durable and robust devices to effectively capture and convert wave energy into usable electricity. The marine environment is harsh, so these devices must withstand rough conditions over long periods.

Currently, no large-scale commercial production of wave energy exists. The technology remains in the experimental and pilot stages, with various designs being explored to determine the most efficient and cost-effective methods of harnessing wave energy (Fig. 4.20). Some of the most efficient designs tested in pilot projects have achieved energy conversion efficiencies of up to 90%, showing promise for future development. However, the widespread deployment of wave energy systems will

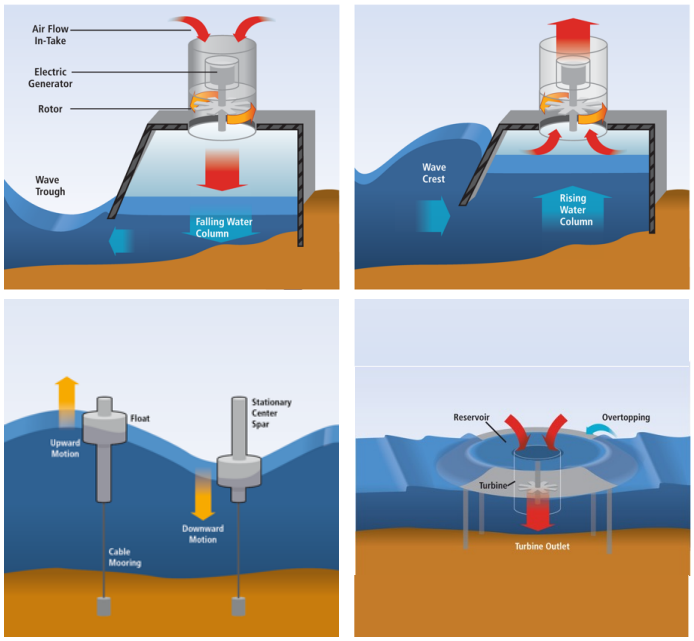


Figure 4.20: Wave energy converters and their operation: (top, left and right) oscillating water column device; (bottom left) oscillating body device; and (bottom right) overtopping device. Designs are by the National Renewable Energy Laboratory (from Lewis et al. (2011) Ocean Energy. In IPCC Special Report on Renewable Energy Sources and Climate Change Mitigation, Cambridge University Press).

depend on further technological advancements, cost reductions, and overcoming the challenges of operating in offshore environments.

4.3.3 Tides and currents

Tidal energy results from the gravitational interaction between the Earth and the Moon, which causes tidal movements. The global tidal power can be estimated by observing the increase in the Earth-Moon distance with time, which is approximately 3.8 cm per year. This increase leads to a transfer of angular momentum from the Earth to the Moon, resulting in a decrease in Earth's rotational energy. The reduction in Earth's rotational energy is around 10^{20} J/yr, which corresponds to a power dissipation of about 3 000 GW. However, this total energy is not fully exploitable for power generation.

Most tidal energy is dissipated through friction at the seafloor or is too diffuse along coastlines to be harnessed efficiently (Fig. 4.21). Tidal energy can only be effectively used in locations where large tidal flows pass through narrow channels, such as estuaries or bays. The global potential for usable tidal energy is estimated at around 80 GW, which is small in comparison to the global energy demand of 20 000 GW. Therefore, while tidal energy is a predictable renewable source, its potential is limited.

There are two primary methods to harness tidal energy: tidal barrages and tidal stream turbines. Tidal barrages function similarly to hydropower dams, where water flows into and out of a large reservoir, passing through turbines to generate electricity (Fig. 4.22). Tidal stream turbines, on the other hand, are similar to underwater wind turbines, capturing the kinetic energy of tidal currents (Fig. 4.23). Both methods are promising but are limited to specific geographic locations where tidal streams are concentrated.

Quite similar to tidal energy is marine current energy. Most marine currents are slow, typically less than 1 m/s, and deep, making them difficult to harness. However, shallow currents near coastlines, faster than 2-3 m/s, could provide opportunities for energy generation. One prominent example is the Gulf Stream, which represents a steady power output of around 250 GW. Out of

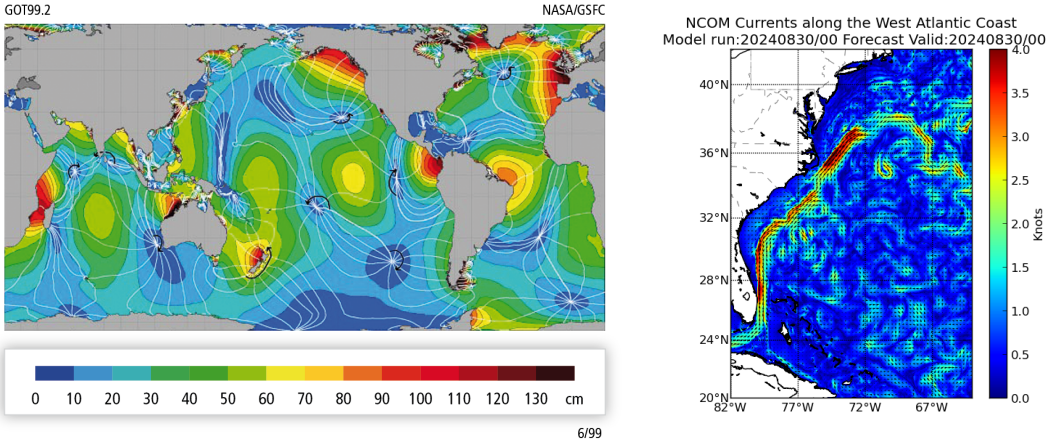


Figure 4.21: (left) World map of M2 tidal amplitude (from Lewis et al. (2011) Ocean Energy. In IPCC Special Report on Renewable Energy Sources and Climate Change Mitigation, Cambridge University Press). M2 is the largest (semidiurnal) tidal constituent, whose amplitude is about 60% of the total tidal range. (right) Currents along the West Atlantic Coast (from National Oceanic and Atmospheric Administration, High Resolution Oceanic Model, https://ocean.weather.gov).

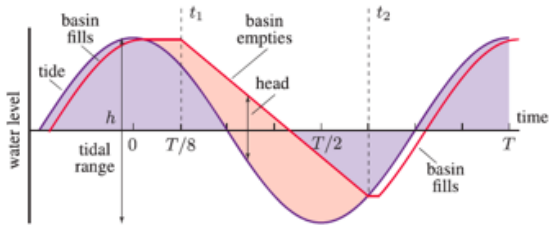


Figure 4.22: Working principle of a tidal barrage. The water level in the tidal barrage (red) is above the water level of the tide (purple) over half of the tidal period during which power is generated (from Jaffe & Taylor (2018) The Physics of Energy. Cambridge University Press).

Figure 4.23: (left) Schematic representation of a horizontal axis twin tidal turbine (from Lewis et al. (2011) Ocean Energy. In IPCC Special Report on Renewable Energy Sources and Climate Change Mitigation, Cambridge University Press). (right) Deployment of an Atlantis AR1000 tidal turbine (from Atlantis Resources Corporation).

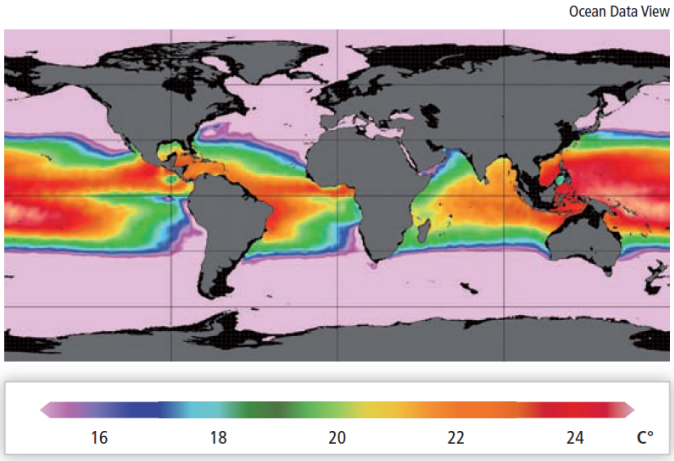


Figure 4.24: Worldwide average ocean temperature differences (°C) between 20 and 1 000 m water depth (from Lewis et al. (2011) Ocean Energy. In IPCC Special Report on Renewable Energy Sources and Climate Change Mitigation, Cambridge University Press).

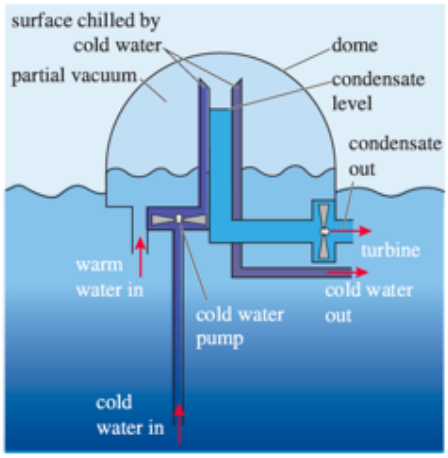


Figure 4.25: Open-cycle OTEC system (from Jaffe & Taylor (2018) The Physics of Energy. Cambridge University Press).

this, only 10 to 20 GW could potentially be harnessed for electricity production. However, like tidal energy, the global potential of marine current energy is quite limited in comparison to the global energy demand.

4.3.4 Ocean thermal energy

Ocean Thermal Energy Conversion (OTEC) is a process that generates electricity by utilizing the temperature difference between the warm surface waters of the ocean and the cold deep waters. This temperature gradient is most pronounced in tropical regions, where the surface water can be significantly warmer than the deep water (Fig. 4.24). OTEC systems use this temperature difference to run a heat engine producing work (see for instance the concept of Fig. 4.25).

The potential of OTEC as a renewable energy source is very large. It represents a much larger energy resource than all other forms of ocean energy combined, with resource estimates reaching thousands of terawatts (TW) of potential power, out of which 5 TW of useful work could be produced. This is far greater than the potential of wave, tidal, or marine current energy. However, despite this vast resource, OTEC's practical potential is constrained by its low efficiency. To

function effectively, an OTEC system requires a minimum temperature difference of around 20°C between surface and deep water, corresponding to a maximum theoretical Carnot efficiency of the heat engine of 6.6%. In practice, real-world OTEC pilots only achieve efficiencies of 2-3%.

Recap of what must be known for the exam

1. Basics of fluid mechanics

At the continuum scale in the Eulerian description, matter at position $\mathbf{x}$ and time t is described by its local intensive state parameters: *mass density* $\rho(\mathbf{x},t)$ and *internal energy density* $\rho(\mathbf{x},t)\,u(\mathbf{x},t)$. According to thermodynamics, these state parameters are conjugated to the *stress tensor* $\underline{\underline{\sigma}}(\mathbf{x},t)$ and *temperature* $T(\mathbf{x},t)$, respectively.

The macroscopic motion of matter is described by the *velocity* $\mathbf{v}(\mathbf{x},t)$. Therefore, the macroscopic *kinetic energy density* is given by $\rho(\mathbf{x},t)\,e_{kin}(\mathbf{x},t)=\frac{1}{2}\rho(\mathbf{x},t)\,|\mathbf{v}(\mathbf{x},t)|^2$.

The *kinetic energy flux* is given by $\mathbf{J}_{e_{kin}}(\mathbf{x},t)=\rho(\mathbf{x},t)\,e_{kin}(\mathbf{x},t)\,\mathbf{v}(\mathbf{x},t)$. The magnitude of this flux $\mathcal{P}=|\mathbf{J}_{e_{kin}}(\mathbf{x},t)|=\frac{1}{2}\rho(\mathbf{x},t)\,|\mathbf{v}(\mathbf{x},t)|^3$ represents the *power density* of the flow of matter that one tries to recover with a wind turbine. The power density of a 30 km/h wind is $\mathcal{P}=1\,740$ W/m^2 is approximately equivalent to that of a 5 km/h water stream, and this amount is of similar magnitude as the solar radiation before entering the atmosphere (1 380 W/m^2).

The time evolution of a continuum system is governed by conservation laws. The conservation of a quantity $A(\mathbf{x},t)$ can be written both in an integral form, and in a local form :

$$\int_\Omega \frac{\partial A}{\partial t}dV+\int_{\partial\Omega}\mathbf{J}_A.\mathbf{n}dA=\int_{\partial\Omega}\mathbf{q}_A.\mathbf{n}dA+\int_\Omega S_A dV$$

$$\frac{\partial A}{\partial t}+\nabla.\mathbf{J}_A=\nabla.\mathbf{q}_A+S_A$$

where $\mathbf{J}_A=A(\mathbf{x},t)\,\mathbf{v}$ is the convective flux of A, $\mathbf{q}_A(\mathbf{x},t)$ is the non-convective flux of A, and $S_A(\mathbf{x},t)$ is the external volumetric source of A.

In fluid mechanics, the time evolution of a continuum system is governed by three conservation laws, presented hereafter in their local form:

Conservation of mass also called the *continuity equation*:

$$\frac{\partial\rho}{\partial t}+\nabla.\mathbf{J}_\rho=0 \text{ with } \mathbf{J}_\rho(\mathbf{x},t)=\rho(\mathbf{x},t)\,\mathbf{v}(\mathbf{x},t) \text{ the mass flux}$$

Conservation of linear momentum also called the *Navier-Stokes equation*:

$$\frac{\partial\mathbf{v}}{\partial t}+\mathbf{v}.\nabla\mathbf{v}=\frac{\nabla.\underline{\underline{\sigma}}}{\rho}+\mathbf{g}$$

where $\mathbf{g}$ is the acceleration due to gravity.

Conservation of energy also called the *energy balance*:

$$\frac{\partial(\rho e)}{\partial t}+\nabla.(\mathbf{J}_e)=\nabla.\left(\underline{\underline{\sigma}}.\mathbf{v}\right)+\rho\mathbf{g}.\mathbf{v}+\nabla.\mathbf{q} \text{ with } \mathbf{J}_e(\mathbf{x},t)=\rho(\mathbf{x},t)\,e(\mathbf{x},t)\,\mathbf{v}(\mathbf{x},t) \text{ the energy flux}$$

where $\mathbf{q}$ is the heat flux.

There is also a conservation of the angular momentum, which requires the stress tensor $\underline{\underline{\sigma}}$ to be symmetrical. The three conservation laws correspond to a total of 5 equations (linear momentum is vectorial), and there are five unknowns $(\rho,T,\mathbf{v})$. To fully set the problem, one

must provide a behavior law relating the internal energy u and stress tensor $\underline{\underline{\sigma}}$ to the unknowns (e.g., ideal gas law) and a heat flux law relating $\mathbf{q}$ to the unknowns (e.g., Fourier's law).

For air and water flows, one can consider a few simplifying assumptions: the fluid is *incompressible*, its evolution is *adiabatic*, and its behavior is *Newtonian* (in the spherical and deviatoric decomposition of the stress tensor $\underline{\underline{\sigma}} = -p\underline{\underline{I}} + \underline{\underline{\tau}}$, the deviatoric part is given by $\underline{\underline{\tau}} = \mu\left(\nabla\mathbf{v} + \nabla\mathbf{v}^T\right)$ with μ the dynamic viscosity). Under these simplifying assumptions, the conservation laws become:

$$\begin{cases} \nabla.\mathbf{v} = 0 & \text{continuity equation} \\ \frac{\partial \mathbf{v}}{\partial t} + \mathbf{v}.\nabla\mathbf{v} = -\frac{\nabla p}{\rho_0} + \frac{\mu}{\rho_0}\Delta\mathbf{v} + \mathbf{g} & \text{Navier-Stokes equation} \\ \frac{\partial e}{\partial t} + \mathbf{v}.\nabla e = -\frac{\mathbf{v}.\nabla p}{\rho_0} + \frac{\nabla.\left(\underline{\underline{\tau}}.\mathbf{v}\right)}{\rho_0} + \mathbf{g}.\mathbf{v} & \text{energy balance} \end{cases}$$

Additionally, sometimes we can assume a stationary evolution ($\frac{\partial X}{\partial t} = 0$), and an ideal fluid (non-viscous fluid, $\mu = 0$).

Another formulation of the conservation of energy is given by *Bernoulli's principle*. For a stationary flow of an incompressible ideal fluid,

$$\text{the quantity } \frac{\mathbf{v}^2}{2} + \frac{p}{\rho_0} + gz \text{ is conserved along the flow}$$

Thus, the pressure of a fluid drops as its velocity (or its altitude) increases. The quantity $p_{dyn} = \frac{1}{2}\rho_0\mathbf{v}^2$ is called the *dynamical pressure* and represents the maximum increase of pressure as the fluid velocity is reduced to zero.

2. Working principle of an airfoil

For a closed curve $\mathcal{C}$, the line integral $\Gamma(\mathcal{C}) = \oint_{\mathcal{C}} \mathbf{v}.d\mathbf{x}$ is called the *circulation*. Applying the Stokes theorem, the circulation can be expressed as the flow of $\boldsymbol{\omega} = \nabla \times \mathbf{v}$, called *vorticity*, through the section delimited by $\mathcal{C}$: $\Gamma(\mathcal{C}) = \int_S \boldsymbol{\omega}.d\mathbf{S}$. Circulation and vorticity characterize the global and local rotation of a fluid. A fluid with no vorticity is said irrotational. For an ideal fluid, the vorticity of a packet of fluid is constant as the packet moves through the flow (Kelvin's circulation theorem). In particular, an ideal fluid that is irrotational will remain irrotational.

The example of a ball moving through the air (Figure 4.2) shows that a lift force appears when the ball is rotating, because, by application of Bernouilli's principle, the circulation of the air velocity around the ball favors lower air pressures on one side and higher air pressures on the other side. The *Kutta-Zhukovskii theorem* provides a formal demonstration of the relationship between lift and circulation: a stationary flow of an incompressible ideal fluid around an object exerts no drag and a lift proportional to the circulation of the fluid around the object:

$$\begin{cases} f_{\text{lift}} = \rho_0 v_\infty \Gamma(\mathcal{C}) \\ f_{\text{drag}} = 0 \end{cases}$$

where v_∞ is the free-stream velocity of the fluid, and $\mathcal{C}$ is any closed curve around the object oriented counterclockwise. The fact that there is no drag is known as d'*Alembert's paradox*. The absence of drag originates from the absence of viscosity of the fluid. Real fluids have non-zero viscosity and there is always a non-zero drag force.

An airfoil is designed to maximize lift while avoiding drag. The shape of the airfoil (in particular the sharp trailing edge) and its *angle of attack* α, induce a non zero circulation of the air around it which causes the lift. Yet, α must be kept small otherwise a region of turbulence develops behind the airfoil and the lift force drops.

By dimensional analysis, one can show that, for a stationary flow around around an object, the lift and drag force are of the form:

$$\begin{cases} f_{drag} = c_d(\alpha) \frac{\rho_0 v_\infty^2}{2} K \\ f_{\text{lift}} = c_l(\alpha) \frac{\rho_0 v_\infty^2}{2} K \end{cases}$$

where v_∞ is the free flow velocity, K is the chord length of the airfoil, and $c_d(\alpha)$ and $c_l(\alpha)$ are dimensionless coefficients called the *drag and lift coefficients*. For a flow a fixed Reynolds number and a object of fixed reduced shape (i.e., same shape upon homothetic expansion or contraction), the drag and lift coefficients depend only on the angle of attack α. Their precise estimation requires computational fluid mechanics or wind tunnel experimental tests. An example of airfoil used for wind energy is given in Fig. 4.6. Optimal lift versus drag force ratio is achieved for angles of attack around 5°-10°.

3. Application to energy

A wind turbine (or a hydrofoil) capturing the kinetic energy from a free flow of air (or water) cannot capture all the kinetic power of the flow. Indeed, as the fluid passes through the turbine, it must keep some of its velocity to continue moving otherwise all the flow would bypass the turbine. The maximum efficiency, known as the *Betz limit*, is $\frac{16}{27} \approx 59.3\%$. It is derived from the conservation of mass, energy and axial momentum applied to the fluid flow passing through the turbine, and it holds irrespective of the nature of the turbine. In the particular case of horizontal axis wind turbines, the Betz limit can be further refined by taking into account the rotation of the fluid in the wake of the turbine (Fig. 4.10).

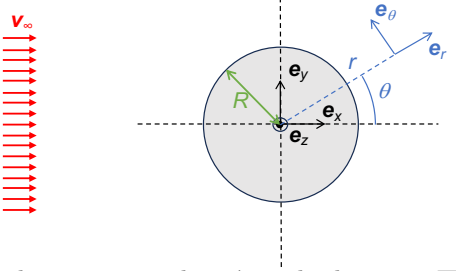
A horizontal axis wind turbine is based on airfoils that are designed to convert a maximum of the fluid kinetic energy into rotational power. As the blades are rotating, the apparent velocity differs from the center of the turbine to the tips of the blades. The orientation and chord length are varying along the blades to reach optimum efficiency for a given *tip-speed ratio* (rotation speed at the tip of a blade divided by the far field velocity of the incoming fluid). Also, one usually considers 3 blades, which is the minimum number of blades with no inertia effects. But sometimes higher number of blades is used, for instance to favor torque over power at low rotation velocities.

Hydropower is not limited by the Betz limit and can reach ideal efficiency, because the flow of the fluid is constrained to go through the turbine and no by-pass is possible. Hydropower converts the potential energy of stored water ($\rho g z$) into mechanical energy.

Exercises

Ex. 4.1 - Effet Magnus

On s'intéresse à l'écoulement stationnaire d'un fluide parfait incompressible autour d'un cylindre. Le cylindre, d'axe (Oz) et de rayon R est supposé de longueur infinie. Loin du cylindre, le fluide a une vitesse uniforme $\mathbf{v} = v_\infty \mathbf{e}_x$ orthogonale à l'axe du cylindre.



On rappelle l'expression des opérateurs gradient et laplacien en coordonnées cylindriques : $\nabla\phi = \left.\frac{\partial\phi}{\partial r}\right|_{\theta,z} \mathbf{e}_r + \frac{1}{r}\left.\frac{\partial\phi}{\partial\theta}\right|_{r,z} \mathbf{e}_\theta + \left.\frac{\partial\phi}{\partial z}\right|_{r,\theta} \mathbf{e}_z$ et $\Delta\phi = \left.\frac{\partial^2\phi}{\partial r^2}\right|_{\theta,z} + \frac{1}{r}\left.\frac{\partial\phi}{\partial r}\right|_{\theta,z} + \frac{1}{r^2}\left.\frac{\partial^2\phi}{\partial\theta^2}\right|_{r,z} + \left.\frac{\partial^2\phi}{\partial z^2}\right|_{r,\theta}$.

1. On définit la vorticité comme $\omega = \nabla \times \mathbf{v}$. Montrer que l'écoulement loin du cylindre est irrotationel ($\omega = 0$).

2. D'après le théorème de Kelvin, un fluide parfait incompressible irrotationel reste irrotationel le long de son écoulement. En conséquence, l'écoulement est irrotationel partout et il est possible d'écrire le champ de vitesse comme le gradient d'un potentiel ϕ, appelé potentiel des vitesses : $\mathbf{v} = \nabla\phi$ (théorème de Poincaré).

   (a) Réécrire l'équation de continuité (conservation de la masse) en fonction de ϕ.

   (b) Cette equation est connue comme l'équation de Laplace, dont les solutions en coordonnées cylindriques (r, θ, z) autour de l'axe (Oz) symétriques autour du plan (Oxz) sont de la forme :

$$\phi(r,\theta,z) = a_0 \ln(r) + b_0 + \sum_{n=1}^{+\infty} \left(a_n r^n + b_n r^{-n}\right) \cos(n\theta)$$

   En considérant le champ de vitesse loin du cylindre ($\mathbf{v} = v_\infty \mathbf{e}_x$) ainsi que l'annulation de la vitesse radiale au contact avec le cylindre ($\mathbf{v}(r = R).\mathbf{e}_r = 0$), montrer que

$$\phi(r,\theta,z) = v_\infty \left(r + \frac{R^2}{r}\right) \cos(\theta)$$

   (c) En déduire le champ de vitesse. Tracer l'allure de ce champ.

3. Si l'on ne se limite plus à un champ de vitesse symétrique autour du plan (Oxz), la solution de l'équation de Laplace devient

$$\phi(r,\theta,z) = v_\infty \left(r + \frac{R^2}{r}\right) \cos(\theta) + R^2\Omega\theta$$

   avec Ω une constante.

   (a) Vérifier que ce potentiel est toujours bien solution du problème (vitesse uniforme loin du cylindre et vitesse radiale nulle au niveau du cylindre)

   (b) A quel mouvement du fluide le terme potentiel supplémentaire correspond-il ? Tracer l'allure du champ de vitesse supplémentaire. Comment peut-on interpréter Ω ?

4. On rappelle le théorème de Bernoulli : pour un fluide parfait incompressible en écoulement stationnaire, la quantité $\frac{\mathbf{v}^2}{2} + \frac{p}{\rho_0} + gz$ se conserve au cours de l'écoulement, avec p la pression, ρ_0 la densité, et g la pesanteur.

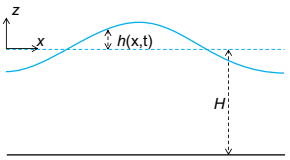
(a) En appliquant le théorème de Bernoulli, déterminer la pression qui s'exerce au contact avec le cylindre. On notera p_∞ la pression loin du cylindre.

(b) Montrer que le fluide exerce sur le cylindre une force par unité de longueur le long de z de la forme $\mathbf{f} = k\left(v_\infty \mathbf{e}_x \times \Omega \mathbf{e}_z\right)$ où l'on donnera l'expression de k.

(c) Ce phénomène est connu comme l'effet Magnus. Citer quelques exemples où cet effet se manifeste.

Ex. 4.2 - Courant des marées

Dans cet exercice, nous nous intéressons aux ondes et courants induits par les marées. On considère une mer infinie d'étendue horizontale infinie, de profondeur H et contenant de l'eau supposée être un fluide parfait et incompressible, de masse volumique ρ_0.



1. On s'intéresse à une onde de gravitation qui se propage à la surface. On admet que, pour une onde de faible amplitude, l'écoulement est irrotationel et le champ de vitesse dérive d'un potentiel : $\mathbf{v} = \nabla\phi$. Par ailleurs, l'équation de conservation du moment se réduit à $\frac{\partial \mathbf{v}}{\partial t} \approx -\frac{\nabla p}{\rho_0} - g\mathbf{e}_z$, où $p$est la pression et g la pesanteur.

   (a) Montrer que le potentiel des vitesses doit vérifier :

   $$\begin{cases} \Delta\phi = 0 \\ \frac{\partial \phi}{\partial t} + \frac{p}{\rho_0} + gz = Cte \end{cases}$$

   (b) On cherche des solutions ondulatoires de la forme $\phi(x,y,z,t) = f(z)\sin(\omega t - kx)$ (onde plane se propageant dans la direction x). En appliquant la conservation de la masse et en tenant compte de la condition au bord au fond de la mer ($v_z(z=-H)=0$) montrer que $f(z)$ est de la forme

   $$f(z) = \alpha\left(\exp(kz) + \exp\left(-k\left(z+2H\right)\right)\right)$$

   (c) Déterminer le champ de vitesse et tracer qualitativement l'allure de la trajectoire du fluide

2. Au niveau de la surface de la mer la pression est uniforme égale à p_0 (atmosphère). On note $h(x,t)$ la position z de cette surface.

   (a) En considérant $h(x,t)$ petit devant H, sa dérivée temporelle peut être approximée par la vitesse verticale en $z=0$. En déduire que la position de la surface est de la forme $h(x,t) = -h_0\cos(\omega t - kx)$, où l'on identifiera l'amplitude h_0.

   (b) En appliquant la conservation du moment à la surface de la mer, montrer que la relation de dispersion entre ω et k est :

   $$\omega^2 = kg\tanh(kH)$$

3. Une onde de marée est une onde très étendue par rapport à la profondeur de la mer (i.e., $kH \ll 1$) et de fréquence angulaire $\omega = \frac{2\pi}{T}$ avec $T \approx 12$ h.

   (a) Dans le régime $kH \ll 1$, que devient la relation de dispersion ? En déduire que le mouvement de l'eau est de vitesse quasi-horizontale et uniforme sur toute la profondeur de la mer. Exprimer l'amplitude de la vitesse en fonction de h_0, H et g

   (b) Quelle vitesse maximale est atteinte pour une marée d'amplitude $h_0 = 5$ m, et une mer de profondeur $H = 30$ m ? A quelle puissance cinétique par unité de surface cela correspond ? On rappelle la densité de l'eau liquide : $\rho_0 = 1000$ kg/m^3.

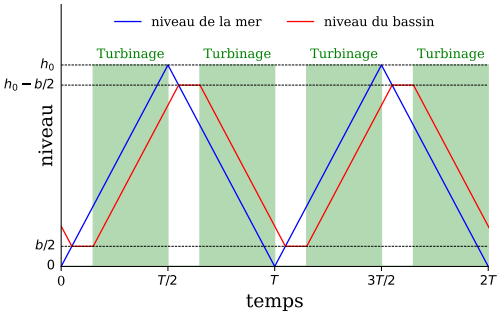
(c) En considérant une hydrolienne 'parfaite' de 10 m de diamètre atteignant la limite de Betz (rendement $C_p^{max} = \frac{16}{27}$), quel travail peut être produit lorsque la marée atteint sa vitesse maximale ?

(d) Pour produire une puissance équivalente avec une éolienne (parfaite également) soumise à un vent de 30 km/h, quel diamètre d'éolienne faudrait-il considérer ? On rappelle la densité de l'air : $\rho_{air} = 1.3$ kg/m^3. En dehors de la densité d'énergie, quel avantage y a-t-il à exploiter l'énergie des marées par rapport à l'énergie du vent ?

Ex. 4.3 - Usine marémotrice

Dans cet exercice, nous nous intéressons à l'énergie des marées. Une façon d'exploiter l'énergie des marées est d'utiliser un barrage séparant un bassin de la mer. Lors de la marée descendante, le bassin se vide et le barrage fonctionne comme un barrage hydroélectrique classique; et lors de la marée montante, le bassin se remplit et le barrage fonctionne dans le sens inverse.

Ce fonctionnement est schématisé ci-contre, où, par simplicité, la forme sinusoïdale de la hauteur de la mer a été approximée par une forme linéaire par morceau. Les turbines du barrage fonctionnent lorsque l'écart de hauteur entre le bassin et la mer atteint une valeur b (phase de turbinage). Dans les autres phases, les turbines sont à l'arrêt et le barrage ne fournit pas de puissance. A marée basse le niveau du bassin ne descend pas en dessous de $b/2$, et à marée haute il n'excède pas $h_0 - b/2$.



1. Déterminer en fonction de b, h_0, et T la durée d'une phase de turbinage.

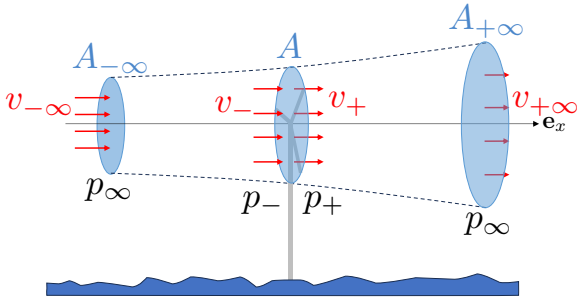
2. On considère un bassin de surface horizontale S (constante avec sur toute la hauteur), et des turbines de rendement η_t. Quelle est la puissance fournie par les turbines lors de la phase de turbinage ?

3. En déduire l'énergie totale E produite lors d'une phase de turbinage. En l'absence de barrage, quelle serait la différence d'énergie potentielle E_p dans le bassin entre la marrée haute et la marée basse ? En déduire que le rendement effectif du barrage vaut $\eta = \frac{E}{E_p} = 2\eta_t \frac{b}{h_0}\left(1 - \frac{3b}{2h_0}\right)$. Quel est le rendement effectif maximum η_{max} et la valeur optimale de b/h_0 ?

4. Le barrage de la Rance (Bretagne) a une superficie $S = 22$ km^2, avec une amplitude de marée moyenne $h_0 = 6$ m. En considérant le rapport b/h_0 optimal et un rendement des turbines de $\eta_t = 90\%$, quelle est la valeur de la puissance moyenne produite lors du turbinage ? Estimer la production totale d'énergie à l'année et comparer à la valeur réelle (500 GWh).

Ex. 4.4 - Limite de Betz

On s'intéresse à un dispositif servant à capter l'énergie cinétique d'un fluide (cf. figure ci-contre). On considère un dispositif de section A, orthogonal à l'écoulement du fluide que l'on supposera incompressible (densité ρ_0), idéal et en écoulement stationnaire dans la direction $\mathbf{e}_x$. Loin en amont du dispositif, la vitesse du fluide est $v_{-\infty}\mathbf{e}_x$ et la pression est la pression ambiante p_∞. Le dispositif captant une partie de l'énergie cinétique, la vitesse est réduite dans le sillage. Dans un premier temps, on suppose que celle-ci ne varie que dans la direction axiale et



est réduite à $v_{+\infty}\mathbf{e}_x = (v_{-\infty} - \Delta v)\,\mathbf{e}_x$ loin en aval, tout en revenant à la pression ambiante p_∞. On note $v_-\mathbf{e}_x$ et $v_+\mathbf{e}_x$ les vitesses, immédiatement en amont et en aval du dispositif. Le schéma ci-dessous représente les lignes de courant délimitant l'écoulement du fluide qui passe par le dispositif. Les section $A_{-\infty}$ et $A_{+\infty}$ représentent les sections occupées par le fluide traversant le dispositif loin en amont et loin en aval.

1. Par application de la conservation de la masse, quelles relations peut-on établir entre les différentes sections et vitesses.

2. Par application de la conservation du moment, déterminer la force $\mathbf{F} = F\mathbf{e}_x$ dans la direction axiale appliquée sur le dispositif en fonction de $A_{\pm\infty}$, $v_{\pm\infty}$ et ρ_0. Factoriser cette expression en exploitent les relations de conservation de la masse de la question précédente.

3. La conservation de l'énergie pour un fluide idéal incompressible impose la conservation de $\frac{\mathbf{v}^2}{2} + \frac{p}{\rho_0} + gz$ (théorème de Bernoulli), avec g la constante de gravitation et z l'altitude. En déduire, la différence de pression $p_- - p_+$ entre les sections immédiatement en amont et en aval du dispositif. Puis déterminer une seconde expression de la force axiale $\mathbf{F} = F\mathbf{e}_x$ appliquée sur le dispositif.

4. En comparant les deux expressions de la force axiale sur le dispositif, en déduire une relation entre les différentes vitesses de fluide du problème.

5. Déterminer la puissance $\mathcal{P}$ fournie au dispositif, et l'exprimer en fonction du facteur d'induction axial $a = \frac{v_{-\infty} - v}{v_{-\infty}}$ et de la vitesse $v_{-\infty}$.

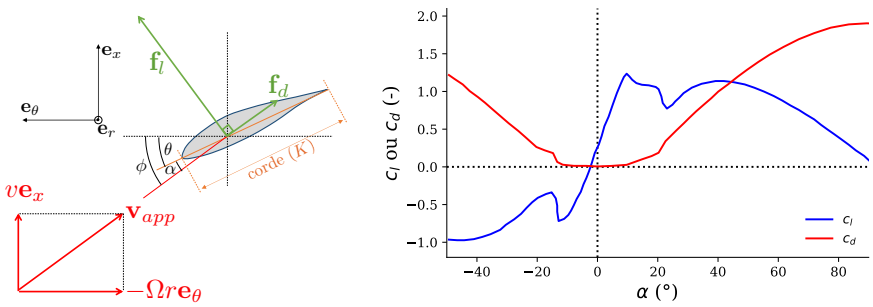
6. Le coefficient de performance C_p du dispositif est défini comme le ratio entre la puissance fournie et la puissance cinétique totale $\mathcal{P}_{-\infty} = \frac{1}{2}\rho_0 A\,(v_{-\infty})^3$ qui serait disponible dans la même section du dispositif si celui-ci n'existait pas. Montrer que C_p ne dépend que du facteur d'induction a. Quel est le coefficient de performance maximum ? (Limite de Betz)

Ex. 4.5 - Design de pale éolienne

On s'intéresse dans cet exercice aux choix de conception d'une éolienne. On s'intéresse pour cela à l'écoulement de l'air autour d'une pale en rotation à une vitesse angulaire Ω. On se place dans le référentiel de la pale à une distance r de l'axe de l'éolienne (cf. Figure ci-dessous, gauche). L'air est supposé être un fluide idéal incompressible (densité ρ_0) en écoulement stationnaire dont la vitesse loin en amont de la pale est $\mathbf{v}_{vent} = v_{-\infty}\mathbf{e}_x$. A proximité immédiate de la pale, la vitesse est $\mathbf{v}_{vent} = v\mathbf{e}_x = (1-a)\,v_{-\infty}\mathbf{e}_x$ avec $a = \frac{v_{-\infty} - v}{v_{-\infty}}$ le facteur d'induction axial. Dans le référentiel attaché à la pale s'ajoute une vitesse tangente au plan de l'éolienne $\mathbf{v}_{rotation} = \Omega r\mathbf{e}_\theta$. La vitesse apparente $\mathbf{v}_{app}$ perçue par la pale est la somme de ces deux contributions, et sa direction forme un angle ϕ avec la direction de la rotation $\mathbf{e}_\theta$. Par ailleurs, le profil de la pale est caractérisé par ses coefficients de portance et de trainée, $c_l\,(\alpha)$ et $c_d\,(\alpha)$, qui déterminent les forces s'exerçant sur la pale résultant de l'écoulement de l'air (forces par unité de longueur le long de l'axe de pale) :

$$\mathbf{f}_\mathrm{l} = c_l \frac{\rho_0 \mathbf{v}_{app}^2}{2} K\mathbf{e}_r \times \mathbf{u} \quad \mathbf{f}_\mathrm{d} = c_d \frac{\rho_0 \mathbf{v}_{app}^2}{2} K\mathbf{u}$$

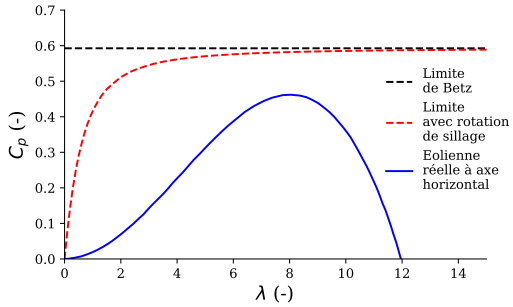
où K est la longueur de corde de la pale, $\mathbf{u} = \frac{\mathbf{v}_{app}}{\|\mathbf{v}_{app}\|}$ est la direction du vent apparent, et $\mathbf{e}_r$ est l'axe de la pale. Les coefficients $c_l = c_l\,(\alpha)$ et $c_d = c_d\,(\alpha)$ ne dépendent que de l'angle d'attaque α, entre la direction de la ligne de corde et la direction du vent apparent. L'allure de c_l et c_d en fonction de α pour un profil de pale d'éolienne type est présenté à la Figure ci-dessous (droite). L'angle θ entre la ligne de corde et la direction de rotation $\mathbf{e}_\theta$ est appelé angle de torsion, et constitue un paramètre de design de la pale.



1. Quel domaine de valeur d'angle d'attaque doit on privilégier ? Pourquoi dans ces conditions peut-on négliger l'une des deux forces ?

2. A partir des composantes du vent apparent et de l'angle de torsion, déterminer l'angle d'attaque α en fonction du rapport des vitesses en pointe de pale $\lambda = \frac{\Omega R}{v_{-\infty}}$. Comment l'angle de torsion doit-il être choisi pour assurer un fonctionnement optimal ?

3. Quelle contrainte le choix d'un angle de torsion optimal impose-t-il sur le fonctionnement de l'éolienne ? Interpréter alors le graphique ci-dessous représentant le coefficient de performance d'une éolienne type à axe horizontal en fonction du rapport des vitesses λ.



4. Déterminer les composantes axiale $f_x\mathbf{e}_x$ et orthoradiale $f_\theta\mathbf{e}_\theta$ de la force qu'exerce l'écoulement du fluide sur la pale (exprimée par unité de longueur de l'axe de pale).

5. Pour une éolienne constituée de N pales, quelle est la force axiale $dF\mathbf{e}_x$ qui s'exerce sur une section annulaire de rayon r et d'épaisseur dr ? L'analyse globale des lois de conservation (cf. exercice sur la limite de Betz) donne par ailleurs $dF = 4\pi\rho_0 a\,(1-a)\,(v_{-\infty})^2\,r dr$. En déduire la contrainte suivante entre les différents paramètres du problème: $\frac{K}{R} = \frac{8\pi a}{N c_l \lambda}\sin(\phi)$.

6. En introduisant cette contrainte dans l'expression de la force orthoradiale, déterminer le couple puis la puissance fournie à l'éolienne. Commenter.

7. A supposer la taille R et le nombre de pales de l'éolienne fixés, comment la longueur de corde évolue-t-elle le long de la pale ? Tracer longueur de corde et angle de torsion sur un même graphique (on considèrera $N = 3$, $c_l = 1$, $\alpha = 5°$, et $\lambda = 8$). Commenter la limite $r \to 0$.

8. Que change le nombre de pales N et dans quelle situation est-il avantageux d'avoir un grand nombre de pales.

9. Dans le cadre d'une hydrolienne, le fluide est trois ordres de grandeur plus dense que l'air. Qu'est ce que cela change ?

Chapter 5

Énergie chimique

Contents

Foreword 121
5.1 Chemical bonds 121
5.1.1 Electronic orbitals 121
5.1.2 Covalent and ionic bonds 124
5.1.3 Metallic bonds 128
5.1.4 Van der Waals interactions 129
5.2 Chemical reactions 129
5.2.1 What is a reaction ? 129
5.2.2 Basics of thermodynamics 133
5.2.3 Thermodynamics of a reaction 136
5.3 Exploitation of chemical energy 139
5.3.1 Main uses of chemical energy 139
5.3.2 Fossil fuels and carbon cycle 142
Key points 144
Exercises 146
Ex. 5.1 - Haut fourneau et réduction du fer 146
Ex. 5.2 - Batteries Li-ion 148

Foreword

This lesson addresses the energy of chemical bonds, the most widespread exploitation of which is the combustion of fossil hydrocarbons and biomass. We discuss the fundamental origin of chemical bonds at the level of the electronic orbitals of atoms and molecules, as well as the different types of chemical bonds. We then introduce chemical reactions and their thermodynamic description after a reminder of essential notions of thermodynamics (enthalpy, entropy, Gibbs free energy). Then, the different forms of exploitation of chemical energy by man are presented. We particularly address combustion and calorific value, electrochemistry, photosynthesis, and refining. Finally, we introduce the carbon cycle, at the heart of climate change issues due, in the first place, to the combustion of fossil fuels.

5.1 Chemical bonds

5.1.1 Electronic orbitals

Chemical reactions find their origin in the structure of electrons within matter. Understanding this structure is a matter of quantum mechanics, and we provide a short summary of it in this section and the next.

According to quantum mechanics, the state of an electron is described by a wave function $\psi(\mathbf{x})$ which is complex-valued and whose square modulus $|\psi(\mathbf{x})|^2$ is the density probability to find the electron at position $\mathbf{x}$ in space (normalization of this probability density requires $\int_{\mathbb{R}^3} |\psi(\mathbf{x})|^2 \mathbf{dx}$). This description is intrinsically probabilistic, and the position of the electron cannot be known with infinite precision, as formulated by the famous Heisenberg uncertainty principle. Instead, the electron is 'distributed' in space according to the wave function, and its uncertainty in position is given by $\Delta x^2 = \langle x^2 \rangle - \langle x \rangle^2 = \int_{\mathbb{R}^3} x^2 |\psi(\mathbf{x})|^2 \mathbf{dx} - \left(\int_{\mathbb{R}^3} x |\psi(\mathbf{x})|^2 \mathbf{dx} \right)^2$ (the brackets $\langle \cdot \rangle$ represent the average value).

The energy of this electron is obtained by applying the Hamiltonian operator $\widehat{H} = -\frac{\hbar^2}{2m_e}\Delta + V(x)$ to this wave function[1]: $\langle E \rangle = \int_{\mathbb{R}^3} \psi^*(\mathbf{x}) \widehat{H} \psi(\mathbf{x}) \mathbf{dx}$, with ψ^* the complex conjugate of ψ. As for the position, the energy may be uncertain, and this expression provides the average energy. The uncertainty in energy is given by $\Delta E^2 = \langle E^2 \rangle - \langle E \rangle^2$, with $\langle E^2 \rangle = \int_{\mathbb{R}^3} \psi^*(\mathbf{x}) \widehat{H}^2 \psi(\mathbf{x}) \mathbf{dx}$, where $\widehat{H}^2$ means that the Hamiltonian operator is applied twice.

Although in the general case, there is an uncertainty in energy, there exist particular states, called energy eigenstates, for which the energy is not uncertain but takes a precise value. These eigenstates satisfy $\widehat{H}\phi(\mathbf{x}) = E_\phi \phi(\mathbf{x})$ with $E_\phi \in \mathbb{R}$ the eigenvalue, which turns out to be the energy since application of the expressions of $\langle E \rangle$ and ΔE^2 gives $\langle E \rangle = E_\phi$ and $\Delta E^2 = 0$. Energy eigenstates are of particular interest as they are stationary. The time evolution in quantum mechanics results from the famous Schrödinger equation $i\hbar \frac{\partial \Psi}{\partial t} = \widehat{H}\Psi$, where $\Psi(t)$ is the wave function at time t. Applied to a system in an energy eigenstate at time zero $\Psi(0) = \phi$, the Schrödinger equation gives $\frac{\partial \Psi}{\partial t} = -\frac{iE_\phi}{\hbar}\Psi$, which can be easily integrated: $\Psi(t) = \exp\left(-\frac{iE_\phi}{\hbar}t\right)\phi$. Accordingly, the time evolution of an energy eigenstate is reduced to a complex phase multiplicative factor, which has no impact on the square modulus of the wave function (i.e., it is stationary). Accordingly, an electron in an energy eigenstate will remain in this state unless it receives enough energy from its environment to move to another eigenstate. Interestingly, when electrons are bound in atoms and molecules, the energy eigenstates of the electrons are quantified, i.e., they take only discrete values. And the difference of energy between neighboring energies is of the order of an eV, i.e., much larger than the typical energy perturbations received from the thermal agitation of the surrounding environment ($kT \sim 25$ meV). As a consequence, electrons in atoms and molecules are usually found in stationary energy eigenstates, called orbitals. How these orbitals evolve when atoms are brought close together is the heart of the chemical reactions.

Let us start with the most simple atom, namely hydrogen. For hydrogen, the electron is confined in the electrostatic potential of the proton located at the origin, so that, in spherical coordinates, the eigenvalue problem of the Hamiltonian is of the form $\widehat{H}\psi = E\psi$ with $\widehat{H} = -\frac{\hbar^2}{2m_e}\Delta - \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r}$. This problem of a single electron around a nucleus can be solved exactly. The energy eigenstates are quantified by three integer quantum numbers n, l, and m and take the form:

$$\psi_{n,l,m}(r,\theta,\varphi) = \frac{R_{n,l}(r)}{r} Y_{l,m}(\theta;\varphi)$$

where $R_{n,l}$ provides the radial part of the wavefunction, and $Y_{l,m}$ provides the angular part. More precisely, $n \in \mathbb{N}^*$ is the principal quantum number and quantifies the energy : $E_n = -\frac{E_{Ryd}}{n^2}$ with $E_{Ryd} = \frac{m_e}{2}\left(\frac{e^2}{4\pi\varepsilon_0\hbar}\right)^2 = 13.6$ eV the Rydberg energy. $l \in \{0, 1, ..., n-1\}$ is the azimuthal quantum number and quantifies the square angular momentum. And $m \in \{-l, -l+1, ..., l\}$ is the magnetic quantum number and quantifies the polar angular momentum.

The radial wave functions $R_{n,l}$ are functions of the radial coordinate r only, and can be expressed in terms of the generalized Laguerre polynomials. Note that the radial wave function depends on two of the three quantum numbers, n and l. $R_{n,l}$ is normalized so that its square modulus $|R_{n,l}(r)|^2$ represents the density probability of finding the electron at a distance r from the proton. This

[1] $\hbar = \frac{h}{2\pi}$ is the reduced Planck constant, m_e is the mass of the electron, $\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is the Laplacian operator, and $V(\mathbf{x})$ is a real-value function corresponding to the energy potential in which the electron evolves (e.g., $V(\mathbf{x}) = -\frac{1}{4\pi\varepsilon_0}\frac{e^2}{\sqrt{x^2+y^2+z^2}}$ for an electron in the hydrogen atom, with the proton at the origin.)

density probability is displayed in Figure 5.1 (top left) for the first orbitals ($n \leq 3$). One can notice that the lower the energy of an orbital is, the closer the electron is to the nucleus. Inner electrons are more tightly bound to the nucleus than outer electrons. The angular part $Y_{l,m}$, called spherical harmonics, are functions of the polar angle θ and azimuthal angle φ only, and can be expressed in terms of the Legendre polynomials. The spherical harmonics depend on two of the three quantum numbers, l and m. The square modulus $|Y_{l,m}|^2(\theta)$ is a function of the polar angle θ only, and represents the density probability of the angular position of the electron. This density probability is displayed in Figure 5.1 (bottom left) for the first orbitals ($l \leq 2$). Interestingly, the orbitals exhibit very distinct angular distributions so that two different orbitals occupy different domains (no overlap).

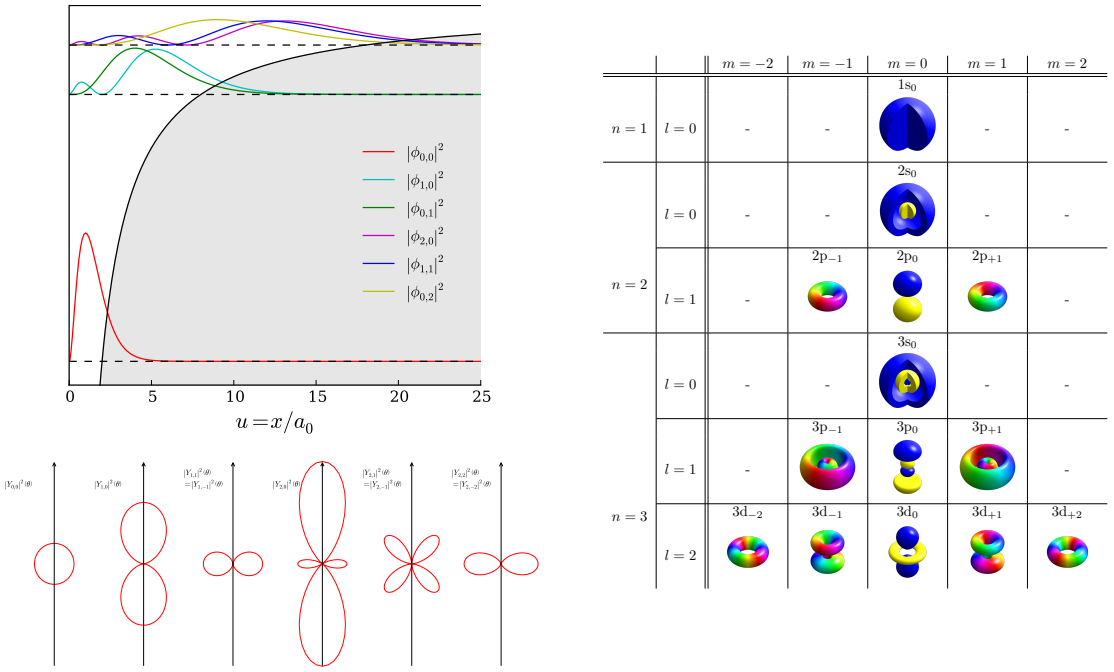


Figure 5.1: (top left) Graphical representation of the square modulus of the radial wavefunction $R_{n,l}(r) = \frac{1}{\sqrt{a_0}}\phi_{n-l-1,l}\left(\frac{r}{a_0}\right)$ of the hydrogen atom. This represents the density probability of finding the electron at a distance r from the proton. (bottom left) Graphical representation of the square modulus of the first spherical harmonics $Y_{l,m}(\theta,\varphi)$ of the hydrogen atom. The square modulus depends only on the polar angle θ and represents the density probability of the angular position of the electron. (right) Graphical representations of the first electronic orbitals ($n \leq 3$) of the hydrogen atom. The volume displayed is where the probability density $|\psi_{n,l,m}|^2$ exceeds a given threshold. The color stands for the complex phase of the wavefunction (blue for real positive, red for imaginary positive, yellow for real negative and green for imaginary negative (pictures from wikimedia commons, CC BY-SA 4.0).

The principal quantum number n determines the electron shells since it mostly controls the distance to the nucleus (Figure 5.1, top left), whereas the azimuthal quantum number l determines the electron subshell since it mostly controls how the electron is distributed within a shell (Figure 5.1, bottom left). The magnetic quantum number m distinguishes the different orbitals within a subshell. The number of states within a subshell is critical to explain the electronic structure of many-electron systems (see hereafter). An atomic orbital is usually identified by its n and l values. n is represented by its numerical value, but l is represented by a letter: 's', 'p', 'd', 'f', and 'g' for $l = 0$, 1, 2, 3, and 4, respectively. For instance, the atomic orbital with $n = 2$ and $l = 1$ is identified as '2p', and the atomic orbital with $n = 3$ and $l = 2$ is identified as '3d'. The magnetic quantum number m can be optionally added as a number in subscript, e.g., '$3d_{-1}$' for $n = 3$, $l = 2$, and

$m = -1$. We display in Figure 5.1 (right) graphical representations of the combined radial and angular distributions for the first atomic orbitals for $n = 3$.

Let us now look at the description of complex atoms with multiple electrons around a nucleus. The periodic table proposed by Mendeleev long before the emergence of quantum mechanics exhibits a structure with surprising similarity with the quantification of the hydrogen atom problem (Fig. 5.2 (left)). In particular, the number of elements in the rows of the periodic table is twice the degeneracy of the energy levels of the hydrogen atom ($2n^2$). This leads to introducing a 4^{th} quantum number s, called spin, that can take only two values ($\pm 1/2$). The spin is a property with no classical analogue and quantifies the projection of an angular momentum of the electron along the polar direction, associated with an intrinsic magnetic dipole moment. The reason why the structure of a complex atom shares similarity with the orbitals of the hydrogen is because, under mean field approximation, the quantum problem of multiple electrons around a nucleus is actually quite similar to that of the hydrogen atom. Each electron can be considered as if submitted to an effective potential different from the coulomb potential of the hydrogen problem, but still with spherical symmetry. The separation between the angular and radial contributions holds irrespective of the nature of the potential, as long as it is radial. The angular problem remains identical and leads to the spherical harmonics $Y_{l,m}$ involving the quantum numbers l, and m. The radial problem is specific, but solving it provides the radial wave function $R_{n,l}$ which, in the general case, can depend on both n and l, but not on m. Since the energy levels arise on the radial problem only, the energies $E_{n,l}$ now depend on both n and l.

Since electrons are fermions, the Pauli principle stipulates that each quantum state can be occupied by a single particle. Taking into account the spin, which can take two values, it means that each orbital can be occupied by two electrons. By convention, the electronic configuration of an atom is named after all the occupied electronic orbitals in the ground state: electron shell and subshell plus an exponent corresponding to the number of electrons in the subshell (limited to $2\,(2l+1)$ ($m \in \{-l, ..., l\}$ and spin $S \in \{-1/2, 1/2\}$)). For instance, the ground state of hydrogen is simply 1s, the ground state of Helium is $1s^2$ (shell $n = 1$, subshell $l = 0$, $m = 0$ and $S = \pm 1/2$), and the ground state of sodium is $1s^2\ 2s^2\ 2p^6\ 3s^1$ (shells $n = 1$ and $n = 2$ fully occupied, last electron in 3s). The ordering of the energies $E_{n,l}$ determines the order in which the shells and subshells are occupied in the ground state. As a first approximation, one can empirically determine this ordering following the Madelung rule:

1. orbitals energy are ordered by increasing value of $n + l$

2. for subshells with the same $n + l$, the energy increases with n

A graphical representation of the Madelung rule is proposed in Figure 5.2 (right). Following this rule, one can readily understand the classification of the periodic table (Fig. 5.2 (left)). The Madelung rule is not universal and many exceptions can be found, in particular because the energy is not limited to n and l but involves also the spin. For instance, the first atom violating the Madelung rule, Chromium, should have the electronic configuration $1s^2$ ... $4s^2\ 3d^4$, but it is actually $1s^2$ ... $4s^1\ 3d^5$.

5.1.2 Covalent and ionic bonds

Chemical bonding arises from the reorganization of the electronic orbitals when two atoms are brought close one to another. Let us consider two neutral atoms. At large distances compared to the extension of their electronic orbitals, the atoms are non-interacting and their respective electronic orbitals are those of independent atoms. When the two atoms are brought close together at distances comparable to the extension of the electronic orbitals, electrostatic interactions modify the electronic orbitals, and the usual atomic orbitals are replaced by molecular orbitals. Some molecular orbitals are 'bonding' as they correspond to a 'constructive' interference of two atomic orbitals leading to a higher probability of presence of the electrons between the nuclei, and thus providing a lower total energy. In contrast, some molecular orbitals are anti-bonding orbitals as they result from a 'destructive' interference of two atomic orbitals leading to a lower probability of presence of the electrons between the nuclei decreasing the electrostatic screening and increasing

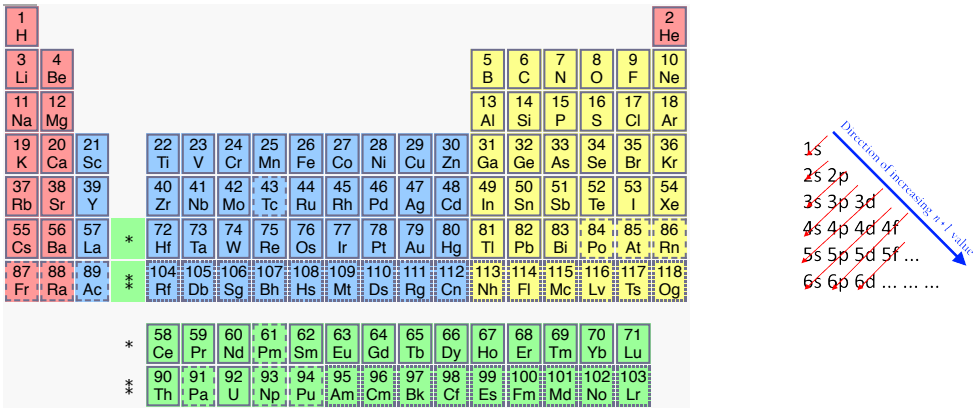


Figure 5.2: (left) Periodic table of the elements. The color provides the correspondence with the last occupied subshell (occupied subshell of highest energy): red for s orbital (first line is 1s), yellow for p orbital (first line is 2p), blue for d orbital (first line is 3d), green for f orbital (first line is 4f). Plain squares are primordial elements (naturally occurring), dashed squares are trace radioisotopes (produced by natural nuclear processes), and dotted squares are synthetic elements (man-made only). (right) Graphical representation of the Madelung rule for the ordering of the energies of the atomic orbitals. (from wikimedia commons (CC BY-SA 4.0))

the energy. The occupancy of these molecular orbitals following the Pauli principle, and the overall energy change of the system determines whether chemical bonding occurs or not.

Let us consider the most simple molecule, namely H_2 (Fig. 5.3, left). At large distances, two hydrogen atoms are in their ground state with their respective electrons in the 1s orbital at -13.6 eV. As the two atoms are brought close together, the two 1s orbitals interfere and form a bonding σ_{1s} orbital of lower energy than the 1s orbitals, and an anti-bonding σ^*_{1s} orbital of higher energy. Following the Pauli principle, the two electrons can occupy the bonding orbital, so that the ground state of the interacting atoms is obtained by fully occupying the σ_{1s} bonding orbital, and the σ^*_{1s} anti-bonding orbital remains empty. As a consequence, the energy of the interacting atoms is lower than that of the two independent atoms and chemical bonding is favored, this is the H_2 molecule.

Considering now the case of helium (Fig. 5.3, right), the structure of the orbitals is very similar. The helium atom contains two electrons, both of which occupy the 1s orbital in the ground state. When two helium atoms are brought close together, two molecular orbitals appear, a σ_{1s} bonding orbital and a σ^*_{1s} anti-bonding orbital. Application of the Pauli principle requires that the four electrons fully occupy both the bonding and the anti-bonding orbitals, resulting in a total energy larger than that of two independent helium atoms. This explains why molecular He_2 does not form.

A qualitative approach to understanding molecular orbitals and chemical interactions is the linear combination of atomic orbitals. In this approach, any pair of 'overlapping' atomic orbitals is likely to form bonding and anti-bonding molecular orbitals. The more complex case of dioxygen (O_2) is illustrated in Figure 5.4 (left), where s and p atomic orbitals combine to form σ and π molecular orbitals. The symbols for the molecular orbital are the Greek analogs (σ, π, δ...) of the symbols for the atomic orbitals (s, p, d) and correspond to the magnitude of the angular momentum of the orbital in the direction of the bond (σ for $m = 0$, π for $|m| = 1$, δ for $|m| = 2$...). In the case of O_2, the 2p orbitals aligned in the direction of the pair of nuclei form σ_{2p} and σ^*_{2p} orbitals, while the 2p orbitals orthogonal to this direction form π_{2p} and π^*_{2p}. How p orbitals combine is represented schematically in Figure 5.4 (right). The linear combination of atomic orbitals provides only a rough estimate of molecular systems, in particular because limiting solutions to linear combinations of atomic orbitals is quite restrictive. More advanced techniques rely on quantum chemistry computational methods.

So far, we have looked at homonuclear diatomic molecules (H_2, O_2) for which the electronic orbitals are symmetric. When heteronuclear atoms are brought close together, the resulting molecular orbitals are no longer symmetric so that the electrons are closer to one of the two atoms. This asymmetry results in a charge redistribution between the atoms, the atom attracting the electrons

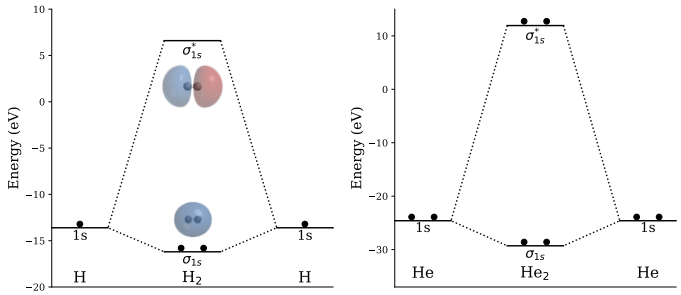


Figure 5.3: Molecular orbital diagrams of molecular H_2 (left) and He_2 (right). The diagram compares the energy levels of the atomic orbitals (H or He) to that of the molecule (H_2 or He_2). In the ground state of H_2, the two electrons occupy the bonding σ_{1s} orbital, so that the total energy is lower than that of two independent atoms. The hypothetical He_2 has a similar diagram as H_2, but contains 4 electrons so that the ground state occupies both the bonding σ_{1s} and anti-bonding σ^*_{1s} orbitals, leading to an energy exceeding that of two independent atoms. Accordingly, two hydrogen atoms form a chemical bond whereas two helium atoms do not form any chemical bond.

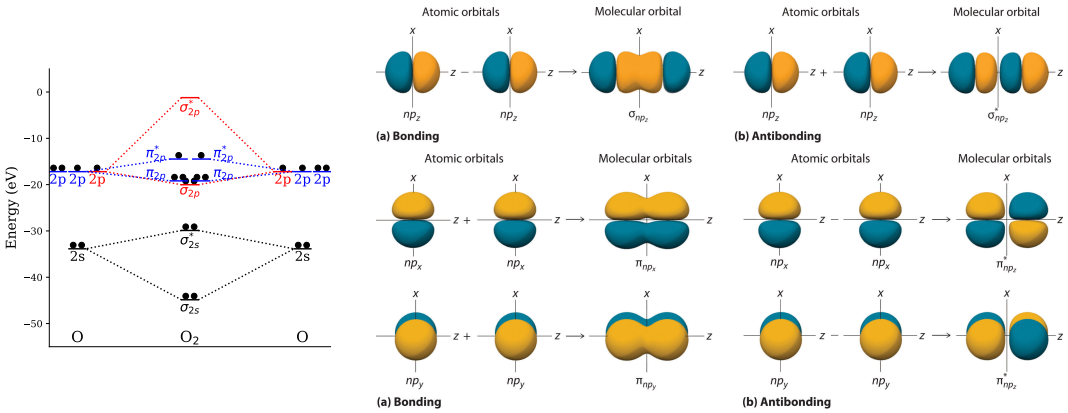


Figure 5.4: (left) Molecular orbital diagrams of O_2. The s and p atomic orbitals combine into σ and π (the case of the 1s orbitals is at much lower energies and not shown here). (right) Schematic representation of the combination of p orbitals into σ and π molecular orbitals (from LibreTexts libraries (CC BY-NC-SA 4.0)).

having a net negative partial charge, while the other atom has a net positive partial charge. As a consequence, the molecule becomes polar and sensitive to the electromagnetic environment. An example of such a molecule is hydrogen fluoride (HF), whose molecular orbital diagram is shown in Figure 5.5. The 1s orbital of the hydrogen atoms has a much higher energy than the 1s orbital of the fluorine. It overlaps the 2p orbital of fluorine instead, resulting in a σ bonding orbital in which the electrons are predominantly moved toward the fluorine nucleus, and a σ^* anti-bonding orbital, centered on the hydrogen nucleus instead. Since only the bonding orbital is occupied according to the Pauli principle, this results in a polar HF molecule in which the fluorine has a net negative partial charge and the hydrogen has the opposite positive partial charge.

The capacity to attract electrons is called electronegativity and is closely related to the position of the atoms in the periodic table. As we have seen, the periodic table classifies atoms according to their electronic configuration. It thus permits the identification of large families of atoms with similar properties. For instance, the first column of the table gathers the alkali metals for which all shells are fully occupied except the outer one which contains a single electron. Due to this peculiar configuration, all these elements have a small electronegativity, i.e., can easily lose the outermost electron. In contrast, elements of the second to last column, such as fluorine, all have a single

place left for an electron in their outer p orbitals, and are thus very electronegative, i.e., can easily capture an additional electron. When alkali metals are combined with halogens, the difference in electronegativity is so large that the outer electron of the alkali metal is fully transferred to the halogen thus forming ions. Historically, the existence of the table was first surmised after observing that such large families of atoms with similar properties existed, and appeared somewhat periodically (whence the name of the table) when atoms are sorted according to their mass. There exist various formal definitions of electronegativity, the first of which was proposed by Pauling. According to this definition, the difference in electronegativity χ between atoms A and B is given by:

$$|\chi_{\mathrm{A}} - \chi_{\mathrm{B}}| = \frac{1}{\sqrt{E_0}} \sqrt{E_d\left(\mathrm{AB}\right) - \frac{E_d\left(\mathrm{AA}\right) - E_d\left(\mathrm{BB}\right)}{2}}$$

where $E_d\left(\mathrm{XY}\right)$ is the dissociation energy of the compound XY (with X and Y $\in \{\mathrm{A}, \mathrm{B}\}$), and $E_0 = 1$ eV is a normalization constant so that electronegativity is a dimensionless quantity. Following this definition, two atoms have a strong difference in electronegativity if the dissociation of their heteronuclear compound requires much more energy than the average dissociation of their respective homonuclear compound. Such a formal definition is based on the observation that strong electronegativity difference favors strong chemical bonding between different elements in comparison with the pure phase of the same elements. The above equation for Pauling electronegativity provides only the difference in electronegativity between atoms. To properly define Pauling electronegativity, hydrogen is considered as a reference and its electronegativity is fixed at $\chi_{\mathrm{H}} = 2.2$. Figure 5.5 (right) shows the values of the Pauling electronegativities for the elements of the periodic table. Values range from 0.79 for cesium to 3.98 for fluorine.

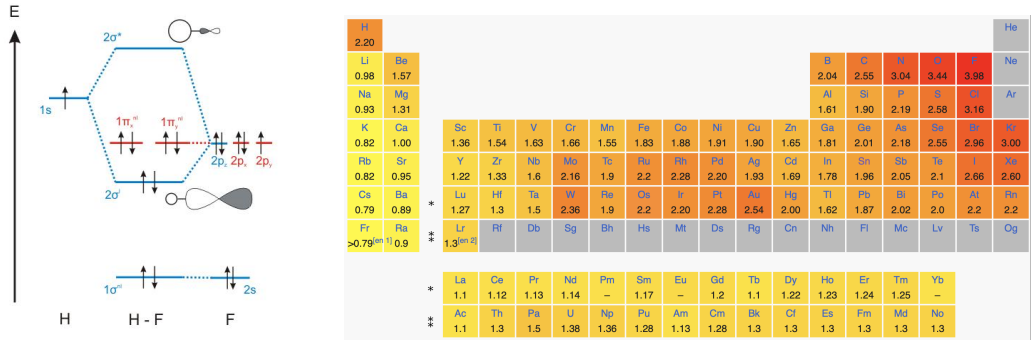


Figure 5.5: (left) Molecular orbital diagram of the hydrogen fluoride molecule (from wikimedia commons, Yohan Loquais, CC BY-SA 3.0). List of the Pauling electronegativities of all the atoms arranged in the periodic table (from wikimedia commons (CC BY-SA 4.0).

Depending on the difference in electronegativities between neighboring atoms in a molecule, we distinguish different types of chemical bonds:

Covalent bonds refer to bonds formed between atoms with small differences in electronegativities. Homonuclear diatomic molecules such as H_2 and O_2 are examples of ideally covalent bonds (no difference in electronegativity). Such bonds have strong directionality and can involve multiple pairs of electrons in molecular orbitals, depending on the combination of atomic orbitals. The resulting electrons are shared equally between the nuclei and do not result in charge imbalance so that the covalent bonds are non-polar.

Polar bonds refer to bonds formed between atoms with intermediate differences in electronegativities, typically around 1.5-1.9, such as the HF molecule. These bonds involve an unequal sharing of the electrons, resulting in a net redistribution of the charges within the molecule, which becomes polar. The polar bonds still hold some directionality.

Ionic bonds refer to bonds formed between atoms with large differences in electronegativity, such as the NaCl salt. The difference is so large that there is a full transfer of one or more electrons between the atoms leading to the formation of ions, such as Na^+ and Cl^- for NaCl. Ionic bonds have little directionality in contrast with covalent bonds, but are often stronger. Moreover, ionic compounds can often dissolve in polar solvents such as water.

5.1.3 Metallic bonds

When atoms form a periodic lattice, the Hamiltonian exhibits a periodicity in space. The associated eigenvalue problem leads to delocalized electronic orbitals of the form $\psi(\mathbf{x}) = \exp(i\mathbf{k}.\mathbf{x})\, u(\mathbf{x})$ with u a periodic function with same periodicity as the Hamiltonian, and $\mathbf{k} \in \mathbb{R}^3$ a wave vector. Interestingly, the energy values of those delocalized orbitals are grouped in bands separated by gaps (Fig. 5.6). The differences of energy within a band are very small compared to the thermal agitation energy (kT) so that an electron can easily move from one orbital to the other if permitted by the Pauli principle. In contrast, the gaps of energy can be large compared to kT so that the transition from one band to another strongly depends on the width of the gap in between. In the ground state (0K limit), the electrons occupy only the states of smallest energy (two electrons by orbitals, following the Pauli principle). The highest energy level occupied in this 0K limit defines the Fermi level. This peculiar electronic structure has some important implications regarding electrical conductivity:

- If the Fermi level lies within a band (i.e., a band is not fully occupied), the material is a conductor, since electrons can easily move by small changes in energy. This is the case in particular of pure metals made of elements of the periodic table with partially filled ns or nd orbitals.
- If the Fermi level lies within a gap, the band just below the Fermi level (valence band) is fully occupied, and the band just above (conduction band) is fully unoccupied. In that case, the electron cannot easily move, and the material is an insulator.
- In the particular case of a 'small' gap, i.e., a gap width comparable to kT, the probability for the electrons to cross the gap becomes non-negligible. In such cases, when the Fermi level lies within a gap, the material can still have a small fraction of electrons moving from the valence band to the conduction providing a small conductivity. And this conductivity can be dramatically enhanced by a small external 'help' such as a photon or an electric field. Such materials are called 'semiconductors' and are central to electronic applications. In particular, semiconductors are the key materials for photovoltaic solar energy (see dedicated lessons).

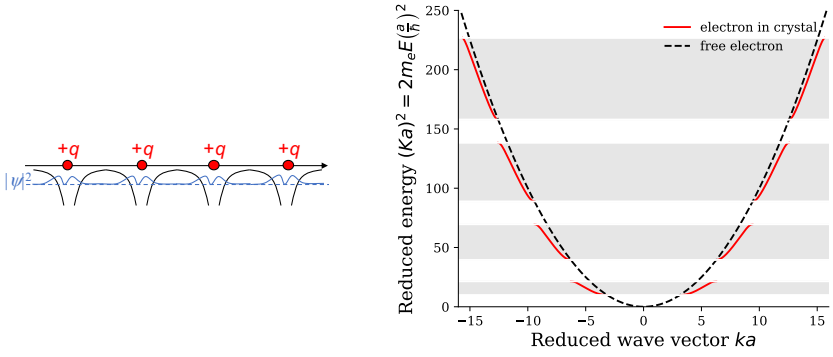


Figure 5.6: (left) Schematic representation of an electron in a periodic lattice. (right) Band and gap structure as a function of the wave vector ka for the Dirac comb problem (spinless particle of mass m_e in 1D submitted to an infinite series of Dirac wells periodically distributed with period a).

Therefore, unlike other materials, conductors contain some peculiar electrons that are delocalized in space and that can easily move. These electrons form metallic bonds that are quite different

in nature compared to the ionic, polar and covalent bonds. In particular, metallic bonds are not attached to pairs of atoms but are shared by many atoms. As consequence, these bonds are strong in tension, but have no directionality and are thus weak in shear, which causes the metals to have a ductile behavior.

5.1.4 Van der Waals interactions

So far, we have discussed the chemical bonds that involve the intra-molecular organization of the electronic orbitals. In a broader sense, 'chemical bond' can also refer to inter-molecular interactions, that do not involve the internal structure of molecules, but the mutual interactions between distinct molecules. All these interactions originate from the electromagnetic forces between the nuclei and electronic orbitals of one molecule with that of the other molecule. And these forces are often referred to as the Van der Waals forces. The typical magnitude of the inter-molecular interactions is of the order of a fraction of an eV, which is significantly less than the intra-molecular interactions (from an eV to tens of eV).

We distinguish the following inter-molecular interactions in decreasing order of energy (see Tab. 5.1):

Hydrogen bonds: This is the strongest inter-molecular interaction, which occurs in particular between water molecules. It forms between a hydrogen atom having a polar bond with a more electronegative atom (e.g., H in HF or in H_2O), and an electronegative atom from another molecule which holds a lone pair of electrons in its outer shell (e.g., F in HF or O in H_2O). The orbital of the lone pair partially deforms and is thus donated by the electronegative atom to the hydrogen. H-bond energy is 0.21 eV for water-water interactions.

Polar interactions: Molecule containing polar bonds exhibit a net electrostatic dipole. This dipole can interact with the surrounding ions and polar molecules. The range and magnitude of electrostatic interactions involving dipoles is significantly reduced with respect to the usual electrostatic interaction between charges: the dipole-charge and dipole-dipole interactions evolve in $1/r^2$ and $1/r^3$, respectively, with r the inter-molecular distance.

Multipolar interactions: In some molecules involving polar bonds the sum of the resulting dipole cancels out, so that the overall dipole of the molecule is zero. Yet, the molecule can exhibit electrostatic moments of higher order (quadrupole, octopole, etc...). For instance, a CO_2 molecule has a quadrupole, and a CH_4 molecule has an octopole. Multipolar interactions are weaker than dipolar interactions. As an example, the interaction between two quadrupoles evolves in $1/r^5$.

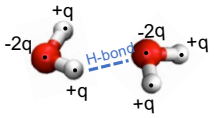
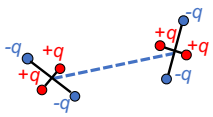
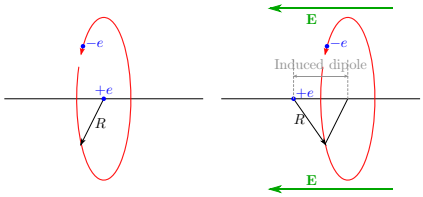
Induced dipole and dispersion interactions: Even if a molecule has no intrinsic dipole, it can be polarized by an external electrostatic field as the electronic orbitals move in the opposite direction as the nuclei. The capacity of a molecule to get polarized by an external field is called polarizability. Since all molecules have a non-zero polarizability, interactions involving induced dipole are ubiquitous between molecules. Even two polarizable molecules have an interaction, called the London dispersion force, which results from the fluctuations of the electronic orbitals. Thus, two Helium atoms, having no net multipole of any order, still exhibit a dispersion interaction whose magnitude is about 1 meV. Dispersion forces are the weakest molecular interactions and evolve in $1/r^6$.

5.2 Chemical reactions

5.2.1 What is a reaction ?

A chemical reaction is a process by which the chemical bonding within or between one or several molecules is modified. This covers the change of molecular orbitals, but also the change in the structure of matter even when there is no change in the molecular orbitals. In addition to molecules, chemical reactions can also involve the exchange of electrons (electro-chemistry) or the exchange

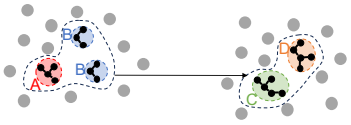
Table 5.1: List of inter-molecular interactions in decreasing order of energy.

Interaction type	**Schematic representation**	**Energy**
Hydrogen bonds		$E_{\text{H-bond}} \sim 0.1 - 1$ eV
Polar interactions		$E_{\text{dipole}} \propto \frac{1}{r^3}$
Multipolar interactions		$E_{\text{quadrupole}} \propto \frac{1}{r^5}$
Induced dipole & dispersion forces		$E_{\text{dispersioin}} \propto \frac{1}{r^6}$

of photons (photo-chemistry). Nuclear reactions can also be considered as a particular form of reaction, but we do not consider them in this lesson, as specific lessons at dedicated to nuclear energy.

A chemical reaction is formally described by a chemical equation which is of the form :

$$\nu_{\text{A}}\text{A} + \nu_{\text{B}}\text{B} \rightleftharpoons \nu_{\text{C}}\text{C} + \nu_{\text{D}}\text{D}$$



where A, B, C, and D are molecular compounds made of one or multiple atoms each and possibly charged, and where the ν_i ($i \in \{\text{A}, \text{B}, \text{C}, \text{D}\}$) are real numbers, called the stoichiometric coefficients, that represent the amount of each compound involved in the reaction. The compounds on the left-hand side (i.e., before the symbol '$\rightleftharpoons$') are called the reactants, and those on the right-hand side are called the products. For simplicity here, we limited ourselves to two reactants (A and B) and two products (C and D), but there is no restriction on the number of reactants and products and a reaction can involve many more compounds. For the fictitious reaction written above ν_{A} molecules A react with ν_{B} molecules B to produce ν_{C} molecules C and ν_{D} molecules D. Importantly, the '$\rightleftharpoons$' in the chemical equation suggests that the reaction can move forward or backward. Indeed, as we shall see later, the thermodynamics of the reaction shows that a reaction can occur both forward or backward, depending on the environmental conditions (temperature and current states of the reactants and products).

The compounds A, B, C, and D contain an integer number of atoms, and the stoichiometric coefficients ν_i must ensure that there is conservation of each type of atom between the reactants and the products. Accordingly, for an atom X that is contained a_{X} times in A, b_{X} times in B, c_{X} times in C, and d_{X} times in D, we must have : $\nu_{\text{A}} a_{\text{X}} + \nu_{\text{B}} b_{\text{X}} = \nu_{\text{C}} c_{\text{X}} + \nu_{\text{D}} d_{\text{X}}$. As a_{X}, b_{X}, c_{X}, and d_{X} are integers, this conservation constraint applied to the different atoms involved in the reaction imposes that the ratios between the stoichiometric coefficients are rational numbers. A similar conservation equation holds for the charge: the total charge of the reactants must be equal to the total charge of the products. Moreover, as the conservation constraints remain unchanged when multiplied by a constant factor, the stoichiometric coefficients are defined up to a multiplicative constant.

Note also that, when phase changes are involved during the reactions, knowing the phase of each

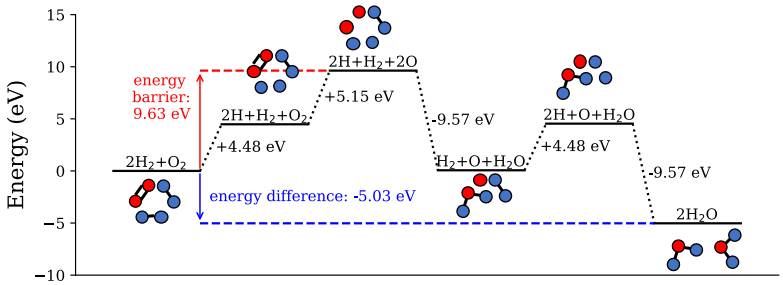


Figure 5.7: Example of a chemical reaction: combustion of dihydrogen. The diagram displays the elementary process of the chemical reaction and the associated energy changes. The overall energy barrier is shown in red, and the overall energy difference is shown in blue.

compound is important for the energy balance of the reaction as phase changes involve significant latent heat. Accordingly, one must precise the phase of each compound in parenthesis at the end of the name of the compound, e.g., A(g) for A in the gas phase. The possible phases are: (g) for gas, (l) for liquid, and (s) for solid. Sometimes, one also refers to (aq) for solutes in aqueous solution, to distinguish from the pure component liquid.

As an illustration, let us consider the case of the combustion of dihydrogen. This chemical reaction involves the reaction of dihydrogen (H_2) with dioxygen (O_2) to form water (H_2O). The chemical equation of this reaction is:

$$2H_2(g) + O_2(g) \rightleftharpoons 2H_2O(g)$$

The stoichiometric coefficients are adjusted to ensure the conservation of the hydrogen and oxygen atoms. Mentioning the phases is important here, as the usual state of water in ambient conditions is liquid, whereas we consider here the case of water vapor as the product of the reaction. This chemical reaction involves several elementary processes at the level of the electronic orbitals, that are detailed in Figure 5.7: first a H_2 and the O_2 molecules dissociate, leading to the formation of a first H_2O molecule. Then, the second H_2 molecule dissociates and combines with the remaining O atom to form the second H_2O molecule. Each elementary step of the reaction involves changes in the energy of the system as shown in Figure 5.7. According to statistical physics, a microscopic state of energy E has a probability of occurrence proportional to the Boltzmann factor $\exp\left(\frac{E}{kT}\right)$. Accordingly, thermal agitation of typical energy kT makes it possible to overcome an energy barrier with some probability. The higher the energy barrier, the lower the probability, and the longer it takes for the process to occur. Therefore, the global energy barrier shown in Figure 5.7 is what controls the kinetics of the chemical reaction. Over long-time scales, the energy barrier is overcome and the system reaches equilibrium. At equilibrium, the relative probability of occurrence of the reactants ($2H_2 + O_2$) and the products (H_2O) depends only on the difference of energy ΔE between the two states ($\frac{P_{\text{products}}}{P_{\text{reactants}}} \sim \exp\left(\frac{\Delta E}{kT}\right)$). Accordingly, this energy difference is what controls the thermodynamic equilibrium of the reaction.

There exists a wide variety of chemical reactions. There is no simple way to classify all chemical reactions, and we present hereafter some important types of reactions regarding energy applications, namely redox reactions and acid-base reactions.

redox: An oxidation is a reaction in which the reactant, called reductant, loses n electrons and becomes an oxidant: $\text{Red} \rightleftharpoons \text{Ox} + ne^-$. For instance $\text{Zn(s)} \rightleftharpoons \text{Zn}^{2+}\text{(aq)} + 2e^-$ is the oxidation of the zinc metal into dissolved zinc ion that can occur at the anode of a Cu-Zn electrochemical battery (see Figure 5.8). Conversely, a reduction is a reaction in which an oxidant receives electrons and becomes a reductant: $\text{Ox} + ne^- \rightleftharpoons \text{Red}$. For instance, at the cathode of the Cu-Zn battery occurs the reduction of the copper metal: $\text{Cu}^{2+}\text{(aq)} + 2e^- \rightleftharpoons \text{Cu(s)}$. Since a lone electron is reactive, an oxidation reaction usually takes place simultaneously as a reduction reaction. This is called a redox reaction which corresponds to the transfer of

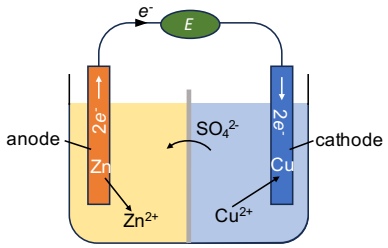


Figure 5.8: Schematic representation of a copper-zinc electro-chemical battery.

electrons from a reductant to an oxidant: $\text{Red1} + \text{Ox2} \rightleftharpoons \text{Ox1} + \text{Red2}$. An electro-chemical battery is a particular case of redox reaction in which the oxidation and the reduction are separated in space and the electrons flow through an electrical circuit, thus converting chemical energy into electrical energy. For the Cu-Zn battery, the overall redox reaction is $\text{Zn(s)} + \text{Cu}^{2+}\text{(aq)} \rightleftharpoons \text{Zn}^{2+}\text{(aq)} + \text{Cu(s)}$. However, in most cases, the transfer of electrons occurs via an electronegative atom (often oxygen) that takes an electron from the reductant to bring it to the oxidant: $\text{Ox1} - \text{O} + \text{Ox2} \rightleftharpoons \text{Ox1} + \text{Ox2} - \text{O}$. For example, an exercise in small class is dedicated to the reduction of iron oxide into iron metal by carbon monoxide, at the heart of the process of steel making in a blast furnace: $3\text{CO(g)} + \text{Fe}_2\text{O}_3\text{(s)} \rightleftharpoons 2\text{Fe(s)} + 3\text{CO}_2\text{(g)}$. The formal definition of a redox reaction according to the IUPAC[2] is a reaction that changes the 'oxidation state' of the atoms. The 'oxidation state' of an atom is the charge of the atom after ionic approximation of its heteronuclear bonds. For instance, the iron atom in the metal Fe(s) has an oxidation state of 0, whereas it has an oxidation state of +3 in the iron oxide Fe_2O_3 (which is $\text{Fe}_2^{3+}\text{O}_3^{2-}$ in the ionic approximation). Following this definition, combustion reactions, widely used in energy applications, are a particular case of redox reaction in which the oxidant O_2 (oxidation state 0) combines with the reductant so that the oxygen state of oxygen becomes -2: $\text{Red} + \frac{n}{2}\text{O}_2 \rightleftharpoons \text{Red}^{2n+} - \text{O}_n^{2-}$. For instance, the combustion of H_2 is a redox reaction: $2\text{H}_2\text{(g)} + \text{O}_2\text{(g)} \rightleftharpoons 2\text{H}_2^{2+}\text{O}^{2-}\text{(g)}$.

acid-base: An acid is a compound that can donate a proton H^+, a reaction called deprotonation in which the acid becomes a base: $\text{Acid} \rightleftharpoons \text{H}^+ + \text{Base}$. An example is the deprotonation of hydrogen chloride: $\text{HCl} \rightleftharpoons \text{H}^+ + \text{Cl}^-$. Conversely, a base is a compound that can receive a proton, a reaction called protonation: $\text{Base} + \text{H}^+ \rightleftharpoons \text{Acid}$. An example is the protonation of water that yields an hydronium ion: $\text{H}_2\text{O} + \text{H}^+ \rightleftharpoons \text{H}_3\text{O}^+$. As for redox reactions, since a lone proton is highly reactive, an acid deprotonation usually occurs simultaneously as a base protonation and the overall reaction is an acid-base reaction: $\text{Acid1} + \text{Base2} \rightleftharpoons \text{Base1} + \text{Acid2}$. For instance, combining hydrogen chloride and water leads to an acid-base reaction producing hydrochloric acid: $\text{HCl} + \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{Cl}^-$. The difference between an acid-base reaction and a redox reaction is that in an acid-base reaction, the oxidation state of the atoms does not change. The transfer of a proton is known as the Brønsted–Lowry definition of acid-base reactions. A broader definition (Lewis) is a reaction that involves a transfer of electron pairs (not necessarily through a transfer of protons). Both definitions are consistent as a proton transferred donates an electron pair to the compound that deprotonates and receives an electron pair from the compound that protonates. But Lewis definition allows other types of transfers (e.g., F^-).

Of course, chemical reactions are not limited to redox and acid-base. Let us mention phase change reactions (and related reactions such as dissolution or precipitation), synthesis or decomposition reactions, photo-chemical reactions, complexation, catalysis, etc. Note that there is no universal classification of chemical reactions, and the same reaction may fall into multiple categories.

[2]International Union of Pure and Applied Chemistry

5.2.2 Basics of thermodynamics

Before describing the thermodynamics of a chemical reaction, we recall in this section some basics of thermodynamics, that we will need to study the particular case of chemical reactions.

Let us first start with the most simple case, the isolated system. We consider a system made of N_i molecules i ($i \in \{\mathrm{A}, \mathrm{B}, ...\}$) in a volume V. Each molecule has a mass and is moving, so that the system has a kinetic energy K. The atoms are also interacting within molecules and between molecules, so that the system has a potential energy V. The total energy $U = K + V$ is called the internal energy. When a system is isolated, it cannot change its amounts of molecules N_i and its volume V. Moreover, according to the first principle of thermodynamics, its internal energy U is conserved as well. Thus N_i, V, and U are fixed and define the state of the system[3].

The second principle of thermodynamics states that there exists an extensive state function S called entropy such that, for an isolated system, thermodynamic equilibrium is reached when this entropy is maximum. 'State function' means that the entropy depends only on the current state of the system, i.e., $S = S(N_i, V, U)$. 'Extensive' means that the entropy of two systems is the sum of the entropy of each of the systems, i.e., $S_{1+2} = S_1 + S_2$. As it is formulated, the second principle is quite mysterious, as the fundamental nature of the entropy remains unclear. Statistical physics provides a microscopic explanation of the origin of the entropy according to the famous Boltzmann's formula: $S = k \ln(\Omega)$, where Ω represents the number of possible 'microstates' having same (N_i, V, U), i.e., the number of ways to arrange the N_i molecules in the volume V while maintaining the same internal energy U and satisfying the uncertainty principle of quantum mechanics. Thus, entropy can be interpreted as a measure of the disorder of the system, e.g., a crystalline solid has a low entropy compared to a gas made of the same molecules.

Since the entropy is a function of N_i, V, and U, one can consider its partial derivatives with respect to these parameters. Doing so, one defines the concepts of temperature T, pressure V, and chemical potential μ_i:

$$\frac{1}{T} = \left.\frac{\partial S}{\partial U}\right|_{N_i,V} \qquad \frac{P}{T} = \left.\frac{\partial S}{\partial V}\right|_{N_i,U} \qquad -\frac{\mu_i}{T} = \left.\frac{\partial S}{\partial N_i}\right|_{V,U}$$

Since S, N_i, V, and U are all extensive properties, the above partial derivatives all provide intensive quantities, i.e., quantities that are independent of the size of a system. Thus, the temperature of the system with state parameter $(2N_i, 2V, 2U)$ is the same as the temperature of the system (N_i, V, U). And pressure and chemical potential are intensive properties as well. These derivatives of the entropy fully define the exact differential of the entropy as follows:

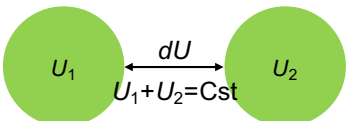
$$dS = \frac{1}{T}dU + \frac{P}{T}dV - \sum_i \frac{\mu_i}{T}dN_i$$

This differential is most often rearranged in the form of the balance of internal energy:

$$dU = \underbrace{TdS}_{\delta Q} \underbrace{-PdV + \sum_i \mu_i dN_i}_{\delta W} = \delta Q + \delta W \tag{5.1}$$

where we have identified two contributions, namely the increment of heat δQ and of work δW. Heat corresponds to the contribution of entropy to a change in internal energy, and work is the remaining part. The second principle constrains the possible exchanges of heat as the total entropy of systems exchanging heat must increase. We come back to this point later, when further discussing the heat.

So far, we have considered the case of an isolated system. But, in practice, most systems of interest, and in particular chemical reactions, are not isolated. So let us see how the thermodynamics change for non-isolated systems. We consider first the case of two

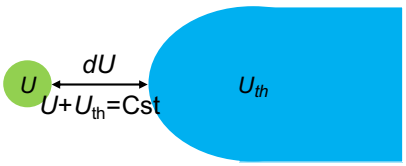


[3]Here we limit ourselves to the volume to describe the space occupied by the system. This is sufficient for fluid, but for solid under non-isotropic loading, one should consider the full strain tensor to fully describe the state of the system. Considering the volume only, we take into account only the spherical behavior of solids and ignore the deviatoric behavior.

systems $((N_i)_1, V_1, U_1)$ and $((N_i)_2, V_2, U_2)$ exchanging energy only,
while the total system (sum of the two systems) remains isolated,
i.e., $U_1 + U_2$ is conserved. One can apply the second principle to the total system: at equilibrium, S_{1+2} is maximum so that U_1 and U_2 must satisfy $\left.\frac{\partial S_{1+2}}{\partial U_1}\right|_{U_1+U_2,(N_i)_1,V_1,(N_i)_2,V_2} = 0$. Taking advantage of the extensivity of the entropy ($S_{1+2} = S_1 + S_2$) and of the conservation of the total energy $dU_1 + dU_2 = 0$, we obtain:

$$\left.\frac{\partial S_1}{\partial U_1}\right|_{(N_i)_1,V_1} = \left.\frac{\partial S_2}{\partial U_2}\right|_{(N_i)_2,V_2} \Rightarrow T_1 = T_2$$

Accordingly, two systems exchanging energy share the same temperature at equilibrium. This is called the thermal equilibrium.



Often, the two systems exchanging energies are of very different sizes. For a chemical reaction, for instance, the system is made of a few molecules exchanging energies with a very large number of other molecules around. In the limit in which the large system is assumed infinite (called thermostat), its state $((N_i)_{th}, V_{th}, U_{th})$ is almost insensitive to a small exchange δU with the small system ($\delta U \ll U_{th} \Rightarrow ((N_i)_{th}, V_{th}, U_{th} + \delta U) \approx ((N_i)_{th}, V_{th}, U_{th})$ (in what follows the subscript '$_{th}$' refers to the thermostat, and we no more use subscript for the small system). Therefore, the temperature of the thermostat is almost unchanged upon exchanges with the small system, and can be considered fixed. Since thermal equilibrium requires that the small system has the same temperature as the thermostat, the small system is thus also at fixed temperature: $T = T_{th} = Cst$. Applying the second principle to this particular case gives:

$$\left.\frac{\partial (S + S_{th})}{\partial U}\right|_{U+U_{th},N_i,V,(N_i)_{th},V_{th}} = 0 \Rightarrow \left.\frac{\partial S}{\partial U}\right|_{N_i,V} = \frac{1}{T}$$

Since the temperature is now a constant, the second principle can be reformulated for a system in thermal equilibrium with a thermostat:

$$\frac{\partial}{\partial U}\left[S - \frac{U}{T}\right] = 0 \Rightarrow S - \frac{U}{T} \text{ maximum at equilbrium}$$

Most often one formulates this new equilibrium condition in terms of the Helmholtz free energy $F = U - TS$:

A system at fixed temperature minimizes its Helmholtz free energy at equilibrium

Note that the Helmholtz free energy F is a function of (N_i, V, T) and not of (N_i, V, U) as for the entropy S. Accordingly, when defining $F = U - TS$, we implicitly assume the Legendre transform $F(N_i, V, T) = \min_U (U - TS(N_i, V, U))$. The Helmholtz free energy balance is deduced from that of the internal energy (Eq. 5.1):

$$dF = -SdT - PdV + \sum_i \mu_i dN_i$$

We have seen how the thermodynamics changes for a system exchanging energy with its environment. But, the molecules of a chemical reaction can also exchange volume with their environment. So let us now consider the case of a system exchanging both energy and volume. The approach is very similar to the approach followed above for thermal equilibrium. We start with two systems 1 and 2, exchanging both energy and volume. As the total system remains isolated, we can apply the second principle to the total system and we obtain two equilibrium conditions:

$$\begin{cases} \left.\frac{\partial S_1}{\partial U_1}\right|_{(N_i)_1,V_1} = \left.\frac{\partial S_2}{\partial U_2}\right|_{(N_i)_2,V_2} \\ \left.\frac{\partial S_1}{\partial V_1}\right|_{(N_i)_1,U_1} = \left.\frac{\partial S_2}{\partial V_2}\right|_{(N_i)_2,U_2} \end{cases} \Rightarrow \begin{cases} T_1 = T_2 \\ P_1 = P_2 \end{cases}$$

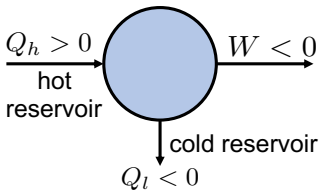
Therefore, two systems exchanging energy and volume share the same temperature and the same pressure at equilibrium. This is called the thermal and mechanical equilibrium. Let us now assume that one of the two systems is infinitely large. Then, its temperature and pressure are insensitive to exchanges with the small system and can be considered fixed. In turn, at thermal and mechanical equilibrium, the small system is also at fixed temperature and pressure. The second principle can then be reformulated as the minimization of the Gibbs free energy (also known as free enthalpy) $G(N_i, P, T) = \min_{U,V}(U - TS(N_i, V, U) + PV)$:

> A system at fixed temperature and pressure minimizes its Gibbs free energy at equilibrium

And the Gibbs free energy balance is:

$$dG = -SdT + VdP + \sum_i \mu_i dN_i \tag{5.2}$$

The most common way chemical energy is used is by transforming it into heat. We defined the concept of heat increment as the part of the internal change that is due to a change of entropy (Eq. 5.1): $\delta Q = TdS$. It is important to note that Q is not a state function like entropy, hence the 'δ' symbol instead of a 'd'. Over a thermodynamic cycle, as the final state is the same as the initial state, entropy comes back to its initial value; but there can be a net exchange of heat if the path followed is not at the same temperatures during entropy increase and decrease. The same remark applies to work.



As the second principle constrains the time evolution of entropy, this leads to efficiency limitations for thermal engines, the most well-known case being the Carnot limit to produce work from heat. Let us consider a thermal engine, called Carnot engine, that performs thermodynamics cycles during which it receives heat from a high-temperature source ($Q_h = T_h \Delta S_h > 0$), donates heat to a low-temperature source ($Q_l = T_l \Delta S_l < 0$), and produces work ($W < 0$). The efficiency of the engine is defined by the ratio of the useful energy produced ($|W|$) over the energy consumed (Q_h): $\eta = |W| / Q_h$. According to the first principle, conservation of energy over a complete cycle of the engine requires that $Q_h + Q_l = -W = |W|$. So the efficiency depends only on the ratio between the heats received and donated: $\eta = 1 + \frac{Q_l}{Q_h}$. An ideal efficiency ($\eta = 1$) would require that $Q_l = 0$, i.e., there is no need to transfer heat to a low-temperature source. It turns out that this is forbidden by the second principle, unless the low-temperature source is at 0K, which is not possible (even deep space is at 2.7K). Indeed, the second principle requires that the total entropy created is positive, that is when the system at temperature T receives a heat increment $\delta Q = TdS$ from a reservoir at temperature T_{res}, the reservoir receives $-\delta Q = TdS_{res}$, and we must have an increase of the total entropy: $dS + dS_{res} \geq 0$. As a consequence, $T_{res} \geq T$ for $\delta Q > 0$, and vice versa for $\delta Q < 0$, i.e., the system receives heat if it is colder than the reservoir and donates heat if it is hotter. Moreover, the lower bound of the entropy change of the engine is $dS \geq -dS_{res} = \frac{\delta Q}{T_{res}}$. Applied to the case of a complete cycle of the Carnot engine, this gives the Clausius inequality: $\Delta S \geq \frac{Q_h}{T_h} + \frac{Q_l}{T_l}$. Since the entropy is a state function, its variation is zero over a complete cycle of the engine (the system comes back to its original state), and therefore: $\frac{Q_h}{T_h} + \frac{Q_l}{T_l} \leq 0 \Rightarrow \frac{Q_l}{Q_h} \leq -\frac{T_l}{T_h} \Rightarrow \eta \leq 1 - \frac{T_l}{T_h}$. The Carnot limit provides an upper bound to the efficiency due to the second principle. This limit is a 'hard' limit, and cannot be overcome no matter the technique used, as it results from the second principle.

So far, the concept of heat has been defined for systems at constant volume (Eq. 5.1). However, for application to chemical reactions, we need to revisit it for systems at constant pressure. According to the definition of heat $\delta Q = dU - \delta W = dU + PdV - \sum_i \mu_i dN_i$, for a system at fixed volume and number of particles all the heat increment is converted into internal energy. In the case of heat exchange at constant pressure, however, the volume is not fixed so that part of the heat is converted into work: $\delta Q = dU|_{N_i,P} - \delta W|_{N_i,P} = dU|_{N_i,P} + P\, dV|_{N_i,P}$. Introducing the enthalpy $H = U + PV$, one readily finds that, at constant pressure, a heat increment is fully converted into enthalpy: $\delta Q = dH|_{N_i,P}$. Therefore, for systems at constant pressure, heat exchanges are associated with variations of enthalpy.

5.2.3 Thermodynamics of a reaction

Let us now apply the concepts of thermodynamics to a chemical reaction. A chemical reaction is a process by which reactants form products while exchanging energy and volume with the surrounding molecules. Accordingly, this is a process in thermal and mechanical equilibrium with a large reservoir, i.e., at fixed temperature and pressure. As we have seen, in such conditions, the thermodynamic equilibrium corresponds to the minimization of the Gibbs free energy $G = U = TS + PV$. We are therefore led to establish the variation of Gibbs free energy due to the progress of the reaction. To do so, we first have to define a measurement of the state of progress of a chemical reaction, namely the extent of the reaction ξ. For the reaction $\nu_\mathrm{A}\mathrm{A} + \nu_\mathrm{B}\mathrm{B} \rightleftharpoons \nu_\mathrm{C}\mathrm{C} + \nu_\mathrm{D}\mathrm{D}$, an extent $\xi = 1$ mol corresponds to the reaction of ν_A moles of A with ν_B moles of B to produce ν_C moles of D and ν_D moles of D. $\xi > 0$ corresponds to a reaction moving forward and $\xi < 0$ corresponds to the reaction moving backward. Based on the extent, we can quantify the change in Gibbs free energy due to the progress reaction of the reaction. We define the Gibbs free energy of reaction $\Delta_r G$ as the change in Gibbs free energy of the system with respect to a small increment of the extent ξ:

$$\Delta_r G = \left.\frac{\partial G}{\partial \xi}\right|_{T,P}$$

If $\Delta_r G < 0$, minimization of the Gibbs free energy requires that $d\xi > 0$, i.e., the reaction moves forward. Conversely, if $\Delta_r G > 0$, equilibrium requires that $d\xi < 0$, i.e., the reaction moves backward. And $\Delta_r G = 0$ means that the system is already at equilibrium and $d\xi = 0$.

The concept of energy of reaction can be applied to all the thermodynamic potentials. Of particular interest are the enthalpy of reaction $\Delta_r H$ and entropy of reaction $\Delta_r S$ which are defined by:

$$\Delta_r H = \left.\frac{\partial H}{\partial \xi}\right|_{T,P} \qquad \Delta_r S = \left.\frac{\partial S}{\partial \xi}\right|_{T,P}$$

As the enthalpy is a measure of the heat exchanged by a system at constant pressure, $\Delta_r H$ quantifies the heat released by the reaction per unit extent of the reaction. $\Delta_r H < 0$ corresponds to an exothermic reaction, i.e., a reaction that releases heat when moving forward. And $\Delta_r H > 0$ corresponds to an endothermic reaction, i.e., a reaction that consumes heat when moving forward. Regarding entropy, it is a measure of the disorder in matter, and it is particularly sensitive to phase changes: the entropy of a solid is significantly smaller than the entropy of a liquid, which in turn is much smaller than the entropy of a gas. Thus, a reaction producing a gas from a solid (e.g., calcination of limestone) has a positive reaction entropy $\Delta_r S > 0$, whereas a reaction producing a solid from a gas (e.g., iron oxidation) has a negative reaction entropy $\Delta_r S < 0$.

These different thermodynamic potentials are not independent as $G = H - TS$. Therefore, the different reaction quantities satisfy:

$$\Delta_r G = \Delta_r H - T\Delta_r S$$

The above relation is convenient as, in many cases, $\Delta_r H$ and $\Delta_r S$ exhibit little variations with temperature, except when crossing a melting or vaporization point of one of the reactants or products. For constant reaction enthalpies and entropies, the Gibbs free energy depends linearly on temperature, so that one can easily estimate the domains of temperature for which the reaction moves forward or backward. Let us illustrate with the case of the calcination of limestone. The chemical equation of this reaction is:

$$\mathrm{CaCO_3(s)} \rightleftharpoons \mathrm{CaO(s)} + \mathrm{CO_2(g)}$$

This reaction is of major industrial importance, as it is a key process in the production of cement, and the associated $\mathrm{CO_2}$ emissions represent 3% of the total anthropic emissions of greenhouse gases. The thermodynamic properties of this reaction at ambient temperature and pressure are:

- $\Delta_r S = 158$ J.K^{-1}.mol^{-1} > 0 because the reaction produces gaseous $\mathrm{CO_2}$ from solid $\mathrm{CaCO_3}$.

- $\Delta_r H = 178$ kJ.mol$^{-1} > 0$, so the reaction is endothermic and consumes a significant amount of heat. The heat needed to calcine 1 ton of limestone corresponds to burning about 35 kg of high-quality coal.
- $\Delta_r G = \Delta_r H - T\Delta_r S$ which gives $\Delta_r G = 131$ kJ.mol$^{-1} > 0$ at $T = 298.15$K. Therefore the reaction does not occur spontaneously at ambient temperature.

As the reaction entropy is positive, increasing the temperature decreases the Gibbs free energy of the reaction so that there is a temperature beyond which the reaction occurs spontaneously:

$$\Delta_r G < 0 \Rightarrow T > \frac{\Delta_r H}{\Delta_r S} = 1127 \text{ K}$$

Therefore, one can estimate that calcination of limestone occurs spontaneously at temperatures larger than about 1127K. This estimation is not far from the precise value of 1163K, considering the variations of enthalpy and entropy with temperature.

In order to estimate the reaction properties, one often refers to so-called 'formation' reactions. The formation reaction of a compound A_aB_b, with A and B different atomic elements, is the reaction that forms 1 mole of this compound from its constituent elements in their reference states A_α and A_β:

$$\frac{a}{\alpha}A_\alpha + \frac{b}{\beta}B_\beta \rightleftharpoons A_aB_b$$

The reference states A_α and A_β are the pure forms of the elements A and B that are the most stable at the reference pressure $P_0 = 1$ bar. A few examples of reference states are: H_2(g) for hydrogen, O_2(g) for oxygen, C(s) (graphite) for carbon, Fe(s) for iron, etc... Note that, by convention, in a formation reaction, the stoichiometric coefficient of the product is 1. For instance, the formation reaction of methane is:

$$C(s) + 2H_2(g) \rightleftharpoons CH_4(g)$$

The reaction properties of the formation reactions are:

$$\Delta_f H^0 = \left.\frac{\partial H}{\partial \xi}\right|_{\text{std. cond.}} \qquad \Delta_f S^0 = \left.\frac{\partial S}{\partial \xi}\right|_{\text{std. cond.}} \qquad \Delta_f G^0 = \left.\frac{\partial G}{\partial \xi}\right|_{\text{std. cond.}}$$

which are called the standard enthalpy of formation, standard entropy of formation, and standard Gibbs free energy of formation, respectively. 'standard' refers to the conditions of the formation reactions (e.g., partial pressure of all gases equal to $P_0 = 1$ bar), and is noted with a superscript '^{0}'. And the subscript '$_r$' is replaced by the subscript '$_f$' to refer to a formation reaction.

As enthalpy, entropy, and Gibbs free energy are state functions, they do not depend on the path followed, so that the properties of any complex reaction in the standard state can be reduced to a linear combination of formation properties, counted negatively for reactants and positively for products. This is known as Hess's law. For the reaction $\sum_{R_i \in \text{reactants}} \nu_i R_i \rightleftharpoons \sum_{P_j \in \text{products}} \nu_j P_j$, the Gibbs free energy in standard condition is given by:

$$\Delta_r G^0 = -\sum_{R_i \in \text{reactants}} \nu_i \Delta_f G^0(R_i) + \sum_{P_j \in \text{products}} \nu_j \Delta_f G^0(P_j)$$

And the same relation applies to enthalpy and entropy. For instance, let us consider again the calcination of limestone $CaCO_3$(s) $\rightleftharpoons$ CaO(s) + CO_2(g). This reaction can be decomposed as the inverse of the formation reaction of limestone $CaCO_3$(s) plus the formation reaction of lime CaO(s) and of carbon dioxide CO_2(g). Using the formation properties listed in the table on the right, one readily finds the reaction

Compound	$\Delta_f G^0$ (kJ/mol)	$\Delta_f H^0$ (kJ/mol)	$\Delta_f S^0$ (J/mol/K)
$CaCO_3$(s)	−1129	−1207	−262
CaO(s)	−604	−635	−104
CO_2(g)	−394	−394	0

properties of calcination in standard conditions: $\Delta_r G^0 = 131$ kJ.mol^{-1}, $\Delta_r H^0 = 178$ kJ.mol^{-1}, and $\Delta_r S^0 = 158$ J.K^{-1}.mol^{-1}.

Thanks to formation properties, one can easily estimate the properties of any reaction in standard conditions. Yet, very often, the conditions of the reaction are not the standard conditions, and one needs to estimate how the non-standard conditions affect the properties of the reaction. To do so, let us come back to the energy balance of the Gibbs free energy (Eq. 5.2). By definition of the extent of the reaction, the changes dN_i in the amounts of the different reactants and products are not independent and satisfy: $dN_i = \nu_i d\xi$ for products and $dN_i = -\nu_i d\xi$ for reactants. Therefore, the energy balance can be written in terms of the extent of the reaction as follows:

$$dG = -SdT + VdP + \left(- \sum_{i \in \text{reactants}} \nu_i \mu_i + \sum_{i \in \text{products}} \nu_i \mu_i \right) d\xi$$

Accordingly, the Gibbs free energy of the reaction is related to the chemical potentials μ_i of the reactants and the products according to:

$$\Delta_r G = \left. \frac{\partial G}{\partial \xi} \right|_{P,T} = - \sum_{i \in \text{reactants}} \nu_i \mu_i + \sum_{i \in \text{products}} \nu_i \mu_i$$

The deviation of a chemical potential from standard conditions quantified by the activity a_i as follows:

$$\mu_i = \mu_i^0 (T) + kT \ln (a_i)$$

where μ_i^0 is the chemical potential in standard conditions that depends on the temperature only. The activity is a dimensionless quantity that varies depending on the conditions (concentration, pressure, etc.) and is equal to 1 in standard conditions. For some particular cases, how the activity relates to conditions is well known. For instance, $a_i = P_i/P_0$ for an ideal gas (P_i the partial pressure), $a_i = C_i/C_0$ for a dilute solute (C_i the molar concentration and $C_0 = 1$ mol/L the standard concentration), or $a_i \approx 1$ for a solvent or a solid. In more complex cases, one uses more advanced models or empirical estimates.

Introducing the activities in the expression of the Gibbs free energy of the reaction, we obtain the deviation from standard conditions:

$$\Delta_r G = \underbrace{- \sum_{i \in \text{reactants}} \nu_i \mu_i^0 + \sum_{i \in \text{products}} \nu_i \mu_i^0}_{\Delta_r G^0 = -kT \ln(K_{eq})} + kT \ln \underbrace{\left(\frac{\prod_{i \in \text{products}} (a_i)^{\nu_i}}{\prod_{i \in \text{reactants}} (a_i)^{\nu_i}} \right)}_{Q_r} = kT \ln \left(\frac{Q_r}{K_{eq}} \right)$$

where we have identified the Gibbs free energy of the reaction in standard conditions $\Delta_r G^0$ which is sometimes also reformulated in terms of the equilibrium constant K_{eq}. The deviation from standard conditions is given by the ratio Q_r between the activities of the products and that of the reactants, called the reaction quotient. If $Q_r > K_{eq}$, then $\Delta_r G > 0$ and the reaction moves backward. If $Q_r < K_{eq}$, then $\Delta_r G < 0$ and the reaction moves forward.

To illustrate how the thermodynamics of the reaction can be applied in practice, let us consider the issue of tin refining. Tin (Sn(s)) is one of the first metals whose metallurgy was mastered during the Bronze Age. In nature, tin ore is made of tin oxide SnO_2(s), and one needs to reduce it, in order to use it as a metal. The simplest reduction reaction is $SnO_2(s) \rightleftharpoons Sn(s) + O_2(g)$, which is the inverse of the formation reaction of SnO_2(s) ($\Delta_f H^0 = -581$ kJ/mol, and $\Delta_f S^0 = -205$ J/mol/K). The activities of the solids (SnO_2(s) and Sn(s)) are 1, and the only activity that varies is that of oxygen: $a_{O_2(g)} = P_{O_2}/P_0$. Therefore, the complete expression of the Gibbs free energy of the reaction is:

$$\Delta_r G = \underbrace{\Delta_r G^0}_{581 \text{ kJ/mol} - T \times 205 \text{ J/mol/K}} + kT \ln \left(\frac{P_{O_2}}{P_0} \right)$$

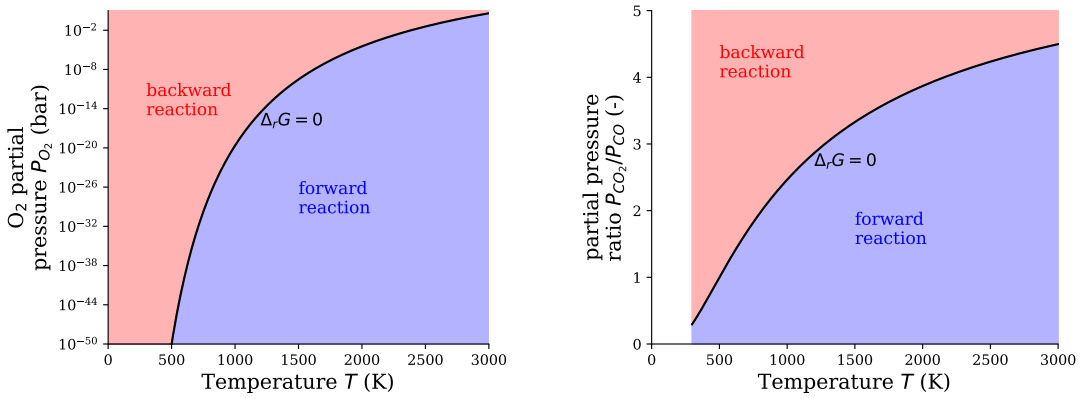


Figure 5.9: Thermodynamic diagrams of tin ore SnO_2(s) reduction in tin metal Sn(s). The conditions for reduction by simple heating (left) are shown in terms of temperature and partial pressure of oxygen O_2(g). The conditions for reduction by carbon monoxide CO(g) (right) are shown in terms of temperature and ratio of partial pressures of CO(g) and CO_2(g). In both cases, the black line represents the thermodynamic equilibrium $\Delta_r G = 0$, the blue area the conditions favorable to the reduction, and the red area the conditions unfavorable to the reduction.

Thanks to this expression, we can draw a diagram of the reaction thermodynamics (Fig. 5.9, left). The diagram represents the temperature along the X-axis and the partial pressure of O_2(g) along the Y-axis. The black line represents thermodynamic equilibrium, i.e., $\Delta_r G = 0$. Any condition above the line (red area) is in favor of the backward reaction ($\Delta_r G > 0$), and any condition below the line (blue area) is in favor of the forward reaction ($\Delta_r G < 0$). One readily observes that the scale of partial pressure of O_2(g) to achieve the reduction (forward reaction), is extremely small. For reasonable partial pressures (e.g., $\sim$%), tin reduction appears possible only at very high temperatures beyond 2500K. Such temperatures were out of reach with the techniques of ancient history.

An alternative to the reduction of tin ore by simple heating consists in adding coke to the furnace while limiting the access to oxygen. Doing so, one forms carbon monoxide CO(g) which can react with SnO_2(s) and reduce it into tin metal: $SnO_2(s) + 2CO(g) \rightleftharpoons Sn(s) + 2CO_2(g)$. To determine the Gibbs free energy of this reaction, one can combine the formation reactions of SnO_2, CO(g), and CO_2(g). And the deviation from standard conditions depends on the partial pressures of CO(g) and CO_2(g). The final expression of the Gibbs free energy of the reaction is:

$$\Delta_r G = \underbrace{\Delta_r G^0}_{15\ \text{kJ/mol} - T \times 30\ \text{J/mol/K}} + kT \ln\left(\left(\frac{P_{\text{CO}_2}}{P_{\text{CO}}}\right)^2\right)$$

Drawing a diagram of the thermodynamics of this reaction (Fig. 5.9, right), one finds that the conditions to achieve tin ore reduction are much easier to achieve: a few hundred °C for similar proportions of CO and CO_2 in the gas. Therefore, adding coke in the furnace made it possible to refine tin ore during the Bronze Age. Reduction by CO(g) is at the heart of many metallurgical processes, in particular for iron reduction in blast furnaces (see exercise in small class).

5.3 Exploitation of chemical energy

5.3.1 Main uses of chemical energy

Chemical energy is the most widespread form of energy used by humans. It started with the control of fire by early humans during prehistory. And nowadays, burning biomass and fossil fuel is still, by far, the main source of energy exploited by modern societies. A reason for this is that chemical energy is very easy to use, especially combustion since the heat of a flame provides enough energy

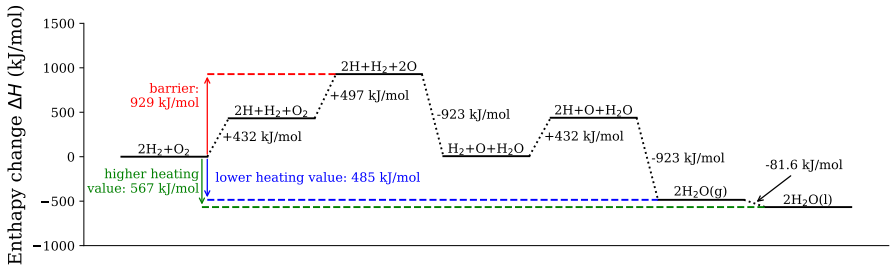


Figure 5.10: Enthalpy diagram of the combustion of dihydrogen: $2H_2 + O_2 \rightleftharpoons 2H_2O$. Depending on the final phase state of water, the enthalpy of the reaction differs. The difference is the latent heat associated with the condensation of water.

to make the reaction self-sustaining. Moreover, chemical energy is a quite concentrated form of energy and can be easily stored and transported: a gasoline reservoir of a few tens of liters provides enough energy to move a car over hundreds of kilometers. Finally, access to chemical energy has been abundant so far, but current consumption has reached unsustainable rates. In particular, the production of biomass energy causes deforestation. And proven fossil fuel reserves, although still large, correspond only to about a century of consumption. An even more serious concern, however, is that deforestation and combustion of fossil fuels have a major impact on the carbon cycle, which is discussed in the last section. Although combustion is by far the most widespread use of chemical energy, many other ways of exploiting chemical energy have been developed, some of which are expected to play major roles in the energy transition (e.g., electro-chemistry, hydrogen, carbon capture and storage).

The current uses of chemical energy are:

Combustion which consists in transforming chemical energy into thermal energy.

Electro-chemistry which consists in transforming chemical energy into electrical energy, and vice versa.

Photosynthesis which consists of transforming the radiation energy of the sun into chemical energy.

Refining and chemical processing which consists of transforming one form of chemical energy into another form of chemical energy.

Chemiluminescence which consists in transforming chemical energy into radiation.

Hereafter, we provide a short description of each of these uses (except chemiluminescence, which is restricted to niche applications).

As already mentioned, combustion is a particular case of redox reaction in which a reductant is oxidized by dioxygen: $\text{Red} + \frac{n}{2}O_2 \rightleftharpoons \text{Red}^{2n+} - O_n^{2-}$. As oxygen is a quite electronegative element, it favors chemical hetero-nuclear chemical bonds as in $\text{Red}^{2n+} - O_n^{2-}$ over homo-nuclear bonds as in O_2. Therefore, combustion reactions are quite exothermic and release significant amounts of heat. Once started it is self-sustained as the heat produced is sufficient to maintain the high temperature needed for the reaction to occur. Combustion is used to provide heat, such as in combustion engines or steam engines. Therefore, the key property is the enthalpy of the reaction, also called the heat of combustion. One distinguishes the lower heating value, which considers the products in the phase they belong to right after the reaction, and the higher heating value, which considers the products in their respective stable phase in standard conditions at 25°C. As combustion occurs at high temperatures the products are in the gas phase right after the reaction, but some are liquid at 25 °C, in particular water. Therefore, the difference between the lower heating value and the higher heating value if the latent heat associated with the condensation of some of the products, in particular water.

Let us consider the example of H_2 combustion whose enthalpy diagram is shown in Figure 5.10. The only product of the reaction is water. Considering water in the gas phase leads to a heat

of combustion of 242 kJ/mol (lower heating value), whereas considering water in the liquid phase leads to a heat of combustion of 283 kJ/mol (higher heating value). The difference is precisely the vaporization enthalpy of water (41 kJ/mol). Note that the enthalpy diagram of Figure 5.10 is for the combustion of two moles of H_2, and therefore the enthalpy differences are twice the heating values.

The high heating value is the heat that could be recovered in the best case. However, in practice, what heating value can one expect depends on the application. Heat of condensation is sometimes recovered in high-efficiency boilers but is never recovered in combustion engines.

Besides heat, combustion often comes with some 'by-products':

- CO_2 which is the main product of the reaction along with H_2O when organic fuels are used; and which is well-known for its contribution to climate change.
- Particle matter (PM) and sulfur oxides (SOx), due to impurities present in the fuels, constitute important issues regarding human health and ecosystems.
- Nitrogen oxide (NO), which results from the reaction of atmospheric nitrogen N_2 with atmospheric oxygen O_2, and is a major precursor of tropospheric ozone pollution. The formation of NO is not spontaneous but occurs marginally at the high temperatures that are reached in combustion chambers. And the higher the temperature the more likely the formation of NO. NO formation inside diesel engines in particular is the origin of the Dieselgate scandal and originates from the fact that diesel engines reach significantly higher temperatures than gasoline engines.
- Carbon monoxide CO, and soot in general, that are due to incomplete combustion of organic fuels. These are issues for human health and ecosystems.

A second major use of chemical energy is electro-chemistry. As explained before, electro-chemistry is a particular case of redox reaction in which the oxidation and the reduction are separated in space, and the transfer of electrons occurs via an electrical circuit. Doing so the chemical energy is converted into electrical energy. The energy recovered is the Gibbs free energy of the reaction $\Delta_r G < 0$ as we are at fixed temperature and pressure. For an electro-chemical reaction involving the transfer of n electrons, the electrical energy is the product of the charge transferred ne by the electric potential V. Therefore, the equation relating electrical energy and chemical energy is:

$$ne \cdot V = -\Delta_r G$$

Accordingly, the electric potential that appears between the anode and the cathode is directly proportional to the Gibbs free energy of the reaction.

Electro-chemistry is widely used for batteries. In particular, Li-ion batteries have emerged as the batteries with some of the highest energy densities (see exercise in small class). Yet, these densities remain an order of magnitude smaller than the heat of combustion of organic fuels.

Electro-chemistry is also widely used for electrolysis, i.e., to convert electrical energy into chemical energy. Electrolysis is used for instance to reduce aluminum oxide into aluminum metal. It is also used for the production of hydrogen from water, considered for the production of green hydrogen. Electrolysis can also be used to produce small organic precursors from CO_2 in the perspective of carbon capture and usage.

Photosynthesis is a biological process, ubiquitous in plants, that converts sunlight into organic matter. It consists of a very complex series of reactions, and it is not unique, but several 'routes' of photosynthesis exist depending on the plants and their environmental conditions. One can roughly describe photosynthesis as the synthesis of a carbohydrate:

$$n\,(h\nu + CO_2 + H_2O) \rightleftharpoons C_nH_{2n}O_n + nO_2$$

where $h\nu$ stands for the absorption of light. In terms of energy transformation, photosynthesis exhibits a low efficiency at transforming solar radiation into chemical energy, typically a fraction

of a %. In comparison, the best efficiency of photovoltaic cells exceeds 45%. Yet, unlike biomass energy, electrical energy cannot be easily stored.

Regarding the exploitation of energy, photosynthesis is used to produce biofuels. Yet, because of the poor efficiency, production of biofuels requires large land areas, in competition with agriculture. And deforestation appears as a negative side-effect to the development of biofuels. Currently, deforestation represents about 10% of the anthropic emissions of greenhouse gases.

Overall, biomass energy (including wood) represents a significant proportion of the energy consumption with 8% of the global primary energy. The current biomass energy consumption is significantly larger than before the industrial revolution.

Refining and chemical processing refers to all the processes by which we change the chemistry. Oil refining aims at converting the raw organic molecules of crude oil, into more usable molecules (e.g., gasoline). The chemical industry industrial energy consumer (according to the International Energy Agency). And about half of the chemicals produced by this industry are dedicated to energy usage: oil refining, coal gasification, steam reforming for hydrogen production, etc... The other half of the chemicals are used as feedstock: polymers and plastics, fertilizers, metals, inorganic chemicals, pharmaceuticals etc... Let us mention a few important processes regarding energy:

Refining (oil, biofuel) consists in transforming a raw fuel into chemicals of interest, e.g., fuels for combustion engines. Refining involves separation by distillation and catalytic cracking to form specific molecules (e.g., alkane, ethanol, etc...).

Haber–Bosch process consists in producing ammonia from atmospheric nitrogen: $N_2(g)+3H_2(g) \rightleftharpoons 2NH_3(g)$. This process is the main source of artificial fertilizers but also represents 3% of CO_2 emissions.

Steam reforming consists of producing dihydrogen from the reaction between hydrocarbons and water. It can be applied to a wide variety of hydrocarbons from coal, $C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$, to methane $CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$. This is currently the main way to produce H_2, which is called 'brown' or 'grey' hydrogen as it relies on fossil fuels.

5.3.2 Fossil fuels and carbon cycle

The issue of climate change is due to an imbalance in the carbon cycle on Earth because of the combustion of fossil fuels. In this section, we provide a short overview of the origin of fossil fuels, the quantitative impact of their exploitation on the carbon budget of the Earth, and the mitigation of climate change by carbon capture and storage.

Fossil fuels are geological deposits of hydrocarbons that originate from the remains of biomass maturated over millions of years in the Earth's crust. A tiny fraction of the biomass that grows at the surface of the Earth does not decompose as in most cases but gets trapped in sedimentary soils. Under the combined effects of temperature and pressure, the composition of this biomass evolves, a process called maturation and described in the so-called Van Krevelen diagram (Fig. 5.11). This diagram summarizes the maturation process in terms of the evolution of the elemental composition of the matter, more precisely the ratios of oxygen over carbon (O/C, X-axis) and of hydrogen of carbon (H/C, Y-axis). Depending on the origin of the biomass, several routes can be identified. Types I and II correspond mostly to marine deposits and have a high hydrogen content, whereas types III and IV correspond mostly to terrestrial deposits and have a high oxygen content. As a deposit gets buried deeper and deeper, the organic matter is exposed to higher pressures and temperatures which triggers the maturation process: the oxygen and hydrogen contents decrease resulting in the formation of small hydrocarbon molecules. During the first stage (diagenesis), the organic matter produces mostly CO_2 and H_2O. In the second stage (catagenesis), liquid hydrocarbons are generated which then migrate in the rock to form oil deposits. Finally, the last stage (metagenesis) produces smaller gaseous hydrocarbons that migrate as well and form gas deposits. The remaining solid organic matter has lost most of its hydrogen and oxygen and is what constitutes coal seams and kerogen in source rocks. The ultimate stage of maturation would be graphite with 100% carbon.

Finally, one is left with different types of geological deposits: solid (coal, kerogen), liquid (oil), and gaseous (natural gas). Another phase is gas hydrates, a crystalline solid made of water and

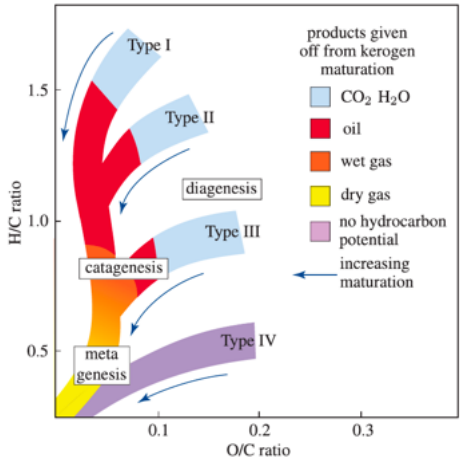


Figure 5.11: Van Krevelen diagram of the maturation of organic matter in geological deposits (from Jaffe & Taylor (2018) The Physics of Energy. Cambridge University Press).

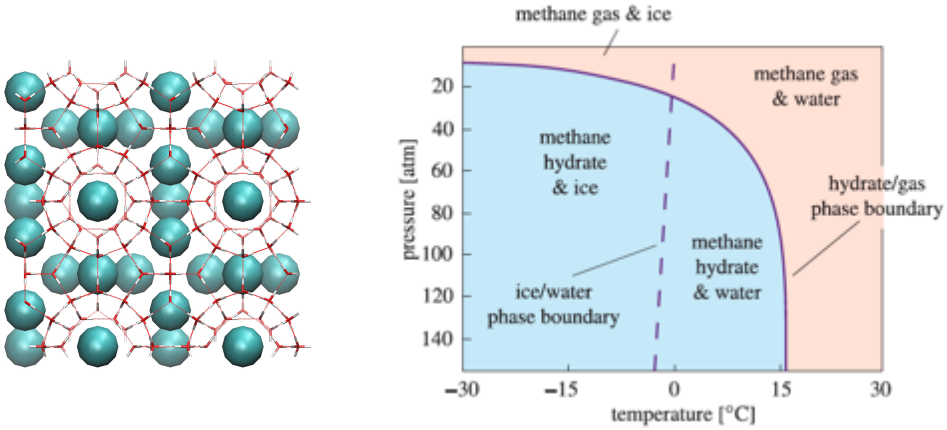


Figure 5.12: (left) Atomic structure of gas hydrate. (right) Phase diagram of water-methane mixtures. The temperature and pressure conditions needed for the formation of hydrate are represented by the domain in blue (from Jaffe & Taylor (2018) The Physics of Energy. Cambridge University Press).

methane ($8CH_4{\cdot}46H_2O$) which forms when a mixture of water and methane is submitted to low temperatures and high pressures (Figure 5.12). Favorable conditions for the formation of gas hydrates are found at the sea bed with pressures of a few tens of atmospheres and temperatures lower than 10°C. Permafrost also offers appropriate conditions with lower pressures but also lower temperatures. Solid, liquid, and gaseous deposits have been extensively exploited by human, and reasonable estimates of their respective reserves are available. In contrast, gas hydrates remain unexploited, and the volume of these deposits is quite debated. Yet, current estimates suggest that the volume of natural gas trapped in gas hydrates would be at least one order of magnitude larger than the proven reserves of oil. Thus, hydrate deposits represent huge volumes, and those deposits are a potential threat to climate change. The threat is not about the combustion of methane, as these deposits are unexploited, but about melting: as permafrost warms, gas hydrates within it may cross the melting temperature of the hydrate which would release large amounts of methane into the atmosphere, a gas with a global warming potential 25 times larger than CO_2.

CO_2 is by far the main product of all the chemical reactions performed by humans: about $3.5 \cdot 10^{10}$ tons per year. From an element perspective, this is a minor impact regarding the consumption of dioxygen as the atmosphere contains more than 10^{15} tons of O_2. However, this is a

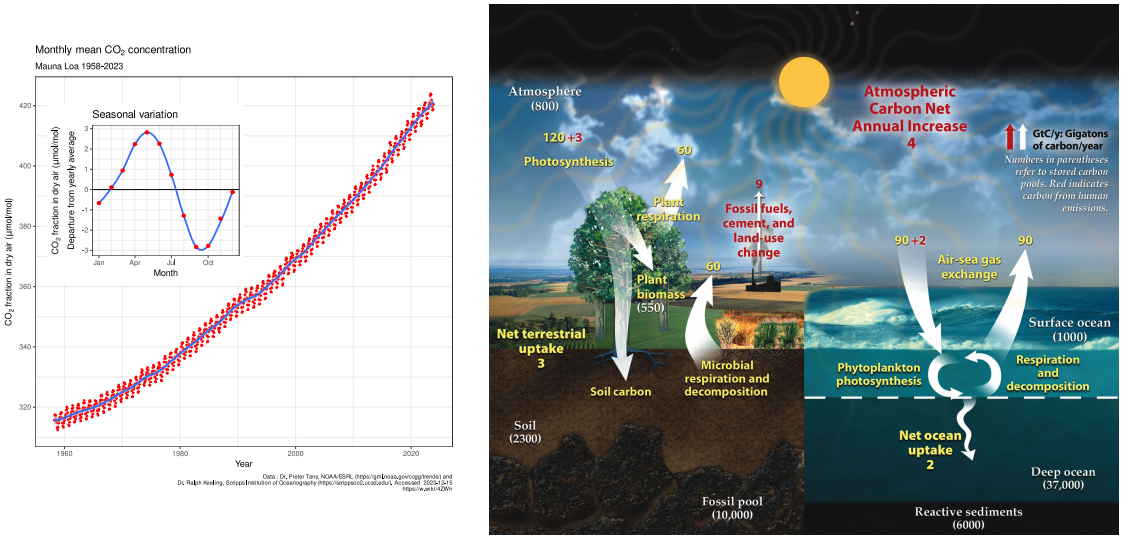


Figure 5.13: (left) CO_2 concentration in the atmosphere (from wikimedia commons, NOAA/ESRL and Scripps Institution of Oceanography, CC BY-SA 4.0). (right) Schematic representation of the carbon cycle of the Earth (from wikimedia commons, NASA Earth observatory, public domain).

significant impact regarding carbon as CO_2 is one of the main forms of atmospheric carbon and its concentration has increased by 50% since the pre-industrial period (Figure 5.13, left). Atmospheric CO_2 is only one aspect of the global carbon cycle of the Earth, which is shown schematically in Figure 5.13 (right). In this Figure, the stocks of carbon (in Gt of C) are represented in white, the flux (in Gt of C per year) in yellow, and the additional flux due to anthropic contributions in red. Anthropic contributions (9 GtC/yr) may seem small, compared to the biggest stocks (deep ocean 37 000 GtC, fossil fuels in 10 000 GtC). But these contributions actually affect primarily stocks that are much smaller, namely the atmosphere (800 GtC) and the surface ocean (1000 GtC). Thus, the annual increase in the atmosphere (+4 GtC/yr) would lead to a doubling of the atmospheric stock in 200 years, which is a major change.

In order to mitigate the perturbation of the carbon cycle, a process called Carbon Capture and Storage (CCS) is being considered. CCS is a process that aims at capturing CO_2 emissions and storing it in order to mitigate anthropic emissions. A schematic representation of the process is shown in Figure 5.14. CO_2 would be separated and captured at concentrated sources (e.g., industrial sites), transported to storage sites, and stored over long times (hundreds of years). There are a variety of techniques for capture (absorption, adsorption, pre-combustion, etc.) and different types of storage (geologic in aquifers, depleted hydrocarbon reservoirs, or coal seams, mineralization, deep ocean, etc.). CCS is considered a key technique to reach carbon neutrality in particular for industry sectors for which complete decarbonation is not possible so far. For instance, the calcination of limestone discussed in the section about reaction thermodynamics, represents 3% of CO_3 emission but remains a necessary process in the production of cement, with no carbon-free alternative. CCS is also considered a temporary solution to allow a faster reduction in carbon emissions of the industry. According to the IPPC, CCS would contribute between 10% and 55% of total greenhouse gas emission reduction depending on the scenarios. Note that carbon storage is also possible without involving capture (e.g., biochar from pyrolysis of biomass).

Key points

- Fundamental origin of chemical bonds (electronic orbitals), different categories of bonding (covalent and ionic, metallic, Van der Waals), and orders of magnitude of bond energies.
- Chemical reaction process, extent of a reaction, and standard reaction properties (enthalpy, entropy, Gibbs free energy).

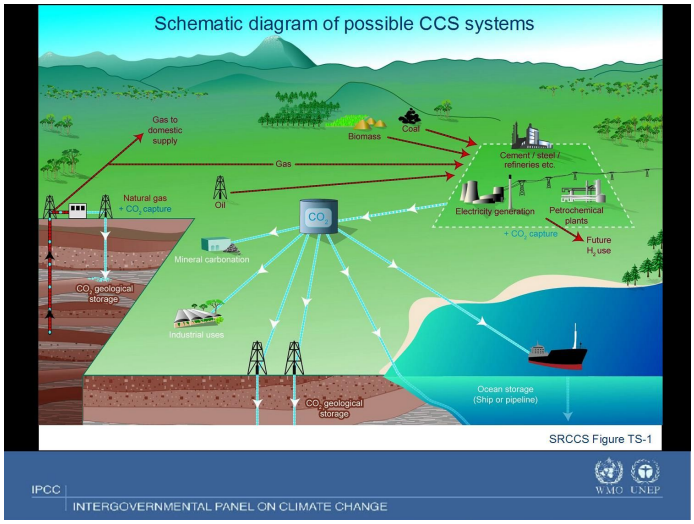


Figure 5.14: Schematic representation of the process of carbone capture and storage (from the special report on CCS of the International Panel on Climate Change, Fig. TS.1).

- Thermodynamics of a reaction, chemical potential and activity, reaction equilibrium.
- Chemical energy: combustion (heating value, by-products), electro-chemistry (batteries and electrolysis), photosynthesis (efficiency of biomass energy), refining and chemical processing.
- Fossil fuels (different types, combustion and carbon cycle, CCS).

Exercises

Ex. 5.1 - Haut fourneau et réduction du fer

Le secteur de la sidérurgie est le premier secteur industriel en terme d'émissions de gaz à effet de serre, avec 7% du total des émissions anthropiques, et le second secteur industriel en terme de consommation d'énergie (après le secteur de la chimie)[4]. Le processus de production le plus impactant est la réduction des oxydes de fer dans les hauts fourneaux lors de la production primaire, i.e., à partir de minerais. Dans cet exercice, on s'intéresse à la chimie de la réduction des oxydes de fer, et comment il est possible de la décarboner.

Les réactions en jeu sont des réactions d'oxydo-réduction impliquant un transfert d'électron via l'oxygène (O) : $\mathrm{A} + \mathrm{BO} \rightleftharpoons \mathrm{AO} + \mathrm{B}$, avec A et B les éléments oxydé et réduit, respectivement. On rappelle que la thermodynamique d'une réaction chimique est déterminée par l'enthalpie et l'entropie de réaction $\Delta_r H^0$ et $\Delta_r S^0$ qui sont la chaleur fournie et l'entropie crée par mole de réaction dans des conditions standards, à savoir pression atmosphérique $P_0 = 1$ atm. Pour les réactions considérées dans cet exercice, on peut raisonnablement considérer l'enthalpie et l'entropie de la réaction indépendantes de la température.

Le sens d'une réaction est déterminé par l'équilibre thermodynamique correspondant à la minimisation de l'enthalpie libre (ou énergie libre de Gibbs) $G = H - TS$. Les dérivées partielles de l'enthalpie libre par rapport aux quantités de matière étant les potentiels chimiques, le bilan d'enthalpie libre à pression et température constante est donné par $dG|_{P,T} = \sum_i \mu_i dn_i$, où μ_i est le potentiel chimique de l'espèce i (exprimé par mole) et n_i est le nombre de moles de l'espèce i. L'avancée de la réaction mène à une variation du nombre de moles des réactifs et des produits si bien que l'enthalpie de réaction est de la forme $\Delta_r G = \sum_i \nu_i \mu_i$ où les ν_i sont les coefficients stœchiométriques ($\nu_i < 0$ pour un réactif et $\nu_i > 0$ pour un produit). Le potentiel chimique μ_i dépend de la pression/concentration/température. L'écart au potentiel chimique μ_i^0 dans les conditions standards et caractérisé par l'activité chimique a_i comme suit : $\mu_i = \mu_i^0 + RT \ln(a_i)$. Dans cet exercice, nous considérons des réactions impliquant des gaz que l'on supposera parfaits et des solides. Pour un gaz parfait l'activité est le rapport entre la pression partielle de l'espèce i et la pression standard : $a_{i(g)} = \frac{P_i}{P_0}$. Pour un solide, l'activité est approximativement constante égale à 1 : $a_{i(s)} \approx 1$. Dans les conditions standards, $\Delta_r G^0 = \sum_i \nu_i \mu_i^0$ et $\Delta_r G^0 = \Delta_r H^0 - T\Delta_r S^0$ si bien que dans le cas général l'enthalpie de réaction vaut :

$$\Delta_r G = \Delta_r H^0 - T\Delta_r S^0 + RT \ln(\Pi_i a_i^{\nu_i})$$

Si $\Delta_r G < 0$, la réaction a lieu et mène à un équilibre chimique en faveur des produits, et inversement si $\Delta_r G > 0$. L'équilibre thermodynamique est atteint pour $\Delta_r G = 0$.

On rappelle enfin que l'enthalpie, l'entropie, et l'enthalpie libre sont des fonctions d'état et ne dépendent pas du chemin suivi pour parvenir à cet état. Une catégorie particulière de réactions chimiques sont les réactions de formation. La réaction de formation d'une espèce composée $\mathrm{A}_a\mathrm{B}_b$ est la réaction menant à la formation d'une mole de ce composé à partir des corps purs simples A_α et B_β qui le constituent sous leur forme stable dans l'état état standard : $\frac{a}{\alpha}\mathrm{A}_\alpha + \frac{b}{\beta}\mathrm{B}_\beta \rightleftharpoons \mathrm{A}_a\mathrm{B}_b$. Pour les composés impliqués dans cet exercice, les formes stables des corps purs simples sont : le dioxygène $\mathrm{O_2(g)}$ pour l'oxygène, le métal Fe(s) pour le fer, le graphite C(s) pour le carbone, et le dihydrogène $\mathrm{H_2(g)}$ pour l'hydrogène.

Données numériques:

- Constante des gaz parfaits : $R = 8.314$ J.mol^{-1}.K^{-1}
- Masses molaires : $M_{\mathrm{CO_2}} = 44$ g/mol, $M_{\mathrm{Fe}} = 55.8$ g/mol.

1. Le fer existe sous forme pure (Fe(s)) et sous trois formes oxydées principales : l'hématite ($\mathrm{Fe_2O_3(s)}$), la magnétite ($\mathrm{Fe_3O_4(s)}$) et la wustite (FeO_x(s) avec $x \approx 1$, composé non stœchimétrique mais que l'on assimilera à FeO(s) dans la suite de cet exercice). Les enthalpies et entropies standards de formation de ces trois oxydes sont données dans le tableau ci-dessous. Quelle

[4] International Energy Agency (2022) Iron and Steel Technology Roadmap, Towards more sustainable steelmaking. Energy Technology Perspectives series. https://www.iea.org/reports/iron-and-steel-technology-roadmap

forme du fer est il le plus probable de trouver dans des gisements de minerai exposés à l'atmosphère ($T \approx 15$°C et teneur en O_2 de 21%) ?

composé	$\Delta_f H^0$ (kJ/mol)	$\Delta_f S^0$ (J.mol^{-1}.K^{-1})
FeO(s)	-265.9	-65.8
Fe_3O_4(s)	-1121.1	-329.3
Fe_2O_3(s)	-824.4	-263.1

2. En considérant les différentes réactions d'oxydation possibles par le dioxygène gazeux (O_2(g)) représenter graphiquement un diagramme de stabilité des différentes formes du fer en fonction de la température T suivant un axe et de la pression en oxygène suivant un autre axe exprimée sous forme d'enthalpie libre (axe $RT \ln\left(\frac{P_{O_2}}{P_0}\right)$ avec $P_0 = 1$ bar la pression de l'état standard), aussi appelé diagramme d'Ellingham. Pourquoi la réduction des oxydes de fer par simple chauffage est-elle compliquée ?

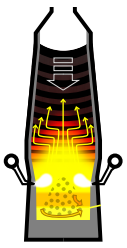
3. La métallurgie du fer qui a débutée dans l'antiquité s'est appuyée sur la réduction par le carbone. Du charbon pyrolysé (coke) était ajouté au minerai de fer pour parvenir à réduire les oxydes en métal. Le coke peut être vue grossièrement comme du graphite solide (C(s)). En présence de dioxygène celui-ci est oxydé par combustion en CO(g) ou CO_2(g). Le tableau ci-dessous donne les enthalpies et entropies standards de formation de ces deux composés. Superposer au diagramme d'Ellingham, un diagramme analogue pour le carbone à la place du fer. Pourquoi la présence de coke facilite-t-elle la réduction des oxydes de fer ? Quelles sont les réactions de réduction en jeu ? Discuter les entropies de formation des différents composés et pourquoi cela joue-t-il un rôle essentiel.

composé	$\Delta_f H^0$ (kJ/mol)	$\Delta_f S^0$ (J.mol^{-1}.K^{-1})
CO_2(g)	-394.9	0
CO(g)	-112.4	88.8

4. On considère des composés ferreux soumis à un gaz ne contenant que du CO(g) et du CO_2(g).
   (a) Pour les différentes réactions de réduction des oxydes de fer, quelle est la relation entre l'enthalpie libre et le rapport des pressions partielles de CO(g) et de CO_2(g).
   (b) En déduire un diagramme de stabilité des différentes phases en fonction de la température T et de la fraction molaire de CO(g) $x_{CO} = \frac{P_{CO}}{P_{CO}+P_{CO_2}}$. Quelles conditions doivent être satisfaites pour la réduction complète en fer métallique ?
   (c) On ajoute à ce système du coke. Comment cela affecte-il l'équilibre entre CO(g) et du CO_2(g) ? Pour quelles températures se produisent les différentes réductions des oxydes de fer ? On considèrera le gaz à une pression totale égale à la pression de l'état standard $P_{CO} + P_{CO_2} = P_0$.

5. On s'intéresse à présent au processus mis en œuvre dans un haut fourneau.

   La figure ci-contre (CC BY-SA 3.0, wikipedia) représente schématiquement un haut fourneau. Le gaz circule à contre courant du minerai (minerai du haut vers le bas, gaz du bas vers le haut) et la température baisse progressivement de bas en haut. A l'intérieur du four, le gaz passe alternativement par des lits de coke et des lits de minerai. Décrire grossièrement l'évolution de l'état du gaz dans le système et les réactions de réduction qu'il induit au fur et à mesure qu'il monte.



6. Faire le bilan en terme de chaleur produite/consommée par le haut fourneau. En déduire la quantité de dioxyde de carbone théorique minimum émise pour la réduction du fer. Comparer à la moyenne estimée pour la production d'acier primaire : 1800 kg CO_2/t (International Energy Agency).

7. Afin de décarboner la production d'acier, il est envisagé une réduction des oxydes de fer par du dihydrogène ($H_2(g)$) qui produit alors de l'eau ($H_2O(g)$). Le tableau ci-dessous donne l'enthalpie et l'entropie standard de formation de l'eau. Procéder comme à la question 4 pour déterminer quelles conditions doivent être satisfaites en terme de température et de fraction molaire de $H_2(g)$ pour la réduction complète du fer. Qu'en est-il du bilan de chaleur? Commenter

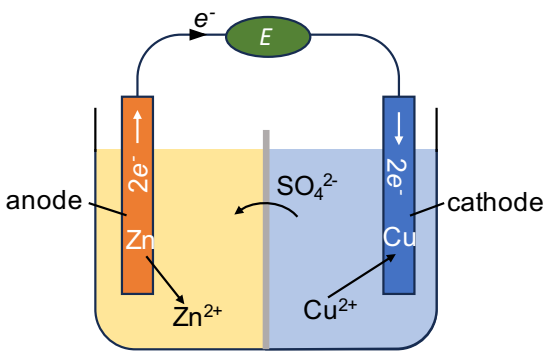
composé	$\Delta_f H^0$ (kJ/mol)	$\Delta_f S^0$ (J.mol^{-1}.K^{-1})
$H_2O(g)$	-249.0	-57.2

Ex. 5.2 - Batteries Li-ion

Apparues dans les années 1970-1980, et objet du prix Nobel de chimie en 2019, les batteries au lithium sont aujourd'hui incontournables notamment dans les applications mobiles et en particulier pour l'électro-mobilité. En 2023, la production annuelle de batteries au lithium est de l'ordre de 50 millions de tonnes. Cet exercice aborde le fonctionnement des batteries permettant de comprendre pourquoi les batteries au lithium sont devenues incontournables.

1. Les batteries électrochimiques produisent de l'énergie électrique en réalisant une réaction d'oxydation (perte d'un électron) séparément d'une réaction de réduction (capture d'un électron). Un cas d'école est celui d'une batterie cuivre-zinc (cellule de Daniel), illustrée ci-dessous:

   - Une électrode en zinc Zn est baignée dans une solution d'ions Zn^{2+} et s'y produit la réaction d'oxydation : $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$ (où e^- symbolise un électron libéré dans le circuit électrique).
   - Une électrode en cuivre Cu est baignée dans une solution d'ions Cu^{2+} et s'y produit la réaction de réduction : $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$ (où l'électron provient du circuit électrique).
   - Afin de garantir l'électro-neutralité des solutions de Zn^{2+} et Cu^{2+}, une membrane semi-perméable permet le transfert de l'anion compensateur (sulfate SO_4^{2-}) entre les deux solutions, tout en empêchant le contact électrique entre les deux électrodes.



   La réaction totale $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$ est une réaction d'oxydoréduction dont l'enthalpie libre de réaction est $\Delta_r G^0 = -212$ kJ/mol pour des conditions standards (température et pression ambiante, concentrations en Zn^{2+} et Cu^{2+} de 1 mol/L).

   A supposer que la batterie est dans les conditions standards, quel potentiel électrique E^0 apparait entre les deux électrodes ? On rappelle que la charge élémentaire d'un électron est $e = 1.6 \times 10^{-19}$ C, et le nombre d'Avogadro vaut $N_A = 6.02 \times 10^{23}$.

2. Afin de comparer les choix possibles de chimie d'une batterie, les potentiels de réduction d'un grand nombre de réactions de réduction sont tabulés (cf. tableau ci-dessous). Le potentiel de

réduction est défini comme le potentiel qui apparait dans une batterie en conditions standards telle que la réaction d'oxydation à l'anode est celle de l'hydrogène ($H_2(g) \rightarrow 2H^+ + 2e^-$) et la réaction à la cathode est la réaction de réduction considérée. Ainsi, la réaction de réduction de l'hydrogène constitue la référence (potentiel de réduction nul). Un potentiel de réduction positif signifie que la réaction de réduction considérée est énergétiquement plus favorable que celle de l'hydrogène, et inversement pour un potentiel négatif.

Retrouver à partir des données du tableau le potentiel électrique de la batterie cuivre-zinc établi à la question précédente. Pourquoi l'oxydation du lithium solide ($Li(s) \rightarrow Li^+ + e^-$) est elle un bon choix pour la réaction à l'anode ?

Réaction de réduction	E^0 (V)
$Au^+ + e^- \rightarrow Au(s)$	1.83
$Pt^{2+} + 2e^- \rightarrow Pt(s)$	1.188
$Hg^{2+} + 2e^- \rightarrow Hg(l)$	0.85
$Ag^+ + e^- \rightarrow Ag(s)$	0.7996
$Cu^{2+} + 2e^- \rightarrow Cu(s)$	0.337
$2H^+ + 2e^- \rightarrow H_2(g)$	0

Réaction de réduction	E^0 (V)
$Fe^{2+} + 2e^- \rightarrow Fe(s)$	-0.44
$Zn^{2+} + 2e^- \rightarrow Zn(s)$	-0.7618
$Al^{3+} + 3e^- \rightarrow Al(s)$	-1.662
$Mg^{2+} + 2e^- \rightarrow Mg(s)$	-2.372
$Na^+ + e^- \rightarrow Na(s)$	-2.71
$Li^+ + e^- \rightarrow Li(s)$	-3.0401

3. Pour que la batterie soit rechargeable, les processus d'oxydation et de réduction doivent être réversibles lorsque le potentiel appliqué aux électrodes est inversé. Par 'processus', il faut comprendre que la réaction elle-même doit être identique et inverse, mais également que la structure solide des électrodes doit revenir à son état avant décharge, et cela pour un grand nombre de cycles. La première batterie rechargeable qui fut inventée est la batterie au plomb, qui a notamment équipée les voitures pendant l'essentiel du 20$^{\text{ème}}$ siècle. La réaction d'oxydation à l'anode est :

$$Pb(s) + H_2SO_4(aq) \rightarrow PbSO_4(s) + 2H^+(aq) + 2e^- \qquad E^0_{\text{anode}} = -0.37V$$

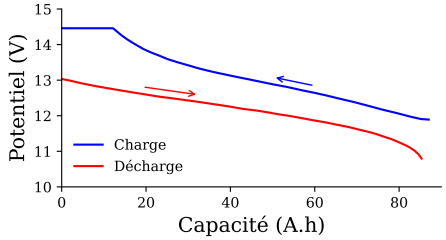
où $Pb(s)$ représente le plomb, $H_2SO_4(aq)$ de l'acide sulfurique en solution, $H^+(aq)$ l'ion hydrogène en solution, et $PbSO_4(s)$ du sulfate de plomb. La réaction de réduction à la cathode est :

$$PbO_2(s) + H_2SO_4(aq) + 2H^+(aq) + 2e^- \rightarrow PbSO_4(s) + 2H_2O(aq) \qquad E^0_{\text{cathode}} = 1.68V$$

où $PbO_2(s)$ représente l'oxyde de plomb, et $H_2O(aq)$ l'eau liquide.

Quel est le potentiel électrique d'une cellule électrochimique au plomb ? Comment fait-on pour réaliser une batterie standard de voiture de 12 V? Au cours de la décharge, comment évolue la composition de la solution aqueuse? Quel est l'état complètement déchargé ?

4. La courbe ci-dessous est la courbe dite de charge-décharge d'une batterie au plomb 12 V typique utilisée pour des voitures. La capacité en abscisse, représente l'intégrale de l'intensité du courant fournit par la batterie en fonction du temps, exprimé en Ampère.heure (A.h). Pour rappel, un ampère correspond au passage d'un coulomb par seconde (1 A = 1 C/s). Quel est le lien entre cette courbe et l'énergie électrique fournie à la décharge ou consommée à la charge ? Quelle est approximativement l'énergie disponible dans cette batterie à pleine charge ? Quel est approximativement la part d'énergie perdue lors d'un cycle complet de charge-décharge ?



5. L'enthalpie libre de réaction standard $\Delta_r G^0$ suppose des conditions standards, à savoir notamment des concentrations de solutés de 1 mol/L. Soit une réaction

$$\sum_{i \in \text{réactifs}} \alpha_i \mathrm{A}_i \rightarrow \sum_{j \in \text{produits}} \beta_j \mathrm{B}_j$$

avec α_i (resp. β_j) le coefficient stœchiométrique associé à l'espèce chimique A_i (resp. B_j). Lorsque les conditions de la réaction évoluent (en particulier les concentrations), l'enthalpie libre évolue selon l'équation de Nernst :

$$\Delta_r G = \Delta_r G^0 - RT \sum_{i \in \text{réactifs}} \alpha_i \ln\left(a_{\mathrm{A}_i}\right) + RT \sum_{j \in \text{produits}} \beta_j \ln\left(a_{\mathrm{B}_j}\right)$$

où R est la constante des gaz parfaits, T la température, et a_{A_i} (resp. a_{B_j}) représente l'activité de l'espèce chimique A_i (resp. B_j). Pour une espèce en solution l'activité augmente avec la concentration, et dans la limite d'une solution idéale l'activité est égale à la concentration divisée par la concentration de l'état standard ($C_0 = 1$ mol/L). Pour une espèce solide comme pour un solvant l'activité est approximativement constante égale à 1.

Au regard de l'équation de Nernst, expliquer qualitativement l'allure de la courbe de charge-décharge de la batterie au plomb.

6. Un facteur déterminant pour les applications mobiles est le poids de la batterie. On appelle densité énergétique, le rapport entre l'énergie récupérable pendant la décharge et la masse de la batterie. En utilisant les masses molaires listées dans le tableau ci-dessous, quelle est la capacité que peut fournir 1 kg de batterie au plomb ? On exprimera le résultat en $\mathrm{A.h.kg^{-1}}$, et on négligera toutes les masses autres que les éléments intervenant dans les réactions électrochimiques. En considérant le potentiel électrique constant égal au potentiel standard E^0 à la décharge, déterminer la densité d'énergie théorique d'une batterie au plomb.

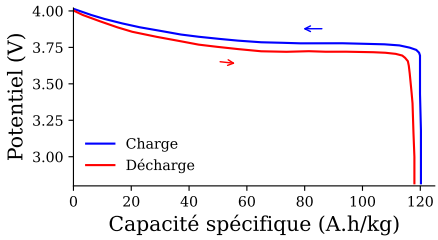
Élément	Hydrogène (H)	Oxygène (O)	Plomb (Pb)	Soufre (S)
Masse molaire (g/mol)	1	16	207	32

En réalité, le poids de la batterie ne se limite pas aux espèces impliquées dans les réactions électrochimiques et il faut tenir compte de l'électrolyte (2/3 d'eau pour 1/3 de $\mathrm{H_2SO_4(aq)}$), des électrodes, de la membrane de séparation, de la structure de la batterie etc. Par ailleurs, comme vue à la question 4, le potentiel électrique décroit significativement lors de la décharge. Sachant que cette batterie au plomb 12 V pèse 20 kg, quelle est approximativement la densité d'énergie réelle ?

7. La première batterie au lithium commercialisée dans les années 90 était une batterie de type LCO (lithium cobalt oxyde), dont les réactions d'oxydation et de réduction sont respectivement :

$$\mathrm{LiC_6(s) \rightarrow C_6(s) + Li^+(aq) + e^-} \qquad E^0 = -2.85\mathrm{V}$$
$$\mathrm{CoO_2(s) + Li^+(aq) + e^- \rightarrow LiCoO_2(s)} \qquad E^0 = 1.06\mathrm{V}$$

où $\mathrm{C_6}$ représente du graphite et $\mathrm{CoO_2}$ de l'oxyde de cobalt. La figure ci-dessous représente la courbe de charge-décharge d'une cellule LCO. Quelle différence qualitative notable apparait et s'explique par l'équation de Nernst ? Pourquoi une courbe de charge-décharge comme celle de la batterie LCO est-elle préférable ?



8. En procédant comme pour la batterie au plomb, déterminer la densité d'énergie théorique d'une batterie LCO à partir des masses molaires listées ci-dessous. La courbe de charge-décharge de la batterie LCO de la question précédente est une courbe par unité de masse de 'cathode + anode' (i.e., excluant la structure, le séparateur, l'électrolyte etc.). La masse réelle est environ 3 fois la masse de 'cathode + anode'. En déduire, la densité d'énergie réelle d'une batterie LCO. Commenter par rapport à la batterie au plomb.

Élément	Carbone (C)	Cobalt (Co)	Lithium (Li)	Oxygène (O)
Masse molaire (g/mol)	12	59	7	16

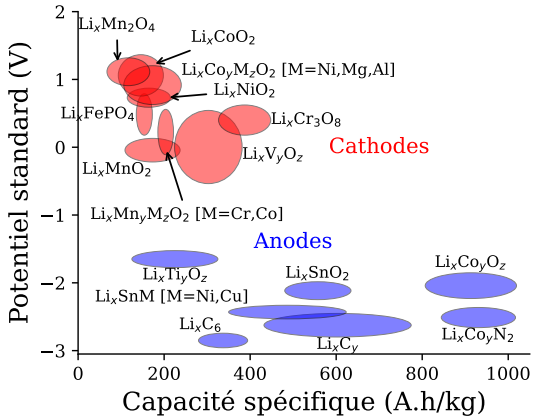
9. Le graphique ci-dessous présente les capacités spécifiques (capacité par unité de masse) et potentiels électriques réels pour divers type d'électrodes de batteries Li-ion (seule la masse de l'électrode, anode ou cathode, est prise en compte dans ce graphique). Les anodes sont toutes des réactions d'oxydation du type :

$$\mathrm{LiM_{anode}(s) \rightarrow M_{anode}(s) + Li^+(aq) + e^-}$$

où $\mathrm{M_{anode}}$ est le matériau de l'anode (e.g., graphite $\mathrm{C_6}$ pour les batterie LCO). Et les cathodes sont toutes des réactions de réduction du type :

$$\mathrm{M_{cathode}(s) + Li^+(aq) + e^- \rightarrow LiM_{cathode}(s)}$$

où $\mathrm{M_{cathode}}$ est le matériau de la cathode (e.g., oxyde de cobalt $\mathrm{CoO_2}$ pour les batteries LCO). En utilisant le graphique, estimer la densité d'énergie réelle d'une batterie lithium de type LCO. Est-ce cohérent avec le résultat de la question précédente ? A partir du graphique, estimer approximativement la densité d'énergie maximum de la meilleure combinaison possible anode/cathode. Les avancées actuelles (batteries dites Li-solides) permettent d'envisager des anodes au Lithium solide pur Li(s) dont la capacité spécifique et le potentiel standard seraient proches de la capacité théorique estimée à partir de la masse molaire, et du potentiel standard du lithium solide. Quel serait le progrès en terme de densité énergétique ?



10. Ces densités d'énergie est ce qui se fait de mieux dans le domaine des batteries. L'hystérésis observable sur les courbes de charge-décharge ainsi que les pertes en chaleur correspondent à une efficacité du stockage électrochimique de l'ordre de 85%. Par comparaison, pour une application à l'automobile, le pouvoir calorifique de l'essence/diesel est de 43 MJ/kg et le rendement d'un moteur thermique de l'ordre de 35%. Quelle quantité de batterie est équivalente à un réservoir de 50L d'essence/diesel ? On considèrera une densité de l'essence/diesel de 0.8 kg/L, et on considèrera que le moteur électrique à un rendement de 100%.

Chapter 6

Exercices supplémentaires

Contents

6.1 Rayonnement 153
Ex. 6.1 - Bilan radiatif simplifié de la Terre 153
Ex. 6.2 - Rayonnement de la Lune 154
6.2 Énergie nucléaire 155
Ex. 6.3 - Décrément logarithmique d'énergie 155
Ex. 6.4 - Produits de fission 156
Ex. 6.5 - Surgénérateur à neutrons rapides 159
6.3 Énergie du mouvement des fluides 161
Ex. 6.6 - Limite de Betz : prise en compte de la rotation de sillage 161
Ex. 6.7 - Panémone 162

6.1 Rayonnement

Ex. 6.1 - Bilan radiatif simplifié de la Terre

Cet exercice développe un modèle rudimentaire du bilan radiatif de la Terre permettant d'estimer grossièrement comment la température de la Terre est influencée par le phénomène d'effet de serre.

On considère la Terre et le soleil comme deux corps noirs sphériques. La puissance rayonnée par un corps noir est prédite par la loi de Stefan-Botlzmann, résultant de l'intégration du spectre d'émission donné par la loi de Planck : $\mathcal{P} = A\sigma T^4$ avec A la surface du corps, $\sigma = \frac{2\pi^5 k^4}{5h^3c^2}$ la constante de Stefan-Boltzmann, et T la température.

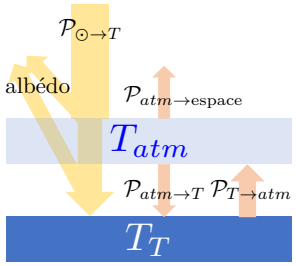
1. Déterminer la puissance du rayonnement solaire reçue par la Terre en fonction de la distance moyenne Terre-Soleil R_{orbite}, de la température de surface du soleil est $T_\odot$, du rayon du soleil $R_\odot$, et du rayon de la Terre R_T.

2. A l'équilibre, la puissance reçue par rayonnement du soleil doit être égale à celle émise par la Terre. A supposer que la Terre absorbe tout le rayonnement reçu (corps noir parfait), qu'elle est dénuée d'atmosphère, et que sa température de surface est uniforme, quelle serait sa température de surface à l'équilibre? Données numériques $R_{orbite} = 1.5 \times 10^{11}$ m, $T_\odot = 5780$ K, et $R_\odot = 7 \times 10^8$ m.

3. La surface de la Terre réfléchie en réalité 16% du rayonnement reçu. En tenant compte de cet albédo $a = 0.16$, quel serait la température de la Terre (toujours sans atmosphère) ?

4. On tient compte à présent d'une couche d'atmosphère qui serait transparente aux longueurs d'ondes visibles (rayonnement reçu du soleil), mais opaque aux longueurs d'ondes infrarouge (rayonnement émis par la Terre). Cette couche est supposée de température uniforme T_{atm}

et rayonne selon la loi de Stefan-Boltzmann à la fois au niveau de sa surface inférieure vers la Terre, et au niveau de sa surface supérieure vers l'espace. Par ailleurs, la présence de nuages dans cette atmosphère augmente significativement l'albédo total ($a = 0.3$). Déterminer les bilans radiatifs à l'équilibre de la Terre et de l'atmosphère et en déduire leur températures respectives.



5. On considère à présent n couches d'atmosphère successives de même nature (transparentes au visible, opaques aux infrarouges). Montrer que la température de surface de la Terre est alors:

$$T_T = \sqrt[4]{\frac{(n+1)\,(1-a)\,R_\odot^2}{4R_{orbite}^2}}T_\odot \approx \sqrt[4]{n+1} \times 255\text{K}$$

Ex. 6.2 - Rayonnement de la Lune

On s'intéresse dans cet exercice au rayonnement de la Lune et son exploitation par un panneau solaire. La lumière que nous recevons de la Lune est la lumière du soleil qui est réfléchie, si bien que son spectre est similaire à celui du soleil, mais en moyenne $\kappa = 500000$ fois moins intense. Le rayonnement issu de la surface de la lune est donc celui d'un corps noir à la température du Soleil $T_\odot$, modulé par un facteur $1/\kappa$, soit une densité spectrale énergétique $\frac{B_{\hbar\omega}(T_\odot)}{\kappa} = \frac{1}{\kappa}\frac{(\hbar\omega)^2}{4\pi^2\hbar^3c^2}\left(\exp\left(\frac{\hbar\omega}{k_B T_\odot}\right)-1\right)^{-1}$ (mêmes notations qu'à l'exercice précédent). On s'intéresse à un panneau solaire thermique recevant le rayonnement d'une pleine Lune (rayon R_L et située à une distance D_{TL} de la Terre).

Données numériques :

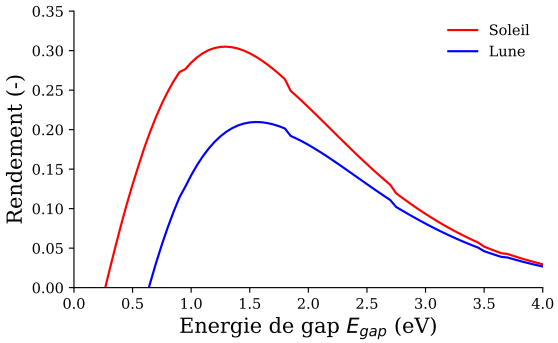
- Température de surface du soleil : $T_\odot = 5800\text{K}$
- Rayon de la Lune : $R_L = 1737$ km
- Distance Terre Lune : $D_{TL} = 384400$ km
- Constante de Stefan-Boltzmann : $\sigma = \frac{\pi^4}{15}\frac{k^4}{4\pi^2\hbar^3c^2} = 5.67$ W.m^{-2}.K^{-4}

1. Estimer la puissance lumineuse par unité surface reçue par le panneau solaire, en condition idéale (orthogonal au rayonnement, pas d'atmosphère).

2. En considérant un panneau solaire thermique de type corps noir, déterminer la puissance du rayonnement qu'il émet en fonction de sa température T_p et en déduire la température d'équilibre, en négligeant toute autre échange d'énergie.

3. Un système de concentration (e.g., miroir parabolique) permet augmenter la puissance reçue. Avec un tel système, l'angle θ_C avec lequel la Lune est vue est significativement plus grand que θ_L. Le facteur de concentration $C = \left(\frac{\sin(\theta_C)}{\sin(\theta_L)}\right)^2$ caractérise le gain du système. Exprimer la puissance reçue en fonction de C et θ_S. Quel est le facteur de concentration maximum ? En déduire la puissance reçue puis la température d'équilibre maximum du panneau avec un système de concentration.

4. On considère à présent un panneau photovoltaïque. Le graphique ci-dessous compare la limite de Shockley-Queisser pour un rayonnement provenant de la Lune à celui d'un rayonnement provenant du Soleil. Pourquoi cette limite dépend-elle de la source du rayonnement ? Et pourquoi le rendement est-il plus mauvais dans le cas de la Lune ? Un panneau réel en silicium (gap d'énergie de 1.124 eV) parvient à un rendement de 25% sous rayonnement du Soleil. A quel rendement peut-on s'attendre lorsqu'il est éclairé par la Lune, et quelle quantité d'énergie électrique peut-on espérer collecter en une nuit de 12 h avec un panneau de 1 m^2 ?



6.2 Énergie nucléaire

Ex. 6.3 - Décrément logarithmique d'énergie

On s'intéresse dans cet exercice au calcul du décrément logarithmique d'énergies dues aux collisions des neutrons avec les noyaux environnant lors de la phase de modération dans un réacteur nucléaire. Le processus en jeu est la diffusion élastique, qui correspond à des collisions entre particules, telles que le moment cinétique total et l'énergie cinétique totale sont conservés. On considère une collision élastique entre une particule de masse m_1 approchant à vitesse $\mathbf{V}_1$ d'une seconde particule de masse m_2 au repos ($\mathbf{V}_2 = 0$). A l'issue de la collision, les deux particules ont des vitesses $\mathbf{V}_1'$ et $\mathbf{V}_2'$. On note $\mathbf{V}_{CM}$ la vitesse du centre de masse des deux particules ($(m_1 + m_2)\,\mathbf{V}_{CM} = m_1\mathbf{V}_1 + m_2\mathbf{V}_2$) et on note $\mathbf{v}_1 = \mathbf{V}_1 - \mathbf{V}_{CM}$ et $\mathbf{v}_2 = \mathbf{V}_2 - \mathbf{V}_{CM}$ les vitesses relatives par rapport à ce centre de masse.

1. Exprimer $\mathbf{V}_1$ et $\mathbf{V}_{CM}$ en fonction de $\mathbf{v}_1$ et des masses m_1 et m_2.
2. Comment évolue $\mathbf{V}_{CM}$ lors de la collision ? En déduire que les vitesse relatives $\mathbf{v}_1'$ et $\mathbf{v}_2'$ par rapport au centre de masse après collision sont colinéaires, puis exprimer $\mathbf{V}_1'$ et $\mathbf{V}_2'$ en fonction de $\mathbf{v}_1'$, $\mathbf{V}_{CM}$, et des masses m_1 et m_2.
3. En appliquant la conservation de l'énergie cinétique, montrer que $|\mathbf{v}_1| = |\mathbf{v}_1'|$.
4. On note θ l'angle entre la direction incidente de la particule 1 relative au centre de masse (direction de $\mathbf{v}_1$) et la direction relative résultante (direction de $\mathbf{v}_1'$). Montrer que les énergies cinétiques E_1 et E_1' de la particule 1 avant et après collision vérifient :

$$\frac{E_1'}{E_1} = \frac{m_1^2 + m_2^2 + 2m_1 m_2 \cos(\theta)}{(m_1 + m_2)^2}$$

5. Dans le cas d'un neutron de nombre de masse 1 et d'un noyau de nombre de masse A, le rapport des énergies cinétiques avant et après collision est donc approximativement :

$$\frac{E'}{E} \simeq \frac{1 + A^2 + 2A\cos(\theta)}{(1 + A)^2}$$

On définit le décrément logarithmique d'énergie comme $\xi(A) = \left\langle -\ln\left(\frac{E'}{E}\right)\right\rangle$, où $\langle\cdot\rangle$ représente la moyenne sur les directions de diffusion θ.

En supposant la diffusion équiprobable dans toutes les directions $\theta \in [0, \pi]$, exprimer $\xi(A)$ comme une intégrale. On se placera en coordonnées sphériques avec θ l'angle polaire.

6. En déduire l'expression suivante du décrément logarithmique d'énergie :

$$\xi(A) = 1 + \frac{(A-1)^2}{2A} \ln\left(\frac{A-1}{A+1}\right)$$

On pourra utiliser l'intégrale suivante : $\int_0^\pi \ln\left(1 + A^2 + 2A\cos(\theta)\right) \sin(\theta)\, d\theta = \frac{(A+1)^2}{A} \ln(A+1) - \frac{(A-1)^2}{A} \ln(A-1) - 2$.

7. Faire un tracé approximatif du décrément logarithmique en fonction du nombre de masse A. Commenter.

Ex. 6.4 - Produits de fission

Dans cet exercice, nous nous intéressons aux produits de la fission induite par l'uranium 235. Les deux tableaux ci-dessous fournissent les données nécessaires à l'exercice. Chaque nucléide, caractérisé par son nombre de charge Z (ligne) et son nombre de masse A (colonne), correspond à une case. En rouge est donné l'excès de masse Δ, en vert le(s) mode(s) de décroissance, et en bleu le temps de demi-vie τ (sauf si le nucléide est stable). Les modes de décroissance possibles sont les décroissances β^- et β^+, la décroissance α, et la capture électronique ε. Parfois la décroissance peut s'accompagner d'une émission de neutron notée (+ n X %), où X % représente la probabilité qu'une décroissance émette un neutron. On précise enfin l'excès de masse de l'${}^{235}_{92}\mathrm{U}$ et du neutron : $\Delta\left({}^{235}_{92}\mathrm{U}\right) = 40.92$ MeV et $\Delta(n) = 8.07$ MeV.

Z\A	87	88	89	90	91	92	93	94	95
35 (Br)	−73.89 MeV β^- (+ n 2.6 %) 55.68 s	−70.72 MeV β^- (+ n 6.62 %) 16.34 s	−68.27 MeV β^- (+ n 13.5 %) 4.348 s	−64 MeV β^- (+ n 25.3 %) 1.91 s	−61.11 MeV β^- (+ n 29.8 %) 0.544 s				
36 (Kr)	−80.71 MeV β^- 4565.4 s	−79.69 MeV β^- 10090.8 s	−76.54 MeV β^- 188.82 s	−74.96 MeV β^- 32.32 s	−70.97 MeV β^- 8.57 s	−68.77 MeV β^- (+ n 0.0332 %) 1.841 s			
37 (Rb)	−84.6 MeV β^- $1.567.10^{18}$ s	−82.61 MeV β^- 1066.5 s	−81.71 MeV β^- 923.4 s	−79.37 MeV β^- 158 s	−77.75 MeV β^- 58 s	−74.77 MeV β^- (+ n 0.0109 %) 4.48 s	−72.62 MeV β^- (+ n 1.46 %) 5.85 s		
38 (Sr)		−87.92 MeV stable	−86.21 MeV β^- $4.368.10^6$ s	−85.95 MeV β^- $9.123.10^8$ s	−83.65 MeV β^- 34848 s	−82.87 MeV β^- 9396 s	−80.09 MeV β^- 445.8 s	−78.85 MeV β^- 75.3 s	
39 (Y)			−87.71 MeV stable	−86.5 MeV β^- $2.306.10^5$ s	−86.35 MeV β^- $5.060.10^6$ s	−84.82 MeV β^- 12744 s	−84.23 MeV β^- 36612 s	−82.35 MeV β^- 1122 s	−81.21 MeV β^- 624 s
40 (Zr)				−88.77 MeV stable	−87.9 MeV stable	−88.46 MeV stable	−87.12 MeV β^- $5.083.10^{13}$ s	−87.27 MeV stable	−85.66 MeV β^- $5.532.10^6$ s
41 (Nb)					−86.64 MeV $\varepsilon + \beta^+$ $2.146.10^{10}$ s	−86.45 MeV $\varepsilon + \beta^+$ $1.095.10^{15}$ s	−87.21 MeV stable	−86.37 MeV β^- $6.432.10^{11}$ s	−86.79 MeV β^- $3.023.10^6$ s
42 (Mo)						−86.81 MeV stable	−86.81 MeV ε $1.262.10^{11}$ s	−88.41 MeV stable	−87.71 MeV stable
43 (Tc)							−83.61 MeV $\varepsilon + \beta^+$ 10008 s	−84.16 MeV $\varepsilon + \beta^+$ 17580 s	−86.02 MeV $\varepsilon + \beta^+$ 69328.8 s

Z\A	137	138	139	140	141	142	143	144	145	146
51 (Sb)	−60.06 MeV β^- (+ n 49 %) 0.506 s	−54.65 MeV β^- (+ n 72 %) 0.314 s	−50.05 MeV β^- (+ n 90 %) 0.182 s	−44.39 MeV β^- (+ n 23 %) 0.171 s						
52 (Te)	−69.3 MeV β^- (+ n 2.77 %) 2.48 s	−65.7 MeV β^- (+ n 6.3 %) 1.46 s	−60.21 MeV β^- 0.724 s	−56.37 MeV β^- 0.351 s	−50.67 MeV β^- 0.193 s					
53 (I)	−76.36 MeV β^- (+ n 7.32 %) 24.13 s	−71.98 MeV β^- (+ n 5.39 %) 6.303 s	−68.47 MeV β^- (+ n 9.64 %) 2.29 s	−63.61 MeV β^- (+ n 8.79 %) 0.38 s	−59.93 MeV β^- (+ n 21.2 %) 0.418 s	−54.8 MeV β^- 0.235 s				
54 (Xe)	−82.38 MeV β^- 229.26 s	−79.97 MeV β^- 848.4 s	−75.64 MeV β^- 39.72 s	−72.99 MeV β^- 13.6 s	−68.2 MeV β^- (+ n 0.044 %) 1.72 s	−65.23 MeV β^- (+ n 0.355 %) 1.227 s	−60.2 MeV β^- (+ n 1 %) 0.511 s	−56.87 MeV β^- (+ n 3 %) 0.388 s		
55 (Cs)	−86.55 MeV β^- $9.469.10^8$ s	−82.89 MeV β^- 1948.8 s	−80.7 MeV β^- 555 s	−77.05 MeV β^- 64 s	−74.48 MeV β^- (+ n 0.036 %) 24.86 s	−70.51 MeV β^- (+ n 0.0912 %) 1.687 s	−67.68 MeV β^- (+ n 1.62 %) 1.795 s	−63.27 MeV β^- (+ n 2.98 %) 0.993 s	−60.05 MeV β^- (+ n 13.01 %) 0.588 s	
56 (Ba)		−88.26 MeV stable	−84.91 MeV β^- 4973.4 s	−83.27 MeV β^- $1.102.10^6$ s	−79.73 MeV β^- 1094.4 s	−77.84 MeV β^- 630.6 s	−73.94 MeV β^- 14.38 s	−71.77 MeV β^- 11.7 s	−67.52 MeV β^- 3.99 s	−64.87 MeV β^- 2.16 s
57 (La)			−87.22 MeV stable	−84.31 MeV β^- $1.450.10^5$ s	−82.93 MeV β^- 14148 s	−80.02 MeV β^- 5508 s	−78.17 MeV β^- 845.4 s	−74.85 MeV β^- 40.8 s	−72.84 MeV β^- 24.8 s	−69.22 MeV β^- 6.1 s
58 (Ce)				−88.07 MeV stable	−85.43 MeV β^- $2.808.10^6$ s	−84.53 MeV stable	−81.61 MeV β^- $1.189.10^5$ s	−80.43 MeV β^- $2.461.10^7$ s	−77.07 MeV β^- 181.2 s	−75.63 MeV β^- 793.2 s
59 (Pr)						−83.79 MeV β^- 99.9836 % ε 0.0164 % 68832 s	−83.07 MeV β^- $1.172.10^6$ s	−80.75 MeV β^- 1036.8 s	−79.63 MeV β^- 21535.2 s	−76.67 MeV β^- 1445.4 s
60 (Nd)							−83.07 MeV stable	−83.75 MeV α $7.219.10^{22}$ s	−81.43 MeV stable	−80.93 MeV stable

1. Rappeler ce qu'est une réaction de fission induite. On s'intéresse aux produits de réaction de la fission induite de $^{235}_{92}$U. Ceux-ci sont aléatoires. Quel est le second nucléide qui se forme lorsque l'on produit :

   - $^{90}_{36}$Kr et 2 neutrons
   - $^{90}_{36}$Kr et 3 neutrons
   - $^{90}_{35}$Br et 3 neutrons
   - $^{94}_{40}$Zr et 2 neutrons

2. On s'intéresse plus spécialement au premier cas ($^{90}_{36}$Kr et 2 neutrons). Calculer l'énergie libérée immédiatement par la réaction

3. $^{90}_{36}$Kr et le second nucléide produit ne sont pas stables. Déterminer leurs chaînes de décroissance respectives, et les temps de demi-vie associés à chaque étape. Ces chaînes de décroissance ont-elles le temps de se produire à l'échelle d'une exploitation du combustible dans un réacteur (3 ans) ?

4. Déterminer l'énergie supplémentaire fournie par la partie des deux chaînes de décroissance qui a lieu pendant l'exploitation dans le réacteur. En déduire l'énergie totale (immédiate et différée) fournie par cette fission induite de $^{235}_{92}$U.

5. On s'intéresse maintenant au cas de la troisième réaction de la question 1 ($^{90}_{35}$Br et 3 neutrons). Qu'y a-t-il de particulier concernant la décroissance de $^{90}_{35}$Br et des autres isotopes de Br ? Quel est l'ordre de grandeurs des temps de demi-vie de ces isotopes ? (BONUS) Pourquoi cela permet-il de contrôler le fonctionnement d'un réacteur ?

6. Déterminer la chaîne de décroissance du $^{90}_{35}$Br.

7. Vérifier que le premier évènement de cette chaîne de décroissance fournit bien une énergie positive.

Ex. 6.5 - Surgénérateur à neutrons rapides

Dans cet exercice, on s'intéresse au principe de surgénération dans un réacteur à neutrons rapides. Le tableau ci-dessous liste les données nécessaires à l'exercice (https://www.nndc.bnl.gov) : sections efficaces de capture radiative σ_c et de fission induite σ_f en présence de neutrons lents (thermiques) ou rapides, nombres moyen de neutrons rapides produits par une réaction de fission (ν), et modes de décroissance radioactive et temps de demi-vie $\tau_{1/2}$ associés. On rappelle qu'un neutron lent (thermique) est un neutron à température ambiante d'énergie ~ 25.3 meV, et un neutron rapide est un neutron venant d'être produit par une réaction nucléaire d'énergie ~ 1 MeV. On rappelle également que 1 barn = 100 fm^2 = 10^{-28} m^2.

Nucléide	Neutrons lents (0.0253 eV)			Neutrons rapides (1 MeV)			Décroissance	
	σ_c (barns)	σ_f (barns)	ν (-)	σ_c (barns)	σ_f (barns)	ν (-)	Mode	$\tau_{1/2}$
$^{16}_{8}$O	1.698×10^{-4}	0	-	2.910×10^{-5}	0	-	stable	-
$^{235}_{92}$U	98.79	586.7	2.43	0.1084	1.204	2.53	α	7.04×10^8 ans
$^{236}_{92}$U	5.134	4.711×10^{-2}	2.37	0.1747	0.3580	2.49	α	2.342×10^7 ans
$^{238}_{92}$U	2.684	1.687×10^{-5}	2.44	0.1278	1.355×10^{-2}	2.53	α	4.468×10^9 ans
$^{239}_{92}$U	22.53	14.23	-	9.817×10^{-2}	0.3782	-	β^-	23.45 min
$^{239}_{93}$Np	45.01	2.801×10^{-2}	2.72	0.1249	0.278382	2.85	β^-	2.3565 jours
$^{239}_{94}$Pu	270.1	747.4	2.87	4.210×10^{-2}	1.7354	3.01	α	24 110 ans

1. On s'intéresse à de l'uranium UO_2 enrichi à $x = 3.5$ %, de densité numérique en uranium $n = 2.44 \times 10^{28}$ m^{-3} et soumis à un flux de neutrons.

   (a) Quelle est l'expression du rapport de reproduction η, c'est à dire le nombre de neutrons produits par fission rapporté au nombre de neutrons consommés par capture radiative ou fission. Pourquoi peut-on négliger l'oxygène ?

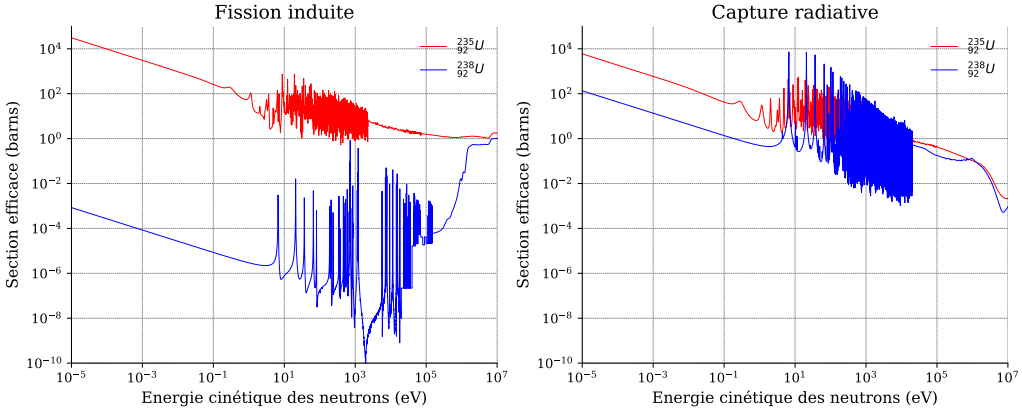
(b) Faire l'application numérique pour un flux incident de neutrons lents et pour un flux incident de neutrons rapides. Discuter les conséquences pour le fonctionnement du réacteur.

(c) Quelle fraction des fissions, en fonction de x, sont dues à $^{235}_{92}$U et $^{238}_{92}$U, respectivement ? Faire l'application numérique pour des neutrons lents et pour des neutrons rapides.

(d) On rappelle que la section efficace macroscopique pour une réaction r et un nucléide i est définie par $\Sigma_r(i) = n(i)\,\sigma_r(i)$, avec $n(i)$ la densité numérique du nucléide i et $\sigma_r(i)$ la section efficace pour la réaction r avec un nucléide i. $\Sigma_r(i)$ représente la probabilité, par unité de longueur traversée par un neutron, que celui-ci produise la réaction r avec un noyau de i. Quelle est la relation entre les sections efficaces macroscopiques et le libre parcours moyen d'un neutron (on ne tient pas compte ici du processus de diffusion) ? Faire l'application numérique pour des neutrons lents et pour des neutrons rapides.

(e) Dans un réacteur nucléaire usuel, les barres de combustible, d'un diamètre de l'ordre de quelques cm, sont placées dans un modérateur (e.g., eau). Pourquoi la fission par neutrons rapides est-elle négligeable ?

2. Le graphique ci-dessous représente les sections efficaces de $^{235}_{92}$U et $^{238}_{92}$U en fonction de l'énergie des neutrons.



(a) Au cours de la modération, l'énergie des neutrons passe progressivement de ~ 1 MeV à ~ 25.3 meV. Dans cette gamme d'énergie, quels sont les deux réactions nucléaires les plus probables après la fission de $^{235}_{92}$U ?

(b) Après décroissance radioactive à l'échelle de quelques jours, quels sont les nucléides issus de ces deux réactions ? Montrer en particulier que l'on aboutit à du $^{239}_{94}$Pu. Que peut-on attendre de la présence de $^{239}_{94}$Pu ?

(c) Le rapport de conversion CR correspond au rapport entre la masse de noyaux fissiles produits et la masse de noyaux fissiles consommés. Un réacteur à neutrons lents a un rapport moyen $CR = 0.6$. Pour un combustible contenant initialement 2.1 t de $^{235}_{92}$U, après 3 ans de fonctionnement, le combustible contient 700 kg de $^{235}_{92}$U et 600 kg de $^{239}_{94}$Pu. Quelle quantité totale de $^{239}_{94}$Pu a été produite ? Quelle quantité de ce $^{239}_{94}$Pu a été consommée ?

(d) Sachant que les fissions des noyaux de $^{235}_{92}$U et de $^{239}_{94}$Pu libèrent des énergies similaires, quelle fraction approximative de l'énergie du réacteur de la question précédente provient de la fission de $^{239}_{94}$Pu ?

3. On considère à présent de l'uranium appauvri contenant une fraction $1 - x$ de $^{238}_{92}$U et un fraction x de $^{239}_{94}$Pu. On néglige dans la suite la fission de l'$^{238}_{92}$U.

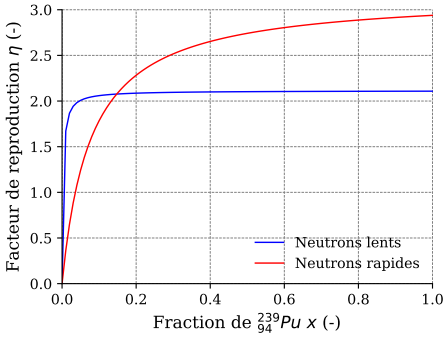
(a) La surgénération est la capacité d'un réacteur à produire plus de noyaux fissiles qu'il n'en consomme ($CR > 1$). Pour ce faire, il faut que la probabilité de capture radiative par le $^{238}_{92}$U excède la probabilité de fission du $^{239}_{94}$Pu. Pour quelles valeurs de la fraction

x cette condition est-elle satisfaite ? Faire l'application numérique pour des neutrons lents et des neutrons rapides.

(b) Pour qu'une réaction en chaîne se produise, il faut par ailleurs que le nombre de neutrons produits excède le nombre de neutrons consommés. Montrer que cette condition correspond à :

$$\eta > 2 \text{ avec } \eta = \frac{x\sigma_f\left(^{239}_{94}\text{Pu}\right)\nu\left(^{239}_{94}\text{Pu}\right)}{(1-x)\,\sigma_c\left(^{238}_{92}\text{U}\right) + x\sigma_a\left(^{239}_{94}\text{Pu}\right)}$$

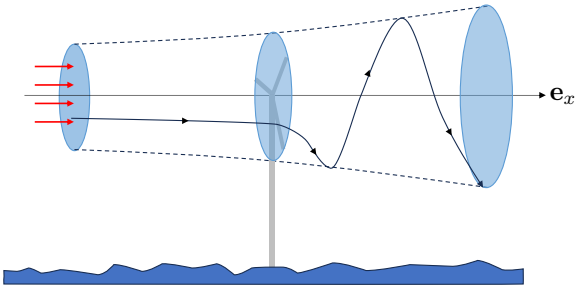
(c) Le graphique ci-dessous représente le rapport de reproduction η en fonction de la fraction x pour des neutrons lents et rapides, respectivement. Pourquoi doit on utiliser des réacteurs à neutrons rapides pour faire de la surgénération à partir d'$^{238}_{92}$U ? Par ailleurs, pourquoi faut-il séparer le combustible en deux parties de fractions x différentes ?



6.3 Énergie du mouvement des fluides

Ex. 6.6 - Limite de Betz : prise en compte de la rotation de sillage

Cet exercice fait suite à celui sur la limite de Betz. La limite de Betz est obtenue par une analyse axiale uniquement, et ignore la nature concrète du dispositif. En pratique, le dispositif le plus courant est celui d'une éolienne à axe horizontal basé sur des pales en rotation, si bien que le fluide en aval est mis en rotation. Une analyse plus détaillée est nécessaire pour tenir compte de l'énergie perdue dans la rotation du sillage.



Pour cette analyse, on considère, une éolienne avec des pales de longueur R telles que la section de l'éolienne est un disque d'aire πR^2. Le plan de rotation de l'éolienne est orthogonal à la vitesse du fluide $v_{-\infty}\mathbf{e}_x$ loin en amont. Le fluide est toujours supposé incompressible (densité ρ_0), idéal et en écoulement stationnaire. Loin en amont et en aval, la pression est la pression ambiante p_∞. Au niveau de l'éolienne, on s'intéresse à un tube annulaire de rayon $r \in [0, R]$, d'épaisseur dr et de section $2\pi r dr$. Pression, vitesse et facteur d'induction peuvent dépendre de r. Les pales sont en rotation à une vitesse angulaire Ω et on se place dans le référentiel d'une pale en rotation. Le fluide immédiatement en amont de l'éolienne a une vitesse $\mathbf{v}_- = v_-\mathbf{e}_x + \Omega\mathbf{e}_\theta$. Le fluide immédiatement après l'éolienne est mis en mouvement de rotation avec une vitesse dans le plan de l'éolienne $\mathbf{v}_+ = v_+\mathbf{e}_x + (\Omega + \omega)\,\mathbf{e}_\theta$.

1. Par application du théorème de Bernoulli et de la conservation de la masse au niveau de l'éolienne, établir la différence de pression $p_- - p_+$ entre les deux cotés de l'éolienne. En déduire la force axiale dF due à la section annulaire et l'exprimer en fonction du facteur d'induction angulaire $a' = \frac{\omega}{2\Omega}$.

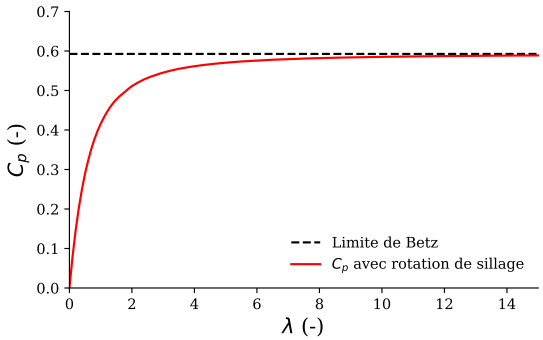
2. En comparant avec l'expression obtenue par conservation du moment dans la direction axiale $F = \rho_0 A v \Delta v_\infty$ (cf. exercice sur la limite de Betz), en déduire que les facteurs d'induction axial et angulaire sont reliés par $\frac{(1-a)a}{(1+a')a'} = \lambda_r^2$, où $\lambda_r = \frac{\Omega r}{v_{-\infty}}$ est le rapport des vitesses angulaire et axiale locales.

3. Par conservation du moment angulaire au niveau de l'éolienne, déterminer le couple dQ reçu par l'éolienne, puis en déduire la puissance fournie.

4. En procédant à un changement de variable de r à λ_r, montrer que le coefficient de performance ne dépend que des facteurs d'induction et du rapport des vitesses en pointe de pale $\lambda = \frac{\Omega R}{v_{-\infty}}$.

5. En utilisant la relation entre les facteurs d'induction (question 2), montrer que le coefficient de performance vaut : $C_p = 8 \int_0^1 \frac{a(1-a)^2}{1+a'} u du$. Commenter le cas limite $a' \to 0$.

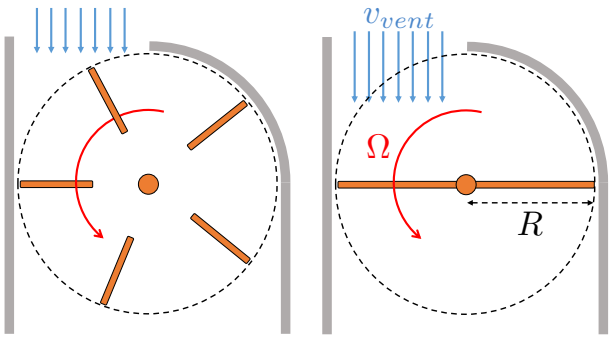
6. Dans le cas général, le coefficient de performance maximum est obtenu en substituant a' par son expression en fonction de a, puis en choisissant pour chaque section annulaire la valeur de a la plus optimale. Le coefficient de performance maximum ne dépend donc que du rapport des vitesses λ en pointe de pale et peut être calculé numériquement. Une représentation graphique est proposée ci-dessous. Commenter.



Ex. 6.7 - Panémone

On s'intéresse dans cet exercice à un panémone, une éolienne ancestrale à axe vertical exploitant la force de trainée et non la force de portance. Un schéma de fonctionnement est proposé ci-dessous (centre): les pales en rotation autour de l'axe vertical sont exposées au vent lorsqu'elles descendent, mais protégées lorsqu'elles remontent le vent de sorte à profiter de la force de trainée lors de la descente et à ne pas la subir lors de la remontée. Une version simplifiée de ce principe de fonctionnement est proposé sur le schéma de droite: la pale est une simple plaque de hauteur h et de largeur $2R$ en rotation à une vitesse Ω autour de son axe central. On rappelle que l'amplitude de la force de trainée exercée par un fluide de vitesse v et de densité ρ sur un objet de section A est $f_d = \frac{1}{2} c_d \rho v^2 A$, avec c_d le coefficient de trainée.

Wikimedia commons (CC BY 4.0)



1. On suppose le vent en écoulement stationnaire et de vitesse uniforme v_{vent} au niveau de la pale 'descendante'. Afin de fournir une borne supérieure de l'efficacité du panémone, on suppose que toute l'impulsion du vent est transférée à la plaque. Montrer que cela correspond à un coefficient de trainée $c_d = 2$.

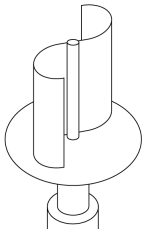
2. On suppose le vent en écoulement stationnaire et de vitesse uniforme v_{vent} au niveau de la pale 'descendante'. On considère la plaque dans la position idéale, i.e., orthogonale au vent. Quelle est la vitesse apparente du vent dans le référentiel de la pale en rotation en fonction de la distance à l'axe r. En déduire la pression qu'exerce le vent sur la pale.

3. Déterminer le couple Q exercé par le vent sur la pale. En déduire, la puissance fournie $\mathcal{P}$ en fonction du rapport $\lambda = \frac{\Omega R}{v_{vent}}$ des vitesses de rotation et de vent.

4. Quelle est la puissance cinétique de référence $\mathcal{P}_0$ qui traverserait la section occupée par le panémone si celui-ci n'était pas présent ? En déduire le coefficient de performance $C_p = \frac{\mathcal{P}}{\mathcal{P}_0}$.

5. Que doit vérifier le rapport des vitesses λ pour que la vitesse de vent apparent sur la pale reste orientée vers le sens descendant ? Quel est le coefficient de performance maximum ? Pour quel rapport λ ? Estimer une borne supérieure du coefficient de performance pour $c_d = 2$. Commenter.

6. Une éolienne de type Savonius (schéma ci-contre) atteint des coefficients de performance de l'ordre de 0.3. Comment cela est-il possible ?



Wikimedia commons
(CC BY-SA 2.0)